CLIFFORD SYSTEMS AND BRAUER OBSTRUCTIONS

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Abstract. Let k be a subring of a commutative ring K and G act on K by k-algebra automorphisms. We identify G-graded Clifford system extensions of a K-central algebra with monoidal functors into its Picard 2-group. This recovers the associativity obstruction to realizing a generalized collective character in third group cohomology with coefficients in the units of K.

For each tensor-closed class of K-central algebras containing K, we give a simplicial construction of the seven-term exact sequence of abelian monoids due to Antony and the second author. Its terms include equivariant invertible modules, invariant Picard classes, and Clifford systems and generalized collective characters modulo compatible Morita equivalences. We realize the latter two monoids as sets of components of homotopy quotients, with explicit boundary formulas. We use quantum tori to give examples of nontrivial third obstructions.

We use a scalar-twisted Duskin–Street complex built from Azumaya algebras, invertible bimodules, and their isomorphisms. Its Postnikov truncation and section spaces extend the sequence to ten terms with an obstruction map to fourth group cohomology. The final obstruction vanishes precisely when the multiplication data admit associativity isomorphisms satisfying the pentagon identity. We classify lifts and their stabilizers.

For finite Galois extensions of fields, we identify the fourth obstruction with a Hochschild–Serre transgression. We give computations for the trivial group, a quaternion-algebra example, and cyclic, biquadratic and nonabelian Galois extensions of nonarchimedean local fields, where reciprocity governs the transgression sign. We combine an exposition of the simplicial homotopy-theoretic connections with new obstruction-theoretic constructions, comparisons, and computations.

2020 *Mathematics Subject Classification.* Primary 16W50; Secondary 16K50, 18M05, 20J06, 55S35.

Key words and phrases. Clifford system extensions, strongly graded algebras, Brauer groups, Picard groups, monoidal categories, group cohomology, obstruction theory.

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1. Introduction

We study two realization problems. A Clifford extension of an algebra R realizes group-indexed invertible R-bimodules as the homogeneous pieces of a graded algebra. The family of isomorphism classes of these bimodules is its *Picard profile*, as defined in Definition 8.3. A Brauer–character system assigns an Azumaya algebra to each group element and isomorphism classes of invertible bimodules implementing multiplication, subject to the associativity condition of Definition 10.2. The family of Brauer classes of these algebras is its *Brauer profile*.

For a prescribed Picard profile, the obstruction to choosing associative multiplication maps on representative bimodules lies in third group cohomology. The obstruction to choosing pentagon-coherent associativity isomorphisms for Brauer–character data lies in fourth group cohomology. For finite Galois extensions of fields, we also ask which fourth classes occur.

1.1. **Picard and Brauer exact sequences.** Let $k \subseteq K$ be commutative rings, put $\Gamma = \mathrm{Aut}_k(K)$, and fix an action $\phi : G \to \Gamma$ of a group G. Write $\mathrm{U}(K)_\phi$ for the unit group with action $g \cdot u = \phi(g)(u)$, and $H^i(G, A)$ for group cohomology of a G-module A. Write A^G, or A^ϕ when the action is specified by ϕ, for its invariant subgroup. The trivial group is denoted by $\{1\}$. The Picard group $\mathrm{Pic}(K)$ consists of isomorphism classes of invertible K-modules, with tensor product. An Azumaya K-algebra is a K-algebra R that is faithful, finitely generated, and projective as a K-module and for which the canonical action map

$$R \otimes_K R^{\mathrm{op}} \to \mathrm{End}_K(R), \qquad x \otimes_K y \longmapsto (r \longmapsto xry),$$

is an isomorphism. The Brauer group $\mathrm{Br}(K)$ consists of Morita equivalence classes of Azumaya K-algebras, with tensor product, and $\mathrm{Br}(K/k) = \ker(\mathrm{Br}(k) \to \mathrm{Br}(K))$.

These groups have familiar interpretations in étale cohomology on $\operatorname{Spec} S$, for a commutative ring S; here we use $S = k$ and $S = K$. Let $\mathbb{G}_m$ denote the étale sheaf of invertible functions. Then there are natural identifications

$$\begin{aligned} H^0_{\text{ét}}(\operatorname{Spec} S, \mathbb{G}_m) &= \mathrm{U}(S), \\ H^1_{\text{ét}}(\operatorname{Spec} S, \mathbb{G}_m) &\cong \mathrm{Pic}(S), \\ H^2_{\text{ét}}(\operatorname{Spec} S, \mathbb{G}_m)_{\mathrm{tors}} &\cong \mathrm{Br}(S). \end{aligned}$$

The degree-one identification is Hilbert 90 [Mil13, Cor. 11.6]; the degree-two identification is Gabber's theorem for affine schemes [dJ, Thm. 1.1]. The subscript denotes the torsion subgroup. Under these identifications, extension of scalars from k to K is pullback along $\operatorname{Spec} K \to \operatorname{Spec} k$.

For a finite G-Galois extension K/k, the Chase–Harrison–Rosenberg sequence is

$$\begin{aligned} \{1\} \to H^1(G, \mathrm{U}(K)) \to \mathrm{Pic}(k) \to \mathrm{Pic}(K)^G \to H^2(G, \mathrm{U}(K)) \to \mathrm{Br}(K/k) \\ \hookrightarrow H^1(G, \mathrm{Pic}(K)) \to H^3(G, \mathrm{U}(K)). \end{aligned} \tag{1}$$

It connects descent of invertible modules, factor sets, the relative Brauer group, and the obstruction to realizing Picard data [CHR65, CR65]. Its cohomological source is the low-degree part of the Hochschild–Serre spectral sequence

$$E_2^{p,q} = H^p(G, H^q_{\text{ét}}(\operatorname{Spec} K, \mathbb{G}_m)) \Longrightarrow H^{p+q}_{\text{ét}}(\operatorname{Spec} k, \mathbb{G}_m)$$

for the Galois cover $\operatorname{Spec} K \to \operatorname{Spec} k$ [Mil13, Thm. 14.9]. The rows $q = 0$ and $q = 1$ are group cohomology with coefficients in $\mathrm{U}(K)$ and $\operatorname{Pic}(K)$; the relative degree-two term gives $\operatorname{Br}(K/k)$. The generalizations of (1) discussed next change the Galois hypothesis, the Picard or Brauer object, or the cohomology theory.

For a K-central algebra R, meaning $Z(R) \cong K$ with a specified identification, let $\boldsymbol{\mathcal{P}ic}_k(R)$ be the monoidal groupoid of invertible R-bimodules on which the two k-actions agree, and let $\operatorname{Pic}_k(R)$ be its group of isomorphism classes. Kanzaki examines whether generalized crossed products can be constructed from a homomorphism $G \to \operatorname{Pic}_k(R)$. In our terminology, these are strongly G-graded algebras and generalized collective characters, respectively. Choosing representative bimodules and multiplication maps produces an associativity defect, whose class lies in third cohomology [Kan68]. Miyashita replaces the commutative Galois extension by a general unital ring extension, using invertible sub-bimodules and generalized crossed products; he also proves comparison with the center and Morita invariance [Miy73, Thms. 2.12, 2.15, 4.1].

Fröhlich–Wall develop the equivariant Brauer version [FW71, FW74]. Its equivariant Picard term is the group of invertible K-modules with semilinear G-action. To describe it, write $K_\phi G = \bigoplus_{g \in G} K u_g$ for the skew group algebra, with $u_g a = \phi(g)(a) u_g$ and $u_g u_h = u_{gh}$. Write $E(\phi, \operatorname{Pic}(K))$ for isomorphism classes of left $K_\phi G$-modules invertible over K, with tensor product over K and diagonal semilinear action. This is the equivariant Picard group [AG26, (7.23)]; see Definition 8.20. A left $K_\phi G$-module structure on L is the same as a compatible family of semilinear maps (Proposition 8.25), which is what makes this the equivariant Picard term. Fröhlich–Wall's later graded monoidal categories organize the coherence data; their categorical grading is distinct from the grading of a strongly graded algebra [FW74].

Caenepeel–Van Oystaeyen–Zhang extend the split equivariant Brauer sequence to arbitrary groups and commutative rings, using Brauer–Taylor groups and Galois objects [CVOZ06, Thm. 2.12]. Huebschmann's equivariant and crossed Brauer groups retain the Teichmüller obstruction: it measures whether a stabilized outer action can be made equivariant [Hue18, Thm. 10.2]. The extension developed below concerns group-indexed Azumaya algebras and their bimodule multiplication data. Crocker–Raeburn–Williams give an equivariant Picard–Brauer sequence for locally compact transformation groups, with Moore cohomology [CRW07, Thm. 4].

El Kaoutit–Gómez-Torrecillas extend Miyashita's constructions to rings with common local units [EKGT10, EKGT12]. Dokuchaev and collaborators develop partial cohomology and the partial-Galois seven-term sequence using partially invertible modules and partial generalized crossed products [DPP19, DPPR21]. Dokuchaev–Rocha treat noncommutative unital extensions and partial representations [DR24, Thm. 5.29]; the comparison with Pinedo recovers the partial-Galois sequence and its relative Brauer quotient [DPR26, Thms. 3.28, 4.2]. We emphasize that our actions are global throughout.

Let R be a K-central algebra and let ϕ be an action of a group G on K. A generalized collective G-character (GCC) of type ϕ over R is a homomorphism $\Phi : G \to \operatorname{Pic}_k(R)$ inducing ϕ on the center K (Definition 3.6). A G-graded Clifford system R-extension is a strongly G-graded algebra $A = \bigoplus_{g \in G} M_g$ with an isomorphism $R \xrightarrow{\cong} M_e$ of R onto the base algebra $M_e \subseteq A$. Each M_g is an invertible R-bimodule, and multiplication gives

isomorphisms

$$\mu_{g,h} : M_g \otimes_R M_h \xrightarrow{\sim} M_{gh}$$

satisfying the associativity and unit identities.

To a G-graded Clifford extension A of R, one associates a GCC $\Phi : G \to \mathrm{Pic}_k(R)$ by

$$\Phi(g) = [M_g], \qquad [M_g][M_h] = [M_{gh}] \quad \text{for all } g, h \in G.$$

The family $([M_g])_{g\in G}$ is the Picard profile of A. We say that A is of type ϕ if its associated GCC is of type ϕ. Thus passing from Clifford data to character data forgets the multiplication maps and retains the bimodule classes (Definition 8.3). Cegarra–Garzón's categorical formulation identifies its associativity obstruction [CG01, CG03], building on Dade's graded-algebra theory [Dad70] and Eilenberg–Mac Lane obstruction theory [EML47].

Fix a class $\mathcal{C}$ of K-central algebras (Definition 8.1), with associated object collection $\mathfrak{U}(\mathcal{C})$ as defined there. Definition 8.3 introduces Clifford data and generalized collective character data of type ϕ with base in $\mathfrak{U}(\mathcal{C})$. In earlier work [Gin22], varying R gives the abelian monoids $\mathrm{Cliff}_k(\phi) = \mathrm{Cliff}_k^{\mathcal{C}}(\phi)$ and $\mathcal{C}_k(\phi) = \mathcal{C}_k^{\mathcal{C}}(\phi)$. They consist respectively of isomorphism classes of Clifford data and of generalized collective characters of type ϕ over $\mathcal{C}$, allowing compatible K-algebra isomorphisms between the base algebras (Definition 8.7). The latter equivalence relation is called *equivariance* in that work. The exact sequence [Gin22, Main Theorem, (1.1)] is

$$\{1\} \to H^2(G, \mathrm{U}(K)_\phi) \xrightarrow{\Sigma_\phi} \mathrm{Cliff}_k(\phi) \xrightarrow{\chi_\phi} \mathcal{C}_k(\phi) \xrightarrow{T_\phi} H^3(G, \mathrm{U}(K)_\phi). \tag{2}$$

Loosely speaking, the map Σ_ϕ assigns a scalar crossed product, χ_ϕ forgets multiplication, and T_ϕ is the associativity obstruction map. Antony and the second author [AG26, Theorem A] establish a seven-term exact sequence for arbitrary actions on commutative rings by passing to central Morita orbits. Using inverse representatives for their constructions [AG26, §§5–6], we take invertible R–S-bimodules P whose central K-actions agree. With the Hom-defined inverse $P^\vee := \mathrm{Hom}_R(P, R)$, the pair $(P, P^\vee)$ induces the monoidal equivalence

$$\Xi_P(M) = P^\vee \otimes_R M \otimes_R P.$$

Definition 8.7 formulates this Morita equivalence relation on Clifford system extensions and on generalized collective characters: the transport datum is a morphism of the groupoid $\mathbf{Mor}^{\mathcal{C}}$ of identity-central bimodules (Definition 8.4), and transporting Clifford data requires compatible multiplication maps as well, while transporting character data requires equality of the induced bimodule classes. Write $\mathrm{Cliff}_\phi^{\mathcal{C}}(K)$ and $\mathrm{Char}_\phi^{\mathcal{C}}(K)$ for the resulting monoids of Morita equivalence classes (Definition 8.9). For each allowed class $\mathcal{C}$ of base algebras, the exact sequence of Antony and the second author is

$$\begin{aligned} &\{1\} \to H^1(G, \mathrm{U}(K)_\phi) \to E(\phi, \mathrm{Pic}(K)) \to \mathrm{Pic}(K)^\phi \to H^2(G, \mathrm{U}(K)_\phi) \\ &\quad \to \mathrm{Cliff}_\phi^{\mathcal{C}}(K) \to \mathrm{Char}_\phi^{\mathcal{C}}(K) \to H^3(G, \mathrm{U}(K)_\phi). \end{aligned} \tag{3}$$

The identities of these monoids are represented by $K_\phi G$ and its character $\mathbf{1}_\phi$. Here we recover this sequence from simplicial mapping spaces and homotopy quotients (Theorem 8.26); Theorem 8.27 identifies its terms and maps with the algebraic ones. For Azumaya base algebras, Theorem 10.42 extends it to a ten-term exact sequence whose final map is the fourth-cohomology obstruction to the pentagon identity.

The two monoids depend on the allowed base algebras; the terms defined over K and the cohomology groups do not. Section 8 realizes these monoids as connected components of simplicial spaces.

1.2. **Brauer–Clifford and Brauer–character data.** For $\eta \in \Gamma$, write R^η for the algebra R with its central K-algebra structure precomposed with η (Definition 9.3). Write ${}_A\mathbf{Mod}_B$ for the category of left-A, right-B bimodules whose two K-actions agree.

A system of Brauer–character data consists of Azumaya K-algebras R_g, with $R_e = K$, and isomorphism classes $[M_{g,h}]$ of invertible bimodules

$$M_{g,h} \in {}_{R_g^{\phi(h)} \otimes_K R_h}\mathbf{Mod}_{R_{gh}}.$$

For every triple g, h, ℓ, these classes satisfy

$$\big[M_{gh,\ell} \circ (M_{g,h}^{\phi(\ell)} \otimes_K 1_{R_\ell})\big] = \big[M_{g,h\ell} \circ (1_{R_g^{\phi(h\ell)}} \otimes_K M_{h,\ell})\big].$$

Both sides are classes in ${}_{R_g^{\phi(h\ell)} \otimes_K R_h^{\phi(\ell)} \otimes_K R_\ell}\mathbf{Mod}_{R_{gh\ell}}$. Here $\circ$ denotes composition by balanced tensor product, with canonical associators and scalar-twist identifications understood. The classes involving e are the specified unit classes; see Definition 10.2.

The Brauer profile $([R_g])_{g\in G}$ satisfies

$$[R_{gh}] = [R_g^{\phi(h)}][R_h] \quad \text{in } \mathrm{Br}(K) \text{ for all } g, h \in G.$$

This parallels the Picard-profile identity $[M_g][M_h] = [M_{gh}]$: multiplication of bimodule classes is realized by bimodule isomorphisms, while multiplication of Brauer classes is realized by invertible bimodules. Brauer–character data retain these multiplication bimodule classes together with their associativity relation.

Brauer–Clifford data additionally choose representatives $M_{g,h}$ and isomorphisms $\alpha_{g,h,\ell}$ from the first composite above to the second, satisfying the pentagon identity. Their unit comparisons are prescribed as in Definition 10.2.

Coherent Morita equivalence between two Brauer–character systems with component algebras R_g and R'_g is specified by an Azumaya K-algebra P and classes of invertible bimodules

$$N_g \in {}_{R_g \otimes_K P}\mathbf{Mod}_{P^{\phi(g)} \otimes_K R'_g},$$

compatible with the multiplication bimodule classes as in (61). For Brauer–Clifford data, one chooses representatives N_g and comparison isomorphisms $\Theta_{g,h}$ satisfying (62). The unit conditions and the full compatibility requirements are given in Definition 10.6.

Write $\mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)$ and $\mathrm{BrChar}_\phi^{\mathcal{A}z}(K)$ for the resulting coherent Morita equivalence classes of Brauer–Clifford and Brauer–character data, respectively (Definition 10.9). Tensor product makes both quotients abelian groups (Propositions 10.15 and 10.16). "Brauer–Clifford" here refers to the systems above; Turull's Brauer–Clifford group [Tur11] instead uses Azumaya algebras with compatible group actions. The forgetful map is

$$\begin{aligned} \mathrm{BrCliff}_\phi^{\mathcal{A}z}(K) &\longrightarrow \mathrm{BrChar}_\phi^{\mathcal{A}z}(K), \\ [(R_g, M_{g,h}, \alpha_{g,h,\ell})] &\longmapsto [(R_g, [M_{g,h}])]. \end{aligned}$$

It parallels the map $\mathrm{Cliff}_\phi^{\mathcal{A}z}(K) \to \mathrm{Char}_\phi^{\mathcal{A}z}(K)$ that forgets the multiplication maps of a Clifford system and retains its Picard profile.

The Brauer three-type is the classical Azumaya complex of Duskin [Dus88], described by Street through the tricategory of Azumaya algebras and invertible bimodules [Str04,

Section 12]; see also [Szy17, Section 4, arXiv v2]. We use all Azumaya K-algebras, with the size convention of Notation 9.4. The one-object Azumaya tricategory $\mathbf{BAz}_K$ has these algebras as its 1-cells, invertible bimodules as its 2-cells, and bimodule isomorphisms as its 3-cells. Its geometric nerve $\mathcal{B}\mathrm{r}(K)$ is the Brauer three-type: edges carry algebras, triangles carry bimodules, tetrahedra carry their isomorphisms, and 4-simplices impose the pentagon. Write $\pi_i(X,x)$ for the homotopy group based at x and $\pi_0(X)$ for the set of components. At the unit basepoint its homotopy groups are

$$\pi_1(\mathcal{B}\mathrm{r}(K)) = \mathrm{Br}(K), \qquad \pi_2(\mathcal{B}\mathrm{r}(K)) = \mathrm{Pic}(K), \qquad \pi_3(\mathcal{B}\mathrm{r}(K)) = \mathrm{U}(K),$$

with our conventions checked in Proposition 9.24. The identifications above express these as $\mathbb{G}_m$-cohomology of $\operatorname{Spec} K$ in degrees 2, 1, and 0, respectively, taking torsion in degree 2.

For a group J, write $B_\bullet J$ for its one-vertex classifying simplicial set, whose n-simplices are strings $[j_1|\cdots|j_n]$ of group elements. Write $E_\bullet J$ for the contractible free right J-simplicial set with n-simplices J^{n+1}; its quotient is $B_\bullet J$. The action ϕ induces a map $B_\bullet\phi : B_\bullet G \to B_\bullet\Gamma$. Scalar twisting gives the semidirect tricategory $\mathbf{BAz}_K^\Gamma$ and its nerve $p_{\mathrm{Br}} : \mathcal{B}\mathrm{r}_\Gamma(K) \to B_\bullet\Gamma$. The fibre is $\mathcal{B}\mathrm{r}(K)$. The map

$$q : \mathcal{B}\mathrm{r}_\Gamma(K) \longrightarrow \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$$

forgets the chosen bimodule isomorphisms and retains isomorphism classes of the triangle bimodules. It models the fibrewise second Postnikov truncation: a truncation $\tau_{\le r}$ preserves homotopy groups in degrees at most r and kills higher ones. Constructions 9.12 and 9.16 specify the algebras, bimodules, and isomorphisms on each simplex, together with the face and degeneracy maps. Our contribution concerns this equivariant construction and its algebraic interpretation: we identify the section-space maps with Clifford and Brauer obstruction maps and compare the fourth map with field transgression.

For a Kan fibration $X \to B_\bullet\Gamma$, write $X_\phi = B_\bullet G \times_{B_\bullet\Gamma} X$ and $\mathrm{Sec}_\phi(X)$ for its simplicial set of sections over $B_\bullet G$. It computes the derived section space of Definition A.32. The unit data specify basepoint sections. Proposition 10.10 identifies $\mathrm{BrCliff}_\phi^{\mathcal{A}\mathrm{z}}(K)$ and $\mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K)$ with the component sets of $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))$ and $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$, respectively. Under these identifications, q induces the forgetful map above. Theorem 10.33 identifies $\mathrm{Cliff}_\phi^{\mathcal{A}\mathrm{z}}(K)$ and $\mathrm{Char}_\phi^{\mathcal{A}\mathrm{z}}(K)$ with the fundamental groups of these same section spaces at their unit sections. The group $E(\phi,\mathrm{Pic}(K))$ consists of isomorphism classes of equivariant invertible K-modules (Definition 8.20); Proposition 10.43 identifies it with $\pi_2(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)))$. Theorems 10.33 and 10.42 identify these algebraic terms and give a sequence of abelian groups, exact through $\mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K)$:

$$\begin{aligned}\{1\} \to H^1(G,\mathrm{U}(K)_\phi) \to E(\phi,\mathrm{Pic}(K)) \to \mathrm{Pic}(K)^\phi \to H^2(G,\mathrm{U}(K)_\phi) \\ \to \mathrm{Cliff}_\phi^{\mathcal{A}\mathrm{z}}(K) \to \mathrm{Char}_\phi^{\mathcal{A}\mathrm{z}}(K) \to H^3(G,\mathrm{U}(K)_\phi) \\ \to \mathrm{BrCliff}_\phi^{\mathcal{A}\mathrm{z}}(K) \to \mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K) \to H^4(G,\mathrm{U}(K)_\phi).\end{aligned} \tag{4}$$

The boundary maps to $H^2(G,\mathrm{U}(K)_\phi)$, $H^3(G,\mathrm{U}(K)_\phi)$, and $H^4(G,\mathrm{U}(K)_\phi)$ are denoted by ∂_2, ∂_3, and ∂_4. Their values are, respectively, the equivariance, associativity, and pentagon obstruction classes.

The map $H^3(G,\mathrm{U}(K)_\phi)\to \mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)$ sends a normalized 3-cocycle u to the class of the unit system, with $R_g=K$ and the canonical multiplication bimodules, whose associators are replaced by

$$\alpha_{g,h,\ell}=\phi(gh\ell)^{-1}\big(u(g,h,\ell)\big)\alpha^0_{g,h,\ell}.$$

Here α^0 denotes the canonical unit associators of Definition 10.8. This is the $H^3(G,\mathrm{U}(K)_\phi)$-action on Brauer–Clifford classes applied to the unit class (Notation 10.17).

To compute ∂_4, choose representatives $M_{g,h}$ and normalized associativity isomorphisms for a Brauer–character system. The discrepancy between the two composites around each pentagon determines its obstruction class in $H^4(G,\mathrm{U}(K)_\phi)$, with the coefficient convention of Proposition 10.47. This class is independent of the choices and vanishes precisely when the Brauer–character class admits a Brauer–Clifford lift (Theorem 10.51).

There is also a map retaining the Brauer profile. Give $\mathrm{Br}(K)_\phi$ the left action ${}^g[R]=[R^{\phi(g)^{-1}}]$. Then

$$\mathrm{BrChar}_\phi^{\mathcal{A}z}(K)\longrightarrow H^1(G,\mathrm{Br}(K)_\phi),$$
$$[(R_g,[M_{g,h}])]\longmapsto \big[g\longmapsto [R_g^{\phi(g)^{-1}}]\big].$$

This map forgets the multiplication bimodule classes and retains the cohomology class determined by the Brauer profile (Proposition 10.22).

Exactness is asserted through the Brauer–character term. The map ∂_4 need not be surjective (Proposition 10.55).

1.3. **Main results and comparisons.**

Prior results and contributions. The third-cohomology obstruction and the fixed-base classification recover the earlier algebraic results of [CG01, Gin22]; the seven-term sequence is due to Antony and the second author [AG26, Theorem A]. Our contributions are the explicit simplicial construction and comparison of these sequences, their Azumaya continuation to the fourth obstruction, and the resulting lift classification. We also give the field-transgression comparison with its sign conventions, local-field computations, and quantum-torus examples.

The fourth obstruction over fields. For a finite Galois extension K/k with group G, Proposition 10.59 identifies the image of the fourth obstruction with the image of

$$d_3^{1,2}: H^1(G,\mathrm{Br}(K))\longrightarrow H^4(G,K^\times).$$

Every Brauer cocycle admits normalized algebraic data, and the proposition fixes the sign relating this transgression to the pentagon convention.

Clifford extensions of an algebra. Sections 2–7 develop the obstruction theory for extensions of a prescribed algebra and give the cochain formulas. Theorem 4.5 identifies Clifford systems over R with monoidal functors $\underline{G}\to \boldsymbol{\mathcal{P}ic}_k(R)$, where $\underline{G}$ is the discrete monoidal category on G. The obstruction theory is that of Eilenberg–Mac Lane and Sính, in the categorical formulations of Baez–Lauda and Cegarra–Garzón [Sín75, BL04, CG03]. Theorem 6.2 identifies its cocycle with the second author's earlier algebraic convention [Gin22]. For a character $\Phi: G\to \mathrm{Pic}_k(R)$ of type ϕ, Theorem 6.4 gives a realization precisely when $T(\Phi)=1$ in $H^3(G,\mathrm{U}(K)_\phi)$. When this obstruction vanishes, the realization classes under graded isomorphisms fixing R form an $H^2(G,\mathrm{U}(K)_\phi)$-torsor. For $R=K$, Proposition 7.11 identifies the descended obstruction map $\overline{T}$ with the Picard descent differential

$$d_2: H^1(G,\mathrm{Pic}(K)_\phi)\longrightarrow H^3(G,\mathrm{U}(K)_\phi).$$

The seven-term sequence. For every class $\mathcal{C}$ as in Definition 8.1, Theorem 8.26 gives a simplicial proof of (3), and Theorem 8.27 identifies its two class monoids and gives algebraic descriptions of all its maps. Proposition 8.29 explains the change from (2): the scalar crossed-product map is injective on isomorphism classes of data, whereas on Morita classes its kernel is $\operatorname{im}\partial_2$. Restricting $\mathcal{C}$ to its Azumaya classes gives the comparison of seven-term exact sequences in Proposition 9.8. For the full Azumaya class, Proposition 10.35 identifies this sequence with the first seven terms of (4).

Write $\partial_3^{\mathcal{C}}$ for the last map in (3). Its image depends on the class. If K is an integral domain, take $\mathcal{C} = \mathcal{C}_{\mathrm{fr},\kappa}(K)$, the class of K-central algebras free as K-modules and of cardinality at most a cardinal $\kappa \geq \max\{|K|, |G|, \aleph_0\}$. Proposition 8.30 proves that $\partial_3^{\mathcal{C}}$ is surjective onto $H^3(G, \mathrm{U}(K)_\phi)$.

The ten-term sequence and the fourth obstruction. The continuation in (4) comes from the homotopy fibre sequence associated with the Postnikov truncation q described above. To state its homotopical form, let $\mathcal{E}_\Gamma(K) \to B_\bullet\Gamma$ be the Borel fibration for the coherent scalar-twist action on $\mathcal{B}\mathrm{r}(K)$, constructed using a strict action on a weakly equivalent Kan model in Proposition 10.31. Its fibre is weakly equivalent to $\mathcal{B}\mathrm{r}(K)$. Let $\mathcal{E}_{\Gamma,\leq 2}(K)$ be its fibrewise second truncation and q_{abs} the truncation map. Pulling back along $B_\bullet\phi$ gives

$$q_\phi : \mathcal{E}_\Gamma(K)_\phi \to \mathcal{E}_{\Gamma,\leq 2}(K)_\phi.$$

For an abelian group A and $n \geq 1$, $K(A,n)$ denotes a pointed connected Kan complex with $\pi_n(K(A,n)) \cong A$ and all other positive homotopy groups trivial; $K(A,0)$ is the constant simplicial set on A. For a G-module A, $K_{B_\bullet G}(A,n) \to B_\bullet G$ is the associated local Eilenberg–Mac Lane group object, with its identity section, as in Definition A.29. Write $\mathrm{Sec}_{B_\bullet G}$ for sections over $B_\bullet G$ and put

$$\mathcal{F}^{(3)}_{\phi,K} = \mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3)\big).$$

Theorem 10.39 gives the homotopy fibre sequence at the unit section

$$\mathcal{F}^{(3)}_{\phi,K} \longrightarrow \mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K)) \xrightarrow{\mathrm{Sec}_{B_\bullet G}(q_\phi)} \mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K)), \tag{5}$$

The associated long exact sequence, beginning at $\pi_2(\mathcal{F}^{(3)}_{\phi,K})$ and extended by the section-lifting obstruction, is

$$\begin{aligned}
\{1\} \longrightarrow \pi_2\big(\mathcal{F}^{(3)}_{\phi,K}\big) &\longrightarrow \pi_2(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) \longrightarrow \pi_2(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K))) \\
&\longrightarrow \pi_1\big(\mathcal{F}^{(3)}_{\phi,K}\big) \longrightarrow \pi_1(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) \longrightarrow \pi_1(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K))) \\
&\longrightarrow \pi_0\big(\mathcal{F}^{(3)}_{\phi,K}\big) \longrightarrow \pi_0(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) \longrightarrow \pi_0(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K))) \xrightarrow{o_4} \\
&\longrightarrow \pi_0\,\mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4)\big).
\end{aligned} \tag{6}$$

The cohomology terms through $H^3(G, \mathrm{U}(K)_\phi)$ come from the fibre:

$$\begin{aligned}
H^0(G, \mathrm{U}(K)_\phi) &\cong \pi_3\big(\mathcal{F}^{(3)}_{\phi,K}\big), & H^1(G, \mathrm{U}(K)_\phi) &\cong \pi_2\big(\mathcal{F}^{(3)}_{\phi,K}\big), \\
H^2(G, \mathrm{U}(K)_\phi) &\cong \pi_1\big(\mathcal{F}^{(3)}_{\phi,K}\big), & H^3(G, \mathrm{U}(K)_\phi) &\cong \pi_0\big(\mathcal{F}^{(3)}_{\phi,K}\big).
\end{aligned} \tag{7}$$

The final term is instead

$$H^4(G, \mathrm{U}(K)_\phi) \cong \pi_0\big(\mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4)\big)\big)\,. \tag{8}$$

Let $\kappa_3(q_\phi)$ denote the Postnikov class classifying the top local $K(\mathrm{U}(K)_\phi, 3)$-fibration. For a truncated section s, the final map is

$$o_4([s]) = s^*\kappa_3(q_\phi) \in H^4(G, \mathrm{U}(K)_\phi). \tag{9}$$

Theorem 10.51 proves that $o_4([s])$ vanishes if and only if s admits a Brauer–Clifford lift. If a lift exists, the homotopy classes of lifts with a fixed identification over s form an $H^3(G, \mathrm{U}(K)_\phi)$-torsor. Forgetting that identification introduces the stabilizer described in Proposition 10.52.

Further examples and comparisons. For a complex algebraic quantum torus whose parameter is not a root of unity, Proposition 8.37 realizes every class in $H^3(C_n, \mathbb{C}^\times)$, with trivial coefficient action, where C_n is the cyclic group of order $n \geq 2$. Proposition 8.40 gives characters on Morita-equivalent endomorphism rings that are not represented by automorphism bimodules, and Proposition 8.42 constructs realizable tensor products whose two factors are individually obstructed.

The affine Hopf line gives a nontrivial second obstruction (Proposition 8.32), and a real group algebra gives a nontrivial third obstruction (Proposition 8.33). For a finite Galois extension K/k of nonarchimedean local fields, with Galois group G and its natural action on $K^\times$, every class in $H^4(G, K^\times)$ is realized as a fourth obstruction (Proposition 10.62). Section 10.10 (Propositions 10.66, 10.68 and 10.70) constructs Brauer–character systems over cyclic, biquadratic, and nonabelian local Galois extensions. Their fourth obstructions distinguish liftable systems from nonliftable ones. The quaternion example (Example 10.61) gives a contrasting case with trivial fourth obstruction. These local-field systems are specified by choices of simple modules and intertwining isomorphisms. Section 8 treats the seven-term sequence for a class of algebras. Section 9 constructs the Azumaya simplicial models, and Section 10 identifies their section spaces and boundary maps. Field applications appear in Subsections 10.8–10.10. Appendix A collects the background.

1.4. **Conventions.**

Algebra. All rings and algebras are associative and unital, and homomorphisms preserve units. Throughout, we fix commutative rings $k \subseteq K$ and a group G with identity element e. Modules and bimodules are unital. Groups, including Brauer and Picard groups, and their cohomology are written multiplicatively, with identity 1. Numerical groups such as $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Q}/\mathbb{Z}$, and $\mathbb{Z}/n$, and cochains and cohomology with these coefficients, retain additive notation; so do ordinary ring and module arithmetic and integral chains. For a general unital ring A, we write $\mathrm{U}(A)$ for its group of units; when the ring is explicitly a field F, we keep the standard notation $F^\times$; the superscript $^\times$ is reserved for unit groups. A superscript naming a class of base algebras records which algebras are allowed: $\mathcal{C}$ for a class as in Definition 8.1, and $\mathcal{A}z$ for the Azumaya class of Notation 9.4, as in $\mathsf{CliffSp}^{\mathcal{C}}_\phi(K)$ and $\mathrm{Cliff}^{\mathcal{A}z}_\phi(K)$.

Set theory. We work in ZFC and fix Grothendieck universes

$$\mathfrak{U} \in \mathfrak{V} \in \mathfrak{W}, \qquad \mathbb{N}, k, K, G \in \mathfrak{U},$$

in the sense of the Bourbaki appendix "Univers" to SGA 4 [AGV72, Exp. I, App.]; for a category-theoretic discussion of fixed and varying universes, see [Low14, §1]. A set is $\mathfrak{U}$-small if it belongs to $\mathfrak{U}$, and similarly for $\mathfrak{V}$ and $\mathfrak{W}$.

Unless specified otherwise, algebraic objects and indexing sets are $\mathfrak{U}$-small; categories and their nerves and homotopy-theoretic constructions are $\mathfrak{V}$-small. Groups and monoids of isomorphism or Morita classes, and groups formed from them, are likewise $\mathfrak{V}$-small.

The ambient categories **Cat**, **Grpd**, and **sSet**, and categories of diagrams in them, are formed in $\mathfrak{W}$. Any additional cardinal bound $\kappa \in \mathfrak{U}$ applies only to base algebras and their invertible Morita bimodules, not to cochain groups or ambient categories; compare [Gin22, §2]. Further size details appear in Appendix B.

Simplicial conventions.

Notation 1.1. For a $\mathfrak{V}$-small groupoid $\mathcal{X}$, we write $\pi_0(\mathcal{X})$ for the set of isomorphism classes of objects. This is functorial: a functor $F : \mathcal{X} \to \mathcal{Y}$ induces $\pi_0(F)([x]) = [F(x)]$, so π_0 is a functor $\mathbf{Grpd}_{\mathfrak{V}} \to \mathbf{Set}_{\mathfrak{V}}$. For a pointed groupoid $(\mathcal{X}, x)$ we write $\pi_1(\mathcal{X}, x) := \mathrm{Aut}_{\mathcal{X}}(x)$. This is functorial for strictly basepoint-preserving functors $(\mathcal{X}, x) \to (\mathcal{Y}, y)$, with values in $\mathbf{Grp}_{\mathfrak{V}}$. For a pointed groupoid $(\mathcal{X}, x)$ we also regard $\pi_0(\mathcal{X})$ as a pointed set with distinguished element $[x]$. If $\mathcal{X}$ is connected and $u : x \to y$ is an isomorphism, conjugation by u identifies $\pi_1(\mathcal{X}, x)$ with $\pi_1(\mathcal{X}, y)$; changing u changes this identification by an inner automorphism. Thus a base object is part of the data whenever π_1 is used. For a 2-group $\mathcal{G}$ the base object is always the monoidal unit $\mathbf{1}$: $\pi_0(\mathcal{G})$ is pointed by $[\mathbf{1}]$, and $\pi_1(\mathcal{G})$ means $\pi_1(\mathcal{G}, \mathbf{1}) = \mathrm{Aut}_{\mathcal{G}}(\mathbf{1})$. A monoidal functor between 2-groups is based through its unit constraint; this is the convention used below for the induced map on π_1. For simplicial sets, π_0 means the set of components and $\pi_n(X, x)$ is based at a chosen vertex; when the basepoint is omitted from a pointed section space, it is the unit section.

All homotopy-theoretic constructions are performed in simplicial sets. Loop spaces are understood as simplicial based loop objects, as in Definition A.24. Geometric realization is used only when invoking an external comparison theorem.

Exactness and cochains. For a sequence of monoid homomorphisms $A \xrightarrow{f} B \xrightarrow{g} C$, exactness at B means

$$\mathrm{im}(f) = \{b \in B \mid g(b) = 1_C\}.$$

For a sequence of pointed maps, the same convention applies with the identity replaced by the distinguished basepoint.

Group cohomology $H^n(G, A)$ of a multiplicatively written G-module A is computed from all inhomogeneous cochains in the cosimplicial abelian group $C^\bullet(G, A)$, as explained in Subsection A.4. A normalized n-cochain is a map $G^n \to A$ taking the value 1 whenever some argument equals e. This is the convention in which the pentagon for a group-indexed monoidal category becomes the inhomogeneous group 3-cocycle equation; compare Kerodon's group-cocycle example [Lur18], specifically Kerodon tag 00CP, Kerodon tag 00CS, and Kerodon tag 0087. For an action $\phi : G \to \mathrm{Aut}_k(K)$ we write $\mathrm{U}(K)_\phi$ for the unit group $\mathrm{U}(K)$, regarded as a G-module by $g \cdot u = \phi(g)(u)$. We use this notation throughout, including in all cohomology groups. The alternating products of the cosimplicial cofaces give the low-degree coboundaries

$$(\partial b)(\sigma, \tau) = {}^{\sigma}b(\tau) \cdot b(\sigma\tau)^{-1} \cdot b(\sigma), \tag{10}$$

$$(\partial \beta)(\sigma, \tau, \nu) = {}^{\sigma}\beta(\tau, \nu) \cdot \beta(\sigma\tau, \nu)^{-1} \cdot \beta(\sigma, \tau\nu) \cdot \beta(\sigma, \tau)^{-1}, \tag{11}$$

$$(\partial t)(\sigma, \tau, \nu, \omega) = {}^{\sigma}t(\tau, \nu, \omega) \cdot t(\sigma\tau, \nu, \omega)^{-1} \cdot t(\sigma, \tau\nu, \omega) \cdot t(\sigma, \tau, \nu\omega)^{-1} \cdot t(\sigma, \tau, \nu). \tag{12}$$

1.5. **Index of notation.**

$k \subseteq K$, G, Γ, ϕ	§1.1
$\mathrm{U}(K)_\phi$	§1.1
$\mathrm{U}(K)^\phi$	Definition 6.7
$\pi_0(\mathcal{X})$, $\pi_1(\mathcal{X}, x)$	Notation 1.1
$\mathcal{C}$, $\mathcal{D}$, $\mathbf{1}$, a, λ, ρ	Definition 2.1
$\lambda^\sharp_X$, $\rho^\sharp_X$	Definition 2.5
$\underline{G}$, $\mathbf{B}G$, $B_\bullet G$, $B_\bullet \phi$	Notation 2.10
$\mathbf{Mon}(\underline{G}, \mathcal{G})$	Definition 2.11
$\boldsymbol{\mathcal{P}ic}_k(R)$, $\mathrm{Pic}_k(R)$	Definition 3.1
η_P, h_R	Definition 3.3
$\mathbf{Cliff}_G(R)$, $\mathrm{Cliff}_k(G, R)$	Definition 4.1
$K^\alpha_\phi G$	Definition 6.12
$T(\Phi)$	Theorem 6.2
$\mathrm{Pic}(K)_\phi$, ξ_Φ	Proposition 7.3
$\mathrm{Pic}(K)^\phi$	Proposition 8.25
Σ_ϕ, $\overline{\chi}_K$, $\overline{T}$	Theorem 7.8
d_2	Proposition 7.11
$\mathcal{C}$, $\mathfrak{U}(\mathcal{C})$	Definition 8.1
$\mathcal{C}_{\mathrm{fr},\kappa}(K)$	Example 8.2(vi)
$\mathrm{Br}(K)$	Definition 9.1
$\mathrm{Br}(K)_\phi$	§10.3
$\mathrm{Br}(K)^\phi$	Proposition 10.22
$\mathbf{Mor}^{\mathcal{C}}$	Definition 8.4
Ξ_P	Construction 8.5
$\mathrm{Cliff}^{\mathcal{C}}_k(\phi)$, $\mathcal{C}^{\mathcal{C}}_k(\phi)$	§1.1
$\mathrm{Cliff}^{\mathcal{C}}_\phi(K)$, $\mathrm{Char}^{\mathcal{C}}_\phi(K)$	Definition 8.9
χ^{Mor}_ϕ	Definition 8.9
$\mathfrak{Cliff}_\phi(K; R)$, $\mathfrak{Char}_\phi(K; R)$	Construction 8.11
$\mathsf{CliffSec}_\phi(K)$, $\mathsf{CharSec}_\phi(K)$	Construction 8.11
$\mathsf{CliffSp}^{\mathcal{C}}_\phi(K)$, $\mathsf{CharSp}^{\mathcal{C}}_\phi(K)$	Construction 8.11
$\partial^{\mathcal{C}}_3$	Definition 8.9
∂_2	Definition 8.23
∂_3	Notation 9.4
∂_4	Definition 10.13
$E(\phi, \mathrm{Pic}(K))$	Definition 8.20
$\mathcal{C} \cap \mathcal{A}\mathrm{z}$, $\mathfrak{U}(\mathcal{C} \cap \mathcal{A}\mathrm{z})$	Definition 9.5
$\mathcal{A}\mathrm{z}$, $\mathfrak{U}(\mathcal{A}\mathrm{z})$	Notation 9.4
R^θ	Definition 9.3
P^θ	Definition 9.9
Tw_θ	Definition A.7
${}_R\mathbf{Mod}_S$, $\mathbf{Az}_K$, $\mathbf{BAz}_K$	Definition 9.9
$\mathbf{BAz}^\Gamma_K$	Construction 9.12
$\mathcal{B}\mathrm{r}(K)$, $\mathcal{B}\mathrm{r}_\Gamma(K)$	Definition 9.15
$\widehat{\mathcal{B}\mathrm{r}}^{(r)}_\Gamma(K)$	Construction 10.30
$\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K)$, q	Construction 9.16
$\mathcal{B}\mathrm{r}_{\Gamma,\leq 1}(K)$, q_1	§10.3
X_ϕ, $\mathrm{Sec}_\phi(X)$	Notation 10.1
R_g, $[R_g]$	Definition 10.2
$\mathbf{Pic}(K)$	Definition 7.9
$\mathcal{E}_\Gamma(K)$, $\mathcal{E}_{\Gamma,\leq 2}(K)$, q_{abs}	Notation 10.36
o_4	Construction 10.38
$\mathrm{BrCliff}^{\mathcal{A}\mathrm{z}}_\phi(K)$, $\mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K)$	Definition 10.9
s_0	Definition 10.8
$\mathcal{F}^{(2)}_{\phi,K}$	Definition 8.18
$\mathcal{F}^{(3)}_{\phi,K}$	Definition 10.37
$d_3^{1,2}$	Notation 10.57
Δ, Δ^n, X_n	Definition A.1
$\mathrm{Map}(X, Y)$	Definition A.12
$N_\bullet \mathcal{X}$, $B_\bullet G$	Definition A.3
$E_\bullet G$	Definition A.4
$\mathrm{Rep}_u([n], \mathcal{T})$	Definition A.9
$N_{\mathrm{St}}\mathcal{C}$, $\Delta\mathcal{T}$	Definition A.8
$\pi_n(X, x)$	Definition A.14
$\mathrm{hofib}_b(p)$	Definition A.20
$\tau_{\leq n}X$	Definition A.21
$\Omega_x X$, ΩX, $\Omega^m X$	Definition A.24
$P^z_B Y$, e_1	Definition A.25
$K(A, n)$	Definition A.28
$K_{B_\bullet G}(A, n)$	Definition A.29
$\mathrm{Sec}_B(E)$, $\mathrm{RSec}_B(E)$	Definition A.32
$\Omega_s \mathrm{Sec}_B(E)$	Definition A.33
X_{hG}, X^{hG}	Definition A.37

Acknowledgments

We used generative AI to assist with checking our work and typesetting, and to perform computations in the examples. It was not used in any other way. All the ideas and proofs in this article are due to the authors alone and not to generative AI. Our use of generative AI was limited to ChatGPT-6.0 Sol only.

2. 2-groups and monoidal functors

We recall 2-groups and monoidal functors, and identify the automorphism groups used in the cocycle calculations. All categories satisfy the size conventions of Subsection 1.4.

2.1. Monoidal categories, 2-groups, and homotopy groups.

Definition 2.1. A *monoidal category* $(\mathcal{C}, \otimes, \mathbf{1}, \mathsf{a}, \lambda, \rho)$ consists of a category $\mathcal{C}$, a functor

$$\otimes\colon \mathcal{C}\times\mathcal{C}\to\mathcal{C},$$

an object $\mathbf{1}$, and natural isomorphisms

$$\mathsf{a}_{X,Y,Z}\colon (X\otimes Y)\otimes Z\to X\otimes(Y\otimes Z),$$
$$\lambda_X\colon \mathbf{1}\otimes X\to X,$$
$$\rho_X\colon X\otimes\mathbf{1}\to X,$$

subject to the pentagon axiom

$$(1_W\otimes \mathsf{a}_{X,Y,Z})\circ \mathsf{a}_{W,X\otimes Y,Z}\circ(\mathsf{a}_{W,X,Y}\otimes 1_Z) \;=\; \mathsf{a}_{W,X,Y\otimes Z}\circ \mathsf{a}_{W\otimes X,Y,Z}, \tag{13}$$

as maps $((W\otimes X)\otimes Y)\otimes Z\to W\otimes(X\otimes(Y\otimes Z))$, and the triangle axiom

$$(1_X\otimes\lambda_Y)\circ \mathsf{a}_{X,\mathbf{1},Y} \;=\; \rho_X\otimes 1_Y, \tag{14}$$

as maps $(X\otimes\mathbf{1})\otimes Y\to X\otimes Y$, see [ML98, Ch. VII]. An object X is *invertible* if there exists an object Y with $X\otimes Y\cong\mathbf{1}\cong Y\otimes X$. A 2-*group*, or *gr-category* in the terminology of Sính [Sín75], is a monoidal category $\mathcal{G}$ which is a groupoid and all of whose objects are invertible.

We freely use the following consequences of the axioms, due to Kelly [Kel64, pp. 397–402]: in any monoidal category,

$$\lambda_{\mathbf{1}}=\rho_{\mathbf{1}}\colon \mathbf{1}\otimes\mathbf{1}\to\mathbf{1}, \tag{15}$$
$$\lambda_{X\otimes Y}=(\lambda_X\otimes 1_Y)\circ \mathsf{a}^{-1}_{\mathbf{1},X,Y}, \tag{16}$$
$$\rho_{X\otimes Y}=(1_X\otimes\rho_Y)\circ \mathsf{a}_{X,Y,\mathbf{1}}. \tag{17}$$

The maps in (16) and (17) have respective sources and targets $\mathbf{1}\otimes(X\otimes Y)\to X\otimes Y$ and $(X\otimes Y)\otimes\mathbf{1}\to X\otimes Y$.

Notation 2.2. Using the groupoid convention of Notation 1.1, for a 2-group $\mathcal{G}$ the symbol $\pi_0(\mathcal{G})$ means the component set of the underlying groupoid and $\pi_1(\mathcal{G})$ means the based automorphism group at the monoidal unit:

$$\pi_1(\mathcal{G})=\pi_1(\mathcal{G},\mathbf{1})=\mathrm{Aut}_{\mathcal{G}}(\mathbf{1}).$$

Since every object is invertible, $\pi_0(\mathcal{G})$ is a group under $[X]\cdot[Y]:=[X\otimes Y]$ (associativity and unitality being witnessed by a,λ,ρ), with distinguished point the identity class $[\mathbf{1}]$.

2.2. **Automorphisms of the unit and of an object.**

Lemma 2.3. *Let $\mathcal{C}$ be a monoidal category and X an invertible object. Then the functors $X \otimes -$ and $- \otimes X$ are equivalences of categories $\mathcal{C} \to \mathcal{C}$.*

Proof. Choose Y and isomorphisms $u\colon X \otimes Y \to \mathbf{1}$, $v\colon Y \otimes X \to \mathbf{1}$. The composite natural isomorphism

$$Y \otimes (X \otimes Z) \xrightarrow{\mathsf{a}^{-1}_{Y,X,Z}} (Y \otimes X) \otimes Z \xrightarrow{v \otimes 1_Z} \mathbf{1} \otimes Z \xrightarrow{\lambda_Z} Z$$

exhibits $(Y \otimes -) \circ (X \otimes -) \cong \mathrm{id}_{\mathcal{C}}$, and symmetrically $(X \otimes -) \circ (Y \otimes -) \cong \mathrm{id}_{\mathcal{C}}$ using u in place of v. Hence $X \otimes -$ is an equivalence with quasi-inverse $Y \otimes -$. The right-tensor case is identical. □

The following is the Eckmann–Hilton argument; see [EH62, pp. 227–255].

Lemma 2.4. *Let $\mathcal{G}$ be a 2-group and $A = \pi_1(\mathcal{G})$. For every $a \in A$,*

$$\lambda_{\mathbf{1}} \circ (a \otimes 1_{\mathbf{1}}) \circ \lambda_{\mathbf{1}}^{-1} = a = \rho_{\mathbf{1}} \circ (1_{\mathbf{1}} \otimes a) \circ \rho_{\mathbf{1}}^{-1}, \tag{18}$$

as maps $\mathbf{1} \to \mathbf{1}$, and A is abelian: for $a, b \in A$ one has

$$a \circ b = \lambda_{\mathbf{1}} \circ (a \otimes b) \circ \lambda_{\mathbf{1}}^{-1} = b \circ a,$$

as maps $\mathbf{1} \to \mathbf{1}$.

Proof. Naturality of the unitors and $\lambda_{\mathbf{1}} = \rho_{\mathbf{1}}$ give

$$\lambda_{\mathbf{1}} \circ (a \otimes 1) \circ \lambda_{\mathbf{1}}^{-1} = a = \lambda_{\mathbf{1}} \circ (1 \otimes a) \circ \lambda_{\mathbf{1}}^{-1},$$

as maps $\mathbf{1} \to \mathbf{1}$, which is (18). Functoriality of tensor product gives

$$a \otimes b = (a \otimes 1) \circ (1 \otimes b) = (1 \otimes b) \circ (a \otimes 1),$$

as maps $\mathbf{1} \otimes \mathbf{1} \to \mathbf{1} \otimes \mathbf{1}$. Conjugating by $\lambda_{\mathbf{1}}$ proves

$$\lambda_{\mathbf{1}} \circ (a \otimes b) \circ \lambda_{\mathbf{1}}^{-1} = a \circ b = b \circ a,$$

as maps $\mathbf{1} \to \mathbf{1}$. □

Definition 2.5. Let $\mathcal{G}$ be a 2-group, $A = \pi_1(\mathcal{G})$, and X an object. Define

$$\lambda^{\sharp}_X\colon A \longrightarrow \mathrm{Aut}_{\mathcal{G}}(X), \quad \lambda^{\sharp}_X(a) := \lambda_X \circ (a \otimes 1_X) \circ \lambda_X^{-1},$$
$$\rho^{\sharp}_X\colon A \longrightarrow \mathrm{Aut}_{\mathcal{G}}(X), \quad \rho^{\sharp}_X(a) := \rho_X \circ (1_X \otimes a) \circ \rho_X^{-1}.$$

Lemma 2.6. *With notation as above:*

(i) *$\lambda^{\sharp}_X$ and $\rho^{\sharp}_X$ are group isomorphisms.*

(ii) (Naturality.) *For every morphism $u\colon X \to Y$ in $\mathcal{G}$ and every $a \in A$,*

$$u \circ \lambda^{\sharp}_X(a) = \lambda^{\sharp}_Y(a) \circ u, \qquad u \circ \rho^{\sharp}_X(a) = \rho^{\sharp}_Y(a) \circ u,$$

as maps $X \to Y$.

(iii) (Monoidality.) *For all objects X, Y and $a \in A$,*

$$\lambda^{\sharp}_{X \otimes Y}(a) = \lambda^{\sharp}_X(a) \otimes 1_Y, \qquad \rho^{\sharp}_{X \otimes Y}(a) = 1_X \otimes \rho^{\sharp}_Y(a),$$

as maps $X \otimes Y \to X \otimes Y$.

(iv) *For all objects X, Y and $a \in A$,*

$$1_X \otimes \lambda^\sharp_Y(a) \;=\; \rho^\sharp_X(a) \otimes 1_Y,$$

as maps $X \otimes Y \to X \otimes Y$.

Proof. (i) Tensoring with X is fully faithful by Lemma 2.3. It therefore identifies $\mathrm{Aut}(\mathbf{1})$ with $\mathrm{Aut}(\mathbf{1} \otimes X)$ and with $\mathrm{Aut}(X \otimes \mathbf{1})$, respectively. Conjugating by the unitors gives the two asserted group isomorphisms.

(ii) Naturality of λ and the interchange law give

$$\lambda^\sharp_Y(a) \circ u = \lambda_Y \circ (a \otimes 1_Y) \circ (1 \otimes u) \circ \lambda_X^{-1} = u \circ \lambda_X \circ (a \otimes 1_X) \circ \lambda_X^{-1} = u \circ \lambda^\sharp_X(a),$$

as maps $X \to Y$. The proof for $\rho^\sharp$ uses ρ in the same way.

(iii) Using (16) and naturality of a in its first variable, namely $\mathsf{a}_{\mathbf{1},X,Y} \circ ((a \otimes 1_X) \otimes 1_Y) = (a \otimes (1_X \otimes 1_Y)) \circ \mathsf{a}_{\mathbf{1},X,Y}$ as maps $(\mathbf{1} \otimes X) \otimes Y \to \mathbf{1} \otimes (X \otimes Y)$, we obtain

$$\begin{aligned}
\lambda^\sharp_{X\otimes Y}(a) &= (\lambda_X \otimes 1_Y) \circ \mathsf{a}^{-1}_{\mathbf{1},X,Y} \circ (a \otimes 1_{X\otimes Y}) \circ \mathsf{a}_{\mathbf{1},X,Y} \circ (\lambda_X^{-1} \otimes 1_Y) \\
&= (\lambda_X \otimes 1_Y) \circ \big((a \otimes 1_X) \otimes 1_Y\big) \circ (\lambda_X^{-1} \otimes 1_Y) \\
&= \lambda^\sharp_X(a) \otimes 1_Y,
\end{aligned}$$

as maps $X \otimes Y \to X \otimes Y$. Dually, using (17) and naturality of a in its third variable, $\rho^\sharp_{X\otimes Y}(a) = 1_X \otimes \rho^\sharp_Y(a)$ as maps $X \otimes Y \to X \otimes Y$.

(iv) By naturality of a in its middle variable, $1_X \otimes (a \otimes 1_Y) = \mathsf{a}_{X,\mathbf{1},Y} \circ ((1_X \otimes a) \otimes 1_Y) \circ \mathsf{a}^{-1}_{X,\mathbf{1},Y}$ as maps $X \otimes (\mathbf{1} \otimes Y) \to X \otimes (\mathbf{1} \otimes Y)$. Hence

$$\begin{aligned}
1_X \otimes \lambda^\sharp_Y(a) &= (1_X \otimes \lambda_Y) \circ \big(1_X \otimes (a \otimes 1_Y)\big) \circ (1_X \otimes \lambda_Y^{-1}) \\
&= \big[(1_X \otimes \lambda_Y) \circ \mathsf{a}_{X,\mathbf{1},Y}\big] \circ \big((1_X \otimes a) \otimes 1_Y\big) \circ \big[(1_X \otimes \lambda_Y) \circ \mathsf{a}_{X,\mathbf{1},Y}\big]^{-1},
\end{aligned}$$

as maps $X \otimes Y \to X \otimes Y$. By the triangle axiom (14), $(1_X \otimes \lambda_Y) \circ \mathsf{a}_{X,\mathbf{1},Y} = \rho_X \otimes 1_Y$ as maps $(X \otimes \mathbf{1}) \otimes Y \to X \otimes Y$, so the right-hand side equals, as an endomorphism of $X \otimes Y$, $(\rho_X \otimes 1_Y) \circ \big((1_X \otimes a) \otimes 1_Y\big) \circ (\rho_X^{-1} \otimes 1_Y) = \rho^\sharp_X(a) \otimes 1_Y$. □

Definition 2.7. Let $\mathcal{G}$ be a 2-group and $A = \pi_1(\mathcal{G})$. For an object X, put

$$\varepsilon_X := (\lambda^\sharp_X)^{-1} \circ \rho^\sharp_X \in \mathrm{Aut}(A).$$

The canonical action is the assignment $[X]\cdot a := \varepsilon_X(a)$; its independence of the representative X and its action law are proved below.

Proposition 2.8. *For the automorphisms of Definition 2.7, we have:*

(i) *ε_X depends only on the isomorphism class $[X] \in \pi_0(\mathcal{G})$;*

(ii) *$\varepsilon_{\mathbf{1}} = \mathrm{id}_A$ and $\varepsilon_{X\otimes Y} = \varepsilon_X \circ \varepsilon_Y$, as maps $A \to A$.*

Consequently the assignment $[X] \cdot a$ is well defined and is a left action of $\pi_0(\mathcal{G})$ on A by group automorphisms. Moreover, for all objects X, Y and all $b \in A$,

$$1_X \otimes \lambda^\sharp_Y(b) \;=\; \lambda^\sharp_{X\otimes Y}\big([X] \cdot b\big), \tag{19}$$

as maps $X \otimes Y \to X \otimes Y$.

Proof. (i) Let $u\colon X \to Y$ be an isomorphism and $a \in A$. By Lemma 2.6(ii),

$$\begin{aligned}\rho^\sharp_Y(a) &= u \circ \rho^\sharp_X(a) \circ u^{-1} \\ &= u \circ \lambda^\sharp_X(\varepsilon_X(a)) \circ u^{-1} = \lambda^\sharp_Y(\varepsilon_X(a)),\end{aligned}$$

as maps $Y \to Y$, whence $\varepsilon_Y(a) = \varepsilon_X(a)$.

(ii) $\varepsilon_{\mathbf{1}} = \mathrm{id}$ is (18). For multiplicativity, let $a \in A$ and put $b := \varepsilon_Y(a)$, so that $\rho^\sharp_Y(a) = \lambda^\sharp_Y(b)$. Then by Lemma 2.6(iii), (iv) and (iii) again,

$$\begin{aligned}\rho^\sharp_{X\otimes Y}(a) &= 1_X \otimes \rho^\sharp_Y(a) = 1_X \otimes \lambda^\sharp_Y(b) \\ &= \rho^\sharp_X(b) \otimes 1_Y = \lambda^\sharp_X(\varepsilon_X(b)) \otimes 1_Y \\ &= \lambda^\sharp_{X\otimes Y}\big(\varepsilon_X(b)\big),\end{aligned}$$

as maps $X \otimes Y \to X \otimes Y$, so $\varepsilon_{X\otimes Y}(a) = \varepsilon_X(\varepsilon_Y(a))$. The identity (19) is contained in this chain: $1_X \otimes \lambda^\sharp_Y(b) = \rho^\sharp_X(b) \otimes 1_Y = \lambda^\sharp_{X\otimes Y}(\varepsilon_X(b)) = \lambda^\sharp_{X\otimes Y}([X]\cdot b)$. □

2.3. **Monoidal functors from groups.**

Definition 2.9. We use the monoidal-functor convention of Mac Lane [ML98, Ch. VII] and Kerodon [Lur18], Kerodon tag 00CC. Let $\mathcal{C}, \mathcal{D}$ be monoidal categories. A (strong) *monoidal functor* $(F, \varphi, \varphi_0)\colon \mathcal{C} \to \mathcal{D}$ is a functor F with natural isomorphisms

$$\varphi_{X,Y}\colon F(X) \otimes F(Y) \longrightarrow F(X \otimes Y), \qquad \varphi_0\colon \mathbf{1}_{\mathcal{D}} \longrightarrow F(\mathbf{1}_{\mathcal{C}}),$$

satisfying the following identities for all objects X, Y, Z of $\mathcal{C}$:

$$\varphi_{X,Y\otimes Z} \circ (1_{F(X)} \otimes \varphi_{Y,Z}) \circ \mathsf{a}^{\mathcal{D}}_{F(X),F(Y),F(Z)} = F(\mathsf{a}^{\mathcal{C}}_{X,Y,Z}) \circ \varphi_{X\otimes Y,Z} \circ (\varphi_{X,Y} \otimes 1_{F(Z)}), \tag{20}$$

$$F(\lambda^{\mathcal{C}}_X) \circ \varphi_{\mathbf{1}_{\mathcal{C}},X} \circ (\varphi_0 \otimes 1_{F(X)}) = \lambda^{\mathcal{D}}_{F(X)}, \tag{21}$$

$$F(\rho^{\mathcal{C}}_X) \circ \varphi_{X,\mathbf{1}_{\mathcal{C}}} \circ (1_{F(X)} \otimes \varphi_0) = \rho^{\mathcal{D}}_{F(X)}. \tag{22}$$

These are equalities of maps with respective sources and targets $(F(X)\otimes F(Y))\otimes F(Z) \longrightarrow F(X \otimes (Y \otimes Z))$, $\mathbf{1}_{\mathcal{D}} \otimes F(X) \longrightarrow F(X)$, and $F(X) \otimes \mathbf{1}_{\mathcal{D}} \longrightarrow F(X)$. A *monoidal natural transformation*

$$\theta\colon (F, \varphi, \varphi_0) \Rightarrow (F', \varphi', \varphi'_0)$$

is a natural transformation with $\theta_{X\otimes Y} \circ \varphi_{X,Y} = \varphi'_{X,Y} \circ (\theta_X \otimes \theta_Y)$ as maps $F(X) \otimes F(Y) \to F'(X \otimes Y)$, and $\theta_{\mathbf{1}} \circ \varphi_0 = \varphi'_0$ as maps $\mathbf{1}_{\mathcal{D}} \to F'(\mathbf{1}_{\mathcal{C}})$.

Notation 2.10. For a group G, let $\underline{G}$ denote the *discrete strict monoidal category* with object set G, only identity morphisms, tensor product given by multiplication in G, and unit object e. It is a 2-group, with

$$\pi_0(\underline{G}) = G \quad \text{as a group pointed by } e, \qquad \pi_1(\underline{G}, e) = \{1\}.$$

The symbol $\mathbf{B}G$ denotes the one-object groupoid with object $*$ and $\mathrm{Aut}_{\mathbf{B}G}(*) = G$. The symbol

$$B_\bullet G := N_\bullet(\mathbf{B}G)$$

denotes its nerve, regarded as the inhomogeneous bar simplicial set, whose simplices, face maps, and degeneracy maps are described in Definition A.3. Our chronological bar label $[g]$ denotes the categorical arrow g^{-1}; this gives the face label gh on the long edge of

$[g|h]$. More generally, $B_\bullet(-)$ denotes the specified simplicial bar construction or nerve. A homomorphism $\phi\colon G \to \Gamma$ induces a functor $\mathbf{B}\phi\colon \mathbf{B}G \to \mathbf{B}\Gamma$ and the simplicial map

$$B_\bullet\phi := N_\bullet(\mathbf{B}\phi)\colon B_\bullet G \longrightarrow B_\bullet\Gamma.$$

Geometric realization is written $|B_\bullet G|$.

Definition 2.11. For 2-groups $\mathcal{G}', \mathcal{G}$ we write $\mathbf{Mon}(\mathcal{G}', \mathcal{G})$ for the category of monoidal functors $\mathcal{G}' \to \mathcal{G}$ and monoidal natural transformations; since $\mathcal{G}$ is a groupoid, then so is $\mathbf{Mon}(\mathcal{G}', \mathcal{G})$. In particular, when $\mathcal{G}' = \underline{G}$, unwinding the definition for this source, an object F of $\mathbf{Mon}(\underline{G}, \mathcal{G})$ is a triple $\big((X_\sigma)_{\sigma\in G}, (\gamma_{\sigma,\tau})_{\sigma,\tau\in G}, \gamma_0\big)$ consisting of objects X_σ of $\mathcal{G}$, isomorphisms

$$\gamma_{\sigma,\tau}\colon X_\sigma \otimes X_\tau \to X_{\sigma\tau},$$
$$\gamma_0\colon \mathbf{1} \to X_e,$$

such that for all $\sigma, \tau, \nu \in G$:

$$\gamma_{\sigma\tau,\nu} \circ (\gamma_{\sigma,\tau} \otimes 1_{X_\nu}) = \gamma_{\sigma,\tau\nu} \circ (1_{X_\sigma} \otimes \gamma_{\tau,\nu}) \circ \mathsf{a}_{X_\sigma,X_\tau,X_\nu}, \tag{23}$$
$$\gamma_{\sigma,e} \circ (1_{X_\sigma} \otimes \gamma_0) = \rho_{X_\sigma}, \qquad \gamma_{e,\sigma} \circ (\gamma_0 \otimes 1_{X_\sigma}) = \lambda_{X_\sigma}. \tag{24}$$

Equation (23) compares maps $(X_\sigma \otimes X_\tau) \otimes X_\nu \to X_{\sigma\tau\nu}$. The two identities in (24) compare maps $X_\sigma \otimes \mathbf{1} \to X_\sigma$ and $\mathbf{1} \otimes X_\sigma \to X_\sigma$, respectively. A morphism $\theta\colon (X, \gamma, \gamma_0) \to (X', \gamma', \gamma'_0)$ is a family of isomorphisms $\theta_\sigma\colon X_\sigma \to X'_\sigma$ with

$$\theta_{\sigma\tau} \circ \gamma_{\sigma,\tau} = \gamma'_{\sigma,\tau} \circ (\theta_\sigma \otimes \theta_\tau) \quad \text{for all } \sigma, \tau \in G, \qquad \theta_e \circ \gamma_0 = \gamma'_0. \tag{25}$$

The common sources and targets in (25) are

$$X_\sigma \otimes X_\tau \to X'_{\sigma\tau},$$
$$\mathbf{1} \to X'_e,$$

respectively. The component map

$$\pi_0(F)\colon \pi_0(\underline{G}) = G \longrightarrow \pi_0(\mathcal{G}), \qquad \sigma \longmapsto [X_\sigma],$$

is the map induced by functoriality of π_0 for groupoids (Notation 1.1). The isomorphisms $\gamma_{\sigma,\tau}$ witness that it is a group homomorphism, and it is invariant under monoidal isomorphism.

3. The Picard 2-group of a K-central algebra

3.1. **Invertible bimodules.**

Definition 3.1. A *K-central k-algebra* is a k-algebra R together with a fixed isomorphism of k-algebras $K \xrightarrow{\sim} Z(R)$, which we treat as an identification $K = Z(R)$; thus $k \subseteq K \subseteq R$ with K central in R. For a left-R, right-S bimodule P with $R \neq S$, we write $r \star_R p$ and $p \star_S s$ for the two actions. When both actions are by the same ring, we write $r \star_\ell p$ and $p \star_\imath r$ for the left and right actions, respectively. For one-sided modules, the subscript names the acting ring. Following [Gin22, §2], an *R-bimodule over k* is an R-bimodule P whose induced left and right k-module structures coincide: $c \star_\ell p = p \star_\imath c$ for all $c \in k$, $p \in P$. Such a bimodule is *invertible* if there is an R-bimodule Q over k with $P \otimes_R Q \cong R \cong Q \otimes_R P$ as R-bimodules.

The *Picard 2-group* $\boldsymbol{\mathcal{P}ic}_k(R)$ is the $\mathfrak{V}$-small category whose objects are the $\mathfrak{U}$-small invertible R-bimodules over k and whose morphisms are the R-bimodule isomorphisms, with

monoidal structure $\otimes_R$, unit object R, and the canonical associativity and unit isomorphisms of the tensor product of bimodules.

If P and Q have matching left and right k-structures, then

$$c \star_\ell (p \otimes_R q) = (p \star_\imath c) \otimes_R q = p \otimes_R (c \star_\ell q) = (p \otimes_R q) \star_\imath c,$$

so $P \otimes_R Q$ has the same property. If Q_1 and Q_2 are inverse bimodules for P_1 and P_2, respectively, then $Q_2 \otimes_R Q_1$ is an inverse for $P_1 \otimes_R P_2$. The unit object R has matching k-structures because $k \subseteq Z(R)$, and the canonical associativity and unit isomorphisms for relative tensor products are bimodule isomorphisms satisfying Mac Lane's pentagon and triangle identities. Hence $\boldsymbol{\mathcal{Pic}}_k(R)$ is a 2-group. By construction, $\pi_0(\boldsymbol{\mathcal{Pic}}_k(R)) = \mathrm{Pic}_k(R)$, the Picard group [Gin22, §2] of isomorphism classes of invertible R-bimodules whose left and right k-module structures are the same. As a pointed set this has distinguished element $[R]$.

3.2. **Central scalars and the type homomorphism.**

Lemma 3.2. *Let P be an invertible R-bimodule over k. The map*

$$K \longrightarrow \mathrm{End}_{R\text{-bimod}}(P), \qquad \alpha \mapsto (p \mapsto \alpha \star_\ell p),$$

is an isomorphism of rings. In particular $\mathrm{End}(P)$ *is commutative,* $\mathrm{Aut}_{R\text{-bimod}}(P) \cong \mathrm{U}(K)$, *and*

$$\pi_1\big(\boldsymbol{\mathcal{Pic}}_k(R)\big) = \mathrm{Aut}_{R\text{-bimod}}(R) \xrightarrow{\sim} \mathrm{U}(K), \qquad u \mapsto u(1). \tag{26}$$

Proof. For $\alpha \in K = Z(R)$, left multiplication by α is a bimodule endomorphism. A bimodule endomorphism u of R satisfies $u(r) = ru(1) = u(1)r$, so evaluation at 1 identifies $\mathrm{End}(R)$ with K as rings. Restriction to units proves (26).

For general invertible P, tensoring on the right by P is an equivalence on the category of $\mathfrak{U}$-small R-bimodules over k, by Lemma 2.3 applied in that ambient monoidal category. It is therefore fully faithful, so

$$\mathrm{End}(R) \to \mathrm{End}(R \otimes_R P), \qquad u \mapsto u \otimes 1_P,$$

is bijective; conjugating by the unit isomorphism

$$\lambda_P \colon R \otimes_R P \to P, \qquad r \otimes_R p \mapsto r \star_\ell p,$$

gives a ring isomorphism $\mathrm{End}(R) \to \mathrm{End}(P)$. Composing with $K \cong \mathrm{End}(R)$ gives the asserted scalar-action map $K \to \mathrm{End}(P)$. Indeed, $\lambda_P \circ \big((\alpha \star_\ell -) \otimes 1_P\big) \circ \lambda_P^{-1}$ sends $p = \lambda_P(1 \otimes_R p)$ to $\lambda_P(\alpha \otimes_R p) = \alpha \star_\ell p$. □

For the type homomorphism, see [Gin22, (2.1)–(2.2)] and [Bas68, Ch. II, §5].

Definition 3.3. For an invertible R-bimodule P over k, its *type* $\eta_P : K \to K$ is the composite of the right scalar-action map $K \to \mathrm{End}_{R\text{-bimod}}(P)$ with the inverse of the left scalar-action isomorphism of Lemma 3.2. Write

$$h_R([P]) := \eta_P.$$

The following lemma shows that this defines a homomorphism $h_R : \mathrm{Pic}_k(R) \to \mathrm{Aut}_k(K)$.

Lemma 3.4. *Let P be an invertible R-bimodule over k, with type as in Definition 3.3.*

(i) *The map $\eta_P \colon K \to K$ is the unique map such that $p \star_\imath \alpha = \eta_P(\alpha) \star_\ell p$ for all $\alpha \in K$, $p \in P$; and $\eta_P \in \mathrm{Aut}_k(K)$.*

(ii) *η_P depends only on the isomorphism class of P; $\eta_R = \mathrm{id}_K$; and $\eta_{P\otimes_R Q} = \eta_P \circ \eta_Q$, as maps $K \to K$.*

Consequently the assignment h_R is a well-defined group homomorphism.

Proof. (i) Right multiplication defines a unital ring homomorphism $K \to \mathrm{End}_{R\text{-bimod}}(P)$, because K is central and commutative. Composing with the inverse of the ring isomorphism in Lemma 3.2 gives the unique unital ring homomorphism η_P satisfying the displayed relation. The matching k-actions give $\eta_P(c) = c$ for $c \in k$. Part (ii), whose proof uses only that relation, shows that an inverse bimodule Q gives $\eta_P \circ \eta_Q = \eta_Q \circ \eta_P = \mathrm{id}$; hence $\eta_P \in \mathrm{Aut}_k(K)$.

(ii) If $u\colon P \to P'$ is a bimodule isomorphism, applying u to $p \star_\iota \alpha = \eta_P(\alpha) \star_\ell p$ gives $u(p) \star_\iota \alpha = \eta_P(\alpha) \star_\ell u(p)$, so $\eta_{P'} = \eta_P$ by uniqueness. For $P = R$: $r \star_\ell \alpha = r\alpha = \alpha r = \alpha \star_\iota r$, so $\eta_R = \mathrm{id}$. Finally, in $P \otimes_R Q$,

$$\begin{aligned}(p \otimes_R q) \star_\iota \alpha &= p \otimes_R (q \star_\iota \alpha) = p \otimes_R \big(\eta_Q(\alpha) \star_\ell q\big)\\ &= \big(p \star_\iota \eta_Q(\alpha)\big) \otimes_R q = \big(\eta_P(\eta_Q(\alpha)) \star_\ell p\big) \otimes_R q\\ &= (\eta_P \circ \eta_Q)(\alpha) \star_\ell (p \otimes_R q).\end{aligned}$$

□

3.3. **The induced action on units.**

Proposition 3.5. *Under the identification (26), the isomorphisms $\lambda^\sharp_P$ and $\rho^\sharp_P$ of $\boldsymbol{\mathcal{P}ic}_k(R)$ are*

$$\lambda^\sharp_P(\alpha)\colon p \mapsto \alpha \star_\ell p, \qquad \rho^\sharp_P(\alpha)\colon p \mapsto p \star_\iota \alpha \quad \text{for all } \alpha \in \mathrm{U}(K) \text{ and } p \in P,$$

and the canonical action of $\pi_0(\boldsymbol{\mathcal{P}ic}_k(R)) = \mathrm{Pic}_k(R)$ on $\pi_1(\boldsymbol{\mathcal{P}ic}_k(R)) = \mathrm{U}(K)$ of Proposition 2.8 is

$$[P] \cdot \alpha \;=\; \eta_P(\alpha) \;=\; h_R([P])(\alpha).$$

In particular, the canonical action factors through the type homomorphism h_R.

Proof. Write $u_\alpha \in \mathrm{Aut}_{R\text{-bimod}}(R)$ for the automorphism $r \mapsto \alpha r$ corresponding to $\alpha \in \mathrm{U}(K)$ under (26). The unit isomorphisms of $\boldsymbol{\mathcal{P}ic}_k(R)$ are $\lambda_P(r \otimes_R p) = r \star_\ell p$ and $\rho_P(p \otimes_R r) = p \star_\iota r$. Then

$$\lambda^\sharp_P(\alpha)(p) = \lambda_P\big((u_\alpha \otimes 1_P)(1 \otimes_R p)\big) = \lambda_P(\alpha \otimes_R p) = \alpha \star_\ell p,$$
$$\rho^\sharp_P(\alpha)(p) = \rho_P\big((1_P \otimes u_\alpha)(p \otimes_R 1)\big) = \rho_P(p \otimes_R \alpha) = p \star_\iota \alpha.$$

Hence $\varepsilon_P(\alpha)$ is the unique $\beta \in \mathrm{U}(K)$ with $\beta \star_\ell p = p \star_\iota \alpha$ for all p, i.e. $\beta = \eta_P(\alpha)$ by Lemma 3.4(i). □

Definition 3.6. For a K-central k-algebra R, a *generalized collective character (GCC) of type ϕ* is a group homomorphism $\Phi : G \to \mathrm{Pic}_k(R)$ satisfying $h_R \circ \Phi = \phi$; see [CG01, p. 152, §2] and [Gin22, §3].

Corollary 3.7. *Let $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$ and let $f\colon G \to \mathrm{Pic}_k(R)$ be a generalized collective character of type ϕ. The induced action on $A = \mathrm{U}(K)$ is*

$$\sigma \cdot \alpha = f(\sigma) \cdot \alpha = \phi(\sigma)(\alpha).$$

Thus A is the G-module $\mathrm{U}(K)_\phi$, written K^*_ϕ in [Gin22, Thm. 5.1(1)], and its action depends only on ϕ.

Proof. By Proposition 3.5,

$$f(\sigma)\cdot\alpha = h_R(f(\sigma))(\alpha) = \phi(\sigma)(\alpha),$$

which is exactly the action defining $\mathrm{U}(K)_\phi$. □

4. Strongly G-graded algebras as monoidal functors

4.1. **Clifford system extensions.** For these definitions, see [Gin22, §4] and [CG01, Def. 2.1].

Definition 4.1. A *G-graded k-algebra* is a k-algebra A with a direct sum decomposition $A = \bigoplus_{g\in G} A_g$ of k-modules such that $A_gA_h \subseteq A_{gh}$ for all g, h; it is *strongly G-graded* if $A_gA_h = A_{gh}$ for all g, h. The component A_e is then a k-subalgebra containing 1_A, and each A_g is an A_e-bimodule.

Let R be a K-central k-algebra. A *G-graded Clifford system extension of R* is a pair (A, j) where A is a strongly G-graded k-algebra and $j\colon R \to A$ is a k-algebra monomorphism with $j(R) = A_e$. A *morphism* $(A, j) \to (A', j')$ is a G-grading-preserving k-algebra isomorphism $\psi\colon A \to A'$ (i.e. $\psi(A_g) = A'_g$ for all g) with $\psi \circ j = j'$ as maps $R \to A'$. We write $\mathbf{Cliff}_G(R)$ for the resulting $\mathfrak{V}$-small groupoid and $\mathrm{Cliff}_k(G, R) := \pi_0(\mathbf{Cliff}_G(R))$ for its set of isomorphism classes, as in [Gin22, Def. 4.2].

A monoidal functor $\underline{G} \to \boldsymbol{\mathcal{P}ic}_k(R)$ consists of invertible bimodules P_g, multiplication isomorphisms

$$P_g \otimes_R P_h \xrightarrow{\sim} P_{gh},$$

and a unit isomorphism, satisfying the associativity and unit axioms. Theorem 4.5 constructs the corresponding Clifford system extension on their direct sum.

If $(A, j) \in \mathbf{Cliff}_G(R)$, each A_g is an R-bimodule via j: $r \star_\ell a := j(r)a$ and $a \star_\iota r := a\, j(r)$. Since A is a k-algebra, $k\,1_A$ is central in A, so the left and right k-structures on A_g agree; thus A_g is an R-bimodule over k.

4.2. **Invertibility of homogeneous components.** For a strongly G-graded k-algebra $A = \bigoplus_{g\in G} A_g$, write

$$m_{g,h}\colon A_g \otimes_{A_e} A_h \to A_{gh}$$

for the map induced by multiplication.

Lemma 4.2. *Every multiplication map $m_{g,h}$ is an isomorphism of A_e-bimodules, and every homogeneous component A_g is an invertible A_e-bimodule, with inverse $A_{g^{-1}}$.*

Proof. Multiplication is A_e-balanced and A_e-bilinear, so $m_{g,h}$ is a well-defined A_e-bimodule map; it is surjective since $A_gA_h = A_{gh}$. For injectivity, use $A_gA_{g^{-1}} = A_e \ni 1_A$ to choose finitely many $a_i \in A_g$, $b_i \in A_{g^{-1}}$ with $\sum_i a_ib_i = 1_A$. Define

$$\psi\colon A_{gh} \to A_g \otimes_{A_e} A_h$$

by $\psi(x) := \sum_i a_i \otimes_{A_e} b_ix$; note $b_ix \in A_{g^{-1}}A_{gh} \subseteq A_h$. Then $m_{g,h} \circ \psi = \mathrm{id}_{A_{gh}}$ as maps $A_{gh} \to A_{gh}$, and for $a \in A_g$, $b \in A_h$,

$$\psi\big(m_{g,h}(a \otimes_{A_e} b)\big) = \sum_i a_i \otimes_{A_e} b_iab = \sum_i a_i(b_ia) \otimes_{A_e} b = \Big(\sum_i a_ib_ia\Big) \otimes_{A_e} b = a \otimes_{A_e} b,$$

where the middle equality holds because $b_ia \in A_{g^{-1}}A_g \subseteq A_e$ may be moved across the tensor sign. By additivity, $\psi \circ m_{g,h} = \mathrm{id}_{A_g\otimes_{A_e}A_h}$ as maps $A_g \otimes_{A_e} A_h \to A_g \otimes_{A_e} A_h$, so

$m_{g,h}$ is bijective. Taking $h = g^{-1}$, and then exchanging the roles of g and g^{-1}, exhibits $A_g \otimes_{A_e} A_{g^{-1}} \cong A_e \cong A_{g^{-1}} \otimes_{A_e} A_g$, so A_g is invertible. □

Remark 4.3. Lemma 4.2 is classical (see, e.g., [Dad70, pp. 236–290] and [NVO04, Ch. I]); the proof is included to keep the paper self-contained.

4.3. **Equivalence with monoidal functors.**

Construction 4.4. Let R be a K-central k-algebra and G a group. The assignment

$$\Theta\colon \mathbf{Cliff}_G(R) \longrightarrow \mathbf{Mon}\big(\underline{G}, \boldsymbol{\mathcal{P}ic}_k(R)\big) \tag{27}$$

is specified as follows. On objects, $\Theta(A, j) := \big((A_g)_g,\ (m_{g,h})_{g,h},\ j\big)$, where each A_g carries the R-bimodule structure via j, the coherence maps are the multiplication isomorphisms of Lemma 4.2, and the unit constraint is $j\colon R \to A_e$. On morphisms, $\Theta(\psi) := (\psi|_{A_g})_{g\in G}$. The associated generalized collective character is

$$\chi_R[A] : G \longrightarrow \mathrm{Pic}_k(R), \qquad g \longmapsto [A_g].$$

Conversely, for a monoidal functor $F = ((P_g)_g, (\gamma_{g,h})_{g,h}, \gamma_0)$, take $A := \bigoplus_{g\in G} P_g$, with summand inclusions $i_g : P_g \to A$. Specify a biadditive multiplication, a unit, and a map j by

$$i_g(x)i_h(y) := i_{gh}\big(\gamma_{g,h}(x \otimes_R y)\big), \qquad 1_A := (i_e \circ \gamma_0)(1_R), \qquad j := i_e \circ \gamma_0 : R \to A.$$

Theorem 4.5. *The assignment Θ of Construction 4.4 is a well-defined equivalence of groupoids. For every (A, j) the homomorphism $\pi_0(\Theta(A, j))\colon G \to \mathrm{Pic}_k(R)$, $g \mapsto [A_g]$, is the associated generalized collective character $\chi_R[A]$ of* [Gin22, (4.2)].

Proof. Θ is well defined on objects. By Lemma 4.2 and the discussion above, each A_g is an invertible R-bimodule over k, i.e. an object of $\boldsymbol{\mathcal{P}ic}_k(R)$, and each $m_{g,h}$ is an isomorphism in $\boldsymbol{\mathcal{P}ic}_k(R)$. The coherence axiom (23) for $(A_g, m_{g,h}, j)$, evaluated on $(a \otimes_R b) \otimes_R c$ with $a \in A_g$, $b \in A_h$, $c \in A_l$, and using that the canonical associator of $\otimes_R$ sends $(a \otimes_R b) \otimes_R c$ to $a \otimes_R (b \otimes_R c)$, reads $(ab)c = a(bc)$, i.e. associativity of A. The unit axioms (24) read: $m_{g,e}(a \otimes_R j(r)) = a\, j(r) = a \star_\iota r = \rho_{A_g}(a \otimes_R r)$ and $m_{e,g}(j(r) \otimes_R a) = j(r)\, a = r \star_\ell a = \lambda_{A_g}(r \otimes_R a)$, which hold by definition of the bimodule structure.

Θ is well defined on morphisms. Since $\psi \circ j = j'$, a graded algebra isomorphism ψ restricts to bimodule isomorphisms on the homogeneous components. Multiplicativity and $\psi \circ j = j'$ are exactly the two equations (25). Restriction commutes with composition and identities.

Θ is essentially surjective. For a monoidal functor F, take A, i_g, and j as in Construction 4.4. The associativity axiom (23) gives associativity of the specified product. The unit axioms (24) give the specified unit and show that $j : R \to A$ is a unital algebra homomorphism. Indeed, $j(r)i_g(x) = i_g(r \star_\ell x)$ and $i_g(x)j(r) = i_g(x \star_\iota r)$; taking $g = e$ and $x = \gamma_0(r')$ gives $j(r)j(r') = j(rr')$. The algebra is strongly graded with $A_g = i_g(P_g)$, since every $\gamma_{g,h}$ is surjective, and j identifies R with A_e. The maps $i_g : P_g \xrightarrow{\sim} A_g$ form a monoidal natural isomorphism $F \xrightarrow{\sim} \Theta(A, j)$, by the definition of multiplication and j. This proves essential surjectivity.

Θ is fully faithful. Let $(A, j), (A', j') \in \mathbf{Cliff}_G(R)$ and let $\theta\colon \Theta(A, j) \to \Theta(A', j')$ be a monoidal natural isomorphism. Define $\psi := \bigoplus_{g\in G} \theta_g\colon A \to A'$. It is a bijective G-grading-preserving k-module map; it is multiplicative on homogeneous elements by the first condition in (25), hence multiplicative; it is unital since $\psi(1_A) = \theta_e(j(1_R)) = j'(1_R) = 1_{A'}$ by the

second condition in (25); and that same condition says $\psi \circ j = j'$ as maps $R \to A'$. Thus ψ is a morphism in $\mathbf{Cliff}_G(R)$ with $\Theta(\psi) = \theta$, and it is the unique such, being determined by its restrictions to the A_g.

The last assertion of the theorem is the definition of χ_R in [Gin22, (4.2)]. □

Corollary 4.6. *For every group homomorphism $\Phi\colon G \to \mathrm{Pic}_k(R)$, the assignment $[(A,j)] \mapsto [\Theta(A,j)]$ is a bijection*

$$\big\{[(A,j)] \in \mathrm{Cliff}_k(G,R) \mid \chi_R[(A,j)] = \Phi\big\} \xrightarrow{\sim} \big\{[F] \mid F \in \mathbf{Mon}(\underline{G}, \boldsymbol{\mathcal{P}ic}_k(R)),\ \pi_0(F) = \Phi\big\},$$

where the right-hand set consists of monoidal isomorphism classes.

Proof. An equivalence of groupoids induces a bijection on isomorphism classes, compatibly with the invariants χ_R and $\pi_0(-)$, which correspond under Θ by Theorem 4.5. □

5. Obstruction theory for monoidal functors from groups

Throughout this section, $\mathcal{G}$ is a 2-group and $A := \pi_1(\mathcal{G})$ is written multiplicatively; it is abelian by Lemma 2.4. We use the canonical action of $\pi_0(\mathcal{G})$ on A from Proposition 2.8: for $x \in \pi_0(\mathcal{G})$ and $a \in A$, choose any object X with $[X] = x$ and set

$$x \cdot a := \varepsilon_X(a), \qquad \text{equivalently} \qquad \rho^\sharp_X(a) = \lambda^\sharp_X(x \cdot a).$$

Proposition 2.8 proves that this is independent of the chosen representative X. Now fix a group homomorphism $f\colon G \to \pi_0(\mathcal{G})$. The induced G-module A_f has underlying group A and action

$$^\sigma a := f(\sigma) \cdot a.$$

In the Picard 2-group $\boldsymbol{\mathcal{P}ic}_k(R)$, Proposition 3.5 identifies this formula with

$$^\sigma\alpha = h_R(f(\sigma))(\alpha) \quad \text{for all } \alpha \in \mathrm{U}(K).$$

Thus, if f has type ϕ, then A_f is exactly $\mathrm{U}(K)_\phi$. The results of this section are, in equivalent formulations, due to Sính's theory of 2-groups (gr-categories) [Sín75]; see also [BL04, §8.3] and [CG03, Thm. 2.2]. For the related classification of braided categorical groups by abelian third cohomology and quadratic maps, see [JS93, §3, pp. 47–51, Thms. 3.2–3.3]. The proofs below use the normalization of [Gin22, §5] and establish the cocycle-level statements applied in §6.

5.1. **Reduced systems and the obstruction cochain.** Throughout, we use the following three rules, valid for all objects X, Y, all morphisms $u\colon X \to Y$ of $\mathcal{G}$, and all $a \in A$; they were established in Lemma 2.6 and Proposition 2.8:

$$\text{(L1)} \qquad u \circ \lambda^\sharp_X(a) = \lambda^\sharp_Y(a) \circ u;$$

$$\text{(L2)} \qquad \lambda^\sharp_X(a) \otimes 1_Y = \lambda^\sharp_{X\otimes Y}(a);$$

$$\text{(L3)} \qquad 1_X \otimes \lambda^\sharp_Y(a) = \lambda^\sharp_{X\otimes Y}\big([X] \cdot a\big).$$

The maps in (L1) and those in (L2)–(L3) have respective sources and targets $X \to Y$ and $X \otimes Y \to X \otimes Y$. By (L1) and (L2), for any isomorphism $w\colon X \otimes Y \to Z$ one has

$$w \circ (\lambda^\sharp_X(a) \otimes 1_Y) = \lambda^\sharp_Z(a) \circ w,$$

as maps $X \otimes Y \to Z$; by (L1) and (L3), we also have

$$w \circ (1_X \otimes \lambda^\sharp_Y(a)) = \lambda^\sharp_Z([X] \cdot a) \circ w,$$

as maps $X \otimes Y \to Z$. The scalar calculations below use these identities, the injectivity of each $\lambda^\sharp_X$, the commutativity of A, and the naturality of a.

Definition 5.1. A *reduced system for* $f\colon G \to \pi_0(\mathcal{G})$ is a pair $\big((X_\sigma)_{\sigma\in G}, (\gamma_{\sigma,\tau})_{\sigma,\tau\in G}\big)$ where each X_σ is an object of $\mathcal{G}$ with $[X_\sigma] = f(\sigma)$, the

$$\gamma_{\sigma,\tau}\colon X_\sigma \otimes X_\tau \to X_{\sigma\tau}$$

are isomorphisms, and

$$X_e = \mathbf{1}, \qquad \gamma_{e,\sigma} = \lambda_{X_\sigma}, \qquad \gamma_{\sigma,e} = \rho_{X_\sigma} \quad \text{for all } \sigma \in G. \tag{28}$$

The map identities in (28) have sources $\mathbf{1} \otimes X_\sigma$ and $X_\sigma \otimes \mathbf{1}$, respectively, and target X_σ. (The two conditions at $(\sigma,\tau) = (e,e)$ are consistent by (15).) Reduced systems exist for every f: choose any X_σ in the class $f(\sigma)$ for $\sigma \neq e$ and set $X_e := \mathbf{1}$ (legitimate, since $f(e) = [\mathbf{1}]$); for $\sigma,\tau \neq e$ choose any isomorphism $\gamma_{\sigma,\tau}$, which exists because $[X_\sigma \otimes X_\tau] = f(\sigma)f(\tau) = f(\sigma\tau) = [X_{\sigma\tau}]$ and $\mathcal{G}$ is a groupoid; and impose (28) otherwise.

Construction 5.2. Let (X,γ) be a reduced system for $f\colon G \to \pi_0(\mathcal{G})$. For $\sigma,\tau,\nu \in G$, both

$$\gamma_{\sigma\tau,\nu} \circ (\gamma_{\sigma,\tau} \otimes 1_{X_\nu}) \qquad \text{and} \qquad \gamma_{\sigma,\tau\nu} \circ (1_{X_\sigma} \otimes \gamma_{\tau,\nu}) \circ \mathsf{a}_{X_\sigma,X_\tau,X_\nu}$$

are isomorphisms $(X_\sigma \otimes X_\tau) \otimes X_\nu \to X_{\sigma\tau\nu}$, so they differ by a unique automorphism of $X_{\sigma\tau\nu}$, which by Lemma 2.6(i) equals $\lambda^\sharp_{X_{\sigma\tau\nu}}(t(\sigma,\tau,\nu))$ for a unique $t(\sigma,\tau,\nu) \in A$:

$$\gamma_{\sigma\tau,\nu} \circ (\gamma_{\sigma,\tau} \otimes 1_{X_\nu}) \;=\; \lambda^\sharp_{X_{\sigma\tau\nu}}\big(t(\sigma,\tau,\nu)\big) \circ \gamma_{\sigma,\tau\nu} \circ (1_{X_\sigma} \otimes \gamma_{\tau,\nu}) \circ \mathsf{a}_{X_\sigma,X_\tau,X_\nu}. \tag{29}$$

We call $t = t_{(X,\gamma)}\colon G^3 \to A$ the *associativity obstruction cochain* of the system. By construction, $t \equiv 1$ if and only if $(X,\gamma,1_{\mathbf{1}})$ satisfies (23); since the unit axioms (24) hold automatically by (28), this happens if and only if $(X,\gamma,1_{\mathbf{1}})$ is a monoidal functor $\underline{G} \to \mathcal{G}$.

Lemma 5.3. *The cochain t is normalized: $t(\sigma,\tau,\nu) = 1$ whenever one of σ,τ,ν equals e.*

Proof. Case $\sigma = e$. The following composites are maps $(\mathbf{1} \otimes X_\tau) \otimes X_\nu \to X_{\tau\nu}$. The left side of (29) is $\gamma_{\tau,\nu} \circ (\lambda_{X_\tau} \otimes 1_{X_\nu})$. On the right, by naturality of λ and then (16),

$$\gamma_{e,\tau\nu} \circ (1_{\mathbf{1}} \otimes \gamma_{\tau,\nu}) \circ \mathsf{a}_{\mathbf{1},X_\tau,X_\nu} = \lambda_{X_{\tau\nu}} \circ (1_{\mathbf{1}} \otimes \gamma_{\tau,\nu}) \circ \mathsf{a} = \gamma_{\tau,\nu} \circ \lambda_{X_\tau \otimes X_\nu} \circ \mathsf{a} = \gamma_{\tau,\nu} \circ (\lambda_{X_\tau} \otimes 1_{X_\nu}),$$

so $t(e,\tau,\nu) = 1$.

Case $\tau = e$. The two sides are maps $(X_\sigma \otimes \mathbf{1}) \otimes X_\nu \to X_{\sigma\nu}$. The left side is $\gamma_{\sigma,\nu} \circ (\rho_{X_\sigma} \otimes 1_{X_\nu})$; the right side is $\lambda^\sharp_{X_{\sigma\nu}}(t) \circ \gamma_{\sigma,\nu} \circ (1_{X_\sigma} \otimes \lambda_{X_\nu}) \circ \mathsf{a}_{X_\sigma,\mathbf{1},X_\nu}$, and the triangle axiom (14) gives $(1 \otimes \lambda) \circ \mathsf{a} = \rho \otimes 1$ as maps $(X_\sigma \otimes \mathbf{1}) \otimes X_\nu \to X_\sigma \otimes X_\nu$; hence $t(\sigma,e,\nu) = 1$.

Case $\nu = e$. The two sides are maps $(X_\sigma \otimes X_\tau) \otimes \mathbf{1} \to X_{\sigma\tau}$. The left side is $\rho_{X_{\sigma\tau}} \circ (\gamma_{\sigma,\tau} \otimes 1_{\mathbf{1}}) = \gamma_{\sigma,\tau} \circ \rho_{X_\sigma \otimes X_\tau}$ by naturality of ρ; the right side is $\lambda^\sharp_{X_{\sigma\tau}}(t) \circ \gamma_{\sigma,\tau} \circ (1_{X_\sigma} \otimes \rho_{X_\tau}) \circ \mathsf{a}_{X_\sigma,X_\tau,\mathbf{1}}$, and (17) gives $(1 \otimes \rho) \circ \mathsf{a} = \rho_{X_\sigma \otimes X_\tau}$ as maps $(X_\sigma \otimes X_\tau) \otimes \mathbf{1} \to X_\sigma \otimes X_\tau$; hence $t(\sigma,\tau,e) = 1$. $\square$

5.2. **The obstruction in third group cohomology and torsor for second group cohomology.** For the following obstruction theorem, see [CG03, Thm. 2.2] and [BL04, pp. 423–491, §8.3].

Lemma 5.4. *For every reduced system for f, the associativity obstruction cochain is a normalized 3-cocycle with coefficients in A_f.*

Proof. Normalization is Lemma 5.3. Fix $\sigma, \tau, \nu, \omega \in G$, and abbreviate the four objects by $X = X_\sigma$, $Y = X_\tau$, $Z = X_\nu$, $W = X_\omega$. In the following pentagon juxtaposition denotes tensor product, and each arrow is the canonical associator indicated by its endpoints:

$$\begin{array}{ccc} V_0 = ((XY)Z)W & \xrightarrow{\;a_0\;} & V_1 = (X(YZ))W \\ {\scriptstyle b_0}\searrow & & \swarrow{\scriptstyle a_1} \\ V_4 = (XY)(ZW) & & V_2 = X((YZ)W) \\ {\scriptstyle b_1}\searrow & & \swarrow{\scriptstyle a_2} \\ & V_3 = X(Y(ZW)) & \end{array}$$

For each vertex let $m_i : V_i \to X_{\sigma\tau\nu\omega}$ be the composite of the γ's following that parenthesization; for example

$$m_4 = \gamma_{\sigma\tau,\nu\omega} \circ (\gamma_{\sigma,\tau} \otimes \gamma_{\nu,\omega}).$$

Write $\lambda^\sharp = \lambda^\sharp_{X_{\sigma\tau\nu\omega}}$. Equation (29), naturality of the associator, and (L1)–(L3) give the five edge comparisons, with respective sources V_0, V_1, V_2, V_0, V_4 and common target $X_{\sigma\tau\nu\omega}$,

$$\begin{aligned} m_0 &= \lambda^\sharp\big(t(\sigma,\tau,\nu)\big) \circ m_1 \circ a_0, \\ m_1 &= \lambda^\sharp\big(t(\sigma,\tau\nu,\omega)\big) \circ m_2 \circ a_1, \\ m_2 &= \lambda^\sharp\big({}^\sigma t(\tau,\nu,\omega)\big) \circ m_3 \circ a_2, \\ m_0 &= \lambda^\sharp\big(t(\sigma\tau,\nu,\omega)\big) \circ m_4 \circ b_0, \\ m_4 &= \lambda^\sharp\big(t(\sigma,\tau,\nu\omega)\big) \circ m_3 \circ b_1. \end{aligned}$$

For the first comparison the scalar is tensored on the right with 1_W, so (L2) leaves it unchanged. In the third comparison it is tensored on the left with 1_X, so (L3) applies the action of σ. In the other comparisons naturality moves the indicated γ through the associator, and (L1) identifies the scalar at the common target. Thus all five comparisons have exactly the displayed sign and action.

The pentagon axiom gives $a_2 \circ a_1 \circ a_0 = b_1 \circ b_0$ as maps $V_0 \to V_3$. Substitute the five comparisons and cancel the common isomorphism $m_3 \circ b_1 \circ b_0$. Injectivity of $\lambda^\sharp$ gives

$$t(\sigma,\tau,\nu)\,t(\sigma,\tau\nu,\omega)\,{}^\sigma t(\tau,\nu,\omega) = t(\sigma\tau,\nu,\omega)\,t(\sigma,\tau,\nu\omega).$$

Equivalently, the product of the five factors

$${}^\sigma t(\tau,\nu,\omega), \quad t(\sigma\tau,\nu,\omega)^{-1}, \quad t(\sigma,\tau\nu,\omega), \quad t(\sigma,\tau,\nu\omega)^{-1}, \quad t(\sigma,\tau,\nu)$$

is 1. This is $(\partial t)(\sigma,\tau,\nu,\omega) = 1$ with the differential (12). □

Construction 5.5. Let (X,γ) and (X',γ') be reduced systems for $f\colon G\to\pi_0(\mathcal{G})$, with associativity obstruction cochains t and t'. Choose isomorphisms $u_\sigma\colon X_\sigma\to X'_\sigma$ for all σ, with $u_e=1_{\mathbf{1}}$ (possible since $[X_\sigma]=f(\sigma)=[X'_\sigma]$ and $X_e=\mathbf{1}=X'_e$), and define $c\colon G^2\to A$ by

$$u_{\sigma\tau}\circ\gamma_{\sigma,\tau}=\lambda^{\sharp}_{X'_{\sigma\tau}}\big(c(\sigma,\tau)\big)\circ\gamma'_{\sigma,\tau}\circ(u_\sigma\otimes u_\tau). \tag{30}$$

This is an equality of maps $X_\sigma\otimes X_\tau\to X'_{\sigma\tau}$.

Lemma 5.6. *The cochain c of Construction 5.5 is normalized, and $t=t'\cdot(\partial c)^{-1}$. In particular $[t]=[t']\in H^3(G,A_f)$.*

Proof. The element $c(\sigma,\tau)$ is well defined by Lemma 2.6(i), both sides of (30) being isomorphisms $X_\sigma\otimes X_\tau\to X'_{\sigma\tau}$. Normalization: for $\sigma=e$, the left side of (30) is $u_\tau\circ\lambda_{X_\tau}=\lambda_{X'_\tau}\circ(1_{\mathbf{1}}\otimes u_\tau)=\gamma'_{e,\tau}\circ(u_e\otimes u_\tau)$ by naturality of λ and $u_e=1_{\mathbf{1}}$, so $c(e,\tau)=1$; the case $\tau=e$ is symmetric, using ρ.

To compare the two cochains t and t', compose the left side of (29) for (X,γ) with $u_{\sigma\tau\nu}$ and rewrite it in terms of the primed system. Using (30) twice and (L1)–(L2) to collect automorphisms of the common target $X'_{\sigma\tau\nu}$ (we write $\lambda^{\sharp}:=\lambda^{\sharp}_{X'_{\sigma\tau\nu}}$), we obtain

$$\begin{aligned}
u_{\sigma\tau\nu}\circ\gamma_{\sigma\tau,\nu}\circ(\gamma_{\sigma,\tau}\otimes 1_{X_\nu}) &= \lambda^{\sharp}\big(c(\sigma\tau,\nu)\big)\circ\gamma'_{\sigma\tau,\nu}\circ\big((u_{\sigma\tau}\circ\gamma_{\sigma,\tau})\otimes u_\nu\big)\\
&= \lambda^{\sharp}\big(c(\sigma\tau,\nu)\big)\circ\gamma'_{\sigma\tau,\nu}\circ\Big(\big[\lambda^{\sharp}_{X'_{\sigma\tau}}(c(\sigma,\tau))\circ\gamma'_{\sigma,\tau}\circ(u_\sigma\otimes u_\tau)\big]\otimes u_\nu\Big)\\
&= \lambda^{\sharp}\big(c(\sigma\tau,\nu)\,c(\sigma,\tau)\big)\circ\gamma'_{\sigma\tau,\nu}\circ(\gamma'_{\sigma,\tau}\otimes 1_{X'_\nu})\circ\big((u_\sigma\otimes u_\tau)\otimes u_\nu\big),
\end{aligned}$$

as maps $(X_\sigma\otimes X_\tau)\otimes X_\nu\to X'_{\sigma\tau\nu}$. Substituting (29) for (X',γ') and then using naturality of a, $\mathsf{a}_{X'_\sigma,X'_\tau,X'_\nu}\circ\big((u_\sigma\otimes u_\tau)\otimes u_\nu\big)=\big(u_\sigma\otimes(u_\tau\otimes u_\nu)\big)\circ\mathsf{a}_{X_\sigma,X_\tau,X_\nu}$ as maps $(X_\sigma\otimes X_\tau)\otimes X_\nu\to X'_\sigma\otimes(X'_\tau\otimes X'_\nu)$, this equals

$$\lambda^{\sharp}\big(c(\sigma\tau,\nu)\,c(\sigma,\tau)\,t'(\sigma,\tau,\nu)\big)\circ\gamma'_{\sigma,\tau\nu}\circ(1_{X'_\sigma}\otimes\gamma'_{\tau,\nu})\circ\big(u_\sigma\otimes(u_\tau\otimes u_\nu)\big)\circ\mathsf{a}_{X_\sigma,X_\tau,X_\nu}.$$

Next, by the interchange law and (30) in the solved form $\gamma'_{\tau,\nu}\circ(u_\tau\otimes u_\nu)=\lambda^\sharp_{X'_{\tau\nu}}\big(c(\tau,\nu)\big)^{-1}\circ u_{\tau\nu}\circ\gamma_{\tau,\nu}$ as maps $X_\tau\otimes X_\nu\to X'_{\tau\nu}$, we obtain

$$\begin{aligned}&(1_{X'_\sigma}\otimes\gamma'_{\tau,\nu})\circ\big(u_\sigma\otimes(u_\tau\otimes u_\nu)\big)\\&\quad=u_\sigma\otimes\big[\gamma'_{\tau,\nu}\circ(u_\tau\otimes u_\nu)\big]\\&\quad=\Big(1_{X'_\sigma}\otimes\lambda^\sharp_{X'_{\tau\nu}}\big(c(\tau,\nu)^{-1}\big)\Big)\circ\big(u_\sigma\otimes(u_{\tau\nu}\circ\gamma_{\tau,\nu})\big),\end{aligned}$$

as maps $X_\sigma\otimes(X_\tau\otimes X_\nu)\to X'_\sigma\otimes X'_{\tau\nu}$, and by (L3) (with $[X'_\sigma]=f(\sigma)$) and (L1), the first factor moves past $\gamma'_{\sigma,\tau\nu}$ as $\lambda^\sharp\big({}^\sigma c(\tau,\nu)^{-1}\big)$. Finally apply (30) once more, in the form

$$\gamma'_{\sigma,\tau\nu}\circ(u_\sigma\otimes u_{\tau\nu})=\lambda^\sharp\big(c(\sigma,\tau\nu)\big)^{-1}\circ u_{\sigma\tau\nu}\circ\gamma_{\sigma,\tau\nu},$$

as maps $X_\sigma\otimes X_{\tau\nu}\to X'_{\sigma\tau\nu}$. Altogether

$$\begin{aligned}&u_{\sigma\tau\nu}\circ\gamma_{\sigma\tau,\nu}\circ(\gamma_{\sigma,\tau}\otimes 1)\\&\quad=\lambda^\sharp\Big(c(\sigma\tau,\nu)c(\sigma,\tau)t'(\sigma,\tau,\nu)\,{}^\sigma c(\tau,\nu)^{-1}c(\sigma,\tau\nu)^{-1}\Big)\\&\qquad\circ u_{\sigma\tau\nu}\circ\gamma_{\sigma,\tau\nu}\circ(1_{X_\sigma}\otimes\gamma_{\tau,\nu})\circ\mathsf{a},\end{aligned}$$

as maps $(X_\sigma\otimes X_\tau)\otimes X_\nu\to X'_{\sigma\tau\nu}$. On the other hand, composing (29) for (X,γ) with $u_{\sigma\tau\nu}$ and using (L1) gives

$$u_{\sigma\tau\nu}\circ\gamma_{\sigma\tau,\nu}\circ(\gamma_{\sigma,\tau}\otimes 1)=\lambda^\sharp\big(t(\sigma,\tau,\nu)\big)\circ u_{\sigma\tau\nu}\circ\gamma_{\sigma,\tau\nu}\circ(1_{X_\sigma}\otimes\gamma_{\tau,\nu})\circ\mathsf{a},$$

as maps $(X_\sigma\otimes X_\tau)\otimes X_\nu\to X'_{\sigma\tau\nu}$. Cancelling the common isomorphism and using injectivity of $\lambda^\sharp$,

$$\begin{aligned}t(\sigma,\tau,\nu)&=t'(\sigma,\tau,\nu)c(\sigma,\tau)c(\sigma\tau,\nu)c(\sigma,\tau\nu)^{-1}\,{}^\sigma c(\tau,\nu)^{-1}\\&=t'(\sigma,\tau,\nu)\cdot(\partial c)(\sigma,\tau,\nu)^{-1}\end{aligned}$$

by (11). □

Definition 5.7. For $f:G\to\pi_0(\mathcal{G})$, put $T(f):=[t]\in H^3(G,A_f)$, where t comes from any reduced system for f. Lemmas 5.4 and 5.6 make this independent of the system. Write $\mathcal{L}(f)$ for the set of monoidal isomorphism classes of monoidal functors $F:\underline{G}\to\mathcal{G}$ with $\pi_0(F)=f$.

Construction 5.8. For a reduced system (X,γ) for f, with obstruction cochain t, and a normalized 2-cochain $\beta:G^2\to A$, define maps $X_\sigma\otimes X_\tau\to X_{\sigma\tau}$ by

$$\big(\lambda^\sharp(\beta)\circ\gamma\big)_{\sigma,\tau}:=\lambda^\sharp_{X_{\sigma\tau}}\big(\beta(\sigma,\tau)\big)\circ\gamma_{\sigma,\tau}.$$

Lemma 5.9. *The system $(X,\lambda^\sharp(\beta)\circ\gamma)$ of Construction 5.8 is reduced and has associativity obstruction cochain $t\cdot(\partial\beta)^{-1}$.*

Proof. Normalization of $\lambda^\sharp(\beta)\circ\gamma$ follows from that of β and γ (and $\lambda^\sharp_X(1)=1_X$). Apply Lemma 5.6 to the systems (X,γ) and $(X',\gamma'):=(X,\lambda^\sharp(\beta)\circ\gamma)$ with $u_\sigma=1_{X_\sigma}$: equation (30) reads $\gamma_{\sigma,\tau}=\lambda^\sharp\big(c(\sigma,\tau)\,\beta(\sigma,\tau)\big)\circ\gamma_{\sigma,\tau}$ as maps $X_\sigma\otimes X_\tau\to X_{\sigma\tau}$, i.e. $c=\beta^{-1}$. Lemma 5.6 then gives $t=t'\cdot\big(\partial(\beta^{-1})\big)^{-1}=t'\cdot\partial\beta$, i.e. $t'=t\cdot(\partial\beta)^{-1}$. □

Construction 5.10. For a monoidal functor $(F,\varphi,\varphi_0):\underline{G}\to\mathcal{G}$, set $X_\sigma := F(\sigma)$ for $\sigma\neq e$ and $X_e := \mathbf{1}$; define $\theta_\sigma := 1_{F(\sigma)}$ for $\sigma\neq e$ and $\theta_e := \varphi_0\colon \mathbf{1}\to F(e)$; and define

$$\gamma_{\sigma,\tau} := \theta_{\sigma\tau}^{-1}\circ\varphi_{\sigma,\tau}\circ(\theta_\sigma\otimes\theta_\tau),$$

as maps $X_\sigma\otimes X_\tau\to X_{\sigma\tau}$.

Lemma 5.11. *Every monoidal functor $(F,\varphi,\varphi_0)\colon\underline{G}\to\mathcal{G}$ is monoidally isomorphic to one of the form $(X,\gamma,1_{\mathbf{1}})$ where (X,γ) is a reduced system for $\pi_0(F)$ with associativity obstruction cochain identically* 1.

Proof. By Construction 5.10, the family $\theta\colon(X,\gamma,1_{\mathbf{1}})\to(F,\varphi,\varphi_0)$ satisfies the conditions (25). The unit condition is

$$\theta_e\circ 1_{\mathbf{1}} = \varphi_0,$$

as maps $\mathbf{1}\to F(e)$. Once we know that $(X,\gamma,1_{\mathbf{1}})$ is a monoidal functor, θ is a monoidal isomorphism. The coherence axiom (23) for γ follows from the one for φ by direct substitution, using naturality of a to move $(\theta_\sigma\otimes\theta_\tau)\otimes\theta_\nu$ across the associator. It remains to verify (28). For $\sigma\neq e$, the unit axioms for (F,φ,φ_0), specialized to the strict discrete source $\underline{G}$ (where F takes unitors to identities), read $\varphi_{e,\sigma}\circ(\varphi_0\otimes 1_{F(\sigma)}) = \lambda_{F(\sigma)}$ and $\varphi_{\sigma,e}\circ(1_{F(\sigma)}\otimes\varphi_0) = \rho_{F(\sigma)}$, as maps $\mathbf{1}\otimes F(\sigma)\to F(\sigma)$ and $F(\sigma)\otimes\mathbf{1}\to F(\sigma)$, respectively. Since $X_\sigma = F(\sigma)$ for $\sigma\neq e$, the same sources and targets apply to the respective identities

$$\gamma_{e,\sigma} = \theta_\sigma^{-1}\circ\varphi_{e,\sigma}\circ(\varphi_0\otimes 1) = \lambda_{X_\sigma},\qquad \gamma_{\sigma,e} = \varphi_{\sigma,e}\circ(1\otimes\varphi_0) = \rho_{X_\sigma}.$$

For $\sigma=\tau=e$, the unit axiom and naturality of λ give

$$\begin{aligned}\gamma_{e,e} &= \varphi_0^{-1}\circ\varphi_{e,e}\circ(\varphi_0\otimes\varphi_0)\\ &= \varphi_0^{-1}\circ\big[\varphi_{e,e}\circ(\varphi_0\otimes 1_{F(e)})\big]\circ(1_{\mathbf{1}}\otimes\varphi_0)\\ &= \varphi_0^{-1}\circ\lambda_{F(e)}\circ(1_{\mathbf{1}}\otimes\varphi_0)\\ &= \varphi_0^{-1}\circ\varphi_0\circ\lambda_{\mathbf{1}} = \lambda_{\mathbf{1}},\end{aligned}$$

as maps $\mathbf{1}\otimes\mathbf{1}\to\mathbf{1}$, and $\lambda_{\mathbf{1}} = \rho_{\mathbf{1}}$ by (15), so both conditions of (28) hold at (e,e). Finally, since $(X,\gamma,1_{\mathbf{1}})$ is a monoidal functor, its associativity obstruction cochain is identically 1 by Construction 5.2. □

Construction 5.12. Let F,F' be monoidal functors with $\pi_0(F)=\pi_0(F')=f$. By Lemma 5.11, choose monoidal isomorphisms to reduced systems (X,γ), (X',γ') with associativity obstruction cochains identically 1; choose (u_σ) as in Construction 5.5 and let c be the resulting normalized cochain (30). Since $t=t'\equiv 1$, Lemma 5.6 gives $\partial c = 1$, i.e. $c\in Z^2(G,A_f)$. Define $\delta(F,F') := [c]\in H^2(G,A_f)$. For a normalized 2-cocycle β, the twisting assignment on classes is $[\beta]\cdot[(X,\gamma)] := [(X,\lambda^\sharp(\beta)\circ\gamma)]$. Independence of representatives for both assignments is proved below.

Theorem 5.13. *Let $A=\pi_1(\mathcal{G})$ and $f\colon G\to\pi_0(\mathcal{G})$ be as above.*

(i) *For every reduced system (X,γ) for f, the associativity obstruction cochain t is a normalized 3-cocycle: $t\in Z^3(G,A_f)$.*
(ii) *The class $T(f)\in H^3(G,A_f)$ does not depend on the choice of reduced system.*
(iii) *There exists a monoidal functor $F\colon\underline{G}\to\mathcal{G}$ with $\pi_0(F)=f$ if and only if $T(f)=1$.*

(iv) *Suppose $T(f) = 1$. The assignment δ of Construction 5.12 is a well-defined map $\mathcal{L}(f) \times \mathcal{L}(f) \to H^2(G, A_f)$ with*

$$\delta(x,z) = \delta(x,y)\cdot\delta(y,z), \qquad \big(\delta(x,y) = 1 \iff x = y\big),$$

and such that for each $y \in \mathcal{L}(f)$ the map $x \mapsto \delta(x,y)$ is a bijection $\mathcal{L}(f) \to H^2(G, A_f)$. Equivalently, $H^2(G, A_f)$ acts simply transitively on $\mathcal{L}(f)$ via the cochain twist of Construction 5.8.

Proof of (i) *and* (ii). These are Lemmas 5.4 and 5.6. □

Proof of Theorem 5.13(iii). If a monoidal functor F with $\pi_0(F) = f$ exists, Lemma 5.11 gives a reduced system with associativity obstruction cochain identically 1, so $T(f) = 1$ by part (ii). Conversely, suppose $T(f) = 1$, and let (X, γ) be a reduced system with obstruction cochain t. Then $t = \partial\beta$ for some 2-cochain β.

We first normalize β. Put $c_0 := \beta(e,e)$. Evaluating $\partial\beta = t$ at (e,τ,ν) and (σ,e,ν) and using (11) gives

$$\beta(e,\tau\nu) = \beta(e,\tau),$$
$$\beta(\sigma,e) = {}^{\sigma}c_0.$$

The first identity says that $\beta(e,-)$ is constant with value c_0. For the constant 1-cochain $b \equiv c_0$, formula (10) gives

$$(\partial b)(\sigma,\tau) = {}^{\sigma}c_0\, c_0^{-1}\, c_0 = {}^{\sigma}c_0.$$

Replacing β by $\beta\cdot(\partial b)^{-1}$ therefore normalizes it without changing its coboundary.

By Lemma 5.9, the reduced system $(X, \lambda^\sharp(\beta)\circ\gamma)$ has obstruction cochain $t\cdot(\partial\beta)^{-1} = 1$. Construction 5.2 then makes it a monoidal functor with component map f. □

Proof of Theorem 5.13(iv). Use the reduced systems and comparison cochain of Construction 5.12.

Independence of comparison maps. If $(\tilde{u}_\sigma)$ is another choice with $\tilde{u}_e = 1_{\mathbf{1}}$, then $\tilde{u}_\sigma = u_\sigma \circ \lambda^\sharp_{X_\sigma}(d(\sigma))$ for a unique map $d\colon G \to A$ with $d(e) = 1$ (Lemma 2.6(i)). Then, by (L1) and (30),

$$\tilde{u}_{\sigma\tau}\circ\gamma_{\sigma,\tau} = \lambda^\sharp_{X'_{\sigma\tau}}\big(d(\sigma\tau)\,c(\sigma,\tau)\big)\circ\gamma'_{\sigma,\tau}\circ(u_\sigma\otimes u_\tau),$$

while by (L2), (L3) and (L1),

$$\begin{aligned}\gamma'_{\sigma,\tau}\circ(\tilde{u}_\sigma\otimes\tilde{u}_\tau) &= \gamma'_{\sigma,\tau}\circ(u_\sigma\otimes u_\tau)\circ\big(\lambda^\sharp_{X_\sigma}(d(\sigma))\otimes 1_{X_\tau}\big)\circ\big(1_{X_\sigma}\otimes\lambda^\sharp_{X_\tau}(d(\tau))\big)\\ &= \lambda^\sharp_{X'_{\sigma\tau}}\big(d(\sigma)\,{}^{\sigma}d(\tau)\big)\circ\gamma'_{\sigma,\tau}\circ(u_\sigma\otimes u_\tau).\end{aligned}$$

Both calculations are equalities of maps $X_\sigma\otimes X_\tau \to X'_{\sigma\tau}$. Comparing with the defining relation (30) for $\tilde{c}$, $\tilde{c}(\sigma,\tau) = c(\sigma,\tau)\,d(\sigma\tau)\,d(\sigma)^{-1}\,{}^{\sigma}d(\tau)^{-1} = c(\sigma,\tau)\cdot(\partial d)(\sigma,\tau)^{-1}$ by (10), so $[\tilde{c}] = [c]$.

Multiplicativity. If (u_σ) compares (X,γ) to (X',γ') with cochain c, and (u'_σ) compares (X',γ') to (X'',γ'') with cochain c', then for $u''_\sigma := u'_\sigma\circ u_\sigma$, using (L1) and the interchange law, we have

$$u''_{\sigma\tau}\circ\gamma_{\sigma,\tau} = u'_{\sigma\tau}\circ\lambda^\sharp_{X'_{\sigma\tau}}\big(c(\sigma,\tau)\big)\circ\gamma'_{\sigma,\tau}\circ(u_\sigma\otimes u_\tau) = \lambda^\sharp_{X''_{\sigma\tau}}\big(c(\sigma,\tau)\,c'(\sigma,\tau)\big)\circ\gamma''_{\sigma,\tau}\circ(u''_\sigma\otimes u''_\tau),$$

as maps $X_\sigma\otimes X_\tau \to X''_{\sigma\tau}$, whence $\delta(F,F'') = \delta(F,F')\cdot\delta(F',F'')$. In particular $\delta(F,F) = 1$ (take $u = \mathrm{id}$), so $\delta(F',F) = \delta(F,F')^{-1}$.

Independence of reduced systems. If $\theta\colon (X,\gamma)\to(X',\gamma')$ is a monoidal isomorphism of reduced systems, then $\theta_e = 1_{\mathbf{1}}$ (by the unit condition in (25), with both unit constraints equal to $1_{\mathbf{1}}$), and taking $u_\sigma := \theta_\sigma$ in (30) yields $c\equiv 1$ directly from the first condition in (25). Together with the multiplicativity and the independence of the maps u_σ proved above, this shows that $\delta(F,F')$ does not depend on the chosen reduced systems of F and F', and that $\delta(F,F')=1$ whenever $F\cong F'$.

Detection of isomorphisms. With the reduced systems, (u_σ), and c chosen above, write $c=\partial d$. Any d with $\partial d = c$ is automatically normalized: $(\partial d)(e,\tau) = {}^{e}d(\tau)\,d(\tau)^{-1}\,d(e) = d(e)$, and $c(e,\tau)=1$ forces $d(e)=1$. Replace u_σ by $\tilde u_\sigma := u_\sigma\circ\lambda^\sharp_{X_\sigma}(d(\sigma))$; by the displayed formula in the independence step, $\tilde c = c\cdot(\partial d)^{-1}=1$, so (30) for $(\tilde u_\sigma)$ says precisely that $\theta := (\tilde u_\sigma)$ satisfies (25) and is a monoidal isomorphism $(X,\gamma)\to(X',\gamma')$; hence $F\cong F'$. Combined with the previous steps, $\delta(x,y)=1 \iff x=y$ on $\mathcal{L}(f)$.

Surjectivity and the action. Given a normalized $\beta\in Z^2(G,A_f)$ and a reduced system (X',γ') representing F' (associativity obstruction cochain identically 1), Lemma 5.9 shows that $(X',\lambda^\sharp(\beta)\circ\gamma')$ is again a monoidal functor with component map f; taking $u_\sigma = 1_{X'_\sigma}$ in (30), from $(X',\lambda^\sharp(\beta)\circ\gamma')$ to (X',γ'), gives $c=\beta$, so $\delta\big([(X',\lambda^\sharp(\beta)\circ\gamma')],\,[F']\big)=[\beta]$. Thus $x\mapsto\delta(x,[F'])$ is surjective; it is injective since $\delta(x,y)=\delta(x,z)$ implies $\delta(y,z)=\delta(y,x)\cdot\delta(x,z)=\delta(x,y)^{-1}\cdot\delta(x,z)=1$, hence $y=z$. Finally, the formula $[\beta]\cdot[(X',\gamma')]:=[(X',\lambda^\sharp(\beta)\circ\gamma')]$ defines the simply transitive action: it is well defined in $[\beta]$, because if $\beta'=\beta\cdot\partial d$ (with $d(e)=1$, automatic as above) then $\theta_\sigma := \lambda^\sharp_{X'_\sigma}(d(\sigma))$ is a monoidal isomorphism $(X',\lambda^\sharp(\beta')\circ\gamma')\to(X',\lambda^\sharp(\beta)\circ\gamma')$ — the verification is the displayed computation in the independence step — and it is well defined in the second argument and simply transitive because $\delta\big([\beta]\cdot x,\,x\big)=[\beta]$ together with the properties of δ already established. □

Construction 5.14. Let $\Xi:\mathcal{G}\to\mathcal{G}'$ be a monoidal functor between 2-groups, $A=\pi_1(\mathcal{G})$, and $A'=\pi_1(\mathcal{G}')$. Write (Ξ,χ,ξ_0) for the monoidal structure, with

$$\chi_{Y,Z}\colon \Xi(Y)\otimes\Xi(Z)\to\Xi(Y\otimes Z),\qquad \xi_0:\mathbf{1}_{\mathcal{G}'}\to\Xi(\mathbf{1}_{\mathcal{G}}).$$

Its induced homomorphism is

$$\Xi_1:=\pi_1(\Xi):A\to A',\qquad \Xi_1(a):=\xi_0^{-1}\circ\Xi(a)\circ\xi_0.$$

For a reduced system (X,γ), the normalized image system is specified as follows. Put $X'_\sigma := \Xi(X_\sigma)$ for $\sigma\neq e$ and $X'_e := \mathbf{1}_{\mathcal{G}'}$, and set

$$\theta_\sigma\colon X'_\sigma\longrightarrow\Xi(X_\sigma),\qquad \theta_\sigma=1\quad\text{for all }\sigma\neq e,\qquad \theta_e=\xi_0.$$

Take

$$\gamma'_{\sigma,\tau}:=\theta^{-1}_{\sigma\tau}\circ\Xi(\gamma_{\sigma,\tau})\circ\chi_{X_\sigma,X_\tau}\circ(\theta_\sigma\otimes\theta_\tau):X'_\sigma\otimes X'_\tau\longrightarrow X'_{\sigma\tau}.$$

Lemma 5.15. *Let $\Xi:\mathcal{G}\to\mathcal{G}'$ and Ξ_1 be as in Construction 5.14. Let $f:G\to\pi_0(\mathcal{G})$ be a group homomorphism and put $f':=\pi_0(\Xi)\circ f$. The coefficient map intertwines the canonical actions: for every object X of $\mathcal{G}$,*

$$\Xi_1\circ\varepsilon_X=\varepsilon_{\Xi(X)}\circ\Xi_1,$$

as maps $A\to A'$. Thus $\Xi_1:A_f\to A'_{f'}$ is a homomorphism of G-modules. If (X,γ) is a reduced system for f with associativity obstruction cochain t, its normalized image system

has obstruction cochain $t' = \Xi_1 \circ t$. *The induced cohomology map satisfies*

$$(\Xi_1)_* : H^3(G, A_f) \longrightarrow H^3(G, A'_{f'}),$$
$$T(f') = (\Xi_1)_* T(f).$$

In particular, if Ξ *is a monoidal autoequivalence of* $\mathcal{G}$ *with* $\pi_1(\Xi) = \mathrm{id}_A$, *then* $A_{f'} = A_f$ *and*

$$T(\pi_0(\Xi) \circ f) = T(f) \in H^3(G, A_f). \tag{31}$$

Proof. Use Construction 5.14. The unit coherence of Ξ makes (X', γ') reduced, including when $\sigma\tau = e$, and θ identifies this reduced system with the image of (X, γ) under Ξ. Applying Ξ to the defining relation (29), then using the associativity coherence of χ and conjugating by the relevant θ's, gives the same associativity comparison for (X', γ') with the automorphism

$$X'_{\sigma\tau\nu} \longrightarrow X'_{\sigma\tau\nu},$$
$$\theta^{-1}_{\sigma\tau\nu} \circ \Xi\big(\lambda^\sharp_{X_{\sigma\tau\nu}}(t(\sigma,\tau,\nu))\big) \circ \theta_{\sigma\tau\nu}.$$

The left unit axiom is $\Xi(\lambda_X) \circ \chi_{\mathbf{1},X} \circ (\xi_0 \otimes 1) = \lambda_{\Xi X}$ as maps $\mathbf{1}_{\mathcal{G}'} \otimes \Xi(X) \to \Xi(X)$. Together with naturality of χ, it gives

$$\Xi(\lambda^\sharp_X(a)) = \lambda^\sharp_{\Xi(X)}(\Xi_1(a)), \qquad \Xi(\rho^\sharp_X(a)) = \rho^\sharp_{\Xi(X)}(\Xi_1(a)),$$

as identities of endomorphisms of $\Xi(X)$; the second follows from the right unit axiom in the same way. By Definition 2.7, for every $a \in A$,

$$\begin{aligned}\lambda^\sharp_{\Xi(X)}\big(\Xi_1(\varepsilon_X(a))\big) &= \Xi\big(\lambda^\sharp_X(\varepsilon_X(a))\big) = \Xi(\rho^\sharp_X(a)) \\ &= \rho^\sharp_{\Xi(X)}(\Xi_1(a)) = \lambda^\sharp_{\Xi(X)}\big(\varepsilon_{\Xi(X)}(\Xi_1(a))\big),\end{aligned}$$

as maps $\Xi(X) \longrightarrow \Xi(X)$. Injectivity of $\lambda^\sharp_{\Xi(X)}$ gives the claimed intertwining identity. Since $[\Xi(X_\sigma)] = f'(\sigma)$, it says exactly that $\Xi_1 : A_f \to A'_{f'}$ is G-equivariant. The scalar of the first displayed automorphism is $\Xi_1(t(\sigma,\tau,\nu))$, so $t' = \Xi_1 \circ t$ as maps $G^3 \longrightarrow A'$. Passing to cohomology gives $T(f') = [t'] = (\Xi_1)_*[t] = (\Xi_1)_* T(f)$. If $\Xi_1 = \mathrm{id}_A$, action compatibility gives $A_{f'} = A_f$ and $t' = t$, proving (31). □

5.3. **Automorphisms of a lift.** Let $\mathbf{Mon}_f(\underline{G}, \mathcal{G})$ be the full groupoid of monoidal lifts of $f\colon G \to \pi_0(\mathcal{G})$, that is

$$\mathbf{Mon}_f(\underline{G}, \mathcal{G}) := \big\{F \in \mathbf{Mon}(\underline{G}, \mathcal{G}) \mid \pi_0(F) = f\big\}.$$

We compute the monoidal automorphisms of a lift F and the subgroup arising from A. Together with Theorem 5.13, this identifies the roles of $H^i(G, A_f)$, for $0 \le i \le 3$, in the lifting problem.

Proposition 5.16. *Let* $F = (X, \gamma, \gamma_0)$ *be a monoidal functor with* $\pi_0(F) = f$. *The assignment* $\theta \mapsto b$, *where* $\theta_\sigma = \lambda^\sharp_{X_\sigma}(b(\sigma))$, *is an isomorphism of groups*

$$\mathrm{Aut}_{\mathbf{Mon}(\underline{G},\mathcal{G})}(F) \xrightarrow{\ \sim\ } Z^1(G, A_f).$$

In particular, $\mathrm{Aut}(F)$ *is abelian and is canonically independent of* F *within* $\mathbf{Mon}_f(\underline{G}, \mathcal{G})$.

Proof. By Lemma 2.6(i), every $\theta_\sigma \in \mathrm{Aut}_{\mathcal{G}}(X_\sigma)$ is uniquely of the form $\lambda^\sharp_{X_\sigma}(b(\sigma))$ for some $b(\sigma) \in A$. The unit condition in (25), $\theta_e \circ \gamma_0 = \gamma_0$ as maps $\mathbf{1} \to X_e$, forces $\theta_e = 1_{X_e}$, i.e. $b(e) = 1$. For the naturality condition in (25), use rules (L2), (L3), (L1) of §5 (the last with $[X_\sigma] = f(\sigma)$, so that $[X_\sigma] \cdot b(\tau) = {}^\sigma b(\tau)$). We have

$$\gamma_{\sigma,\tau} \circ (\theta_\sigma \otimes \theta_\tau) = \gamma_{\sigma,\tau} \circ \big(\lambda^\sharp_{X_\sigma}(b(\sigma)) \otimes 1\big) \circ \big(1 \otimes \lambda^\sharp_{X_\tau}(b(\tau))\big) = \lambda^\sharp_{X_{\sigma\tau}}\big(b(\sigma)\,{}^\sigma b(\tau)\big) \circ \gamma_{\sigma,\tau},$$

while $\theta_{\sigma\tau} \circ \gamma_{\sigma,\tau} = \lambda^\sharp_{X_{\sigma\tau}}(b(\sigma\tau)) \circ \gamma_{\sigma,\tau}$. Both comparisons are equalities of maps $X_\sigma \otimes X_\tau \to X_{\sigma\tau}$. By injectivity of $\lambda^\sharp_{X_{\sigma\tau}}$, condition (25) is equivalent to $b(\sigma\tau) = b(\sigma)\,{}^\sigma b(\tau)$ for all σ, τ, i.e. to $b \in Z^1(G, A_f)$. The assignment is a homomorphism because each $\lambda^\sharp_{X_\sigma}$ is and composition of monoidal automorphisms is componentwise; it is bijective by the uniqueness of b. Independence of F: the target $Z^1(G, A_f)$ does not depend on F, only on f. □

Construction 5.17. For $a \in A(= \pi_1(\mathcal{G}))$, define a family of automorphisms by

$$\theta^a_\sigma := \lambda^\sharp_{X_\sigma}(a)^{-1} \circ \rho^\sharp_{X_\sigma}(a) \in \mathrm{Aut}_{\mathcal{G}}(X_\sigma) \quad \text{for all } \sigma \in G.$$

Notation 5.18. For F and θ^a as in Construction 5.17, write

$$b^a(\sigma) := a^{-1\,\sigma}a, \qquad \iota_F(a) := \theta^a, \qquad A^G := \{a \in A \mid {}^\sigma a = a \text{ for all } \sigma \in G\}.$$

Lemma 5.19. *The family θ^a is a monoidal automorphism of F, corresponding under Proposition 5.16 to b^a. The assignment $\iota_F : A \to \mathrm{Aut}_{\mathbf{Mon}(\underline{G},\mathcal{G})}(F)$ is a homomorphism with image $B^1(G, A_f)$ and kernel $A^G = H^0(G, A_f)$.*

Proof. By Definition 2.7, $\rho^\sharp_{X_\sigma}(a) = \lambda^\sharp_{X_\sigma}(\varepsilon_{X_\sigma}(a)) = \lambda^\sharp_{X_\sigma}({}^\sigma a)$, using $[X_\sigma] = f(\sigma)$. Hence, since $\lambda^\sharp_{X_\sigma}$ is a homomorphism,

$$\begin{aligned}\theta^a_\sigma &= \lambda^\sharp_{X_\sigma}(a)^{-1} \circ \lambda^\sharp_{X_\sigma}({}^\sigma a)\\ &= \lambda^\sharp_{X_\sigma}\big(a^{-1\,\sigma}a\big),\end{aligned}$$

as maps $X_\sigma \to X_\sigma$, so θ^a corresponds to b^a. Regard $a \in A$ as an element of $C^0(G, A_f)$. The degree-zero coboundary is $\partial : C^0(G, A_f) \to C^1(G, A_f)$, and commutativity of A gives

$$(\partial a)(\sigma) = {}^\sigma a \cdot a^{-1} = a^{-1\,\sigma}a = b^a(\sigma).$$

Thus b^a is a principal crossed homomorphism, so $b^a \in Z^1(G, A_f)$ and $\theta^a \in \mathrm{Aut}(F)$ by Proposition 5.16.

The assignment $a \mapsto b^a$ is a homomorphism because

$$\begin{aligned}b^a(\sigma)b^{a'}(\sigma) &= a^{-1\sigma}a\,a'^{-1\sigma}a'\\ &= (aa')^{-1\,\sigma}(aa') = b^{aa'}(\sigma).\end{aligned}$$

Its image is $B^1(G, A_f)$ by definition, and its kernel is

$$\{a \in A \mid a^{-1\,\sigma}a = 1 \text{ for all } \sigma \in G\} = A^G = H^0(G, A_f).$$

□

Theorem 5.20. *Let $f\colon G \to \pi_0(\mathcal{G})$ be realizable, i.e. $T(f) = 1$, and let F be any lift. Then:*

(i) *there is an exact sequence of abelian groups*

$$\{1\} \longrightarrow H^0(G, A_f) \longrightarrow A \xrightarrow{\iota_F} \mathrm{Aut}_{\mathbf{Mon}(\underline{G},\mathcal{G})}(F) \longrightarrow H^1(G, A_f) \longrightarrow \{1\};$$

(ii) *consequently the based lift problem for f is governed by the groups $H^i(G, A_f)$, for $0 \le i \le 3$:*

$T(f) \in H^3(G, A_f)$ *is the obstruction (Theorem 5.13(iii))*;

$\mathcal{L}(f) = \pi_0(\mathbf{Mon}_f(\underline{G}, \mathcal{G}))$ *is a torsor under $H^2(G, A_f)$ (Theorem 5.13(iv))*;

$\mathrm{Aut}(F) \cong Z^1(G, A_f)$, $\quad \mathrm{Aut}(F)/\iota_F(A) \cong H^1(G, A_f)$, $\quad \ker \iota_F \cong H^0(G, A_f)$.

Proof. By Proposition 5.16, $\mathrm{Aut}(F) \cong Z^1(G, A_f)$, and by Lemma 5.19 the composite $A \xrightarrow{\iota_F} \mathrm{Aut}(F) \cong Z^1(G, A_f)$ is $a \mapsto b^a$, with image $B^1(G, A_f)$ and kernel $H^0(G, A_f)$. Exactness of the displayed sequence is then the definition $H^1(G, A_f) = Z^1(G, A_f)/B^1(G, A_f)$ together with $H^0(G, A_f) = \ker(\partial\colon C^0(G, A_f) \to C^1(G, A_f)) = A^G$. This is (i); part (ii) collects (i) with Theorem 5.13(iii)–(iv). □

6. Classification of Clifford extensions

We now specialize $\mathcal{G} := \boldsymbol{\mathcal{P}ic}_k(R)$ for a K-central k-algebra R, and combine §§3–5. Throughout, $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$, and we use the identification $\pi_1(\boldsymbol{\mathcal{P}ic}_k(R)) = \mathrm{U}(K)$ of (26) together with Corollary 3.7: for any GCC $\Phi\colon G \to \mathrm{Pic}_k(R)$ of type ϕ, the induced G-module is $\mathrm{U}(K)_\phi$.

6.1. The associativity obstruction.

Definition 6.1. A G-graded Clifford system extension $(A, j) \in \mathbf{Cliff}_G(R)$ has *type* ϕ if its associated GCC $\chi_R[A]$ has type ϕ, i.e. $\eta_{A_g} = \phi(g)$ for all $g \in G$; equivalently, by Lemma 3.4(i), if

$$a\, j(\alpha) = j\big(\phi(g)(\alpha)\big)\, a \qquad \text{for all } g \in G,\ a \in A_g,\ \alpha \in K. \tag{32}$$

Theorem 6.2. *Let $\Phi\colon G \to \mathrm{Pic}_k(R)$ be a GCC of type ϕ. Choose, as in* [Gin22, §5] *(following* [CG01, §3]*), representatives $P_\sigma \in \Phi(\sigma)$ with $P_e = R$, and bimodule isomorphisms*

$$\gamma_{\sigma,\tau}\colon P_\sigma \otimes_R P_\tau \to P_{\sigma\tau}$$

subject to the normalization [Gin22, (5.1)]*:*

$$\gamma_{\sigma,e}(p \otimes_R r) = p \star_\iota r, \qquad \gamma_{e,\sigma}(r \otimes_R p) = r \star_\ell p \quad \text{for all } \sigma \in G,\ r \in R,\ p \in P_\sigma.$$

Then $\big((P_\sigma), (\gamma_{\sigma,\tau})\big)$ is a reduced system for Φ in the sense of Definition 5.1, and its associativity obstruction cochain t (Construction 5.2) coincides, as a function $G^3 \to \mathrm{U}(K)$, with the cochain T^Φ defined by [Gin22, (5.2)]*:*

$$\gamma_{\sigma\tau,\nu}\big(\gamma_{\sigma,\tau}(p_\sigma \otimes_R p_\tau) \otimes_R p_\nu\big) = T^\Phi_{\sigma,\tau,\nu} \star_\ell \gamma_{\sigma,\tau\nu}\big(p_\sigma \otimes_R \gamma_{\tau,\nu}(p_\tau \otimes_R p_\nu)\big).$$

Consequently the class $T(\Phi) \in H^3(G, \mathrm{U}(K)_\phi)$ defined by Theorem 5.13(ii) *equals the obstruction class $T(\Phi)$ of* [CG01, §3]*. In particular, Theorem 5.13*(i)–(ii) *reprove* [Gin22, Thm. 5.1(1),(2)].

Proof. The conditions [Gin22, (5.1)] say exactly that $\gamma_{\sigma,e} = \rho_{P_\sigma}$ and $\gamma_{e,\sigma} = \lambda_{P_\sigma}$ in $\boldsymbol{\mathcal{P}ic}_k(R)$ (Proposition 3.5 identifies the unit constraints with the two actions), and $P_e = R = \mathbf{1}$; so the data form a reduced system for Φ. By Proposition 3.5, $\lambda^\sharp_{P_{\sigma\tau\nu}}(\alpha)$ is the automorphism $p \mapsto \alpha \star_\ell p$ of $P_{\sigma\tau\nu}$, so the defining relation (29), evaluated on $(p_\sigma \otimes_R p_\tau) \otimes_R p_\nu$ and using that the canonical associator of $\otimes_R$ sends $(p_\sigma \otimes_R p_\tau) \otimes_R p_\nu$ to $p_\sigma \otimes_R (p_\tau \otimes_R p_\nu)$, reads

$$\gamma_{\sigma\tau,\nu}\big(\gamma_{\sigma,\tau}(p_\sigma \otimes_R p_\tau) \otimes_R p_\nu\big) = t(\sigma,\tau,\nu) \star_\ell \gamma_{\sigma,\tau\nu}\big(p_\sigma \otimes_R \gamma_{\tau,\nu}(p_\tau \otimes_R p_\nu)\big),$$

which is [Gin22, (5.2)] in the action notation of Definition 3.1; hence $t = T^{\Phi}$ by the uniqueness in either formulation. The G-module structures on $\mathrm{U}(K)$ agree by Corollary 3.7. Independence of the cohomology class under all choices is Theorem 5.13(ii); the cocycle property is Theorem 5.13(i). □

6.2. **Clifford extensions of R.**

Notation 6.3. Let R be a K-central k-algebra. Write $\mathrm{Cliff}_k(G,R)_\phi$ for classes of Clifford extensions of type ϕ under graded isomorphisms fixing R, and $\mathrm{Hom}_\phi(G, \mathrm{Pic}_k(R))$ for characters of this type.

Theorem 6.4. *For a K-central k-algebra R and a type ϕ, the character map and obstruction map*

$$\mathrm{Cliff}_k(G,R)_\phi \xrightarrow{\chi_R} \mathrm{Hom}_\phi(G, \mathrm{Pic}_k(R)) \xrightarrow{T} H^3(G, \mathrm{U}(K)_\phi)$$

satisfy $\operatorname{Im}\chi_R = T^{-1}(1)$*: a character is realizable exactly when its obstruction vanishes* [Gin22, Thm. 5.1(3)]. *For each such character Φ, the fibre $\chi_R^{-1}(\Phi)$ is a torsor under $H^2(G, \mathrm{U}(K)_\phi)$.*

Proof. The equivalence of Theorem 4.5 identifies $\chi_R^{-1}(\Phi)$ with monoidal-isomorphism classes of lifts of Φ to $\boldsymbol{\mathcal{P}ic}_k(R)$. Apply Theorem 5.13 and the cocycle identification of Theorem 6.2. □

Consider the group

$$Z^1(G, \mathrm{U}(K)_\phi) = \big\{\lambda\colon G \to \mathrm{U}(K) \mid \lambda(gh) = \lambda(g)\cdot \phi(g)(\lambda(h)) \text{ for all } g,h \in G\big\}$$

of crossed homomorphisms under pointwise multiplication.

Construction 6.5. Let $(A,j) \in \mathbf{Cliff}_G(R)$ be of type ϕ. For $\lambda \in Z^1(G, \mathrm{U}(K)_\phi)$, set

$$\psi_\lambda(a) := j\big(\lambda(g)\big)\, a \quad \text{for all } a \in A_g.$$

Proposition 6.6. *The assignment $\lambda \mapsto \psi_\lambda$ of Construction 6.5 gives rise to an isomorphism of groups*

$$Z^1(G, \mathrm{U}(K)_\phi) \xrightarrow{\ \sim\ } \mathrm{Aut}_{\mathbf{Cliff}_G(R)}(A,j).$$

In particular, $\mathrm{Aut}_{\mathbf{Cliff}_G(R)}(A,j)$ *is abelian and is independent, up to canonical isomorphism, of (A,j) within a given type.*

Proof. Theorem 4.5 and Proposition 5.16 give the asserted bijection and group law. Under $\lambda^\sharp_{A_g}(u)(a) = j(u)a$, its formula is ψ_λ. For $a \in A_g$ and $b \in A_h$, the type relation gives

$$\psi_\lambda(a)\psi_\lambda(b) = j\big(\lambda(g)\phi(g)(\lambda(h))\big)ab = j(\lambda(gh))ab = \psi_\lambda(ab).$$

Thus the abstract monoidal automorphism is exactly the graded algebra automorphism displayed here. □

Note that with the above notation $\lambda(e) = 1$, since $\lambda(e) = \lambda(e)\,\phi(e)(\lambda(e)) = \lambda(e)^2$.

Definition 6.7. For $(A,j) \in \mathbf{Cliff}_G(R)$ of type ϕ and $u \in \mathrm{U}(K)$, let $\theta^u(x) := j(u)^{-1}xj(u)$, and write

$$\mathrm{Inn}_{\mathrm{U}(K)}(A,j) := \{\theta^u \mid u \in \mathrm{U}(K)\}, \qquad \mathrm{U}(K)^\phi := \{u \in \mathrm{U}(K) \mid \phi(g)(u) = u \text{ for all } g \in G\}.$$

Proposition 6.8. *For the conjugations of Definition 6.7:*

(i) *θ^u is an automorphism of (A,j) in $\mathbf{Cliff}_G(R)$; it fixes the base $A_e = j(R)$ pointwise, and on A_g it acts by $\theta^u(x) = j\big(u^{-1}\,\phi(g)(u)\big)\,x$.*

(ii) *Under the isomorphism* $\mathrm{Aut}_{\mathbf{Cliff}_G(R)}(A,j) \cong Z^1(G,\mathrm{U}(K)_\phi)$ *of Proposition 6.6, the map* $u \mapsto \theta^u$ *corresponds to the homomorphism* ι_F *of Lemma 5.19, with* $F = \Theta(A,j)$*; it sends* u *to the principal crossed homomorphism* $\sigma \mapsto u^{-1}\,\phi(\sigma)(u)$.

(iii) *There is an exact sequence of abelian groups*

$$\{1\} \longrightarrow \mathrm{U}(K)^\phi \longrightarrow \mathrm{U}(K) \xrightarrow{u\mapsto\theta^u} \mathrm{Aut}_{\mathbf{Cliff}_G(R)}(A,j) \longrightarrow H^1(G,\mathrm{U}(K)_\phi) \longrightarrow \{1\}.$$

Equivalently,

$$\mathrm{Aut}_{\mathbf{Cliff}_G(R)}(A,j)\big/\,\mathrm{Inn}_{\mathrm{U}(K)}(A,j) \;\cong\; H^1(G,\mathrm{U}(K)_\phi).$$

Proof. (i) For $x \in A_g$, using the bimodule structure via j and the type relation (32),

$$\begin{aligned}\theta^u(x) = j(u)^{-1}\,x\,j(u) = u^{-1} \star_\ell (x \star_\iota u) = u^{-1} \star_\ell \big(\phi(g)(u) \star_\ell x\big)\\ = \big(u^{-1}\,\phi(g)(u)\big) \star_\ell x = j\big(u^{-1}\,\phi(g)(u)\big)\,x.\end{aligned}$$

This is $\lambda^\sharp_{A_g}\big(u^{-1}\phi(g)(u)\big)(x)$ by Proposition 3.5, so $\theta^u(A_g) = A_g$; thus θ^u preserves the G-grading, and it is a k-algebra automorphism, being conjugation by a unit. For $r \in R$, since $u \in K = Z(R)$, $\theta^u(j(r)) = j(u)^{-1}j(r)j(u) = j(u^{-1}ru) = j(r)$; hence θ^u fixes $A_e = j(R)$ pointwise and $\theta^u \circ j = j$ as maps $R \to A$, so θ^u is an automorphism of (A,j) in $\mathbf{Cliff}_G(R)$.

(ii) By (i) and Proposition 3.5, θ^u corresponds, under Proposition 6.6, to the crossed homomorphism $\lambda^u(\sigma) = u^{-1}\,\phi(\sigma)(u)$. Comparing with Construction 5.17 and Lemma 5.19 (with $F = \Theta(A,j)$, $a = u$, and the canonical action ${}^\sigma u = \phi(\sigma)(u)$ of Corollary 3.7), $\theta^u = \theta^a = \iota_F(u)$ and $\lambda^u = b^u$.

(iii) This is Theorem 5.20(i) for $\mathcal{G} = \boldsymbol{\mathcal{P}ic}_k(R)$ and $F = \Theta(A,j)$, using the identification in Proposition 6.6: the kernel of $u \mapsto \theta^u$ is $\mathrm{U}(K)^G = \mathrm{U}(K)^\phi$ by Lemma 5.19, the image is $B^1(G,\mathrm{U}(K)_\phi) = \mathrm{Inn}_{\mathrm{U}(K)}(A,j)$, and the cokernel is $H^1(G,\mathrm{U}(K)_\phi)$. □

Remark 6.9. The morphisms of $\mathbf{Cliff}_G(R)$ fix R pointwise. The automorphism groups of its objects of type ϕ are therefore $Z^1(G,\mathrm{U}(K)_\phi)$, not $H^1(G,\mathrm{U}(K)_\phi)$. Proposition 6.8 identifies the principal crossed homomorphisms with conjugations by central units of R, so quotienting by those conjugations gives $H^1(G,\mathrm{U}(K)_\phi)$, with kernel $\mathrm{U}(K)^\phi = H^0(G,\mathrm{U}(K)_\phi)$ on the conjugating units. In the unbased homotopy description, these conjugations are homotopies between paths; see the dictionary before Construction 8.11. They are not additional identifications in $\mathbf{Cliff}_G(R)$ itself.

6.3. **Crossed products over K.** We conclude with the crossed-product computation for $R = K$, recovering the embedding Σ_ϕ of [Gin22, (4.12)] and the classification of [Gin22, Lemmas 4.4 and 4.8] for graded isomorphisms fixing K.

Construction 6.10. For $\eta \in \mathrm{Aut}_k(K)$, let $K_{1,\eta}$ denote the K-bimodule which is K as a left K-module, with right action $s \star_\iota r := s\,\eta(r)$ (cf. [Gin22, (2.3)]). For $\eta, \eta' \in \mathrm{Aut}_k(K)$, use the assignment

$$\gamma^{\eta,\eta'} \colon K_{1,\eta} \otimes_K K_{1,\eta'} \longrightarrow K_{1,\eta\circ\eta'}, \qquad x \otimes_K y \longmapsto x\,\eta(y). \tag{33}$$

Lemma 6.11. *The assignment* $\gamma^{\eta,\eta'}$ *of Construction 6.10 is an isomorphism of* K*-bimodules. Consequently each* $K_{1,\eta}$ *is an invertible* K*-bimodule over* k *with inverse* $K_{1,\eta^{-1}}$*. Furthermore,* $K_{1,\eta}$ *is of type* η *(see Lemma 3.4), that is* $\eta_{K_{1,\eta}} = \eta$.

Proof. Since η fixes k, the two k-actions coincide. The formula $x \otimes_K y \mapsto x\eta(y)$ is balanced because $x\eta(r)\eta(y) = x\eta(ry)$, and its right-linearity follows from $\eta(y\eta'(r)) = \eta(y)(\eta \circ \eta')(r)$; left-linearity is immediate. Its inverse is $z \mapsto z \otimes_K 1$, since

$$x\eta(y) \otimes_K 1 = (x \star_\iota y) \otimes_K 1 = x \otimes_K y.$$

Taking $\eta' = \eta^{-1}$, in both orders, proves invertibility. Finally $s \star_\iota \alpha = s\eta(\alpha) = \eta(\alpha) \star_\ell s$, so the type is η by Lemma 3.4. □

We use the crossed-product convention of [Gin22, §4].

Definition 6.12. Let $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$ and let $\alpha \in Z^2(G, \mathrm{U}(K)_\phi)$ be a normalized 2-cocycle. The *crossed product* $K^\alpha_\phi G$ is the free left K-module on symbols $\{u_\sigma\}_{\sigma \in G}$ with multiplication

$$(x\,u_\sigma)(y\,u_\tau) := x\,\phi(\sigma)(y)\,\alpha(\sigma,\tau)\,u_{\sigma\tau} \quad \text{for all } x, y \in K \text{ and } \sigma, \tau \in G.$$

For the special case in which $\alpha(\sigma,\tau) = 1$ for all σ, τ, we write the resulting skew group algebra as $K_\phi G$. When ϕ is the trivial action this is the group algebra, denoted KG.

Notation 6.13. For $\phi : G \to \mathrm{Aut}_k(K)$, write

$$\mathbf{1}_\phi \colon G \to \mathrm{Pic}_k(K), \qquad \sigma \mapsto [K_{1,\phi(\sigma)}].$$

Lemma 6.14. *$K^\alpha_\phi G$ is a strongly G-graded k-algebra with $(K^\alpha_\phi G)_\sigma = K\,u_\sigma$. Its unit is u_e, and the map $j \colon K \to Ku_e$, $x \mapsto x\,u_e$, makes $(K^\alpha_\phi G,\, j)$ a G-graded Clifford system extension of K of type ϕ, whose associated GCC is $\mathbf{1}_\phi$ of Notation 6.13.*

Proof. Associativity: for $x, y, z \in K$,

$$\begin{aligned}
\big((xu_\sigma)(yu_\tau)\big)(zu_\nu) &= x\,\phi(\sigma)(y)\,\alpha(\sigma,\tau)\,\phi(\sigma\tau)(z)\,\alpha(\sigma\tau,\nu)\,u_{\sigma\tau\nu},\\
(xu_\sigma)\big((yu_\tau)(zu_\nu)\big) &= (xu_\sigma)\big(y\,\phi(\tau)(z)\,\alpha(\tau,\nu)\,u_{\tau\nu}\big)\\
&= x\,\phi(\sigma)(y)\,\phi(\sigma\tau)(z)\,{}^\sigma\alpha(\tau,\nu)\,\alpha(\sigma,\tau\nu)\,u_{\sigma\tau\nu}.
\end{aligned}$$

Since K is commutative, these agree for all x, y, z if and only if

$$\alpha(\sigma,\tau)\,\alpha(\sigma\tau,\nu) = {}^\sigma\alpha(\tau,\nu)\,\alpha(\sigma,\tau\nu),$$

which is the cocycle condition $(\partial\alpha)(\sigma,\tau,\nu) = 1$ of (11). Normalization of α gives $u_e\,(yu_\tau) = y\,\alpha(e,\tau)\,u_\tau = yu_\tau$ and $(xu_\sigma)\,u_e = x\,\phi(\sigma)(1)\,\alpha(\sigma,e)\,u_\sigma = xu_\sigma$, so u_e is a two-sided unit. The structure map $k \to K^\alpha_\phi G$, $c \mapsto c\,u_e$, is central, because $\phi(\sigma)|_k = \mathrm{id}$ gives $(xu_\sigma)(cu_e) = x\,\phi(\sigma)(c)\,u_\sigma = cx\,u_\sigma = (cu_e)(xu_\sigma)$. The G-grading $(K^\alpha_\phi G)_\sigma = Ku_\sigma$ is strong: $Ku_\sigma \cdot Ku_\tau$ is a left K-submodule of $Ku_{\sigma\tau}$ containing $u_\sigma u_\tau = \alpha(\sigma,\tau)\,u_{\sigma\tau}$ with $\alpha(\sigma,\tau) \in \mathrm{U}(K)$, hence equals $Ku_{\sigma\tau}$. The map j is a k-algebra isomorphism onto Ku_e: $j(x)\,j(y) = (xu_e)(yu_e) = xy\,\alpha(e,e)\,u_e = j(xy)$. Finally, the K-bimodule structure of Ku_σ via j is $j(r)(xu_\sigma) = rx\,u_\sigma$ and $(xu_\sigma)j(r) = x\,\phi(\sigma)(r)\,u_\sigma$; hence $xu_\sigma \mapsto x$ is an isomorphism of K-bimodules $Ku_\sigma \xrightarrow{\sim} K_{1,\phi(\sigma)}$. Thus $\chi_K[K^\alpha_\phi G](\sigma) = [K_{1,\phi(\sigma)}] = \mathbf{1}_\phi(\sigma)$, and $\eta_{(K^\alpha_\phi G)_\sigma} = \eta_{K_{1,\phi(\sigma)}} = \phi(\sigma)$ by Lemma 6.11, so the extension has type ϕ. The fact that $\mathbf{1}_\phi$ is a group homomorphism follows from Theorem 4.5 applied to $K_\phi G$, or directly from Lemma 6.11. □

Corollary 6.15. *Let $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$ and let $\mathbf{1}_\phi$ be as in Lemma 6.14. Then:*

(i) $T(\mathbf{1}_\phi) = 1$.

(ii) *The assignment* $[\alpha] \mapsto [K_\phi^\alpha G]$*, using any normalized representative* α*, is a well-defined bijection*

$$H^2(G, \mathrm{U}(K)_\phi) \xrightarrow{\sim} \chi_K^{-1}(\mathbf{1}_\phi) := \big\{[A] \in \mathrm{Cliff}_k(G,K) \mid \chi_K[A] = \mathbf{1}_\phi\big\}.$$

For normalized cocycles α, β *with* $\alpha = \beta\,\partial b$ *for a* 1*-cochain* b*, the comparison is*

$$K_\phi^\alpha G \longrightarrow K_\phi^\beta G, \qquad x u_g^\alpha \longmapsto x b(g) u_g^\beta,$$

where the superscript on u_g *specifies its crossed product. This recovers the crossed-product classification over* K *of* [Gin22, Lemmas 4.4 and 4.8].

Proof. The skew group algebra $K_\phi G$ realizes $\mathbf{1}_\phi$ by Lemma 6.14, so its obstruction vanishes. Twisting its multiplication by α gives exactly $K_\phi^\alpha G$. The torsor statement of Theorem 6.4 proves the bijection. For the comparison map, normalization gives $b(e) = (\partial b)(e,e) = \alpha(e,e)\beta(e,e)^{-1} = 1$, so b is normalized and the map fixes K. The multiplication formula in Definition 6.12 verifies that it is an algebra homomorphism. Since every $b(g)$ is a unit, it is a graded isomorphism. □

7. The exact sequence for extensions of K

We take $R = K$ and pass from generalized collective characters to their classes in $H^1(G, \mathrm{Pic}(K)_\phi)$. The crossed-product map, character map, and associativity obstruction give the exact sequence analogous to [Gin22, (6.1)]. Remark 7.12 compares it with the corresponding portion of the Chase–Harrison–Rosenberg sequence.

We write $\mathrm{Pic}(K) := \mathrm{Pic}_K(K)$ for the Picard group of the commutative ring K: the group of isomorphism classes of invertible K-modules (equivalently, of invertible K-bimodules P over k whose two K-actions coincide, i.e. with $\eta_P = \mathrm{id}_K$), under $\otimes_K$.

7.1. The semidirect decomposition of $\mathrm{Pic}_k(K)$.

Definition 7.1. For $\eta \in \mathrm{Aut}_k(K)$ and $[P] \in \mathrm{Pic}(K)$, put

$$s(\eta) := [K_{1,\eta}], \qquad \eta \cdot [P] := s(\eta)[P]s(\eta)^{-1}.$$

Proposition 7.2. *Let* K *be commutative. The type homomorphism* $h_K \colon \mathrm{Pic}_k(K) \to \mathrm{Aut}_k(K)$ *of Lemma 3.4 is a split surjection, with kernel* $\mathrm{Pic}(K)$ *and with splitting* s *of Definition 7.1. Consequently there is an internal semidirect-product decomposition*

$$\mathrm{Pic}_k(K) \;\cong\; \mathrm{Pic}(K) \rtimes \mathrm{Aut}_k(K),$$

in which $\eta \in \mathrm{Aut}_k(K)$ *acts on* $[P] \in \mathrm{Pic}(K)$ *by the action of Definition 7.1.*

Proof. By Lemma 3.4, h_K is a group homomorphism. Its kernel is $\{[P] \in \mathrm{Pic}_k(K) \mid \eta_P = \mathrm{id}\}$, which is exactly $\mathrm{Pic}(K)$: for an invertible K-bimodule P over k, the condition $\eta_P = \mathrm{id}$ says $p \star_\iota \alpha = \alpha \star_\ell p$ for all $\alpha \in K$, i.e. the two K-actions coincide and P is an invertible K-module; conversely such modules satisfy $\eta_P = \mathrm{id}$. By Lemma 6.11, $\eta_{K_{1,\eta}} = \eta$, so $h_K \circ s = \mathrm{id}_{\mathrm{Aut}_k(K)}$; in particular h_K is surjective and s is a homomorphism, since $s(\eta)s(\eta') = [K_{1,\eta} \otimes_K K_{1,\eta'}] = [K_{1,\eta\circ\eta'}] = s(\eta \circ \eta')$ by Lemma 6.11. The split short exact sequence of groups

$$\{1\} \longrightarrow \mathrm{Pic}(K) \longrightarrow \mathrm{Pic}_k(K) \xrightarrow{h_K} \mathrm{Aut}_k(K) \longrightarrow \{1\}$$

yields the stated internal semidirect-product decomposition. □

We henceforth regard $\mathrm{Pic}(K)$ as a G-module via $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$: for $\sigma \in G$ and $[P] \in \mathrm{Pic}(K)$,

$$^{\sigma}[P] := \phi(\sigma) \cdot [P] = s(\phi(\sigma))\,[P]\,s(\phi(\sigma))^{-1}, \tag{34}$$

and denote this G-module by $\mathrm{Pic}(K)_\phi$. It is abelian since $\mathrm{Pic}(K)$ is, as K is commutative. With this notation, let $\mathrm{Hom}_\phi(G, \mathrm{Pic}_k(K))$ denote the set of generalized collective characters of type ϕ over K. For each $\Phi \in \mathrm{Hom}_\phi(G, \mathrm{Pic}_k(K))$, define

$$\xi_\Phi(\sigma) := \Phi(\sigma)\, s(\phi(\sigma))^{-1} \in \mathrm{Pic}(K), \qquad \sigma \in G.$$

Proposition 7.3. *Let $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$. Then with the above notation $\xi_\Phi : G \to \mathrm{Pic}(K)$ is a crossed homomorphism. Moreover, $\Phi \mapsto \xi_\Phi$ is a bijection from the set $\mathrm{Hom}_\phi(G, \mathrm{Pic}_k(K))$ onto the group $Z^1(G, \mathrm{Pic}(K)_\phi)$.*

Proof. If $h_K \circ \Phi = \phi$, then $h_K(\xi_\Phi(\sigma)) = h_K(\Phi(\sigma))\,\phi(\sigma)^{-1} = \mathrm{id}$, so $\xi_\Phi(\sigma) \in \mathrm{Pic}(K)$. Now

$$\begin{aligned}
\xi_\Phi(\sigma\tau) &= \Phi(\sigma)\Phi(\tau)s(\phi(\tau))^{-1}s(\phi(\sigma))^{-1} \\
&= \Phi(\sigma)\xi_\Phi(\tau)s(\phi(\sigma))^{-1} \\
&= \left[\Phi(\sigma)s(\phi(\sigma))^{-1}\right]\left[s(\phi(\sigma))\xi_\Phi(\tau)s(\phi(\sigma))^{-1}\right] \\
&= \xi_\Phi(\sigma)\,{}^{\sigma}\xi_\Phi(\tau),
\end{aligned}$$

so Φ is a homomorphism if and only if ξ_Φ satisfies the crossed-homomorphism identity, i.e. $\xi_\Phi \in Z^1(G, \mathrm{Pic}(K)_\phi)$ (the normalization $\xi_\Phi(e) = 1$ is automatic). Conversely, given $\xi \in Z^1(G, \mathrm{Pic}(K)_\phi)$, the formula $\Phi(\sigma) := \xi(\sigma)\, s(\phi(\sigma))$ defines a homomorphism $G \to \mathrm{Pic}_k(K)$ of type ϕ with $\xi_\Phi = \xi$, by reversing the same computation. The two assignments are mutually inverse. □

7.2. **Conjugation-invariance of the obstruction.**

Construction 7.4. Choose an invertible module Q representing $[Q]$ and put

$$Q^\vee := \mathrm{Hom}_K(Q, K). \tag{35}$$

The usual evaluation maps, with the symmetry of K-modules used to exchange the factors, identify both $Q^\vee \otimes_K Q$ and $Q \otimes_K Q^\vee$ with K; write

$$\epsilon\colon Q^\vee \otimes_K Q \xrightarrow{\sim} K$$

for the former. The conjugation functor is

$$\Xi_Q := Q^\vee \otimes_K (-) \otimes_K Q : \boldsymbol{\mathcal{P}ic}_k(K) \longrightarrow \boldsymbol{\mathcal{P}ic}_k(K), \tag{36}$$

with the monoidal constraints obtained by evaluation and coevaluation.

Lemma 7.5. *Let $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$, K commutative and let $\Phi : G \to \mathrm{Pic}_k(K)$ be a GCC of type ϕ. Then for every $[Q] \in \mathrm{Pic}(K)$, the conjugate $c_{[Q]} \circ \Phi\colon \sigma \mapsto [Q]\,\Phi(\sigma)\,[Q]^{-1} \in \mathrm{Pic}_k(K)$ is again a GCC of type ϕ. Furthermore,*

(i) *For another GCC $\Phi' : G \to \mathrm{Pic}_k(K)$ of type ϕ, the equality $[\xi_{\Phi'}] = [\xi_\Phi] \in H^1(G, \mathrm{Pic}(K)_\phi)$ holds if and only if there exists a $[Q] \in \mathrm{Pic}(K)$ such that $\Phi' = c_{[Q]} \circ \Phi$.*

(ii) *For every $[Q] \in \mathrm{Pic}(K)$ we have $T(c_{[Q]} \circ \Phi) = T(\Phi) \in H^3(G, \mathrm{U}(K)_\phi)$.*

Proof. Since $[Q] \in \mathrm{Pic}(K)$, Proposition 7.2 gives $\eta_Q = \mathrm{id}$. Lemma 3.4(ii) then gives $\eta_{[Q]\Phi(\sigma)[Q]^{-1}} = \eta_Q \circ \eta_{\Phi(\sigma)} \circ \eta_Q^{-1} = \eta_{\Phi(\sigma)} = \phi(\sigma)$, so $c_{[Q]} \circ \Phi$ has type ϕ.

(i) The claim follows from the fact that

$$\xi_{c_{[Q]}\circ\Phi}(\sigma) = [Q]\,\xi_\Phi(\sigma)\,{}^\sigma[Q]^{-1} = \xi_\Phi(\sigma)\cdot\partial([Q]^{-1})(\sigma),\quad \forall [Q]\in \mathrm{Pic}(K),\ \sigma\in G,$$

as well as from the injectivity of the map $\Phi\mapsto\xi_\Phi$ in Proposition 7.3.

(ii) The conjugation Ξ_Q of Construction 7.4 with its standard duality constraints is a monoidal autoequivalence of $\boldsymbol{\mathcal{Pic}}_k(K)$ with $\pi_0(\Xi_Q) = c_{[Q]^{-1}}$, the inner automorphism of $\mathrm{Pic}_k(K)$. Its effect on $\pi_1(\boldsymbol{\mathcal{Pic}}_k(K)) = \mathrm{U}(K)$ is trivial. Indeed, for $a\in\mathrm{U}(K) = \mathrm{Aut}_K(K)$, $\Xi_Q(a) = 1_{Q^\vee}\otimes a\otimes 1_Q$; since Q and $Q^\vee$ are K-modules, balance moves the scalar a through either tensor factor, and K-linearity of ϵ gives

$$\epsilon\circ(1_{Q^\vee}\otimes a\otimes 1_Q)\circ\epsilon^{-1} = a\colon K\longrightarrow K.$$

This calculation also shows directly that the induced map on $\pi_1(\boldsymbol{\mathcal{Pic}}_k(K))$ is the identity under the canonical identification with $\mathrm{U}(K)$. The induced module is unchanged because $\eta_Q = \mathrm{id}$, so $c_{[Q]^{-1}}\circ\Phi$ has the same type ϕ and module $\mathrm{U}(K)_\phi$. By (31) we infer $T(c_{[Q]^{-1}}\circ\Phi) = T(\Phi)$. Replacing Q by $Q^\vee$ gives (ii).

□

7.3. **The Picard action and exactness.** By Proposition 7.3 and Lemma 7.5, the obstruction $T(\Phi)$ of a generalized collective character over K of type ϕ depends only on the class $[\xi_\Phi]\in H^1(G,\mathrm{Pic}(K)_\phi)$. We may therefore define

$$\overline{T}\colon H^1\big(G,\mathrm{Pic}(K)_\phi\big)\longrightarrow H^3(G,\mathrm{U}(K)_\phi),\qquad \overline{T}\big([\xi_\Phi]\big) := T(\Phi).$$

For a Clifford extension (A,j) of K and $[Q]\in\mathrm{Pic}(K)$, choose a representative Q and use its canonical dual (35). Under the equivalence (27) composed with conjugation (36), the monoidal functor

$$\Xi_Q\circ\Theta(A,j)\colon \underline{G}\longrightarrow \boldsymbol{\mathcal{Pic}}_k(K)$$

corresponds to another Clifford extension of K, called a *Q-conjugate* of (A,j), with character $c_{[Q]^{-1}}\circ\chi_K[A]$, after renormalizing at e by $\Xi_Q(K)\cong K$ as in Construction 5.14. We write

$$\mathrm{Cliff}_k(G,K)_\phi/\,\mathrm{Pic}(K)$$

for the set of classes under the equivalence relation generated by isomorphism over K and Q-conjugation (compare with [HdR99]). An isomorphism $Q\to Q'$ induces the dual isomorphism $(Q')^\vee\to Q^\vee$ and hence a monoidal natural isomorphism $\Xi_Q\cong\Xi_{Q'}$, so the resulting class depends only on $[Q]$. Lemma 7.5 shows that the associated character determines a well-defined map

$$\overline{\chi}_K\colon\ \mathrm{Cliff}_k(G,K)_\phi/\,\mathrm{Pic}(K)\longrightarrow H^1(G,\mathrm{Pic}(K)_\phi),\qquad [A]\longmapsto[\xi_{\chi_K[A]}].$$

Construction 7.6. Put $\eta_g = \phi(g)$. For a K-module L, let ${}^gL$ be its additive group with scalar action $a\star_K\ell = \eta_g^{-1}(a)\ell$. On maps the twist keeps the same additive map. Thus ${}^g({}^hL) = {}^{gh}L$, and the canonical identifications

$${}^g(L\otimes_K M)\cong {}^gL\otimes_K {}^gM,\qquad {}^gK\xrightarrow{\eta_g}K$$

make these twists a coherent symmetric monoidal action. It agrees with the action (34): explicitly

$$K_{1,\eta_g}\otimes_K L\otimes_K K_{1,\eta_g^{-1}}\longrightarrow {}^gL,\qquad x\otimes_K\ell\otimes_K y\longmapsto\eta_g^{-1}(x)y\ell$$

is a K-module isomorphism, compatible with composition of twists by the multiplication formula (33).

A type-η_g bimodule P_g is determined by its underlying left module L_g and the right action $x \star_\iota a = \eta_g(a)x$. Consequently a reduced system over K is equivalently a family of invertible modules L_g, with $L_e = K$, and normalized isomorphisms

$$m_{g,h} : L_g \otimes_K {}^g L_h \xrightarrow{\sim} L_{gh}.$$

Indeed, $x \otimes_K y \mapsto x \otimes_K y$ identifies this tensor product with $P_g \otimes_K P_h$: its balance relation reads $ax \otimes_K y = x \otimes_K \eta_g^{-1}(a)y$ in both descriptions. For the character $\Phi(g) := [P_g]$, one has $[L_g] = \xi_\Phi(g)$, since $P_g \cong L_g \otimes_K K_{1,\eta_g}$. The associativity equation for m is therefore exactly that for the original reduced system, including its scalar defect.

For systems (L, m) and (M, n), use $L_g \otimes_K M_g$ in degree g. Its multiplication is

$$(L_g \otimes_K M_g) \otimes_K {}^g(L_h \otimes_K M_h) \cong (L_g \otimes_K {}^g L_h) \otimes_K (M_g \otimes_K {}^g M_h) \xrightarrow{m_{g,h} \otimes n_{g,h}} L_{gh} \otimes_K M_{gh}.$$

The interchange in this formula is the symmetry of K-modules. The identity system has $L_g = K$ and multiplication $x \otimes_K y \mapsto x\eta_g(y)$, hence corresponds to $K_\phi G$.

The dual system uses $L_g^\vee = \mathrm{Hom}_K(L_g, K)$ and

$$m_{g,h}^\vee : L_g^\vee \otimes_K {}^g L_h^\vee \cong (L_g \otimes_K {}^g L_h)^\vee \xrightarrow{(m_{g,h}^*)^{-1}} L_{gh}^\vee,$$

where $m_{g,h}^*$ is the dual of $m_{g,h}$. At e, evaluation at 1 identifies $L_e^\vee$ with K and normalizes the unit maps. The twisted dual is identified with the dual of the twist by sending f to $\ell \mapsto \eta_g(f(\ell))$.

For an invertible K-module Q, with $Q^\vee = \mathrm{Hom}_K(Q, K)$, define the normalized system $D(Q)$ by

$$D(Q)_g = Q \otimes_K {}^g Q^\vee \quad (g \neq e), \qquad D(Q)_e = K.$$

Before normalization, its multiplication

$$(Q \otimes_K {}^g Q^\vee) \otimes_K {}^g(Q \otimes_K {}^h Q^\vee) \longrightarrow Q \otimes_K {}^{gh} Q^\vee$$

contracts the adjacent factors ${}^g Q^\vee \otimes_K {}^g Q$. Use evaluation $Q \otimes_K Q^\vee \to K$ for the identity component and conjugate the affected multiplication maps by that identification.

Lemma 7.7. *Degreewise tensor product of the underlying left K-modules gives* $\mathrm{Cliff}_k(G, K)_\phi$ *a canonical abelian group structure, with identity* $[K_\phi G]$. *Picard conjugation is translation by a subgroup, so the quotient* $\mathrm{Cliff}_k(G, K)_\phi / \mathrm{Pic}(K)$ *is an abelian group. The crossed-product map,* $\overline{\chi}_K$, *and* $\overline{T}$ *are homomorphisms for these structures. The assignment* $[Q] \mapsto [D(Q)]$ *is a homomorphism* $\mathrm{Pic}(K) \to \mathrm{Cliff}_k(G, K)_\phi$; *its character cocycle is* $\partial([Q]^{-1})$, *hence is trivial in* $H^1(G, \mathrm{Pic}(K)_\phi)$.

Proof. Use the scalar twists and product of Construction 7.6. All tensor products here are over K. The interchange is the ordinary symmetry of modules. On three factors, the two multiplication routes separate into the respective routes for m and n, so their defects multiply:

$$t_{m \otimes n}(g, h, \ell) = t_m(g, h, \ell) t_n(g, h, \ell).$$

In particular associative systems remain associative. Module associators, unitors, and symmetry give associativity, identity, and commutativity on isomorphism classes. The identity is the system corresponding to $K_\phi G$.

For the dual system of Construction 7.6, dualizing the associativity equation and inverting it proves associativity of $m^\vee$ when m is associative. Evaluation $L_g \otimes_K L_g^\vee \to K$ identifies their product with the identity system: for pure tensors this is the identity

$$m^\vee_{g,h}(f \otimes_K v)\big(m_{g,h}(x \otimes_K y)\big) = f(x)\eta_g(v(y)).$$

Thus the isomorphism classes form an abelian group.

For an invertible module Q the $Q^\vee$-conjugate system has modules $Q \otimes_K L_g \otimes_K {}^gQ^\vee$. By symmetry it is the product of (L, m) with the system $D(Q)$ of Construction 7.6, whose multiplication evaluates the adjacent factors ${}^gQ^\vee \otimes_K {}^gQ$. These evaluations also give $D(Q \otimes_K Q') \cong D(Q) \otimes_K D(Q')$ and $D(K) \cong K_\phi G$. The classes of the $D(Q)$ therefore form a subgroup; its cosets are exactly the Picard-conjugacy classes. This proves the quotient assertion. The character cocycle of $D(Q)$ is

$$[Q]\left(g \cdot [Q]^{-1}\right) = \partial([Q]^{-1})(g),$$

which also proves the stated relation with the cohomological character map in the base-K seven-term sequence.

The character cocycle of a product is the pointwise product $[L_g][M_g]$, so $\overline{\chi}_K$ is a homomorphism. For arbitrary character systems, the displayed defect formula gives multiplicativity of T; Lemma 7.5 then gives multiplicativity of $\overline{T}$ on $H^1(G, \mathrm{Pic}(K)_\phi)$. Finally the product of the scalar systems for α and β has multiplication scalar $\alpha(g,h)\beta(g,h)$, proving $\Sigma_\phi([\alpha][\beta]) = \Sigma_\phi([\alpha])\Sigma_\phi([\beta])$. □

Theorem 7.8. *Let K be commutative and $\phi \in \mathrm{Hom}(G, \mathrm{Aut}_k(K))$.*

(i) *$\overline{T}$ is a well-defined group homomorphism, with $\overline{T}(1) = 1$.*

(ii) *A generalized collective character Φ over K of type ϕ is realizable by a G-graded Clifford system extension of K if and only if $\overline{T}([\xi_\Phi]) = 1$. Thus $\mathrm{im}\,\overline{\chi}_K = \ker \overline{T} := \overline{T}^{-1}(1)$.*

(iii) *The fibre over $\mathbf{1}_\phi$, before Picard conjugation, is the $H^2(G, \mathrm{U}(K)_\phi)$-torsor identified in Corollary 6.15. Its map to the quotient is Σ_ϕ.*

(iv) *There is an exact sequence of abelian groups (see* [AG26, §8]*)*

$$H^2(G, \mathrm{U}(K)_\phi) \xrightarrow{\Sigma_\phi} \mathrm{Cliff}_k(G, K)_\phi / \mathrm{Pic}(K) \xrightarrow{\overline{\chi}_K} H^1(G, \mathrm{Pic}(K)_\phi) \xrightarrow{\overline{T}} H^3(G, \mathrm{U}(K)_\phi). \tag{37}$$

Proof. (i) Well-definedness and the equality $\overline{T}(1) = 1$ are Lemma 7.5 and Corollary 6.15(i) (which gives $T(\mathbf{1}_\phi) = 1$, and $\xi_{\mathbf{1}_\phi} = 1$ since $\mathbf{1}_\phi(\sigma) = [K_{1,\phi(\sigma)}] = s(\phi(\sigma))$). Multiplicativity is Lemma 7.7.

(ii) By Theorem 6.4 (with $R = K$), Φ is realizable by a G-graded Clifford system extension of K if and only if $T(\Phi) = 1$, and $T(\Phi) = \overline{T}([\xi_\Phi])$ by definition of $\overline{T}$. Conjugation by $[Q] \in \mathrm{Pic}(K)$ does not affect realizability, as $c_{[Q]} \circ \Phi$ is realized by the $Q^\vee$-conjugate extension whenever Φ is; hence the condition descends to $[\xi_\Phi]$, consistently with (i).

(iii) This is Corollary 6.15, followed by the quotient map.

(iv) Lemma 7.7 gives the Clifford-extension term its canonical abelian group structure and proves that all three maps are homomorphisms. The map $\overline{\chi}_K$ is well defined by Lemma 7.5(i), and exactness at $H^1(G, \mathrm{Pic}(K)_\phi)$ is (ii). Every crossed product lies in $\ker(\overline{\chi}_K)$, so $\mathrm{im}\,\Sigma_\phi \subseteq \ker(\overline{\chi}_K)$. Conversely, suppose $\overline{\chi}_K[A] = 1$. Then $\xi_{\chi_K[A]} = \partial[Q]$ for some $[Q] \in \mathrm{Pic}(K)$. By Lemma 7.5(i), conjugation by this $[Q]$ multiplies the cocycle by $\partial([Q]^{-1}) = (\partial[Q])^{-1}$, so the $Q^\vee$-conjugate of A has associated character exactly $\mathbf{1}_\phi$. Part (iii)

identifies that conjugate with a crossed product, proving exactness at the Clifford-extension term.

□

Definition 7.9. The *symmetric Picard groupoid* $\mathbf{Pic}(K)$ has invertible K-modules as objects and K-module isomorphisms as morphisms. Its product is $\otimes_K$, its unit is K, and its symmetry is the flip of tensor factors. Its group of isomorphism classes is $\mathrm{Pic}(K)$.

The coherent action induced by ϕ sends an invertible module L to

$$ {}^{g}L := K_{1,\phi(g)} \otimes_K L \otimes_K K_{1,\phi(g^{-1})}. $$

An *equivariant invertible module* is a pair

$$ \big(L, (\lambda_g)_{g\in G}\big), \qquad \lambda_g : {}^{g}L \xrightarrow{\sim} L, $$

such that

$$ \lambda_e : K \otimes_K L \otimes_K K \to L $$

is the canonical unit map $x \otimes_K \ell \otimes_K y \mapsto xy\ell$, and

$$ \lambda_g \circ {}^{g}\lambda_h = \lambda_{gh} \qquad \text{for all } g, h \in G, $$

as maps ${}^{gh}L \to L$, after the canonical identification ${}^{g}({}^{h}L) \cong {}^{gh}L$. A morphism from $(L, (\lambda_g))$ to $(L', (\lambda'_g))$ is a K-module isomorphism $u : L \to L'$ satisfying

$$ u \circ \lambda_g = \lambda'_g \circ {}^{g}u \qquad \text{for all } g \in G. $$

This is an equality of maps ${}^{g}L \to L'$.

Construction 7.10. The nerve $N_\bullet\mathbf{Pic}(K)$ of the symmetric Picard groupoid is a grouplike E_∞-space: its tensor product is commutative up to all higher coherences, and its component monoid is a group. The infinite loop space functor gives an equivalence of ∞-categories between connective spectra and grouplike E_∞-spaces [Lur17, Rem. 5.2.6.26]. Applying its inverse gives a connective spectrum E with

$$ \Omega^\infty E \simeq N_\bullet\mathbf{Pic}(K), \qquad \pi_0(E) = \mathrm{Pic}(K), \quad \pi_1(E) = \mathrm{U}(K), $$

and no other non-trivial homotopy groups; compare [JO12, Thm. 1.5]. The scalar twists of Construction 7.6, together with their symmetric monoidal comparisons, give a $B_\bullet G$-indexed diagram of grouplike E_∞-spaces. Applying the same inverse functor to this entire diagram gives E its homotopy-coherent G-action. Thus E is an ordinary spectrum with G-action in the sense of Subsection A.8. The categorical arrow g acts by the twist $L \mapsto {}^{g}L$; on $\pi_0(E)$ this is the action on $\mathrm{Pic}(K)_\phi$, and on $\pi_1(E)$ it sends a to $\phi(g)(a)$.

For its first delooping, use the normal geometric nerve of the one-object bicategory associated to $\mathbf{Pic}(K)$, with tensor products in chronological order. An edge carries an invertible module L_{ij}, and a triangle carries an isomorphism

$$ f_{ijk} : L_{ij} \otimes_K L_{jk} \longrightarrow L_{ik}. $$

A tetrahedron exists exactly when

$$ f_{023} \circ (f_{012} \otimes 1) = f_{013} \circ (1 \otimes f_{123}) \circ \mathsf{a}, $$

as maps $(L_{01} \otimes_K L_{12}) \otimes_K L_{23} \longrightarrow L_{03}$. The unit maps specify the degenerate simplices. This is a Kan 2-type model of $\Omega^\infty\Sigma E$, functorial for the scalar twists; compare [CCG10, Rems. 4.4–4.6].

Proposition 7.11. *The homotopy fixed point spectral sequence of Construction 7.10 is concentrated in the rows*

$$E_2^{s,0} = H^s(G, \mathrm{Pic}(K)_\phi), \qquad E_2^{s,1} = H^s(G, \mathrm{U}(K)_\phi),$$

and its only possible differentials are

$$d_2 : H^s(G, \mathrm{Pic}(K)_\phi) \longrightarrow H^{s+2}(G, \mathrm{U}(K)_\phi).$$

With the convention of Definition A.44, the differential at $s = 1$ is the obstruction map of Theorem 7.8:

$$d_2([\xi_\Phi]) = \overline{T}([\xi_\Phi]) = T(\Phi).$$

Proof. The spectral sequence and its indexing follow from Subsection A.8. The delooping of the symmetric Picard groupoid is the infinite loop space of the suspension of its associated spectrum. Its first Postnikov stage is $K(\mathrm{Pic}(K), 1)$, and its second homotopy group is $\mathrm{U}(K)$. A cocycle ξ_Φ therefore specifies a section of the first stage of the Borel construction. The next lifting obstruction represents $d_2([\xi_\Phi])$ by Definition A.44.

Choose invertible modules L_g with $[L_g] = \xi_\Phi(g)$ and $L_e = K$, and normalized isomorphisms

$$m_{g,h} : L_g \otimes_K {}^g L_h \xrightarrow{\sim} L_{gh}.$$

To see the corresponding skeletal lift explicitly, pull back to the homogeneous bar construction with its diagonal left action (Definition A.29). On an edge (g_0, g_1) use ${}^{g_0}L_{g_0^{-1}g_1}$. On a triangle (g_0, g_1, g_2) put $g = g_0^{-1}g_1$ and $h = g_1^{-1}g_2$ and use

$${}^{g_0}m_{g,h} : {}^{g_0}(L_g \otimes_K {}^g L_h) \longrightarrow {}^{g_0}L_{gh}.$$

The canonical twist comparison identifies its source with the tensor product of the two consecutive edge modules. With the monoidal and unit comparisons of Construction 7.6, these assignments are homotopy-coherently G-equivariant and descend to a lift over the 2-skeleton. Whenever an edge is unit-valued, normalize it by

$${}^{g_0}K \xrightarrow{\phi(g_0)} K.$$

The same comparisons prescribe the maps on degenerate simplices; the formulas below use these unit identifications.

The homogeneous tetrahedron $(1, g, gh, gh\ell)$ projects to $[g|h|\ell]$. Its consecutive edge modules are $L_g, {}^g L_h, {}^{gh}L_\ell$, and its four triangle maps are $m_{g,h}, {}^g m_{h,\ell}, m_{gh,\ell}, m_{g,h\ell}$. Write

$$a_{g,h,\ell} : (L_g \otimes_K {}^g L_h) \otimes_K {}^{gh}L_\ell \xrightarrow{\sim} L_g \otimes_K {}^g(L_h \otimes_K {}^h L_\ell)$$

for the canonical reassociation and twist comparison. The two pastings are

$$A_{g,h,\ell} = m_{gh,\ell} \circ (m_{g,h} \otimes 1),$$
$$B_{g,h,\ell} = m_{g,h\ell} \circ (1 \otimes {}^g m_{h,\ell}) \circ a_{g,h,\ell},$$

as maps $(L_g \otimes_K {}^g L_h) \otimes_K {}^{gh}L_\ell \longrightarrow L_{gh\ell}$. By the tetrahedral equation in Construction 7.10, this boundary fills exactly when $A_{g,h,\ell} = B_{g,h,\ell}$. Pasting the four faces and cancelling the common edge comparisons identifies its boundary sphere with the automorphism

$$A_{g,h,\ell} \circ B_{g,h,\ell}^{-1} : L_{gh\ell} \longrightarrow L_{gh\ell}.$$

Every automorphism of an invertible module is multiplication by a unique unit; write this one as multiplication by $t(g, h, \ell)$. This is the identification of the boundary class in π_2 with $\mathrm{U}(K)$, with the orientation of Definition A.44. All values are in the initial-vertex

trivialization above; translating the initial vertex by g_0 applies $\phi(g_0)$ to the scalar. In particular the first coface uses $\phi(g)$, not its inverse.

The pentagon on four consecutive edges gives

$$\begin{aligned} &t(g,h,\ell)t(g,h\ell,r)\phi(g)\big(t(h,\ell,r)\big) \\ &\quad = t(gh,\ell,r)t(g,h,\ell r), \end{aligned}$$

which is $\partial t = 1$ with our left-module coboundary. The unit maps make t normalized. If the triangle maps are replaced by $m'_{g,h} = b(g,h)m_{g,h}$ for a normalized 2-cochain b, the two pastings acquire, respectively, the factors $b(gh,\ell)b(g,h)$ and $b(g,h\ell)\phi(g)(b(h,\ell))$. Thus

$$t' = t(\partial b)^{-1}.$$

Replacing the edge representatives by isomorphic modules and transporting the triangle maps conjugates the displayed automorphism by the long-edge isomorphism, which leaves its scalar unchanged.

The next Postnikov transition has fibre $K(\mathrm{U}(K),2)$. Its obstruction over a 3-simplex is precisely the boundary-sphere class of this skeletal lift. The calculation therefore identifies $[t]$ with the pulled-back Postnikov class, including its coefficient action, orientation, and independence of choices. Finally, Construction 7.6 identifies both pastings with the bimodule composites in (29), preserving their scalar ratio. Hence $[t] = T(\Phi) = \overline{T}([\xi_\Phi])$, as required. □

Remark 7.12. The realization criterion and exactness in Theorem 7.8 and its abelian group structure are proved algebraically above. The sequence is parallel to the corresponding portion of the Chase–Harrison–Rosenberg sequence [CHR65, pp. 15–33], where the kernel of the crossed-product map $H^2(G,\mathrm{U}(K)_\phi) \to \mathrm{Br}(K/k)$ is the image of the preceding map from $\mathrm{Pic}(K)^\phi$; see (1).

8. Clifford extensions and the seven-term exact sequence for a class of algebras

For a class of K-central algebras, we identify Clifford extensions and generalized collective characters under compatible Morita equivalences. The resulting monoids of Morita classes occur in the seven-term sequence of Antony and the second author, recovered in Theorem 8.26. The affine Hopf line gives a nontrivial equivariance obstruction. A real group algebra gives an explicit third obstruction. The quantum-torus examples then exhibit cyclic obstruction classes, their invariance under Morita equivalence, and cancellation under tensor product.

8.1. **Classes of algebras.**

Definition 8.1. A *class of K-central algebras* is a tensor-closed set (necessarily $\mathfrak{V}$-small) $\mathcal{C}$ of isomorphism classes of K-central k-algebras in the universe $\mathfrak{U}$ (see Definition 3.1), containing $[K]$. Here isomorphisms are K-algebra isomorphisms respecting the specified identifications of the centres with K. Tensor closure means that whenever $[R],[S] \in \mathcal{C}$, the algebra $R \otimes_K S$ is K-central and $[R \otimes_K S] \in \mathcal{C}$. Its associated *object collection* is

$$\mathfrak{U}(\mathcal{C}) := \{R \mid [R]_{\cong} \in \mathcal{C}\},$$

where R ranges over all $\mathfrak{U}$-small K-central k-algebras and $[R]_{\cong}$ denotes the isomorphism class of R, also written $[R]$ when no confusion is possible. The collection $\mathfrak{U}(\mathcal{C})$ is $\mathfrak{V}$-small and closed under $\otimes_K$.

Example 8.2. The following examples illustrate different classes.

(i) When K-centrality is preserved by $\otimes_K$, for instance when K is a field, the isomorphism classes of K-central algebras of cardinality at most a fixed infinite cardinal $\kappa \in \mathfrak{U}$ with $\kappa \geq |K|$ form a class $\mathcal{C}_{\mathrm{all},\kappa}(K)$.

(ii) The singleton class $\mathcal{C}_0 = \{[K]\}$ recovers the sequence for extensions of K, together with the three preceding terms.

(iii) The split matrix class consists of $[M_d(K)]$, $d \geq 1$. Its objects are all Morita equivalent to K.

(iv) More generally, for a subgroup $B \leq \mathrm{Br}(K)$ (see Definition 9.1 in the sequel), take all Azumaya algebras whose Brauer classes lie in B. The full Azumaya class is the case $B = \mathrm{Br}(K)$; we denote it by $\mathcal{C}_{\mathrm{Az}}(K)$ (Notation 9.4).

(v) If every tensor power of a K-central algebra A is again K-central, one may take the tensor class $\{[K], [A], [A^{\otimes_K 2}], \ldots\}$. One may also include all algebras Morita equivalent to those powers. Over $K = \mathbb{C}$, the latter class for a quantum torus will be used below. It contains the endomorphism rings appearing in the application below. For a field K of characteristic zero, another example is the first Weyl algebra $A_1(K) = K\langle x, \partial\rangle/(\partial x - x\partial - 1)$. Its tensor powers $A_n(K) := A_1(K)^{\otimes_K n}$ have centre K and are infinite dimensional for $n \geq 1$, so they are not Azumaya.

(vi) Fix an infinite cardinal $\kappa \in \mathfrak{U}$ with $\kappa \geq |K|$. The free-module class $\mathcal{C}_{\mathrm{fr},\kappa}(K)$ consists of the isomorphism classes of K-central algebras of cardinality at most κ that are free as K-modules, not necessarily free as algebras. Its object collection is $\mathfrak{U}(\mathcal{C}_{\mathrm{fr},\kappa}(K))$.

The cardinality bound in (i) and (vi) is preserved by tensor products: every element of $R \otimes_K S$ is a finite sum of pure tensors $r \otimes_K s$, so $|R \otimes_K S| \leq \kappa$ when $|R|, |S| \leq \kappa$ and κ is infinite. The last class is tensor closed. Indeed, tensoring bases gives a basis of $R \otimes_K S$. If $x = \sum_j r_j \otimes_K s_j$ is central, with the s_j distinct basis elements, commutation with $R \otimes_K 1$ implies $r_j \in Z(R) = K$ for every j. Thus $x = 1 \otimes_K s$. Since S is free over K, tensoring the injection $K \hookrightarrow R$ with S shows that the map

$$S \longrightarrow R \otimes_K S, \qquad s \longmapsto 1 \otimes_K s,$$

is injective. Commutation with $1 \otimes_K S$ therefore gives $s \in Z(S) = K$.

Let $R \in \mathfrak{U}(\mathcal{C})$ and let $\Phi : G \to \mathrm{Pic}_k(R)$ be a generalized collective character of type ϕ (Definition 3.6). Choose representatives M_g and normalized bimodule isomorphisms

$$\mu_{g,h} : M_g \otimes_R M_h \to M_{gh}.$$

The obstruction convention of Theorem 6.2 is

$$\mu_{gh,\ell} \circ (\mu_{g,h} \otimes 1) = \lambda^{\sharp}_{M_{gh\ell}}(t_{(M,\mu)}(g,h,\ell)) \circ \mu_{g,h\ell} \circ (1 \otimes \mu_{h,\ell}) \circ \mathsf{a},$$

as maps $(M_g \otimes_R M_h) \otimes_R M_\ell \to M_{gh\ell}$, where a is the canonical reassociation. The cohomology class $T(\Phi) = [t_{(M,\mu)}] \in H^3(G, \mathrm{U}(K)_\phi)$ is independent of these choices. By Theorems 4.5 and 6.4, $T(\Phi) = 1$ precisely when the M_g admit an associative multiplication, hence form a strongly G-graded algebra with identity component R. When nonempty, the set of realizations up to graded isomorphisms restricting to the identity on R is a torsor under $H^2(G, \mathrm{U}(K)_\phi)$.

Definition 8.3. Let $\mathcal{C}$ be a class of K-central algebras (Definition 8.1). A system of *Clifford data of type ϕ over $\mathcal{C}$* consists of:

(i) a base algebra R, that is, a K-central algebra with $[R] \in \mathcal{C}$;

(ii) for every $g \in G$, an invertible R-bimodule over k

$$M_g \in \boldsymbol{\mathcal{P}ic}_k(R), \qquad \eta_{M_g} = \phi(g);$$

(iii) for every $g, h \in G$, a bimodule isomorphism

$$\mu_{g,h} : M_g \otimes_R M_h \xrightarrow{\sim} M_{gh},$$

such that for every $g, h, \ell \in G$ the two composites agree:

$$\mu_{gh,\ell} \circ (\mu_{g,h} \otimes 1_\ell) = \mu_{g,h\ell} \circ (1_g \otimes \mu_{h,\ell}) \circ \mathsf{a}_{g,h,\ell}. \tag{38}$$

Both sides of (38) are isomorphisms $(M_g \otimes_R M_h) \otimes_R M_\ell \longrightarrow M_{gh\ell}$. Here $1_g = \mathrm{id}_{M_g}$, and

$$\mathsf{a}_{g,h,\ell} : (M_g \otimes_R M_h) \otimes_R M_\ell \to M_g \otimes_R (M_h \otimes_R M_\ell)$$

is the canonical associator. The data are *normalized*: $M_e = R$, and $\mu_{e,g}$, $\mu_{g,e}$ are the specified left and right unit constraints (28). We call R the *base algebra*, M_g the *g-component bimodule*, $[M_g] \in \mathrm{Pic}_k(R)$ the *g-component Picard class*, and the family $\big([M_g]\big)_{g\in G}$ the *Picard profile* of the system. Thus the identity component bimodule is R and its component Picard class is the neutral class $[R]$. By Theorem 4.5, passing to $\bigoplus_{g\in G} M_g$ identifies Clifford data based at R with the G-graded Clifford system extensions of R of type ϕ (Definitions 4.1 and 6.1).

A system of *character data of type ϕ over $\mathcal{C}$* consists of a base algebra R as in (i) and, for every $g \in G$, an isomorphism class $[M_g]$ of invertible R-bimodules over k with $\eta_{M_g} = \phi(g)$, such that for every $g, h \in G$

$$[M_g \otimes_R M_h] = [M_{gh}] \in \mathrm{Pic}_k(R), \tag{39}$$

in particular $[M_e] = [R]$. Equality (39) says exactly that $g \mapsto [M_g]$ is a group homomorphism $\Phi : G \to \mathrm{Pic}_k(R)$ with $h_R \circ \Phi = \phi$, so character data with base R are the generalized collective characters of type ϕ with base R (Definition 3.6). We write character data as (R, Φ) with $\Phi(g) = [M_g]$. Forgetting the chosen isomorphisms $\mu_{g,h}$ and retaining only the classes $[M_g]$ sends Clifford data to character data.

8.2. **Invertible bimodules and Morita classes.**

Definition 8.4. The $\mathfrak{V}$-small *Morita groupoid* $\mathbf{Mor}^{\mathcal{C}}$ of $\mathcal{C}$ has objects $R \in \mathfrak{U}(\mathcal{C})$. A morphism $R \to S$, also called a *transport* from R to S, is the isomorphism class of an invertible R–S-bimodule P identifying the two centers through their specified identifications with K; equivalently,

$$a \star_R p = p \star_S a \qquad \text{for all } a \in K \text{ and } p \in P. \tag{40}$$

For $P : R \to S$ and $Q : S \to T$, composition is

$$[Q] \circ [P] = [P \otimes_S Q].$$

For an invertible R–S-bimodule P, take

$$P^\vee := \mathrm{Hom}_R(P, R), \qquad (s \star_S f \star_R r)(p) := f(p \star_S s)r.$$

Here the homomorphisms are left R-linear, and the displayed formula gives $P^\vee$ its S–R-bimodule structure. The two bimodule isomorphisms are

$$\epsilon_P : P \otimes_S P^\vee \longrightarrow R, \quad p \otimes_S f \longmapsto f(p),$$
$$\delta_P : P^\vee \otimes_R P \longrightarrow S, \quad f \otimes_R p \longmapsto s_{f,p},$$

where $s_{f,p}$ is the unique element of S satisfying

$$x \star_S s_{f,p} = f(x) \star_R p \qquad \text{for every } x \in P.$$

Existence and uniqueness follow from the right-action isomorphism $S^{\mathrm{op}} \cong \mathrm{End}_R(P)$ for a Morita equivalence. The dual-basis isomorphism for the finitely generated projective left R-module P gives the second map; evaluation gives the first, which is an isomorphism because P is a generator. The relation $x \star_S s_{f,p} = f(x) \star_R p$ gives the triangle identities, with the inverse of the second map used as coevaluation. Thus $P^\vee$ represents the inverse morphism. Given morphisms $R \to S$ and $R' \to S'$ represented by bimodules P and Q, the external product of morphisms is $[P] \boxtimes [Q] = [P \otimes_K Q]$, a morphism $R \otimes_K R' \to S \otimes_K S'$, and the symmetry is induced by the flip of tensor factors. The automorphism group of the object K of this groupoid is $\mathrm{Pic}(K)$.

Construction 8.5. For a morphism $R \to S$ in $\mathbf{Mor}^{\mathcal{C}}$, choose an invertible bimodule P representing it and define

$$\Xi_P : \boldsymbol{\mathcal{P}ic}_k(R) \longrightarrow \boldsymbol{\mathcal{P}ic}_k(S), \qquad M \longmapsto P^\vee \otimes_R M \otimes_R P.$$

On a bimodule map $u : M \to N$, it acts by

$$\Xi_P(u) := 1_{P^\vee} \otimes u \otimes 1_P : \Xi_P(M) \longrightarrow \Xi_P(N).$$

Its monoidal comparison

$$\xi_{P;M,N} : \Xi_P(M) \otimes_S \Xi_P(N) \longrightarrow \Xi_P(M \otimes_R N)$$

contracts the adjacent factors $P \otimes_S P^\vee$ using ϵ_P. Its unit comparison

$$\xi_{P;0} : S \longrightarrow \Xi_P(R)$$

is induced by δ_P^{-1}. For composable morphisms $P : R \to S$ and $Q : S \to T$, the canonical identification $(P \otimes_S Q)^\vee \cong Q^\vee \otimes_S P^\vee$ and tensor associators give the monoidal comparison

$$c_{Q,P} : \Xi_Q \circ \Xi_P \xrightarrow{\sim} \Xi_{P \otimes_S Q}$$

between functors $\boldsymbol{\mathcal{P}ic}_k(R) \to \boldsymbol{\mathcal{P}ic}_k(T)$. If $f : P \to Q$ is an isomorphism of representatives of the same morphism $R \to S$, its dual $f^\vee : Q^\vee \to P^\vee$ is precomposition by f. The induced monoidal natural comparison is

$$d_f : \Xi_P \Rightarrow \Xi_Q, \qquad \text{as functors } \boldsymbol{\mathcal{P}ic}_k(R) \longrightarrow \boldsymbol{\mathcal{P}ic}_k(S),$$

with components

$$d_{f,M} = (f^{-1})^\vee \otimes 1_M \otimes f : \Xi_P(M) \longrightarrow \Xi_Q(M).$$

Given a reduced system (M, μ) over R, put $M'_e = S$ and $M'_g = \Xi_P(M_g)$ for $g \neq e$. Let

$$j_g : M'_g \longrightarrow \Xi_P(M_g)$$

be the identity for $g \neq e$, and let j_e be the inverse of the canonical unit isomorphism $\Xi_P(R) \cong S$ induced by δ_P. Define

$$\mu'_{g,h} = j_{gh}^{-1} \circ \Xi_P(\mu_{g,h}) \circ \xi_{P;M_g,M_h} \circ (j_g \otimes j_h) \qquad \text{for all } g, h \in G.$$

These are maps $M'_g \otimes_S M'_h \to M'_{gh}$.

Lemma 8.6. *The functor Ξ_P is a monoidal equivalence preserving types and central scalar automorphisms. The normalized system of Construction 8.5 is a reduced system with the same obstruction cocycle. The construction gives an equivalence between the groupoids of realizations.*

Proof. The inverse equivalence is $\Xi_{P^\vee}$. Evaluations and tensor associators supply the monoidal and unit comparisons; their coherence follows from the triangle identities and tensor-product coherence. For $[P] : R \to S$ and $[Q] : S \to T$, these comparisons give $\Xi_Q \circ \Xi_P \cong \Xi_{P \otimes_S Q}$. Equation (40) lets a central scalar pass through the two outside factors without changing its label. Hence $h_S(\Xi_P(M)) = h_R(M)$ and $\Xi_P(\lambda^\sharp_M(u)) = \lambda^\sharp_{\Xi_P(M)}(u)$ as endomorphisms of $\Xi_P(M)$, for $u \in \mathrm{U}(K)$. Applying Ξ_P to the defining associativity equation gives the same scalar for the resulting system. Applying the inverse bimodule shows that the induced functor on realizations is also an equivalence. □

Definition 8.7. Let (R, M, μ) and (S, N, ν) be Clifford data, and (R, Φ) and (S, Ψ) character data, of type ϕ over $\mathcal{C}$ (Definition 8.3).

The Clifford data are *isomorphic* if there are a K-algebra isomorphism $f : R \to S$ and S-bimodule isomorphisms

$$\theta_g : f_* M_g \xrightarrow{\sim} N_g,$$

with unit component

$$\theta_e : f_* R \xrightarrow{\sim} S, \qquad r \longmapsto f(r),$$

such that for all $g, h \in G$

$$\theta_{gh} \circ f_* \mu_{g,h} = \nu_{g,h} \circ (\theta_g \otimes \theta_h), \tag{41}$$

as S-bimodule maps $f_* M_g \otimes_S f_* M_h \to N_{gh}$. Here $f_* M$ denotes the S-bimodule with the same underlying k-module as M and actions transported along f^{-1}; it is invertible with $\eta_{f_* M} = \eta_M$, because f is the identity on K. The canonical identification $f_*(M_g \otimes_R M_h) \cong f_* M_g \otimes_S f_* M_h$ is understood. The character data are *isomorphic* if there is a K-algebra isomorphism $f : R \to S$ with

$$\Psi(g) = [f_* M_g] \qquad \text{for all } g \in G, \tag{42}$$

where $\Phi(g) = [M_g]$.

The Clifford data are *Morita equivalent* if there are a morphism $P : R \to S$ in $\mathbf{Mor}^{\mathcal{C}}$ (Definition 8.4) and bimodule isomorphisms

$$\theta_g : \Xi_P(M_g) \to N_g,$$

with θ_e the canonical unit isomorphism $\Xi_P(R) \cong S$ of Construction 8.5, such that for all $g, h \in G$

$$\theta_{gh} \circ \Xi_P(\mu_{g,h}) \circ \xi_{P;M_g,M_h} = \nu_{g,h} \circ (\theta_g \otimes \theta_h). \tag{43}$$

This is an equality of S-bimodule maps $\Xi_P(M_g) \otimes_S \Xi_P(M_h) \to N_{gh}$. The character data are *Morita equivalent* if there is such a morphism P with

$$\Psi(g) = [\Xi_P(M_g)] \qquad \text{for all } g \in G. \tag{44}$$

Denote the two Morita relations by $\sim_{\mathrm{Mor}}$.

All four relations are equivalence relations: K-algebra isomorphisms and the morphisms of $\mathbf{Mor}^{\mathcal{C}}$ compose and invert, and the monoidal comparisons ξ of Construction 8.5 make the transported coherence isomorphisms compose. Isomorphism is the special case of Morita

equivalence in which P is the class of the invertible bimodule S with left action through a K-algebra isomorphism $f : R \to S$ and regular right S-action, for which $\Xi_P \cong f_*$; so $\sim_{\mathrm{Mor}}$ is coarser than isomorphism. For Clifford data it can be strictly coarser even with fixed base K: the two crossed products constructed in §8.4 are graded Morita equivalent but are not isomorphic as graded extensions fixing K.

Construction 8.8. For Clifford or character systems over R and S, respectively, if M and N have the same type η, define

$$M \otimes_\eta N = M \otimes_k N / \langle (m \star_\iota a) \otimes_k n - m \otimes_k (\eta(a) \star_\ell n) \mid m \in M, n \in N, a \in K \rangle.$$

The left and right actions of $R \otimes_K S$ are componentwise. This gives the equal-type tensor functor on the full subgroupoid of $\boldsymbol{\mathcal{P}ic}_k(R) \times \boldsymbol{\mathcal{P}ic}_k(S)$ consisting of pairs of the same type. Its monoidal comparison interchanges the middle factors in successive balanced tensor products; its unit comparison is the canonical identification of the product of the regular bimodules with the regular $(R \otimes_K S)$-bimodule. In degree g the product is $M_g \otimes_{\phi(g)} N_g$; its multiplication uses the middle-factor interchange followed by the two multiplication maps. Normalize the unit by this unit comparison.

Definition 8.9. Write

$$\mathrm{Cliff}^{\mathcal{C}}_\phi(K), \qquad \mathrm{Char}^{\mathcal{C}}_\phi(K)$$

for the sets of Morita equivalence classes of Clifford, respectively character, data of type ϕ over $\mathcal{C}$ (Definitions 8.3 and 8.7). In Proposition 8.16 we show that under the external products of Construction 8.8 they are abelian monoids. Moreover, we show that Morita equivalence is compatible with these external products via (48).

Meanwhile, forgetting the chosen multiplications gives the *character map*

$$\chi^{\mathrm{Mor}}_\phi : \mathrm{Cliff}^{\mathcal{C}}_\phi(K) \longrightarrow \mathrm{Char}^{\mathcal{C}}_\phi(K), \qquad [(R, M, \mu)] \longmapsto [(R, \Phi)], \quad \Phi(g) = [M_g], \tag{45}$$

a homomorphism of monoids. These are the fifth and sixth terms of the seven-term sequence (3). The isomorphism classes of the same data likewise form abelian monoids under the external product; they map onto $\mathrm{Cliff}^{\mathcal{C}}_\phi(K)$ and $\mathrm{Char}^{\mathcal{C}}_\phi(K)$ with fibres the Morita equivalence classes, by construction. The third-obstruction assignment is

$$\partial_3^{\mathcal{C}} : \mathrm{Char}^{\mathcal{C}}_\phi(K) \longrightarrow H^3(G, \mathrm{U}(K)_\phi), \qquad [(R, \Phi)] \longmapsto T(\Phi).$$

It is well defined by Lemma 8.6 and Theorem 5.13.

Classifying objects and paths. The notation $B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)$ denotes the normal geometric nerve of the one-object bicategory associated to the monoidal groupoid $\boldsymbol{\mathcal{P}ic}_k(R)$. Its edges are invertible bimodules, its triangles are bimodule multiplication isomorphisms, and its tetrahedral equations are associativity. It is the connected classifying 2-type with

$$\begin{aligned} \pi_1(B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)) &= \mathrm{Pic}_k(R), \\ \pi_2(B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)) &= \mathrm{U}(K), \\ \pi_i(B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)) &= \{1\} \quad (i > 2). \end{aligned}$$

Proposition 3.5 identifies the action of $\pi_1(B_\bullet \boldsymbol{\mathcal{P}ic}_k(R))$ on $\pi_2(B_\bullet \boldsymbol{\mathcal{P}ic}_k(R))$ with the type action $[P] \cdot u = h_R([P])(u)$. It is not the ordinary nerve of the underlying groupoid $\boldsymbol{\mathcal{P}ic}_k(R)$, whose components are indexed by $\mathrm{Pic}_k(R)$ and whose automorphism groups occur in degree 1.

Put $\Gamma = \mathrm{Aut}_k(K)$. For maps $X \to B$ and $Y \to B$, let

$$\mathrm{Map}_B(X,Y)_n = \mathrm{Hom}_{\mathbf{sSet}/B}(X \times \Delta^n, Y),$$

where the structural map of $X \times \Delta^n$ is the composite of its projection to X with $X \to B$. We form this mapping space using the given maps to B, without fibrant replacement.

Lemma 8.10. *For every K-central algebra R, the type maps factor as Kan fibrations*

$$B_\bullet \boldsymbol{\mathcal{P}ic}_k(R) \longrightarrow B_\bullet(\mathrm{Im}\, h_R), \qquad B_\bullet \,\mathrm{Pic}_k(R) \longrightarrow B_\bullet(\mathrm{Im}\, h_R).$$

For either target Y, the space $\mathrm{Map}_{B_\bullet\Gamma}(B_\bullet G, Y)$ is empty when $\phi(G) \not\subseteq \mathrm{Im}\, h_R$. When $\phi(G) \subseteq \mathrm{Im}\, h_R$, it is canonically

$$\mathrm{Map}_{B_\bullet(\mathrm{Im}\, h_R)}(B_\bullet G, Y),$$

where the structural map of $B_\bullet G$ is induced by $\phi : G \to \mathrm{Im}\, h_R$. This space is Kan and computes the derived mapping space over $B_\bullet(\mathrm{Im}\, h_R)$. For $R = K$, one has $\mathrm{Im}\, h_K = \Gamma$.

Proof. The one-object bicategory associated to $\boldsymbol{\mathcal{P}ic}_k(R)$ is a bigroupoid, so its normal geometric nerve is Kan [CCG10, p. 242, Rem. 4.4]. The ordinary group nerve $B_\bullet \,\mathrm{Pic}_k(R)$ is also Kan. For either type map to $B_\bullet(\mathrm{Im}\, h_R)$, a relative horn of dimension one can be filled by choosing an invertible bimodule of the prescribed type. In dimensions at least two, first fill the horn in the source. Its image is the prescribed base simplex because a group nerve has unique horn fillers in those dimensions. This proves the two fibration assertions.

Every edge in either source has type in $\mathrm{Im}\, h_R$, so a map over $B_\bullet\phi$ requires $\phi(G) \subseteq \mathrm{Im}\, h_R$. When this inclusion holds, the displayed identification of mapping spaces follows degreewise from the inclusion $B_\bullet(\mathrm{Im}\, h_R) \hookrightarrow B_\bullet\Gamma$. In the simplicial slice model structure of Definition A.15, the source is cofibrant and the target is fibrant over $B_\bullet(\mathrm{Im}\, h_R)$. Thus this mapping space is Kan and computes the derived mapping space there. Finally, Lemma 6.11 provides the bimodule $K_{1,\eta}$ of type η for every $\eta \in \Gamma$, proving $\mathrm{Im}\, h_K = \Gamma$. □

Vertices of $\mathrm{Map}_{B_\bullet\Gamma}(B_\bullet G, B_\bullet \boldsymbol{\mathcal{P}ic}_k(R))$ are normalized monoidal functors of type ϕ, hence Clifford systems by Theorem 4.5. A path from A to B determines an invertible R-bimodule Q of identity type and isomorphisms

$$\theta_g : A_g \otimes_R Q \xrightarrow{\sim} Q \otimes_R B_g$$

satisfying the multiplication and unit equations. Indeed, the two triangles over each edge $[g]$ compare these two composites with a common diagonal, and the tetrahedra over $[g|h]$ impose multiplication compatibility. Conversely, choose the diagonal over $[g]$ to be $Q \otimes_R B_g$ for $g \neq e$, using θ_g and the identity for the two triangle maps. For $g = e$, take the diagonal to be Q and use the canonical unitors. The multiplication and unit equations give a simplicial map

$$B_\bullet G \times \Delta^1 \to B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)$$

over $B_\bullet\Gamma$. These comparisons are pseudonatural equivalences. Their modifications are bimodule isomorphisms $v : Q \to Q'$ with

$$(v \otimes 1_{B_g}) \circ \theta_g = \theta'_g \circ (1_{A_g} \otimes v);$$

these are maps $A_g \otimes_R Q \to Q' \otimes_R B_g$. The same prism construction realizes a modification as a homotopy of comparisons. For the character mapping space, only the character and the conjugating class $[Q]$ remain.

The mapping spaces are unbased: the edge at the vertex of $B_\bullet G$ may be any invertible bimodule of identity type. For $Q = R$, scalar modifications identify the principal 1-cocycles of Proposition 6.8, giving the distinction between $Z^1(G, \mathrm{U}(K)_\phi)$ and $H^1(G, \mathrm{U}(K)_\phi)$ in Remark 6.9. The homotopy quotient below also permits comparisons between different algebras. Only its components enter the seven-term sequence.

Construction 8.11. Use the mapping spaces of Lemma 8.10, with the structural map $B_\bullet\phi$ fixed in every simplicial degree. Passing to component groups gives truncation maps

$$r_R : B_\bullet \boldsymbol{\mathcal{Pic}}_k(R) \longrightarrow B_\bullet \operatorname{Pic}_k(R).$$

For a fixed $\phi : G \to \Gamma$, define

$$\mathfrak{Cliff}_\phi(K; R) := \operatorname{Map}_{B_\bullet\Gamma}(B_\bullet G, B_\bullet \boldsymbol{\mathcal{Pic}}_k(R)),$$
$$\mathfrak{Char}_\phi(K; R) := \operatorname{Map}_{B_\bullet\Gamma}(B_\bullet G, B_\bullet \operatorname{Pic}_k(R)).$$

At the distinguished base K, put $\mathsf{CliffSec}_\phi(K) = \mathfrak{Cliff}_\phi(K; K)$ and $\mathsf{CharSec}_\phi(K) = \mathfrak{Char}_\phi(K; K)$. Their basepoints x_ϕ^{Cliff} and x_ϕ^{char} are represented by the skew group algebra $K_\phi G$ and the character $g \mapsto [K_{1,\phi(g)}]$.

For a morphism $P : R \to S$, normalize Ξ_P at the unit before taking its normal geometric nerve: replace its value at R by S and conjugate its comparison maps using the unit isomorphism $\Xi_P(R) \cong S$, as in Lemma 8.6. Postcomposition with this normalized nerve gives the map associated with P. Since Ξ_P is a type-preserving equivalence, $\operatorname{Im} h_R = \operatorname{Im} h_S$. Its comparisons and modifications are over $B_\bullet\Gamma$. Lemma 8.13 therefore gives coherent diagrams of the mapping spaces just defined, compatibly with truncation.

Define the homotopy quotients

$$\mathsf{CliffSp}_\phi^{\mathcal{C}}(K) := \left(\coprod_{R \in \mathfrak{U}(\mathcal{C})} \mathfrak{Cliff}_\phi(K; R) \right)_{h\mathbf{Mor}^{\mathcal{C}}},$$
$$\mathsf{CharSp}_\phi^{\mathcal{C}}(K) := \left(\coprod_{R \in \mathfrak{U}(\mathcal{C})} \mathfrak{Char}_\phi(K; R) \right)_{h\mathbf{Mor}^{\mathcal{C}}}.$$

The subscript denotes the homotopy colimit over $\mathbf{Mor}^{\mathcal{C}}$. Both homotopy quotients are $\mathfrak{V}$-small simplicial sets. Lemma 8.13 supplies the coherent diagrams on this isomorphism-class groupoid. We rectify them as in Definition A.43 before applying the strict bar formula. Only the components of the resulting spaces enter the seven-term sequence.

Concretely, their component relations are generated by paths in the mapping spaces for each $R \in \mathfrak{U}(\mathcal{C})$ and the maps induced by Morita equivalences. Lemma 8.14 below identifies these relations with (43) and (44). The truncations induce

$$q_\phi^{\mathcal{C}} : \mathsf{CliffSp}_\phi^{\mathcal{C}}(K) \to \mathsf{CharSp}_\phi^{\mathcal{C}}(K).$$

Remark 8.12. The terms $E(\phi, \operatorname{Pic}(K))$ and $\operatorname{Pic}(K)^\phi$ in (51) are the fundamental groups of $\mathsf{CliffSec}_\phi(K)$ and $\mathsf{CharSec}_\phi(K)$, respectively, not of their homotopy quotients. Those quotients also include loops arising from Morita self-equivalences. For example, when $G = \{1\}$ and the only base class is $[K]$, the character space is $B_\bullet \operatorname{Pic}(K)$, while the group of automorphisms of K in $\mathbf{Mor}^{\mathcal{C}}$ is another copy of $\operatorname{Pic}(K)$ acting by conjugation, hence trivially. Its ordinary action-groupoid model therefore has fundamental group $\operatorname{Pic}(K) \times \operatorname{Pic}(K)$, rather

than the single $\mathrm{Pic}(K)$ required in the sequence. Only the component sets of these homotopy quotients are used in the sequence.

The next lemma makes the categorical levels in this dictionary explicit: bimodule isomorphisms give comparisons, and their central scalar discrepancies give modifications. Its conclusion permits the rectification used in Definition A.43. The cubical extension argument is recalled in Subsection A.7.

Lemma 8.13. *The maps induced by Ξ_P in Construction 8.11 extend to homotopy-coherent diagrams of spaces on $\mathbf{Mor}^{\mathcal{C}}$, compatibly with truncation. Choices of representing bimodules and composition isomorphisms give equivalent diagrams.*

Proof. Morita transport preserves $H := \operatorname{Im} h_R$ (Lemma 8.6), so work in one Morita component and put $B = B_\bullet H$. Write $Y_R = B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)$, with its type map to B. These are Kan 2-types fibrant over B (Lemma 8.10). The enriched mapping space

$$\mathcal{H}_{R,S} := \mathrm{Map}_B(Y_R, Y_S)$$

is a Kan 2-type: it is the fibre over the type map of the Kan fibration

$$\mathrm{Map}(Y_R, Y_S) \longrightarrow \mathrm{Map}(Y_R, B),$$

whose source is a 2-type and whose target is a 1-type. The corresponding mapping spaces for the character nerves are Kan 1-types. All comparisons below lie over B.

For each arrow $x : R \to S$ choose a representative P_x, taking $P_{1_R} = R$, and for $x : R \to S$, $y : S \to T$ choose an isomorphism

$$\theta_{y,x} : P_x \otimes_S P_y \to P_{y\circ x},$$

with the canonical unit values. At each indexing identity 1_R, use the identity map of Y_R, transferring the transport comparisons along the canonical monoidal isomorphism $\Xi_R \cong \mathrm{id}_{\boldsymbol{\mathcal{P}ic}_k(R)}$. Identity-containing cubes are then prescribed by the unit rules of the cubical resolution, including faces where nonidentity arrows compose to an identity. All transport comparisons below use these unit identifications. Evaluations and $\theta_{y,x}$ give monoidal comparisons

$$\Xi_{P_y} \circ \Xi_{P_x} \Rightarrow \Xi_{P_{y\circ x}} \qquad \text{as functors } \boldsymbol{\mathcal{P}ic}_k(R) \longrightarrow \boldsymbol{\mathcal{P}ic}_k(T).$$

Their normalized nerves give the interval data of the cubical resolution, with endpoint maps $Y_R \to Y_T$ over B.

Every automorphism of a representing bimodule is multiplication by a unique $u \in \mathrm{U}(K)$, by Morita equivalence with the identity bimodule. For an isomorphism $f : P \to Q$, Construction 8.5 gives

$$d_{uf,M} = \eta(u)u^{-1}d_{f,M},$$

as maps $\Xi_P(M) \longrightarrow \Xi_Q(M)$, when M has type η. Regard these monoidal functors as functors of one-object bicategories. Multiplication by u on the identity bimodule gives a modification $d_{uf} \Rrightarrow d_f$: its naturality equation is $u\, d_{uf,M} = \eta(u) d_{f,M}$ between maps with the displayed source and target. The reverse modification has label u^{-1}. Taking nerves gives a homotopy between the comparison homotopies. This modification lies over the identity of $B_\bullet\Gamma$; no condition fixes its value at the source vertex, since the mapping spaces here are unbased.

For three composable arrows $x : R_0 \to R_1$, $y : R_1 \to R_2$, $z : R_2 \to R_3$, fix the ratio by

$$\theta_{z\circ y,x} \circ (1 \otimes \theta_{z,y}) \circ \mathsf{a} = u(z,y,x)\, \theta_{z,y\circ x} \circ (\theta_{y,x} \otimes 1),$$

as maps $(P_x \otimes_{R_1} P_y) \otimes_{R_2} P_z \longrightarrow P_{z\circ y\circ x}$. The modification just described supplies the square comparing the corresponding pasted transport comparisons. It is normalized by the chosen unit values of θ. For four composable arrows x, y, z, w, tensor associativity gives

$$u(z,y,x)u(w,z\circ y,x)u(w,z,y) = u(w\circ z,y,x)u(w,z,y\circ x).$$

No action on these units occurs, because all representing bimodules are identity-central. Modifications multiply as their scalar labels do, so the square data for triples have the required compatibility. More explicitly, four arrows prescribe the boundary of a 3-cube in $\mathcal{H}_{R_0,R_4}$. Five faces are the triple modifications indexed by the five terms of the displayed identity. The sixth is the interchange square for composition of the pair comparisons, which commutes by functoriality. Pasting this boundary cancels the common comparisons and leaves the modification labelled by

$$\frac{u(z,y,x)u(w,z\circ y,x)u(w,z,y)}{u(w\circ z,y,x)u(w,z,y\circ x)} = 1.$$

The identity modification represents the trivial class in $\pi_2(\mathcal{H}_{R_0,R_4})$. Thus this boundary 2-sphere is nullhomotopic and has a 3-cube filler. This is the last possibly nontrivial obstruction.

For $m \geq 5$ composable arrows, all previously chosen data prescribe a map from $\partial(\Delta^1)^{m-1}$ to $\mathcal{H}_{R_0,R_m}$. Its obstruction lies in $\pi_{m-2}(\mathcal{H}_{R_0,R_m}) = \{1\}$. The cubical extension argument in Subsection A.7 therefore supplies the remaining fillers, with contractible choices relative to the data already fixed; the same holds for the 3-cube fillers once their boundaries have been shown nullhomotopic. Apply truncation to the entire coherent diagram to obtain the character diagram. This supplies the compatibility with truncation, including its higher comparisons. If $\phi(G) \subseteq H$, enriched precomposition by $B_\bullet G \to B$ gives the two diagrams of mapping spaces in the statement. Otherwise their values in this Morita component are empty by Lemma 8.10, so its diagrams are unique.

For another choice P'_x, choose isomorphisms $a_x : P_x \to P'_x$, with $a_{1_R} = 1_R$. Transport the pair comparisons by

$$\widetilde{\theta}_{y,x} = a_{y\circ x} \circ \theta_{y,x} \circ (a_x^{-1} \otimes a_y^{-1}),$$

as maps $P'_x \otimes_S P'_y \longrightarrow P'_{y\circ x}$. The scalar for these transported comparisons is still u. Write $\theta'_{y,x} = b(y,x)\widetilde{\theta}_{y,x}$ for the unique normalized unit-valued 2-cochain b on composable pairs. Substituting into the defining ratio gives

$$\frac{u'(z,y,x)}{u(z,y,x)} = \frac{b(z\circ y,x)b(z,y)}{b(z,y\circ x)b(y,x)}.$$

This mixed equation states that the arrow comparisons induced by a_x and the pair modifications induced by b agree with the triple modifications u, u'. Their mixed 3-cube boundaries are therefore nullhomotopic; the unmixed boundaries satisfy the quadruple identities already proved. Using the relative cubical extension argument on $\mathbf{Mor}^{\mathcal{C}} \times [1]$ extends these data to a coherent comparison of the two diagrams. Its object components are identity maps, hence it is an equivalence of coherent diagrams [CP86, Thm. 1.1]. Applying truncation to this comparison gives the compatible comparison for characters. □

Lemma 8.14. *Two Clifford systems represent the same component of* $\mathsf{CliffSp}^{\mathcal{C}}_{\phi}(K)$ *if and only if they are related by a single identity-central morphism and a comparison satisfying* (43)*. The analogous assertion holds for characters, with* (44)*. In particular, when both endpoints have base* K*, the morphism can be represented by an invertible* K*-module.*

Proof. First consider a path between Clifford systems A and B over the same base R. Its vertical edge is an invertible R-bimodule Q. Since the path lies over the constant homotopy of ϕ, this bimodule has identity type. The triangle comparisons give

$$A_g \otimes_R Q \xrightarrow{\sim} Q \otimes_R B_g.$$

Tensoring on the left with $Q^\vee = \mathrm{Hom}_R(Q, R)$ gives $\Xi_Q(A_g) \cong B_g$. The remaining face equations are precisely the multiplication compatibility (43). Thus a path in $\mathfrak{Cliff}_\phi(K; R)$ gives a compatible invertible R-bimodule.

Now let $P : R \to S$ and $Q : S \to T$ be morphisms, with comparisons $\theta : \Xi_P(A) \to B$ and $\psi : \Xi_Q(B) \to C$. The monoidal comparison from Lemma 8.6 is

$$c_{Q,P} : \Xi_Q \circ \Xi_P \xrightarrow{\sim} \Xi_{P \otimes_S Q}.$$

The composite morphism $P \otimes_S Q : R \to T$ therefore carries the comparison

$$\Xi_{P\otimes_S Q}(A) \xrightarrow{c^{-1}_{Q,P,A}} (\Xi_Q \circ \Xi_P)(A) \xrightarrow{\Xi_Q(\theta)} \Xi_Q(B) \xrightarrow{\psi} C.$$

Because these comparisons are monoidal, their composite preserves multiplication and units. A reversed step can be replaced using the inverse bimodule and the inverse comparison. Induction therefore replaces any finite zigzag by one morphism and one comparison.

Conversely, a compatible morphism gives a path by the definition of the homotopy quotient. For characters, the same argument uses bimodule isomorphism classes and (44). □

Construction 8.15. Apply the normal geometric nerve to the normalized equal-type product of Construction 8.8. The resulting map has domain

$$B_\bullet \boldsymbol{\mathcal{P}ic}_k(R) \times_{B_\bullet \Gamma} B_\bullet \boldsymbol{\mathcal{P}ic}_k(S).$$

Two maps over the prescribed map $B_\bullet\phi$ determine a map into this fibre product in every simplicial degree. Composing with the normalized nerve gives the Clifford product below; passing to isomorphism classes gives the character product. Thus there are external products

$$\mathfrak{Cliff}_\phi(K; R) \times \mathfrak{Cliff}_\phi(K; S) \longrightarrow \mathfrak{Cliff}_\phi(K; R \otimes_K S), \tag{46}$$

$$\mathfrak{Char}_\phi(K; R) \times \mathfrak{Char}_\phi(K; S) \longrightarrow \mathfrak{Char}_\phi(K; R \otimes_K S). \tag{47}$$

We denote their values on pairs of inputs in the same simplicial degree by

$$(F, F') \longmapsto F \otimes_\phi F', \qquad (\Phi, \Psi) \longmapsto \Phi \otimes_\phi \Psi,$$

respectively.

Proposition 8.16. *The relations in Definition 8.7 make the two sets of Morita classes of Definition 8.9 abelian monoids. Their identity elements are represented by* $K_\phi G$ *and its character. The crossed-product, forgetful and obstruction maps are monoid homomorphisms, and*

$$T(\Phi \otimes_\phi \Psi) = T(\Phi)T(\Psi).$$

Proof. Use Construction 8.8. For $M \in \boldsymbol{\mathcal{P}ic}_k(R)$ and $N \in \boldsymbol{\mathcal{P}ic}_k(S)$ of type η, $M \otimes_\eta N$ is the tensor product of the left K-modules M and N, with componentwise left and right actions of $R \otimes_K S$. The defining relation makes the right action balanced, and the resulting bimodule has type η. Evaluations for the inverse bimodules show that it is invertible. The interchange and unit comparisons specified in Construction 8.8 satisfy the monoidal coherence identities, since they are induced by the associativity, unit, and interchange isomorphisms of balanced tensor products. Thus the specified functor is strong monoidal. Evaluations give

$$\Xi_{P\boxtimes Q}(M \otimes_\eta N) \cong \Xi_P(M) \otimes_\eta \Xi_Q(N). \tag{48}$$

The product therefore descends to the two sets of Morita classes of Definition 8.9. Associators, unitors, and flips of tensor factors give associativity, the unit law, and commutativity on equivalence classes, making both class monoids abelian.

For arbitrary reduced systems representing Φ and Ψ, their associativity comparisons multiply factorwise, so the obstruction cochain of the product system is the pointwise product of their obstruction cochains. For crossed products the same calculation gives

$$[K_\phi^\alpha G][K_\phi^\beta G] = [K_\phi^{\alpha\beta} G].$$

These identities prove the homomorphism assertions. □

Remark 8.17. The two coherent diagrams used here are $R \mapsto B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)$ and $R \mapsto B_\bullet \operatorname{Pic}_k(R)$, with maps induced by Ξ_P and truncation. For each R, the truncation has homotopy fibre weakly equivalent to $K(\mathrm{U}(K), 2)$. Section 9 restricts to Azumaya algebras and deloops the resulting Azumaya bicategory; its top fibre is then $K(\mathrm{U}(K), 3)$.

8.3. **The seven-term sequence and its algebraic terms.**

Definition 8.18. Put

$$\mathcal{F}^{(2)}_{\phi,K} := \operatorname{Sec}_{B_\bullet G} K_{B_\bullet G}(\mathrm{U}(K)_\phi, 2).$$

Proposition 8.19. *The truncation r_R has homotopy fibre $K(\mathrm{U}(K), 2)$. Pullback along a character of type ϕ gives the local coefficient system $\mathrm{U}(K)_\phi$. If that character has a chosen lift, its space of lifts is equivalent to the section space of the corresponding local $K(\mathrm{U}(K), 2)$-object. At the unit character this gives*

$$\mathcal{F}^{(2)}_{\phi,K} \longrightarrow \mathsf{CliffSec}_\phi(K) \longrightarrow \mathsf{CharSec}_\phi(K). \tag{49}$$

Proof. The simplicial set $B_\bullet \boldsymbol{\mathcal{P}ic}_k(R)$ has first homotopy group $\operatorname{Pic}_k(R)$, second homotopy group $\mathrm{U}(K)$, and no higher homotopy groups. Truncation retains the first group and removes the second. The action on the latter is the type action by Proposition 3.5. A chosen lift trivializes the corresponding lifting torsor, giving the asserted equivalence with $\operatorname{Sec}_{B_\bullet G} K_{B_\bullet G}(\mathrm{U}(K)_\phi, 2)$ by Proposition A.35. The homotopy-group calculation is Theorem A.31. □

Cohomology comparison. By Theorem A.31,

$$\pi_i(\mathcal{F}^{(2)}_{\phi,K}) \cong H^{2-i}(G, \mathrm{U}(K)_\phi) \qquad (0 \le i \le 2).$$

Definition 8.20. The group $E(\phi, \mathrm{Pic}(K))$ consists of isomorphism classes of left $K_\phi G$-modules whose underlying K-modules are invertible, with tensor product and diagonal semilinear action. Antony and the second author denote this group by $\mathrm{Ext}(\phi, \mathrm{Pic}(K))$ in [AG26, (7.23)]. Its unit is K with $u_g x = \phi(g)(x)$, and the inverse of L is $\mathrm{Hom}_K(L, K)$ with

$$(u_g f)(\ell) = \phi(g)\big(f(u_{g^{-1}}\ell)\big).$$

Construction 8.21. For an invariant class $[L] \in \mathrm{Pic}(K)^\phi$, choose isomorphisms $\lambda_g : {}^gL \to L$, normalized at e, with ${}^gL = K_{1,\phi(g)} \otimes_K L \otimes_K K_{1,\phi(g^{-1})}$. Using the canonical identification ${}^g({}^hL) \cong {}^{gh}L$, define the scalars $a_L(g,h)$ by

$$\lambda_g \circ {}^g\lambda_h = a_L(g,h)\lambda_{gh},$$

as maps ${}^{gh}L \to L$. The equivalent semilinear maps $v_g : L \to L$ are specified by

$$v_g(\ell) = \lambda_g(1 \otimes_K \ell \otimes_K 1), \qquad \lambda_g(x \otimes_K \ell \otimes_K y) = x\phi(g)(y)v_g(\ell).$$

Thus $v_g(a\ell) = \phi(g)(a)v_g(\ell)$.

Lemma 8.22. *The cochain a_L is a normalized 2-cocycle whose class is independent of the choices and of the representative L.*

Proof. Normalization gives $a_L(e,g) = a_L(g,e) = 1$. Compare the two ways of successively composing the three λ-maps from ${}^g({}^h({}^\ell L))$ to L. The route which first combines g and h gives the scalar $a_L(g,h)a_L(gh,\ell)$, while the route which first combines h and ℓ gives $\phi(g)(a_L(h,\ell))a_L(g,h\ell)$. Therefore

$$a_L(g,h)a_L(gh,\ell) = \phi(g)(a_L(h,\ell))a_L(g,h\ell),$$

which is $\partial a_L = 1$ in the convention of (11). If $\lambda'_g = b(g)\lambda_g$, then the new defect is

$$a'_L(g,h) = \phi(g)(b(h))\, b(gh)^{-1}\, b(g)\, a_L(g,h) = (\partial b)(g,h)a_L(g,h).$$

Thus $[a_L]$ is independent of the choices. An isomorphism $f : L \to L'$ gives the same cocycle after replacing λ_g by $f \circ \lambda_g \circ ({}^g f)^{-1} : {}^gL' \to L'$. Tensoring choices gives $a_{L\otimes_K M} = a_L a_M$. □

Definition 8.23. Define $\partial_2 : \mathrm{Pic}(K)^\phi \to H^2(G, \mathrm{U}(K)_\phi)$ by $\partial_2([L]) := [a_L]$. For $b \in Z^1(G, \mathrm{U}(K)_\phi)$, write K_b for K with semilinear action $u_g x = b(g)\phi(g)(x)$.

Remark 8.24. For $\mathcal{C}_0 = \{[K]\}$ the groupoid $\mathbf{Mor}^{\mathcal{C}_0}$ is equivalent to the one-object groupoid with object K and automorphism group $\mathrm{Pic}(K)$. Thus, after choosing K as representative, a transport is a class in $\mathrm{Pic}(K)$, and the two class monoids of Definition 8.9 are the base-K orbit sets

$$\mathrm{Cliff}_\phi^{\mathcal{C}_0}(K) = \mathrm{Cliff}_k(G,K)_\phi/\,\mathrm{Pic}(K), \qquad \mathrm{Char}_\phi^{\mathcal{C}_0}(K) = H^1(G, \mathrm{Pic}(K)_\phi),$$

by the conjugacy calculation of Section 7. Thus the tail beginning with $H^2(G, \mathrm{U}(K)_\phi)$ is exactly (37); the seven-term sequence supplies its three preceding terms. In particular, the Morita classes in the Clifford term cannot generally be replaced by strict graded-isomorphism classes.

Proposition 8.25. *For the singleton class $\mathcal{C}_0$ of Example 8.2(ii), the homotopy groups at the unit sections and the component monoids are*

i	$\pi_i(\mathsf{CliffSec}_\phi(K))$	$\pi_i(\mathsf{CharSec}_\phi(K))$
2	$\mathrm{U}(K)^\phi$	$\{1\}$
1	$E(\phi, \mathrm{Pic}(K))$	$\mathrm{Pic}(K)^\phi$
0	$\mathrm{Cliff}_\phi^{\mathcal{C}_0}(K)$	$\mathrm{Char}_\phi^{\mathcal{C}_0}(K)$.

(50)

Both spaces have trivial higher homotopy groups. The two class monoids in the last row are the base-K orbit sets computed in Section 7; see Remark 8.24. The identifications in degree one send the induced map to forgetting the action and the connecting map to ∂_2.

Proof. The space $\mathsf{CharSec}_\phi(K)$ is a 1-type. Hence the fibre sequence (49) gives $\pi_2(\mathsf{CliffSec}_\phi(K)) \cong \mathrm{U}(K)^\phi$ and the vanishing of the higher homotopy groups of both section spaces. A character loop over the constant homotopy at ϕ is an element of $\mathrm{Pic}(K)$ centralizing $g \mapsto [K_{1,\phi(g)}]$. These are exactly the invariant classes, by Proposition 7.2. A Clifford loop additionally specifies the compatible maps λ_g. They are the same data as semilinear maps $v_g : L \to L$ with $v_g \circ v_h = v_{gh}$: explicitly

$$v_g(\ell) = \lambda_g(1 \otimes_K \ell \otimes_K 1), \qquad \lambda_g(x \otimes_K \ell \otimes_K y) = x\phi(g)(y)v_g(\ell).$$

Here $v_g(a\ell) = \phi(g)(a)v_g(\ell)$ for $a \in K$. This semilinearity makes the formula for λ_g compatible with both tensor-balancing relations and with the left and right K-actions. Its inverse is $\ell \mapsto 1 \otimes_K v_{g^{-1}}(\ell) \otimes_K 1$. Homotopies of loops are the equivariant module isomorphisms, giving the first identification. Without the compatibility condition, the defect is a_L. In the fibre sequence (49), the connecting map measures this failure to choose the λ_g compatibly, so it sends $[L]$ to $[a_L]$. Tensoring choices proves $a_{L\otimes_K M} = a_L a_M$. The component formulas follow from Theorem 4.5 and the conjugacy calculation of Section 7. □

Theorem 8.26. *Let $\mathcal{C}$ be a class as in Definition 8.1. There is an exact sequence of abelian monoids due to Antony and the second author* [AG26, Theorem A]*:*

$$\{1\} \longrightarrow H^1(G, \mathrm{U}(K)_\phi) \longrightarrow E(\phi, \mathrm{Pic}(K)) \longrightarrow \mathrm{Pic}(K)^\phi \xrightarrow{\partial_2} H^2(G, \mathrm{U}(K)_\phi) \longrightarrow \mathrm{Cliff}_\phi^{\mathcal{C}}(K) \longrightarrow \mathrm{Char}_\phi^{\mathcal{C}}(K) \xrightarrow{\partial_3^{\mathcal{C}}} H^3(G, \mathrm{U}(K)_\phi). \tag{51}$$

The first line consists of abelian groups. Only the Clifford and character terms involve the Morita relation of Definition 8.7. The map joining the two lines is

$$H^2(G, \mathrm{U}(K)_\phi) \longrightarrow \mathrm{Cliff}_\phi^{\mathcal{C}}(K), \qquad [\alpha] \longmapsto [K_\phi^\alpha G].$$

Explicitly, $\partial_3^{\mathcal{C}}([R, \Phi]) = T(\Phi)$, and

$$\partial_3^{\mathcal{C}}([R, \Phi]) = 1 \quad \Longleftrightarrow \quad \Phi \text{ has a Clifford system realization over } R.$$

The same criterion holds for every character obtained from (R, Φ) by applying Ξ_P for a morphism P. The value $T(\Phi)$ is intrinsic to the character; the domain of $\partial_3^{\mathcal{C}}$, and hence its image, depends on the class.

Proof. The long exact sequence of (49), together with Proposition 8.25, gives the first line and its continuation into $\pi_0(\mathsf{CliffSec}_\phi(K))$ and $\pi_0(\mathsf{CharSec}_\phi(K))$.

We check the effect of passing to Morita classes. Suppose $K_\phi^\alpha G$ represents the unit Morita class. Lemma 8.14 represents the comparison by an automorphism $L : K \to K$ in $\mathbf{Mor}^{\mathcal{C}}$. Its class $[L]$ is ϕ-invariant, and the compatibility of its multiplication comparisons says precisely that $[\alpha]$ is in $\operatorname{im}\partial_2$. The reverse inclusion holds because the comparisons λ_g of Proposition 8.25 identify the multiplication transported from $K_\phi G$ by $\Xi_{L^\vee}$ with that of $K_\phi^{a_L} G$. Thus exactness at $H^2(G, \mathrm{U}(K)_\phi)$ survives the passage to Morita classes.

If a Clifford class maps to the unit character class, Lemma 8.14 supplies a single morphism witnessing this equality. Its associated functor gives a representative Clifford system over K with unit character. By Corollary 6.15 it is a crossed product $K_\phi^\alpha G$. This proves exactness at $\mathrm{Cliff}_\phi^{\mathcal{C}}(K)$. Conversely every such crossed product has unit character.

By Lemma 8.6, T is well defined on $\mathrm{Char}_\phi^{\mathcal{C}}(K)$. Theorem 6.4 gives its vanishing criterion and hence exactness there. Proposition 8.16 proves all remaining monoid-homomorphism assertions. □

Comparison convention for the Picard boundary. Let $\alpha_3^{\mathrm{AG}} : \mathrm{Pic}(K)^\phi \to H^2(G, \mathrm{U}(K)_\phi)$ be the boundary of [AG26, §2.3, (2.10)]. The semilinear maps of Construction 8.21 satisfy $v_g \circ v_h = a_L(g,h) v_{gh}$, so

$$v_{gh} \circ v_h^{-1} \circ v_g^{-1} = a_L(g,h)^{-1} 1_L,$$

as maps $L \longrightarrow L$. Consequently $\alpha_3^{\mathrm{AG}}([L]) = \partial_2([L])^{-1}$. The comparison with that sequence uses inversion on $H^1(G, \mathrm{U}(K)_\phi)$, equivariant duality on $E(\phi, \mathrm{Pic}(K))$, and $[L] \mapsto [L^\vee]$ on $\mathrm{Pic}(K)^\phi$, with the identity identifications from $H^2(G, \mathrm{U}(K)_\phi)$ onward.

Theorem 8.27. *For the class $\mathcal{C}$ of Theorem 8.26, the following identifications give the algebraic meaning of the sequence.*

(i) *There are isomorphisms of abelian monoids*

$$\begin{aligned} \pi_0(\mathsf{CliffSp}_\phi^{\mathcal{C}}(K)) &\cong \mathrm{Cliff}_\phi^{\mathcal{C}}(K), \\ \pi_0(\mathsf{CharSp}_\phi^{\mathcal{C}}(K)) &\cong \mathrm{Char}_\phi^{\mathcal{C}}(K). \end{aligned} \tag{52}$$

(ii) *The arrows, in order, have the following formulas:*

$$\begin{aligned} [b] &\longmapsto [K_b], \quad u_g x = b(g)\phi(g)(x), \\ [L, \{v_g\}] &\longmapsto [L], \\ [L] &\longmapsto [a_L], \\ [\alpha] &\longmapsto [K_\phi^\alpha G], \\ [A] &\longmapsto [g \mapsto [A_g]], \\ [R, \Phi] &\longmapsto T(\Phi). \end{aligned} \tag{53}$$

With the equivalence relations of Definition 8.7, these formulas recover the seven-term sequence of Antony and the second author [AG26, Theorem A] *for the chosen class $\mathcal{C}$, with the comparison convention above.*

Proof. The path description following Lemma 8.10 identifies comparisons over each R with identity-type invertible bimodules and compatible isomorphisms. The homotopy quotient also allows identity-central Morita morphisms between algebras. Lemma 8.14 combines every resulting zigzag into one such morphism with a compatible comparison. Conversely, each Morita equivalence of data gives such a zigzag. The component relations are therefore exactly those of Definition 8.7, proving (i). The first three arrow formulas follow from semilinear descent for invertible modules; the fourth is the crossed-product construction;

the fifth follows from Theorem 4.5; and the sixth is Theorem 6.2. Applying Ξ_P preserves the fifth formula and the scalar cocycle by Lemma 8.6. Tensor compatibility was proved above. □

Notation 8.28. Let $q^{\mathcal{C}}_{\mathrm{Cliff}}$ and $q^{\mathcal{C}}_{\mathrm{Char}}$ be the maps from isomorphism classes to Morita classes of Definition 8.9. Denote by $\Sigma^{\mathcal{C}}_{\phi}$, $\chi^{\mathcal{C}}_{\phi}$, and $T^{\mathcal{C}}_{\phi}$ the maps of (2) with their Clifford and character terms restricted to bases in $\mathfrak{U}(\mathcal{C})$. Write

$$\Sigma^{\mathrm{Mor},\mathcal{C}}_{\phi} : H^2(G, \mathrm{U}(K)_{\phi}) \to \mathrm{Cliff}^{\mathcal{C}}_{\phi}(K)$$

for the crossed-product map.

Proposition 8.29. *The maps of Notation 8.28 satisfy*

$$\begin{aligned} q^{\mathcal{C}}_{\mathrm{Cliff}} \circ \Sigma^{\mathcal{C}}_{\phi} &= \Sigma^{\mathrm{Mor},\mathcal{C}}_{\phi}, \\ q^{\mathcal{C}}_{\mathrm{Char}} \circ \chi^{\mathcal{C}}_{\phi} &= \chi^{\mathrm{Mor}}_{\phi} \circ q^{\mathcal{C}}_{\mathrm{Cliff}}, \\ \partial^{\mathcal{C}}_{3} \circ q^{\mathcal{C}}_{\mathrm{Char}} &= T^{\mathcal{C}}_{\phi}. \end{aligned} \tag{54}$$

These equalities have respective types $H^2(G, \mathrm{U}(K)_{\phi}) \to \mathrm{Cliff}^{\mathcal{C}}_{\phi}(K)$, $\mathrm{Cliff}^{\mathcal{C}}_{k}(\phi) \to \mathrm{Char}^{\mathcal{C}}_{\phi}(K)$, and $\mathcal{C}^{\mathcal{C}}_{k}(\phi) \to H^3(G, \mathrm{U}(K)_{\phi})$. The cohomology maps are identities. Nevertheless, the strict $\Sigma^{\mathcal{C}}_{\phi}$ is injective, whereas

$$\ker \Sigma^{\mathrm{Mor},\mathcal{C}}_{\phi} = \mathrm{im}\big(\partial_2 : \mathrm{Pic}(K)^{\phi} \to H^2(G, \mathrm{U}(K)_{\phi})\big). \tag{55}$$

Proof. All three identities follow on representatives from (53). Strict injectivity is the crossed-product classification, while the kernel formula is exactness at $H^2(G, \mathrm{U}(K)_{\phi})$ in Theorem 8.26. □

Proposition 8.30. *Suppose K is an integral domain and $\kappa \in \mathfrak{U}$ is a cardinal with $\kappa \geq \max\{|K|, |G|, \aleph_0\}$. Take $\mathcal{C} = \mathcal{C}_{\mathrm{fr},\kappa}(K)$. Then the obstruction map $\partial^{\mathcal{C}}_{3}$ is surjective onto $H^3(G, \mathrm{U}(K)_{\phi})$. In fact, a single K-central group algebra, depending only on K and G and of cardinality at most κ, carries characters of type ϕ with every prescribed obstruction class.*

Proof. For $G = \{e\}$ the cohomology group is trivial and the base algebra K suffices. Otherwise, the construction in [Gin22, proof of Thm. 5.5] uses a group H depending only on G. For $|G| > 2$, it is the free group on $(G \setminus \{e\})^2$. For $G = C_2$, it is the semidirect product of the free abelian group on $\mathbb{Z}$-indexed generators with an infinite cyclic group whose generator shifts the indices by 2. In both cases,

$$|H| \leq \max\{|G|, \aleph_0\}.$$

The group algebra $R = K[H]$ is K-central and free as a K-module. Its elements are finite K-linear combinations of elements of H, so

$$|R| \leq \max\{|K|, |H|, \aleph_0\} \leq \kappa.$$

The cited theorem constructs collective characters in $\mathrm{Out}_k(R)$ with every prescribed obstruction on this fixed algebra. For $\eta \in \mathrm{Aut}_k(R)$, let $[\eta]$ denote its outer class and let $R_{1,\eta}$ have its regular left R-action and right action $x \star_{\iota} r = x\eta(r)$. The embedding

$$\mathrm{Out}_k(R) \longrightarrow \mathrm{Pic}_k(R), \qquad [\eta] \longmapsto [R_{1,\eta}],$$

is that of [Gin22, (2.4)] and [Bas68, Ch. II, §5]. Theorem 6.2 identifies our obstruction with the second author's for the resulting generalized collective character; [Gin22, Lemma 5.4]

identifies it with the outer-action obstruction. Lemma 8.6 shows that these prescribed obstructions are unchanged on passing to Morita classes. □

Example 8.31. The image of the third-obstruction map depends on the class of algebras. For $k = K = \mathbb{R}$ and the trivial C_2-action, the class of finite-dimensional central simple real algebras has obstruction image $\{1\}$, although $H^3(C_2, \mathbb{R}^\times) \cong \{\pm 1\}$. Indeed, Morita equivalence and Skolem–Noether give $\mathrm{Pic}_{\mathbb{R}}(A) = \{1\}$ for every such algebra A; see [GS06, Thm. 2.7.2].

A proper, nontrivial image occurs over $k = K = \mathbb{R} \times \mathbb{R}$. Let $G = C_2$ act trivially, and take $\kappa = 2^{\aleph_0}$, which satisfies the bound of Proposition 8.30 for $\mathbb{R}$. Take the class

$$\mathcal{C}_{\mathrm{mix}} = \big\{ [A \times M_n(\mathbb{R})] \mid [A] \in \mathcal{C}_{\mathrm{fr},\kappa}(\mathbb{R}),\ n \geq 1 \big\},$$

with object collection $\mathfrak{U}(\mathcal{C}_{\mathrm{mix}})$. This class contains $[K]$ and is tensor closed, since

$$(A \times M_m(\mathbb{R})) \otimes_K (B \times M_n(\mathbb{R})) \cong (A \otimes_{\mathbb{R}} B) \times M_{mn}(\mathbb{R}).$$

The two central idempotents of K split every K-central invertible bimodule, giving

$$\mathrm{Pic}_K(A \times M_n(\mathbb{R})) \cong \mathrm{Pic}_{\mathbb{R}}(A) \times \mathrm{Pic}_{\mathbb{R}}(M_n(\mathbb{R})).$$

The multiplication maps split as well, so the associativity obstruction is computed coordinatewise. The second coordinate is always trivial, because $\mathrm{Pic}_{\mathbb{R}}(M_n(\mathbb{R})) = \{1\}$. Conversely, Proposition 8.30 realizes both first coordinates; pairing a realizing character on A with the trivial character on $\mathbb{R}$ gives a character on $A \times \mathbb{R}$. Thus, under the coordinatewise identification

$$H^3(C_2, \mathrm{U}(K)) \cong \{\pm 1\}^2,$$

the obstruction image is exactly

$$\mathrm{im}(\partial_3^{\mathcal{C}_{\mathrm{mix}}}) = \{(1,1), (-1,1)\}.$$

8.4. **The equivariance obstruction of the affine Hopf line.** Let $k = \mathbb{R}$ and

$$K = \mathbb{C}[x,y,z]/(x^2 + y^2 + z^2 - 1).$$

Let $G = C_2 = \{1, s\}$ act by the $\mathbb{R}$-algebra automorphism

$$\phi(s)(\mathrm{i}) = -\mathrm{i}, \qquad \phi(s)(x) = -x, \qquad \phi(s)(y) = -y, \qquad \phi(s)(z) = -z.$$

The affine Hopf line is the module $L = eK^2$, where

$$e = \frac{1}{2}\begin{pmatrix} 1+z & x - \mathrm{i}y \\ x + \mathrm{i}y & 1 - z \end{pmatrix} \in M_2(K).$$

The quadric relation gives $e^2 = e$. Since $\mathrm{tr}(e) = 1$, its image has rank one at every prime, so L is invertible.

Proposition 8.32. *The Hopf line L has $[L] \in \mathrm{Pic}(K)^\phi$, and*

$$\partial_2([L]) = [a] \neq 1 \quad \text{in } H^2(C_2, \mathrm{U}(K)_\phi), \qquad a(s,s) = -1,$$

where a is normalized. It represents the nontrivial class under $H^2(C_2, \mathrm{U}(K)_\phi) \cong \{\pm 1\}$. Consequently L admits no coherent C_2-equivariant structure.

Proof. The units of K are the constant units $\mathbb{C}^\times$: after compactifying the affine quadric to a smooth projective quadric, a unit has divisor supported on the irreducible hyperplane section at infinity, say nD; since $D^2 = 2$ and every principal divisor has zero intersection with D, one gets $n = 0$. Thus the induced action on $\mathrm{U}(K)$ is complex conjugation.

Let

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

A direct multiplication gives

$$A\,\phi(s)(e) = eA.$$

Therefore A restricts to a K-linear isomorphism

$$\lambda_s : {}^sL \cong \phi(s)(e)K^2 \xrightarrow{\sim} eK^2 = L.$$

Together with $\lambda_1 = \mathrm{id}_L$, this proves $[L] \in \mathrm{Pic}(K)^\phi$. Since A has real entries, identifying ${}^{s^2}L$ with L gives

$$\lambda_s \circ {}^s\lambda_s = \big(A\,\phi(s)(A)\big)|_L = A^2|_L = -\,\mathrm{id}_L : L \longrightarrow L.$$

Thus the normalized Picard-boundary cocycle has $a(s,s) = -1$ and all normalized values involving the identity equal to 1. This class is not a coboundary. Indeed, if $a = \partial b$ for a normalized 1-cochain $b : G \to \mathrm{U}(K) = \mathbb{C}^\times$, then

$$a(s,s) = (\partial b)(s,s) = b(s)\,\phi(s)(b(s)) = b(s)\overline{b(s)} \in \mathbb{R}_{>0},$$

which cannot equal -1. Equivalently,

$$H^2(G, \mathrm{U}(K)_\phi) = H^2(C_2, \mathbb{C}^\times_{\mathrm{conj}}) \cong \mathbb{R}^\times / N_{\mathbb{C}/\mathbb{R}}(\mathbb{C}^\times) \cong \{\pm 1\},$$

and a represents the nontrivial class. Hence

$$\partial_2([L]) = [a] \neq 1 \qquad \text{in } H^2(G, \mathrm{U}(K)_\phi).$$

The invariant Hopf line L therefore gives a nontrivial value of the Picard obstruction map in Theorem 8.26. □

The associated crossed products. For the normalized cocycle a above, put $B_- := K^a_\phi C_2$ and $B_+ := K_\phi C_2$. These are the free left K-modules with multiplication specified by

$$B_- = K \oplus Ku, \quad ub = \phi(s)(b)u, \quad u^2 = -1,$$
$$B_+ = K \oplus Kv, \quad vb = \phi(s)(b)v, \quad v^2 = 1,$$

for $b \in K$, with K in degree 1 and u, v in degree s. In terms of the generators of K, the first presentation includes

$$u\mathrm{i} = -\mathrm{i}u, \qquad ux = -xu, \qquad uy = -yu, \qquad uz = -zu.$$

The two crossed products are not isomorphic as graded extensions fixing K. Indeed, such an isomorphism would send u to cv for some $c \in \mathrm{U}(K) = \mathbb{C}^\times$, forcing

$$-1 = (cv)^2 = c\phi(s)(c) = c\bar{c} \in \mathbb{R}_{>0},$$

a contradiction.

Nevertheless, $\partial_2([L]) = [a]$ identifies B_- with the normalized $L^\vee$-conjugate of B_+, by the boundary calculation in Theorem 8.26. Thus they are Picard-conjugate and hence graded Morita equivalent, already without changing the base algebra K:

$$[B_-] = [B_+] \quad \text{in } \mathrm{Cliff}_k(C_2, K)_\phi / \mathrm{Pic}(K).$$

For $\mathcal{C}_0 = \{[K]\}$, Theorem 8.27 and Remark 8.24 therefore give

$$[B_-] = [B_+] = 1 \quad \text{in } \pi_0(\mathsf{CliffSp}_\phi^{\mathcal{C}_0}(K)).$$

Thus the nontrivial class $[a] \in H^2(C_2, \mathrm{U}(K)_\phi)$ lies in the kernel described in Proposition 8.29.

8.5. **A third obstruction over a real group algebra.** We give an explicit example over $\mathbb{R}$ of the group-algebra realization in Proposition 8.30. We will use the usual kernel calculation for automorphism bimodules. For $\alpha \in \mathrm{Aut}_k(R)$, write $R_{1,\alpha}$ for R with its regular left action and right action $x \star_\iota r = x\alpha(r)$. If $f : R \to R_{1,\alpha}$ is a bimodule isomorphism and $u = f(1)$, then u is a unit and $ru = u\alpha(r)$ for all $r \in R$. Thus $\alpha(r) = u^{-1}ru$. Conversely, for $u \in \mathrm{U}(R)$ with $\alpha(r) = u^{-1}ru$ for all $r \in R$, the formula $f(r) = ru$ defines a bimodule isomorphism $f : R \to R_{1,\alpha}$, with inverse $x \mapsto xu^{-1}$. Thus the kernel of $\mathrm{Aut}_k(R) \to \mathrm{Pic}_k(R)$ consists exactly of inner automorphisms.

Let $k = K = \mathbb{R}$, $G = C_2 = \{1, s\}$, and let ϕ be trivial. Let $T^{\mathbb{Z}}$ be the free abelian group generated by p_i, $i \in \mathbb{Z}$, and put

$$H = T^{\mathbb{Z}} \rtimes \langle w \rangle, \qquad wp_iw^{-1} = p_{i+2}, \qquad R = \mathbb{R}[H].$$

This is the group used in the C_2 case of the earlier construction [Gin22, proof of Thm. 5.5], following Eilenberg–Mac Lane [EML47]. Define $\rho : H \to H$ and $\chi : H \to \{\pm 1\}$ by

$$\rho(p_i) = p_{i+1}, \quad \rho(w) = w, \qquad \chi(p_i) = 1, \quad \chi(w) = -1,$$

and extend $\psi(h) = \chi(h)\rho(h)$ linearly to R. Then $\psi^2 = \mathrm{Inn}(w)$ as maps $R \to R$, where $\mathrm{Inn}(w)(r) = wrw^{-1}$.

Put $P_1 = R$ and let P_s be R with regular right action and left action $r \star_\ell x = \psi(r)x$. Here we use the left-twisted model: $x \mapsto \psi^{-1}(x)$ identifies P_s with $R_{1,\psi^{-1}}$, whose regular left action and twisted right action are $r \star_\ell x = rx$ and $x \star_\iota r = x\psi^{-1}(r)$. Since ψ^2 is inner, this represents the same Picard class as $R_{1,\psi}$. Thus

$$\Phi(1) = [R], \qquad \Phi(s) = [P_s]$$

is a character of type id. Since H is countable, $|R| = 2^{\aleph_0}$. Set $\mathcal{C} = \mathcal{C}_{\mathrm{fr},\kappa}(\mathbb{R})$ with $\kappa = 2^{\aleph_0} \in \mathfrak{U}$, or any larger cardinal in $\mathfrak{U}$.

Proposition 8.33. *The algebra R is $\mathbb{R}$-central and free as an $\mathbb{R}$-module. For the character Φ just defined,*

$$\partial_3^{\mathcal{C}}([R, \Phi]) = [t] \neq 1 \quad \text{in } H^3(C_2, \mathbb{R}^\times) \cong \{\pm 1\},$$
$$t(s, s, s) = -1,$$

where t is normalized. The same formula holds for any class of Definition 8.1 containing $[R]$. Consequently Φ admits no Clifford-system realization.

Proof. Order the finitely supported exponent vectors of $T^{\mathbb{Z}}$ lexicographically by their largest nonzero index. The shift $i \mapsto i + 2$ preserves this order. Ordering H first by its w-exponent and then by this order on the kernel gives a two-sided invariant total order. A nonidentity element of $T^{\mathbb{Z}}$ has infinitely many conjugates under powers of w, because its exponent vector has finite support. An element tw^m with $m \neq 0$ has infinitely many conjugates under powers of any p_i: the exponent vectors differ by multiples of $p_i - p_{i+2m}$. Thus every nonidentity conjugacy class is infinite. A central element of $\mathbb{R}[H]$ has finite conjugation-invariant support, so it is a scalar. This proves $Z(R) = \mathbb{R}$.

The two-sided order also shows that the least and greatest terms of a product of nonzero group-ring elements are the products of their respective least and greatest terms, with nonzero coefficients. If the product is 1, those two terms coincide, forcing each factor to have singleton support. Hence

$$\mathrm{U}(R) = \mathbb{R}^\times \times H.$$

The group basis makes R free over $\mathbb{R}$, as required.

The defining relation of H is preserved by ρ, which is invertible by shifting indices back by one, and $\rho^2 = \mathrm{Inn}(w)$ as maps $H \to H$. The character χ is ρ-invariant, so ψ is an algebra automorphism with the same square. Since $\psi(w) = -w$, it is not inner: conjugation by a unit ch sends group elements to group elements without changing their signs. Its outer class, and hence its Picard class, has exact order two by the kernel calculation above. This also verifies directly the character assertion above.

Choose

$$\mu_{s,s}(x \otimes_R y) = w^{-1}\psi(x)y, \qquad \mu_{1,s}(r \otimes_R p) = \psi(r)p, \qquad \mu_{s,1}(p \otimes_R r) = pr.$$

The respective sources and targets are

$$P_s \otimes_R P_s \to R,$$
$$R \otimes_R P_s \to P_s,$$
$$P_s \otimes_R R \to P_s.$$

Products on the right are products in R. The balancing identity is $\psi(xr)y = \psi(x)\psi(r)y$; left linearity uses $w^{-1}\psi^2(r) = rw^{-1}$. Thus these are bimodule maps, and the latter two are precisely the unitors. The first has inverse $r \mapsto 1 \otimes_R wr$, since $x \otimes_R y = 1 \otimes_R \psi(x)y$ in $P_s \otimes_R P_s$. The two triple products are maps $(P_s \otimes_R P_s) \otimes_R P_s \to P_s$, with the canonical associator inserted in the second. Evaluate them on $1 \otimes_R 1 \otimes_R 1 \in P_s \otimes_R P_s \otimes_R P_s$. The route

$$\mu_{1,s} \circ (\mu_{s,s} \otimes \mathrm{id})$$

gives $\psi(w^{-1}) = -w^{-1}$, while the route

$$\mu_{s,1} \circ (\mathrm{id} \otimes \mu_{s,s})$$

gives w^{-1}. Therefore the scalar in Theorem 6.2 is

$$t(s,s,s) = -1,$$

with all normalized values involving the identity equal to 1. This is a normalized 3-cocycle. It is not a coboundary: for a normalized 2-cochain $b : G^2 \to \mathbb{R}^\times$,

$$(\partial b)(s,s,s) = b(s,s)b(1,s)^{-1}b(s,1)b(s,s)^{-1} = 1.$$

Consequently

$$\partial_3^{\mathcal{C}}([R,\Phi]) = [t] \neq 1 \qquad \text{in } H^3(C_2, \mathbb{R}^\times) \cong \{\pm 1\}.$$

Thus $[R,\Phi]$ has nontrivial third obstruction and admits no Clifford-system realization. □

8.6. **Cyclic symmetries of a quantum torus.** In this and the next subsection, $k = K = \mathbb{C}$ and the coefficient action is trivial. Fix $q \in \mathbb{C}^\times$ that is not a root of unity, and let

$$A_q = \mathbb{C}\langle X^{\pm 1}, Y^{\pm 1} \mid XY = qYX\rangle.$$

The monomials X^aY^b form a basis as (a, b) ranges over $\mathbb{Z}^2$. Commutation with X and Y shows that $Z(A_q) = \mathbb{C}$. The same argument applies to every tensor power, with its independent pairs of generators. These algebras are infinite dimensional for positive tensor powers and hence are not Azumaya over their center.

Let $\mathcal{C}_q$ consist of the isomorphism classes of $\mathfrak{U}$-small $\mathbb{C}$-algebras Morita equivalent over $\mathbb{C}$ to some $A_q^{\otimes_{\mathbb{C}} d}$, $d \geq 0$, where $A_q^{\otimes_{\mathbb{C}} 0} = \mathbb{C}$. Its object collection is $\mathfrak{U}(\mathcal{C}_q)$. Tensoring invertible bimodules proves that this class is tensor closed. The Morita-invariant center remains $\mathbb{C}$.

We use the automorphism-bimodule convention and kernel calculation from Subsection 8.5.

Construction 8.34. Let $C_n = \langle s\rangle$ act trivially on an abelian group D. Write $\overline{a+b}$ for reduction modulo n, with representatives in $\{0, \dots, n-1\}$, and put $\epsilon(a, b) = \lfloor (a+b)/n \rfloor$. Write $D[n] := \{u \in D \mid u^n = 1\}$ and $\mu_n := \mathbb{C}^\times[n]$. For $u \in D[n]$, define the cochain

$$\omega_u(s^a, s^b, s^c) := u^{a\,\epsilon(b,c)}.$$

Lemma 8.35. *The cochain ω_u of Construction 8.34 is a normalized 3-cocycle. Under the cyclic periodic-resolution identification $H^3(C_n, D) \cong D[n]$, its class corresponds to u. In particular,*

$$H^3(C_n, \mathbb{C}^\times) \cong \mu_n, \qquad H^2(C_n, \mathbb{C}^\times) = \{1\}.$$

Proof. The carry identity is

$$\epsilon(a, b) + \epsilon(\overline{a+b}, c) = \epsilon(b, c) + \epsilon(a, \overline{b+c}).$$

Substitution in the inhomogeneous differential gives

$$(\partial\omega_u)(s^a, s^b, s^c, s^d) = u^{n\epsilon(a,b)\epsilon(c,d)} = 1.$$

The periodic resolution gives $H^3(C_n, D) = D[n]$ and $H^2(C_n, D) = D/D^n$; see [Bro82, p. 58]. Its degree-three comparison with the normalized bar resolution sends the generator to $\sum_{a=0}^{n-1}[s|s^a|s]$. Evaluation therefore sends a cocycle t to $\prod_{a=0}^{n-1} t(s, s^a, s)$, which is u for ω_u. For a coboundary the same product telescopes to 1. Finally every nonzero complex number has an nth root. □

Construction 8.36. Fix $n \geq 2$, choose $r \in \mathbb{C}^\times$ with $r^n = q$, and put $\zeta = \exp(2\pi\mathrm{i}/n)$. For $m \in \mathbb{Z}/n$, define

$$\sigma_m(X) = \zeta^{-m}X, \qquad \sigma_m(Y) = rY.$$

The bimodules $P_a^{(m)} = (A_q)_{1,\sigma_m^a}$ give the assignment

$$\Phi_m : C_n \longrightarrow \mathrm{Pic}_{\mathbb{C}}(A_q), \qquad s^a \longmapsto [P_a^{(m)}].$$

The multiplication comparison is the assignment

$$\mu_{a,b} : P_a^{(m)} \otimes_{A_q} P_b^{(m)} \longrightarrow P_{\overline{a+b}}^{(m)}, \qquad x \otimes_{A_q} y \longmapsto x\sigma_m^a(y)X^{\epsilon(a,b)}. \tag{56}$$

For $m = 0$, consider the graded algebra

$$A_r = \mathbb{C}\langle Z^{\pm 1}, Y^{\pm 1} \mid ZY = rYZ\rangle, \qquad X = Z^n, \quad \deg Z = s, \quad \deg Y = e.$$

Proposition 8.37. *For the data of Construction 8.36, $\sigma_m^n = \mathrm{Ad}_X$ as maps $A_q \to A_q$, where $\mathrm{Ad}_X(x) = XxX^{-1}$. The assignment Φ_m is a character. Its obstruction, with the preceding cyclic identification, is*

$$\partial_3^{\mathcal{C}_q}([A_q, \Phi_m]) = \zeta^m \quad \text{in } H^3(C_n, \mathbb{C}^\times) \cong \mu_n. \tag{57}$$

Thus the obstruction map for $\mathcal{C}_q$ is surjective, already on the characters over A_q. The character Φ_m is realizable if and only if $m = 0$. For $m = 0$ its realization is unique up to graded isomorphisms restricting to the identity on A_q, and is represented by A_r of Construction 8.36.

Proof. Scaling the two generators preserves the relation. The nth power fixes X and sends Y to qY, which is conjugation by X. Set $v(a,b) = X^{\epsilon(a,b)}$. Then

$$\sigma_m^a \circ \sigma_m^b = \mathrm{Ad}_{v(a,b)} \circ \sigma_m^{\overline{a+b}} : A_q \longrightarrow A_q.$$

Consequently the maps $\mu_{a,b}$ of Construction 8.36 are normalized bimodule isomorphisms $P_a^{(m)} \otimes_{A_q} P_b^{(m)} \to P_{\overline{a+b}}^{(m)}$. For instance, right linearity is the displayed inner-automorphism identity, and balancing follows from the right twist of $P_a^{(m)}$.

The carry identity cancels the powers of X in the two triple products. The remaining scalar comes from

$$\sigma_m^a(X^{\epsilon(b,c)}) = \zeta^{-ma\,\epsilon(b,c)} X^{\epsilon(b,c)}.$$

In the convention "left composite equals t times right composite," this gives

$$t_m(s^a, s^b, s^c) = \zeta^{ma\,\epsilon(b,c)}.$$

Lemma 8.35 proves (57). The vanishing criterion follows from Theorem 8.26, and uniqueness in the stated sense follows from $H^2(C_n, \mathbb{C}^\times) = \{1\}$.

In the displayed realization A_r, the degree-e subalgebra is generated by $Z^{\pm n}$ and $Y^{\pm 1}$ and is A_q. The other components are $A_q Z^a$, all containing homogeneous units. The relation $Zx = \sigma_0(x)Z$ identifies them with $P_a^{(0)}$, and $Z^n = X$ gives exactly (56). Thus the grading is strong and realizes the specified character. □

Example 8.38. For $n = 2$ and $m = 1$, the automorphism $\sigma(X) = -X$, $\sigma(Y) = \sqrt{q}\,Y$ has $\sigma^2 = \mathrm{Ad}_X$. Its bimodule P satisfies $P \otimes_{A_q} P \cong A_q$, but its associativity defect for the triple product is -1. Thus $A_q \oplus P$ admits no strongly C_2-graded algebra structure with these components and the specified base action.

8.7. **Morita equivalences and cancellation of obstructions.** Let I be a nonzero finitely generated projective right ideal of A_q, and put $S = \mathrm{End}_{A_q}(I)$ and $I^\vee = \mathrm{Hom}_{A_q}(I, A_q)$, where the Hom is taken in right A_q-modules. The quantum torus is simple, so the nonzero trace ideal of I is all of A_q; hence I is a generator. It defines an identity-central morphism $I^\vee : A_q \to S$ with inverse I. Noncyclic projective ideals exist, and their classification is described in [BRT13, §§2–4].

Construction 8.39. For each m, use the canonical identification $(I^\vee)^\vee \cong I$ and set

$$Q_a^{(m)} := I \otimes_{A_q} P_a^{(m)} \otimes_{A_q} I^\vee, \qquad \Psi_m(s^a) := [Q_a^{(m)}].$$

Multiplication comparisons evaluate $I^\vee \otimes_S I$ and then apply $\mu_{a,b}$. For $m = 0$, the candidate graded realization is

$$\bigoplus_{a=0}^{n-1} I \otimes_{A_q} (A_q Z^a) \otimes_{A_q} I^\vee,$$

with multiplication by evaluation and multiplication in A_r.

Proposition 8.40. *The assignment Ψ_m of Construction 8.39 is a character with $T(\Psi_m) = \zeta^m$. It is realizable precisely for $m = 0$; in that case its realization is unique up to graded isomorphism fixing S. If I is noncyclic and $m \neq 0$, the generator class $[Q_1^{(m)}]$ is not represented by an automorphism bimodule $S_{1,\theta}$.*

Proof. The multiplication comparisons on $Q_a^{(m)}$ evaluate the middle $I^\vee \otimes_S I$ and then apply $\mu_{a,b}$. Lemma 8.6 preserves their scalar cocycle, proving the obstruction formula and realization criterion. For $m = 0$, use the graded algebra in Construction 8.39. Its identity component is S. Uniqueness again follows from $H^2(C_n, \mathbb{C}^\times) = \{1\}$.

For a noncyclic ideal, [BRT13, Thm. 2] gives $\mathrm{Aut}_{A_q}(I) = \mathbb{C}^\times$, with the canonical scalar action. Therefore $S^\times = \mathbb{C}^\times$ and $\mathrm{Inn}(S) = \{1\}$. If $[Q_1^{(m)}] = [S_{1,\theta}]$, then the fact that its nth power is the identity in $\mathrm{Pic}_{\mathbb{C}}(S)$ would imply that θ^n is inner, hence equal to the identity. The strict cyclic action generated by θ would then give a skew group algebra realizing Ψ_m. This contradicts $T(\Psi_m) = \zeta^m \neq 1$. □

Thus, for noncyclic I and $m \neq 0$, the classes $[Q_1^{(m)}] \in \mathrm{Pic}_{\mathbb{C}}(S)$ have finite order and are not represented by automorphism bimodules. They are obtained from the displayed automorphism bimodules over A_q by applying $\Xi_{I^\vee}$, which preserves the obstruction.

Construction 8.41. Set $R = A_q \otimes_{\mathbb{C}} A_q$, $\alpha = \sigma_m \otimes \sigma_{-m}$, and $U = X \otimes_{\mathbb{C}} X$. Consider the cyclic graded algebra

$$B = R[z^{\pm 1}; \alpha]/(z^n - U), \qquad zx = \alpha(x)z \quad \text{for all } x \in R, \qquad \deg z = s.$$

Here $R[z^{\pm 1}; \alpha]$ is the skew Laurent extension defined by the displayed commutation relation.

Proposition 8.42. *For the diagonal C_n-character on $A_q \otimes_{\mathbb{C}} A_q$,*

$$T(\Phi_m \otimes_{\mathbb{C}} \Phi_\ell) = \zeta^{m+\ell}.$$

It is realizable if and only if $m + \ell = 0$ in $\mathbb{Z}/n$. In particular, for $m \neq 0$, neither factor of $\Phi_m \otimes_{\mathbb{C}} \Phi_{-m}$ is realizable, but their product is.

The algebra B of Construction 8.41 realizes $\Phi_m \otimes_{\mathbb{C}} \Phi_{-m}$.

Proof. Obstruction multiplication was proved in Proposition 8.16; applying it to (57) gives the formula and exactness gives the criterion. For the presentation, directly compute

$$\alpha^n = \mathrm{Ad}_U : R \longrightarrow R, \qquad \alpha(U) = U.$$

These identities make the relation $z^n = U$ compatible with both R and z. More explicitly, the free left R-module $\bigoplus_{a=0}^{n-1} Rz^a$ has multiplication

$$(xz^a)(yz^b) = x\alpha^a(y)U^{\epsilon(a,b)} z^{\overline{a+b}}.$$

Its associativity follows from the carry identity, $\alpha^n = \mathrm{Ad}_U$, and $\alpha(U) = U$, exactly as in the preceding cocycle calculation with scalar defect 1. This constructs the quotient and proves

that no additional relations collapse its normal form. Since U is a unit, z is invertible, and the grading is strong. Its degree-s^a bimodule is $R_{1,\alpha^a} = P_a^{(m)} \otimes_{\mathbb{C}} P_a^{(-m)}$, as required. □

The base $A_q \otimes_{\mathbb{C}} A_q$ is not Morita equivalent to $\mathbb{C}$: a unital algebra Morita equivalent to $\mathbb{C}$ is a finite matrix algebra, whereas this tensor product is infinite dimensional. Thus the product character is not the identity of the character monoid, although it lies in the kernel of $\partial_3^{\mathcal{C}_q}$ and has the explicit realization constructed above.

Remark 8.43. The Picard and Morita theory of algebraic quantum tori is developed in [BRT13]. In the operator-algebra setting, Jones proves that for every $n \geq 1$, each class in $H^3(\mathbb{Z}/n\mathbb{Z}, \mathrm{U}(1))$ occurs as an anomaly on each irrational rotation C^*-algebra [Jon20, Cor. 3.6]; here $\mathrm{U}(1)$ is the complex unit circle with trivial $\mathbb{Z}/n\mathbb{Z}$-action. For $q = \exp(2\pi \mathrm{i}\theta)$ with θ irrational and a unit-modulus choice of r, our scaling automorphisms also preserve the involution on the unitary generators and extend to that completion.

9. The Brauer three-type of Azumaya algebras

We construct the Brauer 3-type $\mathcal{B}\mathrm{r}(K)$ from Azumaya K-algebras, invertible bimodules, and their isomorphisms. Scalar twisting by $\Gamma = \mathrm{Aut}_k(K)$ gives the projection $p_{\mathrm{Br}} : \mathcal{B}\mathrm{r}_\Gamma(K) \to B_\bullet\Gamma$. We construct its second Postnikov truncation, prove the Kan fibration property, and compute the homotopy fibre. These constructions will give the obstruction sequence in Section 10.

9.1. **Azumaya algebras and Morita invertibility.**

Definition 9.1. An Azumaya K-algebra A is faithful, finitely generated and projective over K, and the map

$$A \otimes_K A^{\mathrm{op}} \longrightarrow \mathrm{End}_K(A), \qquad a \otimes_K b^{\mathrm{op}} \longmapsto (x \mapsto axb),$$

is an isomorphism. Two K-algebras R, S are *Morita equivalent over* K if there are bimodules $P \in {}_R\mathbf{Mod}_S$ and $Q \in {}_S\mathbf{Mod}_R$, with coincident central K-actions, and bimodule isomorphisms

$$\epsilon_R : P \otimes_S Q \xrightarrow{\sim} R, \qquad \epsilon_S : Q \otimes_R P \xrightarrow{\sim} S,$$

satisfying the triangle identities

$$\epsilon_R \otimes 1_P = 1_P \otimes \epsilon_S,$$
$$\epsilon_S \otimes 1_Q = 1_Q \otimes \epsilon_R,$$

as maps $P \otimes_S Q \otimes_R P \to P$ and $Q \otimes_R P \otimes_S Q \to Q$, respectively, with canonical reassociations and unit maps understood. Equivalently, the categories of left modules are equivalent over K: the functors $P \otimes_S - : {}_S\mathbf{Mod} \to {}_R\mathbf{Mod}$ and $Q \otimes_R - : {}_R\mathbf{Mod} \to {}_S\mathbf{Mod}$ are inverse equivalences [Bas68, Ch. II]. The resulting equivalence classes of K-algebras are called *Morita classes*. The Brauer group $\mathrm{Br}(K)$ consists of the Morita classes of Azumaya K-algebras, with product $[A][B] = [A \otimes_K B]$, identity $1 = [K]$, and inverse $[A]^{-1} = [A^{\mathrm{op}}]$; see [AG60].

Lemma 9.2. *Let R be a unital K-central algebra. Then R is Azumaya over K if and only if there exists a unital K-central algebra S such that*

$$R \otimes_K S \simeq_{\mathrm{Mor}} K,$$

where the Morita equivalence is over K in the preceding sense. When these conditions hold, one may take $S = R^{\mathrm{op}}$.

Proof. Apply [Joh14, arXiv version, Thm. 1.6 and §2.1] to the bicategory of unital K-algebras and unital K-central bimodules. The existence of a tensor inverse up to Morita equivalence implies that R is faithful, finitely generated and projective over K and that the canonical map

$$R \otimes_K R^{\mathrm{op}} \longrightarrow \mathrm{End}_K(R), \qquad r \otimes_K s^{\mathrm{op}} \longmapsto (x \mapsto rxs),$$

is an isomorphism. These are precisely the conditions in Definition 9.1.

Conversely, if R is Azumaya, this isomorphism and the fact that R is a K-progenerator give

$$R \otimes_K R^{\mathrm{op}} \cong \mathrm{End}_K(R) \simeq_{\mathrm{Mor}} K. \qquad \square$$

The requirement that the algebras have identity elements is essential here. Brauer–Taylor theory also allows algebras without identity; a Taylor–Azumaya algebra with identity is an ordinary Azumaya algebra [CVOZ06, §1.1].

Definition 9.3. We denote composition in Γ by $\circ$, so $(\eta \circ \theta)(a) = \eta(\theta(a))$. For a K-central algebra R and $\eta \in \Gamma$, the right twist R^{η} is the underlying k-algebra R with its specified central map $K \to Z(R)$ precomposed with η. Thus

$$(R^{\eta})^{\theta} = R^{\eta\circ\theta}.$$

Notation 9.4. Let $\mathcal{C}_{\mathrm{Az}}(K)$ be the full class in Example 8.2(iv): it consists of the isomorphism classes of all Azumaya K-algebras, equivalently all Morita-invertible K-central algebras by Lemma 9.2. We abbreviate this class by $\mathcal{A}\mathrm{z}$, and use it as the superscript naming the Azumaya case. The associated object collection $\mathfrak{U}(\mathcal{A}\mathrm{z})$ consists of all $\mathfrak{U}$-small Azumaya K-algebras. It is closed under $\otimes_K$, $(-)^{\mathrm{op}}$, and the scalar twists $(-)^{\eta}$ for $\eta \in \Gamma = \mathrm{Aut}_k(K)$.

The one-object Azumaya tricategory, its Brauer 3-type $\mathcal{B}\mathrm{r}(K)$, and the total simplicial sets $\mathcal{B}\mathrm{r}_{\Gamma}(K)$ and $\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K)$ below are formed from $\mathfrak{U}(\mathcal{A}\mathrm{z})$. Thus the projections p_{Br} and $p_{\mathrm{Br},\leq 2}$ to $B_{\bullet}\Gamma$ have first fibre homotopy groups (Definition A.20)

$$\pi_1\big(\mathrm{hofib}_*(p_{\mathrm{Br}})\big) \cong \pi_1\big(\mathrm{hofib}_*(p_{\mathrm{Br},\leq 2})\big) \cong \mathrm{Br}(K),$$

where $*$ is the unique vertex of $B_{\bullet}\Gamma$ and $\mathrm{Br}(K)$ is the full Brauer group; see the proof of Proposition 9.24. Write $\partial_3 := \partial_3^{\mathcal{A}\mathrm{z}}$ for the corresponding algebraic obstruction map.

Definition 9.5. Let $\mathcal{C}$ be a class of K-central algebras in the sense of Definition 8.1. Its *Morita-invertible part* is the sub-class

$$\mathcal{C} \cap \mathcal{A}\mathrm{z} := \{[R] \in \mathcal{C} \mid [R] \text{ is invertible under } \otimes_K \text{ up to Morita equivalence}\},$$

where invertibility means that there is a unital K-central algebra S such that $R \otimes_K S$ is K-linearly Morita equivalent to K; the algebra S need not represent a class in $\mathcal{C}$. By Lemma 9.2 these are precisely the classes in $\mathcal{C}$ represented by Azumaya K-algebras, so the

notation agrees with the Azumaya class $\mathcal{A}\mathrm{z}$ of Notation 9.4. The associated object collection is

$$\mathfrak{U}(\mathcal{C}\cap\mathcal{A}\mathrm{z}) = \{R \in \mathfrak{U}(\mathcal{C}) \mid [R] \in \mathcal{C}\cap\mathcal{A}\mathrm{z}\}.$$

The Morita classes of its members lie in the Brauer group $\mathrm{Br}(K)$ of Definition 9.1. Since $\mathcal{C}$ is closed under tensor product and Morita-invertibility is preserved by tensor product, $\mathfrak{U}(\mathcal{C}\cap\mathcal{A}\mathrm{z})$ is again tensor closed and contains K. We define $\mathbf{Mor}^{\mathcal{C}\cap\mathcal{A}\mathrm{z}}$ as the full subgroupoid of $\mathbf{Mor}^{\mathcal{C}}$ on these objects. The inclusion induces a monoidal functor

$$\mathbf{Mor}^{\mathcal{C}\cap\mathcal{A}\mathrm{z}} \longrightarrow \mathbf{Mor}^{\mathcal{C}}.$$

Remark 9.6. The singleton class $\{[K]\}$ gives the sequence for extensions of K. The split matrix class gives the same Morita-class terms as the singleton class $\{[K]\}$, since every $M_n(K)$ is identity-centrally Morita equivalent to K. For a subgroup $B \le \mathrm{Br}(K)$, restricting the base Brauer classes to B gives a group-like construction over $B_\bullet G$ when B is $\phi(G)$-stable, and over $B_\bullet\Gamma$ when it is Γ-stable. The seven-term sequence applies to a tensor-closed class of K-central algebras without requiring its members to be Morita-invertible; the Brauer three-type continuation uses that additional hypothesis.

Construction 9.7. The inclusion $\mathfrak{U}(\mathcal{C}\cap\mathcal{A}\mathrm{z}) \subseteq \mathfrak{U}(\mathcal{C})$ gives the assignments on Morita classes

$$\iota^{\mathcal{C}}_{\mathrm{Cliff}} : \mathrm{Cliff}^{\mathcal{C}\cap\mathcal{A}\mathrm{z}}_{\phi}(K) \longrightarrow \mathrm{Cliff}^{\mathcal{C}}_{\phi}(K), \qquad \iota^{\mathcal{C}}_{\mathrm{Char}} : \mathrm{Char}^{\mathcal{C}\cap\mathcal{A}\mathrm{z}}_{\phi}(K) \longrightarrow \mathrm{Char}^{\mathcal{C}}_{\phi}(K).$$

The same inclusion induces simplicial maps on the homotopy quotients of Construction 8.11; their effect on components is identified with these assignments by Theorem 8.27.

Proposition 9.8. *Together with the identity maps on the terms defined over K and on $H^i(G, \mathrm{U}(K)_\phi)$ for $1 \le i \le 3$, the maps of Construction 9.7 give a map of the two seven-term sequences. On the class monoids it is*

$$\begin{array}{ccccc} \mathrm{Cliff}^{\mathcal{C}\cap\mathcal{A}\mathrm{z}}_{\phi}(K) & \to & \mathrm{Char}^{\mathcal{C}\cap\mathcal{A}\mathrm{z}}_{\phi}(K) & \to & H^3(G, \mathrm{U}(K)_\phi) \\ \downarrow & & \downarrow & & \| \\ \mathrm{Cliff}^{\mathcal{C}}_{\phi}(K) & \to & \mathrm{Char}^{\mathcal{C}}_{\phi}(K) & \to & H^3(G, \mathrm{U}(K)_\phi). \end{array}$$

Proof. The inclusion of the groupoids of invertible bimodules respects the functors Ξ_P and their comparisons, and commutes with truncation. It therefore induces the displayed maps. The section spaces over K are unchanged, and Ξ_P preserves central scalar labels by Lemma 8.6. Hence the cohomology maps are identities and the obstruction maps commute. □

9.2. **Azumaya tricategories and their nerves.** We form the Azumaya tricategory $\mathbf{BAz}_K$ of Azumaya K-algebras and its scalar-twisted version, then describe their nerves. We use the inhomogeneous bar simplicial sets $B_\bullet G$ and $B_\bullet\Gamma$ of Definition A.3, with the map $B_\bullet\phi[g_1|\cdots|g_n] = [\phi(g_1)|\cdots|\phi(g_n)]$. The tensor-product convention below twists earlier factors by the degrees to their right.

Construction 9.12 uses the Azumaya bicategory associated with a fixed Γ-label. In the resulting Street nerve, its objects label edges, its 1-cells label triangles, and its 2-cells label tetrahedra, as made explicit in Proposition 9.14. The scalar-twist automorphisms at the end of the definition assemble these fixed-label bicategories into the semidirect tricategory of Definition A.7.

Definition 9.9. For K-central algebras R and S, we write $P \in {}_R\mathbf{Mod}_S$ when P is a $\mathfrak{U}$-small left-R, right-S-bimodule for which the induced K-actions agree:

$$a \star_R p = p \star_S a \quad \text{for all } a \in K \text{ and } p \in P,$$

where a acts on the left through the central map $K \to Z(R)$ and on the right through the central map $K \to Z(S)$. A map in ${}_R\mathbf{Mod}_S$ is an R–S-bimodule map. This is also the convention for morphisms $P : R \to S$ in Definition 8.4. An *inverse bimodule pair* consists of

$$P \in {}_R\mathbf{Mod}_S, \qquad Q \in {}_S\mathbf{Mod}_R,$$

and bimodule isomorphisms

$$\epsilon : P \otimes_S Q \xrightarrow{\sim} R, \qquad \delta : Q \otimes_R P \xrightarrow{\sim} S.$$

Here R and S are the identity bimodules. Under the canonical associativity and unit identifications, the isomorphisms must satisfy the triangle identities

$$(\epsilon \otimes 1_P) \circ (1_P \otimes \delta^{-1}) = 1_P,$$

$$(1_Q \otimes \epsilon) \circ (\delta^{-1} \otimes 1_Q) = 1_Q,$$

as maps $P \longrightarrow P$ and $Q \longrightarrow Q$, respectively. The coevaluation is δ^{-1}. A bimodule P is *Morita-invertible* if it occurs in such a pair. For the specific Hom inverse, constructed as in Definition 8.4, we write

$$P^\vee := \mathrm{Hom}_R(P, R),$$

where the maps are left R-linear, with the resulting S–R-bimodule structure. Vertical composition is relative tensor product: if $M \in {}_R\mathbf{Mod}_S$ and $N \in {}_S\mathbf{Mod}_T$, then $M \otimes_S N \in {}_R\mathbf{Mod}_T$. The identity bimodule on R is R itself.

In the sense of Definition A.5, let $\mathbf{Az}_K$ denote the $\mathfrak{V}$-small bicategory whose objects are the algebras in $\mathfrak{U}(\mathcal{A}\mathrm{z})$ of Notation 9.4; whose 1-cells are Morita-invertible bimodules; whose 2-cells are bimodule isomorphisms; and whose composition and units are as above. Its associativity and unit constraints are the canonical balanced-tensor isomorphisms

$$(M \otimes_S N) \otimes_T U \cong M \otimes_S (N \otimes_T U), \qquad R \otimes_R M \cong M \cong M \otimes_S S.$$

For $\theta \in \Gamma$ and $P \in {}_R\mathbf{Mod}_S$, write $P^\theta \in {}_{R^\theta}\mathbf{Mod}_{S^\theta}$ for the same underlying bimodule, viewed between the scalar-twisted central algebras. This gives a bicategory automorphism of $\mathbf{Az}_K$ for every θ. Scalar twisting changes the specified K-actions but not the underlying rings or the actions of R and S on P. For composable bimodules $P : R \to S$ and $Q : S \to T$, the canonical comparisons

$$(P \otimes_S Q)^\theta \cong P^\theta \otimes_{S^\theta} Q^\theta, \qquad (R \otimes_K S)^\theta \cong R^\theta \otimes_K S^\theta$$

send each tensor class to the same underlying tensor class. The first comparison leaves the balancing relations unchanged, and the second does so because θ is surjective and twists both K-actions simultaneously. The second formula also holds for external tensor products of bimodules. With the usual quotient construction, these comparisons are identity maps on the underlying quotients. Likewise $(P^\theta)^\nu = P^{\theta\circ\nu}$, including its ring actions. Tensor product over K, on algebras, bimodules, and their isomorphisms, equips $\mathbf{Az}_K$ with a symmetric monoidal structure. We write $\mathbf{BAz}_K$ for its one-object delooping: its endomorphism bicategory is $\mathbf{Az}_K$, and horizontal composition is this tensor product.

Remark 9.10. For $\mathcal{M} = \mathrm{Mod}(K)$, the $\mathfrak{V}$-small category of $\mathfrak{U}$-small K-modules, Street's monoidal bicategory of algebras, restricted to Azumaya algebras, Morita equivalences, and bimodule isomorphisms, is precisely $\mathbf{Az}_K$ [Str04, arXiv v1, §12, pp. 38–39]. His arrow $R \to S$ is also a left-R, right-S bimodule. The comparison is the identity on these three kinds of data; relative tensor product, tensor product over K, and their canonical comparisons agree. It is a monoidal biequivalence using the same $\mathfrak{U}$-small objects and cells. Taking the one-object deloopings of these monoidal bicategories gives a triequivalence between $\mathbf{BAz}_K$ and Street's Azumaya tricategory. Street's construction replaces this delooped tricategory by a triequivalent Gray-category model [GPS95] before taking its nerve. This optional comparison is made in $\mathbf{sSet}_{\mathfrak{V}}$. The nerve comparison [CH14, arXiv v2, Thms. 4.2.4(ii), 5.3.1] gives a weak equivalence between the nerve of $\mathbf{BAz}_K$ and the nerve of Street's invertible Gray-category. Szymik's account identifies the latter nerve with Duskin's Azumaya complex [Dus88]; our comparison with that complex uses [Szy17, arXiv v2, §4, pp. 16–17].

The identity Γ-labels in Construction 9.12 recover this untwisted tricategory and its nerve $\mathcal{B}\mathrm{r}(K)$. Scalar twisting acts on the same algebras, bimodules, and maps, with canonical comparisons $(P \otimes_S Q)^\eta \cong P^\eta \otimes_{S^\eta} Q^\eta$ and $(R \otimes_K S)^\eta \cong R^\eta \otimes_K S^\eta$. The semidirect projection p_{Br} records this coherent action. Here we use the Duskin–Street Azumaya complex with its scalar action and algebraic section data.

Street's relative nine-term sequence is induced by

$$\mathrm{Br}(F) : \mathrm{Br}(\mathcal{M}) \to \mathrm{Br}(\mathcal{N}),$$

for a right-exact braided strong-monoidal functor F, in particular scalar extension $- \otimes_K L$ for a commutative ring map $K \to L$. The sequence here instead uses the map on sections over $B_\bullet G$ induced by the second Postnikov truncation

$$q : \mathcal{B}\mathrm{r}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K).$$

For an algebra R in $\mathfrak{U}(\mathcal{A}\mathrm{z})$, let $\boldsymbol{\mathcal{P}ic}_K(R)$ denote the full monoidal subgroupoid of $\boldsymbol{\mathcal{P}ic}_k(R)$ consisting of invertible bimodules whose two K-actions agree. This is the endomorphism 2-group of R in $\mathbf{Az}_K$. By Lemma 3.4,

$$\pi_0\big(\boldsymbol{\mathcal{P}ic}_K(R)\big) = \ker\big(h_R : \mathrm{Pic}_k(R) \longrightarrow \mathrm{Aut}_k(K)\big)\,.$$

Bimodules of arbitrary type enter through scalar twisting. Indeed, an invertible R-bimodule P over k has type $\eta \in \Gamma$ precisely when

$$p \star_\iota a = \eta(a) \star_\ell p \qquad \text{for all } a \in K \text{ and } p \in P.$$

By Definition 9.3, this is exactly the condition that the same underlying bimodule, regarded as a left-R^η, right-R bimodule, belongs to ${}_{R^\eta}\mathbf{Mod}_R$. Thus type-$\phi$ data are recovered using these twisted bimodules, with $\eta = \phi(g)$ for each $g \in G$.

Remark 9.11. We use $\mathfrak{U}(\mathcal{A}\mathrm{z})$ from Notation 9.4 and the nested universes of the conventions. These algebras, together with their $\mathfrak{U}$-small bimodules and isomorphisms, give a $\mathfrak{V}$-small set of labels on each finite ordinal $[n]$.

Put $\eta_g := \phi(g)$. For a K-module L, let L^η denote the corresponding right twist: it has the same underlying abelian group and scalar action $a \star_K^\eta x := \eta(a) \star_K x$. Thus $(L^\eta)^\theta = L^{\eta\circ\theta}$, and we define the induced *left* action by ${}^g L := L^{\eta_g^{-1}}$. Thus ${}^{gh}L = {}^g({}^h L)$ canonically. Under the standard identification $\mathrm{Aut}_K(L) \cong \mathrm{U}(K)$ for invertible L, this left action sends a scalar a to $\eta_g(a)$; the inverse in the definition is necessary for this left-action convention. It agrees

with the conjugation action of (34): for $\eta \in \Gamma$, the map $x \otimes_K \ell \otimes_K y \mapsto \eta^{-1}(x)\ell y$ identifies $K_{1,\eta} \otimes_K L \otimes_K K_{1,\eta^{-1}}$ with $L^{\eta^{-1}}$.

Convention dictionary. The conventions needed for the nerve and its coefficient cochains are:

Right scalar twist	$(R^\eta)^\theta = R^{\eta\circ\theta}$
Induced left action	${}^gR = R^{\eta_g^{-1}}$, ${}^gL = L^{\eta_g^{-1}}$
Arrow in $\mathbf{Mor}^{\mathcal{C}}$, $\mathbf{Az}_K$	$P : R \to S$ means $P \in {}_R\mathbf{Mod}_S$
Bar label	$[\eta]$ labels the categorical arrow η^{-1}
Coefficient maps along $[\eta]$	forward $a \mapsto \eta^{-1}(a)$, backward $a \mapsto \eta(a)$
Scalar on a p-simplex	$u = \eta_{g_1\cdots g_p}(c)$

In the last row, c multiplies the underlying bimodule map, while u is its initial-fibre coordinate in a left-module group cochain. Thus rescaling by the cochain value u multiplies that map by $\eta^{-1}_{g_1\cdots g_p}(u)$. Over a fixed base this is the convention of Section 5: u is the $\lambda^\sharp$-coordinate in (29) and c the $\rho^\sharp$-coordinate, since $\rho^\sharp_P(c) = \lambda^\sharp_P(\eta_P(c))$ by Proposition 3.5. In $\mathbf{Az}_K$ every 1-cell is Morita-invertible and every 2-cell is an isomorphism, and in the deloopings $\mathbf{BAz}_K$ and $\mathbf{BAz}^\Gamma_K$ every cell is likewise invertible, strictly in the top dimension and up to higher cells below it.

Street-nerve notation used in this section. For a tricategory $\mathcal{T}$ and an ordinal category $[n]$, write

$$\mathrm{Rep}_u([n], \mathcal{T})$$

for the set of unitary representations of $[n]$ in $\mathcal{T}$, equivalently the normal lax functors of Definition A.9 in the sense of Cegarra–Heredia's unitary lax functors [CH14, pp. 2005–2006]; see also Definition A.8. The subscript u stands for *unitary*: degenerate edges and higher degenerate simplices are sent to the specified unit cells. Concretely, such a representation assigns objects to vertices, 1-cells to edges, comparison 2-cells to triangles, comparison 3-cells to tetrahedra, and satisfies the five-vertex coherence equation. If $\mathcal{B}$ is only a bicategory, we regard it as a tricategory with identity 3-cells; then the tetrahedral comparison is an equality of pasted 2-cells, and the five-vertex coherence is then an equality between identity 3-cells. With the inverse labelling of Definition A.3, $\mathrm{Rep}_u([n], \Gamma) = (B_\bullet\Gamma)_n$ when Γ is regarded as the one-object strict tricategory associated to the group Γ.

Construction 9.12. Let $\mathbf{BAz}^\Gamma_K$ be the one-object tricategory whose 1-cells are pairs (η, R), where $\eta \in \Gamma$ and R is an Azumaya K-algebra; whose 2-cells between (η, R) and (η, S) are invertible R–S-bimodules; and whose 3-cells are bimodule isomorphisms. This is the semidirect construction of Definition A.7, applied to the monoidal Azumaya bicategory with its coherent right scalar-twist action. Equivalently, its endomorphism bicategory is the coproduct

$$\coprod_{\eta\in\Gamma} \eta\,\mathbf{Az}_K,$$

where $\eta\,\mathbf{Az}_K$ is a labelled copy of Definition 9.9: its objects are (η, R), its 1-cells $(\eta, R) \to (\eta, S)$ are the R–S-bimodules, and there are no 1-cells between different Γ-components. The word "tricategory" is used in the Gordon–Power–Street sense recalled in Definition A.6. Horizontal composition of 1-cells is

$$(\eta, R) \odot (\theta, S) := \big(\eta \circ \theta, R^\theta \otimes_K S\big).$$

Here $\odot$ lists the factors in chronological order: if $f = (\eta, R)$ is followed by $g = (\theta, S)$, then $g \circ f = f \odot g$. On 2-cells, if

$$P \in {}_R\mathbf{Mod}_{R'}, \qquad Q \in {}_S\mathbf{Mod}_{S'},$$

then

$$P \odot Q := P^{\theta} \otimes_K Q \in {}_{R^{\theta}\otimes_K S}\mathbf{Mod}_{(R')^{\theta}\otimes_K S'},$$

and on 3-cells it is the tensor product of bimodule isomorphisms. The unit 1-cell is (id_K, K). The associator for three 1-cells (η, R), (θ, S), (ν, T) is the canonical bimodule implementing the rebracketing

$$(R^{\theta} \otimes_K S)^{\nu} \otimes_K T = R^{\theta\circ\nu} \otimes_K S^{\nu} \otimes_K T \longrightarrow R^{\theta\circ\nu} \otimes_K (S^{\nu} \otimes_K T),$$

and the left and right unitors are implemented by the following algebra isomorphisms, where $a \in K$, $r \in R$, and scalars act through the specified central map of R:

$$K^{\eta} \otimes_K R \xrightarrow{\sim} R, \quad a \otimes_K r \longmapsto \eta^{-1}(a)r,$$
$$R^{\mathrm{id}} \otimes_K K \xrightarrow{\sim} R, \quad r \otimes_K a \longmapsto ra.$$

The interchange comparison between vertical composition of 2-cells and horizontal tensor product is the canonical balanced-tensor isomorphism

$$\big(P \otimes_{R'} P'\big)^{\theta} \otimes_K \big(Q \otimes_{S'} Q'\big) \cong \big(P^{\theta} \otimes_K Q\big) \otimes_{(R')^{\theta}\otimes_K S'} \big((P')^{\theta} \otimes_K Q'\big).$$

On pure tensors it is

$$((p \otimes_{R'} p') \otimes_K (q \otimes_{S'} q')) \longmapsto ((p \otimes_K q) \otimes_{(R')^{\theta}\otimes_K S'} (p' \otimes_K q')).$$

Together with the canonical comparison 3-cells supplied by the associativity and unit constraints for relative tensor products, these are the Gordon–Power–Street tricategory data; the coherence check is given in the proof below. In this sense $\mathbf{BAz}_K^{\Gamma}$ is the semidirect product of the Azumaya tricategory $\mathbf{BAz}_K$ of Definition 9.9 by the coherent scalar twist action of Γ. Projection is the trihomomorphism

$$\mathrm{pr}_{\Gamma} : \mathbf{BAz}_K^{\Gamma} \longrightarrow \Gamma, \qquad (\eta, R) \longmapsto \eta^{-1},$$

which sends every 2-cell and 3-cell to the appropriate identity higher cell. The arrow η^{-1} has bar label η, so the induced nerve map retains the displayed labels. Here Γ is regarded as a one-object strict tricategory with only identity higher cells.

For every $n \geq 0$, define

$$\mathcal{B}\mathrm{r}_{\Gamma}(K)_n := \mathrm{Rep}_u([n], \mathbf{BAz}_K^{\Gamma}),$$

the set of unitary representations, equivalently normal lax functors, of Definition A.9 on the ordinal category $[n]$ in this tricategory. For a monotone map $\alpha : [m] \to [n]$, define

$$\alpha^* : \mathcal{B}\mathrm{r}_{\Gamma}(K)_n \longrightarrow \mathcal{B}\mathrm{r}_{\Gamma}(K)_m, \qquad F \longmapsto F \circ \alpha.$$

The projection to Γ gives the maps

$$p_{\mathrm{Br},n} := (\mathrm{pr}_{\Gamma})_* : \mathcal{B}\mathrm{r}_{\Gamma}(K)_n \longrightarrow \mathrm{Rep}_u([n], \Gamma) = (B_{\bullet}\Gamma)_n, \qquad F \longmapsto \mathrm{pr}_{\Gamma} \circ F.$$

Construction 9.13. Over a bar simplex $[\eta_1|\cdots|\eta_n]$, put

$$\eta_{ij} := \eta_{i+1} \circ \cdots \circ \eta_j \qquad \text{for all } 0 \leq i < j \leq n.$$

A normalized labelling over this simplex consists of the following data:

(i) for every $i < j$, an Azumaya K-algebra R_{ij};

(ii) for every $i<j<k$, an invertible bimodule

$$M_{ijk} \in {}_{R_{ij}^{\eta_{jk}}\otimes_K R_{jk}}\mathbf{Mod}_{R_{ik}};$$

(iii) for every $i<j<k<\ell$, put

$$\mathsf{L}_{ijk\ell} := M_{ik\ell}\circ\big(M_{ijk}^{\eta_{k\ell}}\otimes_K \mathrm{id}_{R_{k\ell}}\big),$$
$$\mathsf{R}_{ijk\ell} := M_{ij\ell}\circ\big(\mathrm{id}_{R_{ij}^{\eta_{j\ell}}}\otimes_K M_{jk\ell}\big).$$

Both are bimodules in

$${}_{R_{ij}^{\eta_{j\ell}}\otimes_K R_{jk}^{\eta_{k\ell}}\otimes_K R_{k\ell}}\mathbf{Mod}_{R_{i\ell}},$$

with the canonical associativity and twist identifications understood. The data include a specified isomorphism

$$\alpha_{ijk\ell} : \mathsf{L}_{ijk\ell} \xrightarrow{\sim} \mathsf{R}_{ijk\ell};$$

With this chronological tensor order, $\alpha_{ijk\ell}$ is the inverse of the normal-lax structure 3-cell of Definition A.9.

(iv) for every $i<j<k<\ell<m$, the five isomorphisms

$$\alpha_{ijk\ell},\quad \alpha_{ijkm},\quad \alpha_{ij\ell m},\quad \alpha_{ik\ell m},\quad \alpha_{jk\ell m}$$

satisfy the following pentagon. Put

$$X_{ijk\ell m} := R_{ij}^{\eta_{jm}}\otimes_K R_{jk}^{\eta_{km}}\otimes_K R_{k\ell}^{\eta_{\ell m}}\otimes_K R_{\ell m}.$$

Define the five composite bimodules in ${}_{X_{ijk\ell m}}\mathbf{Mod}_{R_{im}}$ by

$$\mathsf{C}_1 := M_{i\ell m}\circ\big(M_{ik\ell}^{\eta_{\ell m}}\otimes_K \mathrm{id}\big)\circ\big(M_{ijk}^{\eta_{km}}\otimes_K \mathrm{id}\otimes_K \mathrm{id}\big),$$
$$\mathsf{C}_2 := M_{i\ell m}\circ\big(M_{ij\ell}^{\eta_{\ell m}}\otimes_K \mathrm{id}\big)\circ\big(\mathrm{id}\otimes_K M_{jk\ell}^{\eta_{\ell m}}\otimes_K \mathrm{id}\big),$$
$$\mathsf{C}_3 := M_{ijm}\circ\big(\mathrm{id}\otimes_K M_{j\ell m}\big)\circ\big(\mathrm{id}\otimes_K M_{jk\ell}^{\eta_{\ell m}}\otimes_K \mathrm{id}\big),$$
$$\mathsf{C}_4 := M_{ijm}\circ\big(\mathrm{id}\otimes_K M_{jkm}\big)\circ\big(\mathrm{id}\otimes_K \mathrm{id}\otimes_K M_{k\ell m}\big),$$
$$\mathsf{C}_5 := M_{ikm}\circ\big(M_{ijk}^{\eta_{km}}\otimes_K M_{k\ell m}\big).$$

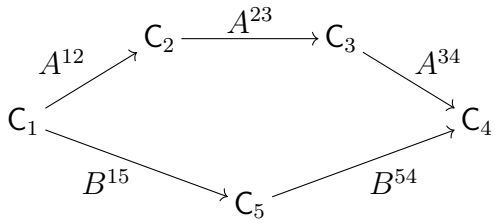
Each unlabelled id is the identity bimodule on the adjacent algebra factor forced by the displayed tensor product, and the canonical associativity and twist identifications are suppressed. Let

$$A^{12} : \mathsf{C}_1 \to \mathsf{C}_2,\quad A^{23} : \mathsf{C}_2 \to \mathsf{C}_3,\quad A^{34} : \mathsf{C}_3 \to \mathsf{C}_4,$$

be the isomorphisms obtained by tensoring $\alpha_{ijk\ell}^{\eta_{\ell m}}$, $\alpha_{ij\ell m}$, and $\alpha_{jk\ell m}$, respectively, with the identity bimodules displayed above and then composing with the indicated outer M's. Similarly let

$$B^{15} : \mathsf{C}_1 \to \mathsf{C}_5,\qquad B^{54} : \mathsf{C}_5 \to \mathsf{C}_4$$

be induced by $\alpha_{ik\ell m}$ and α_{ijkm}. The required condition is commutativity of the labelled pentagon



that is, $A^{34} \circ A^{23} \circ A^{12} = B^{54} \circ B^{15}$ as maps $\mathsf{C}_1 \to \mathsf{C}_4$.

For an injective α, α^* restricts this data to the selected vertices. When α identifies vertices, α^* inserts the unit algebra and the specified unit Morita data.

Proposition 9.14. *The sets and maps of Construction 9.12 define a simplicial set $\mathcal{B}\mathrm{r}_\Gamma(K)$ and a simplicial map*

$$p_{\mathrm{Br}} : \mathcal{B}\mathrm{r}_\Gamma(K) \longrightarrow B_\bullet\Gamma$$

whose degree-n map is $p_{\mathrm{Br},n}$. Its simplices are exactly the labellings of Construction 9.13, with the stated face and degeneracy maps.

Proof. First, the asserted semidirect Azumaya tricategory is well defined. Its endomorphism bicategory is the coproduct of labelled copies of the bicategory $\mathbf{Az}_K$ from Definition 9.9. Horizontal composition is a homomorphism of bicategories because scalar twist preserves relative tensor products and bimodule isomorphisms, and because the interchange comparison displayed in Construction 9.12 is natural in all four bimodules. The tricategorical associators and unitors are the canonical maps between rebracketed iterated tensor products of algebras and bimodules. The Gordon–Power–Street coherence modifications π, μ, λ, ρ are the corresponding canonical comparison isomorphisms between such rebracketings. On pure tensors, associativity comparisons rebracket, interchange comparisons permute the indicated independent factors, and unit comparisons apply the scalar multiplication formulas displayed in Construction 9.12; the unit maps are not literal identities on pure tensors. In every tricategorical coherence diagram, the two composites therefore have the same values on pure tensors, which generate the balanced tensor products. The scalar-twist comparisons of Definition 9.9 are identities on the underlying tensor quotients, preserve these maps, and satisfy $(R^\eta)^\theta = R^{\eta\circ\theta}$. Thus the semidirect composition has the required hom-bicategory, associators, unitors, interchange maps, and coherence, and projection to Γ preserves them.

Precomposition preserves unitary representations [CH14, p. 2010, §2.2], and for $\alpha : [m] \to [n]$ and $\beta : [\ell] \to [m]$, $\beta^* \circ \alpha^* = (\alpha \circ \beta)^*$ as maps $\mathcal{B}\mathrm{r}_\Gamma(K)_n \to \mathcal{B}\mathrm{r}_\Gamma(K)_\ell$, and $(\mathrm{id}_{[n]})^* = \mathrm{id}$ on $\mathcal{B}\mathrm{r}_\Gamma(K)_n$. Remark 9.11 gives the required smallness. Projection to Γ commutes with precomposition, so p_{Br} is simplicial.

We now describe the simplices. The vertex data of a representation are trivial because $\mathbf{BAz}_K^\Gamma$ has one object. Its nonidentity arrows give a 1-cell on every edge, hence a pair (η_{ij}, R_{ij}). Its structure 2-cell on a triangle $i < j < k$ is an invertible bimodule from the composite of the edges ij and jk to the edge ik; the composition law in Construction 9.12 makes its source

$$R_{ij}^{\eta_{jk}} \otimes_K R_{jk}.$$

The structure 3-cell on each tetrahedron is the inverse of the isomorphism in Construction 9.13(iii), and the representation coherence axiom is equivalent to its pentagon in (iv).

Unitarity forces the data on identity arrows to be the specified unit constraints. This proves the description of the simplices and their faces and degeneracies. □

Definition 9.15. For every $n \geq 0$, let

$$1_n = [\mathrm{id}_K|\cdots|\mathrm{id}_K] \in (B_\bullet\Gamma)_n$$

be the totally degenerate n-simplex; for $n = 0$ this is the unique vertex. Define

$$\mathcal{B}\mathrm{r}(K)_n := p_{\mathrm{Br},n}^{-1}(1_n) = \mathrm{Rep}_u([n], \mathbf{BAz}_K). \tag{58}$$

The face and degeneracy maps are the restrictions of those of $\mathcal{B}\mathrm{r}_\Gamma(K)$. They preserve these subsets because every face or degeneracy of 1_n is again an all-identity bar simplex. Equivalently, $\mathcal{B}\mathrm{r}(K)$ is the degreewise pullback

$$\begin{array}{ccc} \mathcal{B}\mathrm{r}(K) & \longrightarrow & \mathcal{B}\mathrm{r}_\Gamma(K) \\ \downarrow & & \downarrow{\scriptstyle p_{\mathrm{Br}}} \\ \Delta^0 & \longrightarrow & B_\bullet\Gamma, \end{array}$$

where the lower map selects the unique vertex. We call $\mathcal{B}\mathrm{r}(K)$ the *simplicial Brauer* 3*-type of* K.

Its simplices are described in Proposition 9.14 with every Γ-label equal to the identity. In particular, degeneracies insert the specified unit algebra K, identity bimodules, and canonical unit comparisons.

The action and the homotopy fibre. Scalar twisting gives a coherent action on the Azumaya bicategory. To obtain the induced maps on its unitary nerve, one must identify the twisted unit K^η with K by $a \mapsto \eta^{-1}(a)$. Write Tw_η for the induced map after normalizing units. The tensor and unit comparisons are part of this construction. The corresponding coherent left action of η is

$$\mathrm{Tw}_{\eta^{-1}} : \mathcal{B}\mathrm{r}(K) \longrightarrow \mathcal{B}\mathrm{r}(K).$$

The action identities hold with these specified comparisons. Proposition 10.31 constructs a strict action on a weakly equivalent Kan complex $\widehat{\mathcal{B}\mathrm{r}}_\Gamma^{(3)}(K)$. Homotopy fixed points of $\mathcal{B}\mathrm{r}(K)$ below mean those of this coherent action, computed in that strict model. Under the canonical identification of the twisted unit algebra with K, $\mathrm{Tw}_{\eta^{-1}}$ sends multiplication by $u \in \mathrm{U}(K)$ to multiplication by $\eta(u)$. The pullback square displayed above says precisely that $\mathcal{B}\mathrm{r}(K)$ is the degreewise fibre of p_{Br} over the constant simplicial map $\Delta^0 \to B_\bullet\Gamma$ selecting the unique vertex. Here "degreewise fibre" means this literal pullback of simplicial sets, in contrast with the homotopy fibre of Definition A.20. Proposition 9.19 proves that p_{Br} is a Kan fibration; therefore this pullback is Kan and also represents $\mathrm{hofib}_*(p_{\mathrm{Br}})$. Since $\mathcal{B}\mathrm{r}(K)$ has one vertex, it has one component. Proposition 9.24 computes its homotopy groups and proves that they vanish above degree 3, which justifies the name "3-type."

The suffix twist in Construction 9.12 is forced by homogeneous multiplication: if degrees η and θ are composed, the first central algebra occurs as R^θ; compare the type relation in Definition 6.1. The general Street-nerve definition and the meaning of its simplices are recalled in Definitions A.8 and A.9.

9.3. **Truncation and the Kan fibration property.** Let $\mathbf{BAz}^{\Gamma}_{K,\le 2}$ denote the bicategory obtained from $\mathbf{BAz}^{\Gamma}_{K}$ by taking its homotopy 2-truncation. The unitary Street nerve of $\mathbf{BAz}^{\Gamma}_{K,\le 2}$ is the concrete simplicial model for the second Postnikov truncation of the unitary Street nerve of $\mathbf{BAz}^{\Gamma}_{K}$. In the present concrete case this bicategory has one object, with the same 1-cells (η, R) as $\mathbf{BAz}^{\Gamma}_{K}$, and with 2-cells

$$(\eta, R) \Longrightarrow (\eta, S)$$

given by isomorphism classes $[P]$ of invertible R–S-bimodules $P \in {}_{R}\mathbf{Mod}_{S}$. There are no 2-cells between different Γ-labels. Vertical and horizontal composition are induced from relative tensor product and from the horizontal product

$$[P] \odot [Q] = [P^{\theta} \otimes_{K} Q]$$

when Q has Γ-label θ. This is well defined because bimodule isomorphisms are precisely the 3-cells of $\mathbf{BAz}^{\Gamma}_{K}$. The truncation homomorphism is

$$\mathrm{tr}_{\le 2} : \mathbf{BAz}^{\Gamma}_{K} \longrightarrow \mathbf{BAz}^{\Gamma}_{K,\le 2}.$$

It is the identity on the unique object and on 1-cells, sends an invertible bimodule P to its isomorphism class $[P]$, and forgets the bimodule isomorphisms.

Construction 9.16. Using the notation just fixed, define

$$\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)_n := \mathrm{Rep}_u([n], \mathbf{BAz}^{\Gamma}_{K,\le 2}).$$

Thus an n-simplex is a unitary representation of $[n]$ in this bicategory: it has the same edge algebras as an n-simplex of $\mathcal{B}\mathrm{r}_{\Gamma}(K)$, but on each triangle it remembers only the isomorphism class of the invertible bimodule, and every tetrahedron asserts equality of the two pasted triangle classes. Precomposition defines its simplicial operators. The truncation homomorphism induces maps

$$q_n : \mathcal{B}\mathrm{r}_{\Gamma}(K)_n \longrightarrow \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)_n, \qquad F \longmapsto \mathrm{tr}_{\le 2} \circ F.$$

Proposition 9.17. *Construction 9.16 defines a simplicial set over $B_{\bullet}\Gamma$ with structure map*

$$p_{\mathrm{Br},\le 2} : \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K) \longrightarrow B_{\bullet}\Gamma.$$

The maps q_n define a simplicial map over $B_{\bullet}\Gamma$,

$$q : \mathcal{B}\mathrm{r}_{\Gamma}(K) \longrightarrow \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K).$$

An n-simplex of $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ retains the edge algebras R_{ij} and only the isomorphism classes $[M_{ijk}]$ of the triangle bimodules; on each quadruple, the two composite bimodule classes are equal.

Proof. Tensor product and composition are well-defined on isomorphism classes, and the tricategorical coherence 3-cells become equalities; hence $\mathbf{BAz}^{\Gamma}_{K,\le 2}$ is a bicategory over Γ. The same precomposition calculation as in Proposition 9.14 proves the simplicial identities, and naturality of the truncation homomorphism proves $q_m \circ \alpha^{*} = \alpha^{*} \circ q_n$ for every $\alpha : [m] \to [n]$. A unitary representation in this bicategory consists of the algebras and bimodule classes just described. This is also the unitary geometric nerve of the bicategory in the sense of [CCG10, Def. 4.1 and Lem. 4.2]. □

Remark 9.18. The tensor products in $\mathbf{BAz}_K^\Gamma$ are not strictly associative. Proposition 9.14 avoids choosing a Gray-category strictification: tricategorical associators and unitors are retained as cells, while the Street nerve itself is a strict simplicial set. Thus the simplicial identities follow from the definition of the nerve, independently of the formulas for the scalar twists.

Horn fillers for the Morita projections. The property being checked is the Kan-fibration condition of Definition A.11(iii). Thus, for every $n \geq 1$, every $0 \leq i \leq n$, and every bar simplex $b = [\eta_1|\cdots|\eta_n] \in (B_\bullet\Gamma)_n$, we must solve the following problem. For each visible face $d_j b$, $j \neq i$, we are given a simplex of $\mathcal{B}\mathrm{r}_\Gamma(K)$ over $d_j b$, and these visible simplices agree after restriction to each common subface $(d_r \circ d_j)(b) = (d_{j-1} \circ d_r)(b)$. We must construct a simplex of $\mathcal{B}\mathrm{r}_\Gamma(K)$ over b whose jth face is the prescribed one for every $j \neq i$.

Using the description in Proposition 9.14, we construct the missing algebra, bimodule, or coherence isomorphism from the prescribed faces, and then verify the required coherence equation. The following proof treats each dimension separately.

Proposition 9.19. *The simplicial projections*

$$p_{\mathrm{Br}} : \mathcal{B}\mathrm{r}_\Gamma(K) \longrightarrow B_\bullet\Gamma, \qquad p_{\mathrm{Br},\leq 2} : \mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K) \longrightarrow B_\bullet\Gamma$$

are Kan fibrations. For either projection, every horn-lifting problem over a prescribed bar simplex has a solution, in every dimension $n \geq 1$ and for every missing face $0 \leq i \leq n$.

Proof. We first prove the assertion for p_{Br}. The bar simplex in the base fixes every η_{ij}.

For $n = 1$, choose the edge algebra K over the prescribed label $[\eta_1]$; this fills either endpoint horn.

Now let $n = 2$ and let $b = [\eta_1|\eta_2]$. A 2-horn is a pair of two edges of the triangle, agreeing at their common endpoint; filling it means choosing the third edge algebra and then choosing the triangle bimodule M_{012} which identifies the composite edge with the long edge. Thus exactly one of R_{01}, R_{12}, R_{02} is missing. If R_{02} is missing (the horn missing the middle face d_1), take

$$R_{02} = R_{01}^{\eta_{12}} \otimes_K R_{12}$$

and use the identity Morita equivalence. If R_{12} is missing (the horn missing d_0), take

$$R_{12} = (R_{01}^{\eta_{12}})^{\mathrm{op}} \otimes_K R_{02}.$$

Then

$$R_{01}^{\eta_{12}} \otimes_K R_{12} = R_{01}^{\eta_{12}} \otimes_K (R_{01}^{\eta_{12}})^{\mathrm{op}} \otimes_K R_{02}$$

is Morita equivalent to R_{02}. If R_{01} is missing (the horn missing d_2), take

$$R_{01} = (R_{02} \otimes_K R_{12}^{\mathrm{op}})^{\eta_{12}^{-1}},$$

so that $R_{01}^{\eta_{12}} \otimes_K R_{12}$ is again Morita equivalent to R_{02}. All three choices of missing edge algebra belong to $\mathfrak{U}(\mathcal{A}\mathrm{z})$ by the closure conditions in Notation 9.4. Choosing an implementing invertible bimodule supplies M_{012}.

For a 3-horn, the abstract property is this: three triangular faces of a tetrahedron are prescribed and agree on their edges, and we must construct the missing triangular face together with the tetrahedral comparison 3-cell. All edge algebras are therefore known, and exactly one triangle bimodule may be missing. In coordinates the comparison to be supplied is

$$M_{023} \circ (M_{012}^{\eta_{23}} \otimes_K \mathrm{id}) \cong M_{013} \circ (\mathrm{id} \otimes_K M_{123}),$$

between invertible vertical bimodules lying in

$$R_{01}^{\eta_{13}} \otimes_K R_{12}^{\eta_{23}} \otimes_K R_{23} \mathbf{Mod}_{R_{03}}.$$

Let $\mathcal{C}_{0123}$ be the groupoid of such composite bimodules. Scalar twisting and composition with an invertible vertical bimodule are equivalences of bimodule groupoids. Tensoring with the identity bimodule of an Azumaya algebra A is also an equivalence on these groupoids: tensor with A^{op} and use the Morita equivalence $A \otimes_K A^{\mathrm{op}} \simeq_{\mathrm{Mor}} K$ to obtain a quasi-inverse. Thus for each triangular face ijk, the functor F_{ijk} is an equivalence from the groupoid of invertible bimodules in ${}_{R_{ij}^{\eta_{jk}} \otimes_K R_{jk}}\mathbf{Mod}_{R_{ik}}$ and their isomorphisms to $\mathcal{C}_{0123}$, with the following formula:

$$\begin{aligned} F_{012}(N) &= M_{023} \circ (N^{\eta_{23}} \otimes_K \mathrm{id}), \\ F_{023}(N) &= N \circ (M_{012}^{\eta_{23}} \otimes_K \mathrm{id}), \\ F_{013}(N) &= N \circ (\mathrm{id} \otimes_K M_{123}), \\ F_{123}(N) &= M_{013} \circ (\mathrm{id} \otimes_K N). \end{aligned}$$

For the missing face I, choose M_I whose image under F_I is isomorphic to the prescribed composite. Essential surjectivity supplies M_I, and the chosen isomorphism, oriented from left to right, supplies α_{0123}. This fills each of the four horn cases without changing the three prescribed faces.

For a 4-horn Λ_i^4, every triangle omits two vertices. At least one of them, say j, differs from i, so the triangle lies in the visible facet opposite j. Thus the edge and triangle data are already known, and precisely one tetrahedral isomorphism $\alpha_I : \mathsf{L}_I \to \mathsf{R}_I$ is missing. Let F_I be the functor that inserts this tetrahedron into the pentagon of Construction 9.13(iv): it applies the prescribed scalar twist, tensors with the displayed identity bimodules, and composes with the outer M's. Each operation is an equivalence of bimodule groupoids by the preceding argument. In particular, F_I is fully faithful and induces a bijection

$$\mathrm{Iso}(\mathsf{L}_I, \mathsf{R}_I) \xrightarrow{\sim} \mathrm{Iso}(F_I\mathsf{L}_I, F_I\mathsf{R}_I).$$

The map $F_I(\alpha_I)$ occurs exactly once among the five arrows of the pentagon. The other four arrows are invertible, so they determine the required value Θ_I of that map uniquely. For example, if $I = 0123$, the missing arrow is A^{12} and its required value is

$$\Theta_{0123} = (A^{34} \circ A^{23})^{-1} \circ B^{54} \circ B^{15} : \mathsf{C}_1 \longrightarrow \mathsf{C}_2.$$

The displayed bijection gives the unique tetrahedral isomorphism α_I with $F_I(\alpha_I) = \Theta_I$. Thus the required isomorphism between the composite bimodules comes from an actual tetrahedral cell and fills the horn.

For a 5-horn Λ_i^5, every tetrahedron likewise omits two vertices, at least one of which differs from i. It therefore lies in a visible 4-face, so all cell data are present. Five visible 4-faces have pentagon scalar equal to 1; the boundary identity in Lemma 9.23 below forces the sixth pentagon scalar to be 1. Finally, for $n \geq 6$, each face of dimension at most four is contained in a visible facet of the horn, so all cell data and all pentagon equations are already specified.

We now prove the assertion for $p_{\mathrm{Br},\leq 2}$. In dimensions one and two, use the same edge algebras and take the isomorphism classes of the triangle bimodules constructed above. For a 3-horn, choose representatives of the three visible triangle classes. The 3-horn construction supplies a fourth bimodule and an isomorphism between the two tetrahedral composites.

The class of the fourth bimodule gives the missing face, and the isomorphism gives the required equality of composite classes.

For a 4-horn, all edge algebras and triangle classes are prescribed. Choose one representative for each triangle class, shared by its incident tetrahedra. On each of the four visible tetrahedra, the prescribed equality of composite classes allows us to choose an isomorphism between the two composites of these representatives. These four tetrahedra form a 4-horn in $\mathcal{B}\mathrm{r}_\Gamma(K)$ over the same bar simplex. Fill it by the construction above and apply q. The result has exactly the prescribed visible faces and supplies the missing equality of composite classes. For $n \geq 5$, every tetrahedron is contained in a visible facet, so all edge and triangle data and all equalities on tetrahedra are already specified. Proposition 9.17 then supplies the simplex without further choices. This proves the horn-lifting property for both projections. □

Remark 9.20. Proposition 9.19 proves that the two projections to $B_\bullet\Gamma$ are Kan fibrations. We use a fibrant replacement of q in Proposition 9.24. The global loop identifications are supplied independently by Theorem 10.33.

Remark 9.21. For

$$q : \mathcal{B}\mathrm{r}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K),$$

the lifting process chooses representatives of the triangle-bimodule classes in dimension two and isomorphisms between the two tetrahedral composites in dimension three. In dimensions four and five, the lifting argument uses the pentagon equation and its boundary identity. This gives a local description of the map whose global homotopy fibre is identified in Proposition 9.24.

9.4. **The pentagon cocycle and the homotopy fibre.**

Definition 9.22. Fix the labels R_{ij}, M_{ijk}, and $\alpha_{ijk\ell}$ of Construction 9.13(i)–(iii) over $[\eta_1|\cdots|\eta_5]$, without imposing the pentagon equations. For an ordered 4-face $I = [i\,j\,k\,\ell\,m]$, define $c_I \in \mathrm{U}(K)$ by the equality

$$B^{54} \circ B^{15} = c_I\,(A^{34} \circ A^{23} \circ A^{12}),$$

as maps $\mathsf{C}_1 \longrightarrow \mathsf{C}_4$ in ${}_{X_{ijk\ell m}}\mathbf{Mod}_{R_{im}}$, for the face I, in the notation of Construction 9.13(iv), and put

$$\Omega_I = \eta_{im}(c_I) \in \mathrm{U}(K).$$

The scalar c_I is the short-over-long scalar in the terminal coefficient fibre, and Ω_I is its initial-fibre coordinate. Write $\widehat{r}$ for the face omitting vertex r.

Lemma 9.23. *For the face scalars of Definition 9.22, we have*

$$\eta_{01}(\Omega_{\widehat{0}})\,\Omega_{\widehat{1}}^{-1}\,\Omega_{\widehat{2}}\,\Omega_{\widehat{3}}^{-1}\,\Omega_{\widehat{4}}\,\Omega_{\widehat{5}}^{-1} = 1 \qquad \text{in } \mathrm{U}(K). \tag{59}$$

In particular, five commuting pentagons force the sixth to commute.

Proof. Put $A_i = R_{i-1,i}^{\eta_{i5}}$ for all $1 \leq i \leq 5$, with $\eta_{55} = 1$. The fourteen parenthesizations of $A_1 \cdots A_5$ give composite invertible bimodules X_v with source $A_1 \otimes_K \cdots \otimes_K A_5$ and target R_{05}. The edges of the associahedron K_5 are the isomorphisms $f_e : X_v \to X_w$ obtained from the α's by tensoring and composition. Its boundary has six pentagons and three interchange squares.

A proper consecutive block $[a, b]$ containing at least two factor indices determines the facet of parenthesizations in which $A_a, \ldots, A_b$ form one subtree. The pentagon orientation

corresponding to c_I follows the short path $\mathsf{C}_1 \to \mathsf{C}_5 \to \mathsf{C}_4$ and then the reverse of the long path $\mathsf{C}_1 \to \mathsf{C}_2 \to \mathsf{C}_3 \to \mathsf{C}_4$. Orient the boundary of K_5 so that the face $\widehat{0}$ has this orientation. The six pentagonal facets are then as follows; the last column gives their oriented scalar in the terminal fibre at vertex 5:

I	fixed block	sign	terminal scalar
$\widehat{0} = [12345]$	$[2,5]$	$+$	$c_{\widehat{0}}$
$\widehat{1} = [02345]$	$[1,2]$	$-$	$c_{\widehat{1}}^{-1}$
$\widehat{2} = [01345]$	$[2,3]$	$+$	$c_{\widehat{2}}$
$\widehat{3} = [01245]$	$[3,4]$	$-$	$c_{\widehat{3}}^{-1}$
$\widehat{4} = [01235]$	$[4,5]$	$+$	$c_{\widehat{4}}$
$\widehat{5} = [01234]$	$[1,4]$	$-$	$\eta_{45}^{-1}(c_{\widehat{5}}^{-1})$

The other three facets, with fixed blocks $[1,3]$, $[2,4]$, and $[3,5]$, are interchange squares. The listed pentagon orientations cancel on their common edges; their remaining edges form the three square boundaries, with the opposite induced orientations.

Only $\widehat{5}$ has terminal vertex 4 rather than 5. Its comparison is first twisted by η_{45}, tensored with the identity bimodule of R_{45}, and then composed with M_{045} to reach R_{05}. An underlying scalar map $c \in \mathrm{U}(K)$ on a bimodule P becomes the scalar $\eta_{45}^{-1}(c)$ on $P^{\eta_{45}}$, since its new K-action is precomposed with η_{45}. All other faces already end at 5; tensoring and composition with identity-central bimodules preserve their scalar labels. This explains the last column of the table.

Choose a root v_* and composite isomorphisms $\lambda_v : X_v \to X_{v_*}$ along a spanning tree. All X_v have the same source and target and are identity-central, so these bimodule isomorphisms commute with the common K-scalar action. Transport by λ_v therefore preserves scalar labels. Each oriented edge gives a scalar

$$z_e = \lambda_w \circ f_e \circ \lambda_v^{-1},$$

as maps $X_{v_*} \longrightarrow X_{v_*}$, identified with an element of $\mathrm{Aut}(X_{v_*}) = \mathrm{U}(K)$. Along any closed path the λ's cancel, so the product of the z_e is the scalar of the original composite isomorphism. In the product over all oriented faces, each edge occurs twice with opposite orientations. The product is therefore 1. The three squares contribute 1 by interchange, leaving the six pentagonal scalars in the table. In terminal coordinates this reads

$$c_{\widehat{0}} c_{\widehat{1}}^{-1} c_{\widehat{2}} c_{\widehat{3}}^{-1} c_{\widehat{4}} \eta_{45}^{-1}(c_{\widehat{5}}^{-1}) = 1.$$

Apply η_{05} and use $\eta_{05} = \eta_{01} \circ \eta_{15} = \eta_{04} \circ \eta_{45}$ to obtain (59). The reference maps λ_v are chosen only on this associahedron. □

Proposition 9.24. *The map q of Construction 9.16 presents the fibrewise second Postnikov truncation of $\mathcal{B}\mathrm{r}_\Gamma(K)$ over $B_\bullet\Gamma$. Its functorial fibrant replacement over $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ has pointwise homotopy fibre $K(\mathrm{U}(K),3)$, and its local coefficient group object is the pullback of $K_{B_\bullet\Gamma}(\mathrm{U}(K),3) \to B_\bullet\Gamma$ along*

$$p_{\mathrm{Br},\le 2} : \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K) \to B_\bullet\Gamma.$$

Thus every pointwise homotopy fibre is $K(\mathrm{U}(K),3)$, and forward monodromy along the bar edge $[\eta]$ acts on its third homotopy group by

$$a \longmapsto \eta^{-1}(a) \qquad \text{for all } a \in \mathrm{U}(K),$$

in the sense of Definition A.18. The backward coefficient map along this edge is $a \mapsto \eta(a)$. *At the unit basepoint,* $\pi_1(\mathcal{B}\mathrm{r}(K)) = \mathrm{Br}(K)$, $\pi_2(\mathcal{B}\mathrm{r}(K)) = \mathrm{Pic}(K)$, $\pi_3(\mathcal{B}\mathrm{r}(K)) = \mathrm{U}(K)$, *and* $\pi_i(\mathcal{B}\mathrm{r}(K)) = \{1\}$ *for* $i > 3$.

The terms "fibrewise Postnikov truncation," "Kan-fibration model," and "homotopy fibre" are recalled in Subsections A.2–A.3.

Proof. The categorical fibre of $\mathbf{BAz}_K^{\Gamma} \to \Gamma$ over the identity is a one-object 3-groupoid: its 1-cells are Azumaya K-algebras, its 2-cells are invertible bimodules, and its 3-cells are bimodule isomorphisms. This is the one-object delooping of the monoidal bicategory $\mathbf{Az}_K$. We use the homotopy-group description in [CH14, arXiv v2, §5.7, p. 44]: for a monoidal bicategory whose objects are tensor-invertible up to equivalence, whose 1-cells are equivalences, and whose 2-cells are isomorphisms, its geometric nerve is a connected Kan 3-type. Its fundamental group consists of equivalence classes of objects, with tensor product; its second homotopy group consists of isomorphism classes of autoequivalences of the unit object, with horizontal composition; and its third consists of automorphisms of the identity 1-cell of the unit object, with vertical composition. Applying this description to $\mathbf{Az}_K$ identifies the homotopy groups of its Street nerve $\mathcal{B}\mathrm{r}(K)$ from Definition 9.15 as

$$\pi_1(\mathcal{B}\mathrm{r}(K)) = \mathrm{Br}(K), \qquad \pi_2(\mathcal{B}\mathrm{r}(K)) = \mathrm{Pic}(K), \qquad \pi_3(\mathcal{B}\mathrm{r}(K)) = \mathrm{Aut}_{K\text{-bimod}}(K) = \mathrm{U}(K),$$

and there are no higher homotopy groups. The first group consists of Azumaya algebras modulo Morita equivalence, with tensor product over K, hence is the Brauer group. The second consists of isomorphism classes of invertible K-bimodules in ${}_K\mathbf{Mod}_K$, that is, invertible K-modules, from the unit 1-cell to itself, with product induced by tensor product, and is therefore $\mathrm{Pic}(K)$. The third group can be computed directly: every K-bimodule endomorphism $f : K \to K$ satisfies $f(a) = af(1)$ for all $a \in K$, and invertibility is equivalent to $f(1) \in \mathrm{U}(K)$. Composition corresponds to multiplication of these units.

Passing to $\mathbf{BAz}_{K,\leq 2}^{\Gamma}$ leaves the equivalence classes of 1-cells and 2-cells unchanged and identifies all parallel 2-cells joined by a 3-cell. Consequently q induces isomorphisms on the first two homotopy groups of the fibres of $\mathcal{B}\mathrm{r}_{\Gamma}(K) \to B_\bullet\Gamma$ and $\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K) \to B_\bullet\Gamma$, while the latter fibres have no homotopy groups above degree 2. It therefore presents the fibrewise second Postnikov truncation. Its homotopy fibre is connected and has the single nontrivial homotopy group $\mathrm{U}(K)$ in degree 3, hence is $K(\mathrm{U}(K), 3)$. To identify its coefficient system, an intrinsic loop represented by an Azumaya algebra acts on a scalar automorphism by tensoring with identity bimodules and then applying K-linear Morita equivalences. Both operations preserve multiplication by that scalar, including the evaluation equivalences for the inverse algebra. Thus the $\mathrm{Br}(K)$-part of the fundamental group acts trivially on $\mathrm{U}(K)$, and the coefficient action factors through Γ. Finally, forward motion along $[\eta]$ is the right twist Tw_η. Identifying its unit K^η with K by η^{-1} sends multiplication by a to multiplication by $\eta^{-1}(a)$. Its inverse is the left action $\mathrm{Tw}_{\eta^{-1}}$, which gives the backward coefficient map $a \mapsto \eta(a)$. Applying the functorial replacement of Definition A.19 to q as defined in Proposition 9.17 gives the stated Kan fibration without altering these fibre homotopy groups or this monodromy action. □

10. The ten-term exact sequence for Azumaya algebras

Fix an action $\phi : G \to \Gamma$ and write $\eta_g = \phi(g)$. We identify sections of the simplicial families constructed in Section 9 with Brauer–Clifford and Brauer–character systems. The

main results are the obstruction sequence of Theorem 10.42 and the lift classification of Theorem 10.51: a Brauer–character system admits pentagon-coherent associators precisely when its fourth obstruction vanishes. Proposition 10.59 then identifies this obstruction over fields with the inverse transgression.

10.1. **Brauer–Clifford and Brauer–character systems.** We pull $\mathcal{B}\mathrm{r}_\Gamma(K)$ and $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ back along $B_\bullet\phi : B_\bullet G \to B_\bullet\Gamma$.

Notation 10.1. Let $p_X : X \to B_\bullet\Gamma$ be a simplicial set over $B_\bullet\Gamma$. Its type-ϕ pullback is the simplicial set over $B_\bullet G$

$$X_\phi := B_\bullet G \times_{B_\bullet\Gamma} X, \qquad p_{X,\phi} : X_\phi \longrightarrow B_\bullet G, \qquad (b,x) \longmapsto b,$$

where the fibre product is formed using $B_\bullet\phi$ and p_X. We write

$$\mathrm{Sec}_\phi(X) := \mathrm{Sec}_{B_\bullet G}(p_{X,\phi})$$

for its simplicial set of sections in the sense of Definition A.32. Thus a vertex may equivalently be written as a simplicial map $s : B_\bullet G \to X$ satisfying $p_X \circ s = B_\bullet\phi$ as maps $B_\bullet G \to B_\bullet\Gamma$.

If $f : X \to Y$ is a simplicial map over $B_\bullet\Gamma$, its pullback and the induced map on sections are

$$f_\phi := \mathrm{id}_{B_\bullet G} \times f : X_\phi \longrightarrow Y_\phi$$

over $B_\bullet G$ and

$$\mathrm{Sec}_\phi(f) : \mathrm{Sec}_\phi(X) \longrightarrow \mathrm{Sec}_\phi(Y), \qquad s \longmapsto f_\phi \circ s.$$

Definition 10.2. Throughout this section and Section 9 the algebras R_g, R_{ij} and P are taken from $\mathfrak{U}(\mathcal{A}\mathrm{z})$ of Notation 9.4. A system of *Brauer–character data of type* ϕ consists of:

(i) an Azumaya K-algebra R_g for every $g \in G$;

(ii) for every $g,h \in G$, an isomorphism class $[M_{g,h}]$ of invertible bimodules

$$M_{g,h} \in {}_{R_g^{\eta_h}\otimes_K R_h}\mathbf{Mod}_{R_{gh}};$$

(iii) for every $g,h,\ell \in G$, equality of the two composite bimodule classes

$$\begin{aligned} \mathsf{L}_{g,h,\ell} &:= \big[M_{gh,\ell} \circ (M_{g,h}^{\eta_\ell} \otimes_K \mathrm{id}_{R_\ell})\big], \\ \mathsf{R}_{g,h,\ell} &:= \big[M_{g,h\ell} \circ (\mathrm{id}_{R_g^{\eta_{h\ell}}} \otimes_K M_{h,\ell})\big], \end{aligned} \tag{60}$$

$$\mathsf{L}_{g,h,\ell} = \mathsf{R}_{g,h,\ell}.$$

Both sides of (60) are classes of invertible bimodules in

$${}_{R_g^{\eta_{h\ell}}\otimes_K R_h^{\eta_\ell}\otimes_K R_\ell}\mathbf{Mod}_{R_{gh\ell}};$$

the canonical associators and twist identifications are understood.

The data are *normalized*: $R_e = K$, and all bimodule classes involving e are the classes of the specified left and right unit constraints. We call R_g the *g-component algebra*, $[R_g] \in \mathrm{Br}(K)$ the *g-component Brauer class*, and the family $\big([R_g]\big)_{g\in G}$ the *Brauer profile* of the system. Thus the identity component algebra is K and its component Brauer class is the neutral class $[K]$. A system of *Brauer–Clifford data of type* ϕ consists of representatives $M_{g,h}$ of such classes together with, for every triple, a bimodule isomorphism

$$\alpha_{g,h,\ell} : M_{gh,\ell} \circ (M_{g,h}^{\eta_\ell} \otimes_K \mathrm{id}) \xrightarrow{\sim} M_{g,h\ell} \circ (\mathrm{id} \otimes_K M_{h,\ell}),$$

such that the α's satisfy the pentagon of Construction 9.13(iv) for every quadruple g, h, ℓ, m. The representatives $M_{e,g}$ and $M_{g,e}$ are the specified unit bimodules. If any argument is e, the isomorphism $\alpha_{g,h,\ell}$ is prescribed by the unit constraints. Thus the data on degenerate simplices are those of Definition A.9.

Brauer–character systems are not generalized collective characters: they are points of the truncated section space, whereas generalized collective characters occur one homotopy degree lower, as loops at its unit section; see Lemma 10.20(iii) and Theorem 10.33.

Remark 10.3. The Morita-invertibility of the algebras R_g is already forced by the multiplication data and the normalization $R_e = K$. Indeed, the pair (g, g^{-1}) gives

$$R_g^{\phi(g^{-1})} \otimes_K R_{g^{-1}} \simeq_{\mathrm{Mor}} K.$$

Thus $R_g^{\phi(g^{-1})}$ is tensor-invertible up to Morita equivalence, and scalar twisting gives the same conclusion for R_g.

Construction 10.4. Given Brauer–character data, label the simplex $[g_1|\cdots|g_n]$ by

$$g_{ij} := g_{i+1}\cdots g_j, \qquad R_{ij} := R_{g_{ij}}, \qquad [M_{ijk}] := [M_{g_{ij},g_{jk}}].$$

For Brauer–Clifford data, use the chosen representatives $M_{ijk} := M_{g_{ij},g_{jk}}$ and the tetrahedral maps

$$\alpha_{ijk\ell} := \alpha_{g_{ij},g_{jk},g_{k\ell}}.$$

Repeated indices use the prescribed unit data.

Proposition 10.5. *Restriction to the nondegenerate bar simplices in dimensions at most three gives a bijection between sections of $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ over $B_\bullet\phi$ and the Brauer–character data of Definition 10.2. Restriction to the nondegenerate simplices through dimension four gives the analogous bijection between sections of $\mathcal{B}\mathrm{r}_\Gamma(K)$ and Brauer–Clifford data. Under these bijections, q sends $(R_g, M_{g,h}, \alpha_{g,h,\ell})$ to $(R_g, [M_{g,h}])$.*

Proof. On $[g]$, $[g|h]$, and $[g|h|\ell]$, a section of the truncated nerve gives respectively R_g, $[M_{g,h}]$, and the equality (60), by Proposition 9.17. Degeneracies give the normalized unit data.

Conversely, use the labelling of Construction 10.4. For $i < j < k < \ell$, the tetrahedral equality is (60) for $(g_{ij}, g_{jk}, g_{k\ell})$; these are all the required coherence conditions in the truncated nerve. Restriction to a face replaces each interval product by the corresponding shorter interval product, and a degeneracy inserts e and the prescribed units. Hence the formulas define a section, inverse to restriction.

For the full nerve, use the representatives and tetrahedral maps of the same construction. The five-vertex condition is exactly the pentagon for $(g_{ij}, g_{jk}, g_{k\ell}, g_{\ell m})$, by Proposition 9.14. There are no further independent data above dimension four. The same restriction argument gives the second bijection, and q forgets the α's and retains $[M]$. □

Definition 10.6. Let (R, M) and (R', M') be systems of Brauer–character data of type ϕ (Definition 10.2). A *coherent Morita transport* $(R, M) \Rightarrow (R', M')$ consists of:

(i) an Azumaya K-algebra P;
(ii) for every $g \in G$, an isomorphism class $[N_g]$ of invertible bimodules

$$N_g \in {}_{R_g \otimes_K P}\mathbf{Mod}_{P^{\phi(g)} \otimes_K R'_g};$$

(iii) for every $g, h \in G$, the equality of bimodule classes

$$\big[N_{gh} \circ (M_{g,h} \odot 1_P)\big] = \big[(1_P \odot M'_{g,h}) \circ (N_g \odot 1_{R'_h}) \circ (1_{R_g} \odot N_h)\big]. \tag{61}$$

Here $\odot$ is the horizontal composition of Construction 9.12: on 1-cells $(\eta, R) \odot (\theta, S) = (\eta \circ \theta, R^\theta \otimes_K S)$, and on 2-cells $X \odot Y = X^\theta \otimes_K Y$ when the second factor has label θ. Thus

$$M_{g,h} \odot 1_P = M_{g,h} \otimes_K P, \qquad 1_{R_g} \odot N_h = R_g^{\phi(h)} \otimes_K N_h,$$
$$N_g \odot 1_{R'_h} = N_g^{\phi(h)} \otimes_K R'_h, \qquad 1_P \odot M'_{g,h} = P^{\phi(gh)} \otimes_K M'_{g,h},$$

and both sides of (61) are classes of invertible bimodules in

$${}_{R_g^{\phi(h)} \otimes_K R_h \otimes_K P}\mathbf{Mod}_{P^{\phi(gh)} \otimes_K R'_{gh}};$$

the associators and unit constraints of $\odot$ are understood. The transport is *normalized*: $[N_e]$ is the class of the canonical unit bimodule implementing $K \otimes_K P \cong P^{\mathrm{id}} \otimes_K K$, and (61) is the identity whenever g or h equals e.

A coherent Morita transport of *Brauer–Clifford data* $(R, M, \alpha) \Rightarrow (R', M', \alpha')$ consists of P and representatives N_g as in (i)–(ii), taking N_e to be the specified unit bimodule, together with:

(iv) for every $g, h \in G$, a bimodule isomorphism

$$\Theta_{g,h} : N_{gh} \circ (M_{g,h} \odot 1_P) \xrightarrow{\sim} (1_P \odot M'_{g,h}) \circ (N_g \odot 1_{R'_h}) \circ (1_{R_g} \odot N_h),$$

prescribed by the unit constraints when g or h equals e;

(v) for every $g, h, \ell \in G$, put

$$\mathsf{C}_L := M_{gh,\ell} \circ (M_{g,h} \odot 1_{R_\ell}),$$
$$\mathsf{C}_R := M_{g,h\ell} \circ (1_{R_g} \odot M_{h,\ell}).$$

Let $\mathsf{C}'_L, \mathsf{C}'_R$ be the same composites formed from R', M', and put

$$\mathsf{N} := (N_g \odot 1_{R'_h} \odot 1_{R'_\ell}) \circ (1_{R_g} \odot N_h \odot 1_{R'_\ell}) \circ (1_{R_g} \odot 1_{R_h} \odot N_\ell),$$
$$\mathsf{X}_s := N_{gh\ell} \circ (\mathsf{C}_s \odot 1_P),$$
$$\mathsf{Y}_s := (1_P \odot \mathsf{C}'_s) \circ \mathsf{N} \quad (s = L, R).$$

Put

$$\mathsf{A} := R_g^{\phi(h\ell)} \otimes_K R_h^{\phi(\ell)} \otimes_K R_\ell \otimes_K P,$$
$$\mathsf{B} := P^{\phi(gh\ell)} \otimes_K R'_{gh\ell}.$$

All four bimodules $\mathsf{X}_L, \mathsf{X}_R, \mathsf{Y}_L, \mathsf{Y}_R$ lie in ${}_{\mathsf{A}}\mathbf{Mod}_{\mathsf{B}}$. Let $\vartheta_L : \mathsf{X}_L \to \mathsf{Y}_L$ be the pasting of $\Theta_{g,h}$ with $\Theta_{gh,\ell}$, and let $\vartheta_R : \mathsf{X}_R \to \mathsf{Y}_R$ be the pasting of $\Theta_{h,\ell}$ with $\Theta_{g,h\ell}$, whiskered with the remaining factors. Let $\upsilon_X : \mathsf{X}_L \to \mathsf{X}_R$ be $\alpha_{g,h,\ell} \odot 1_P$ whiskered with $N_{gh\ell}$, and let $\upsilon_Y : \mathsf{Y}_L \to \mathsf{Y}_R$ be $1_P \odot \alpha'_{g,h,\ell}$ whiskered with N. The canonical associators, twist comparisons, and interchange maps are understood. The required condition is commutativity of

$$\begin{array}{ccc} \mathsf{X}_L & \xrightarrow{\vartheta_L} & \mathsf{Y}_L \\ \downarrow{\scriptstyle \upsilon_X} & & \downarrow{\scriptstyle \upsilon_Y} \\ \mathsf{X}_R & \xrightarrow{\vartheta_R} & \mathsf{Y}_R. \end{array} \tag{62}$$

Two systems of Brauer–character data, respectively of Brauer–Clifford data, are *coherently Morita equivalent* if there is a coherent Morita transport between them.

Proposition 10.7. *Coherent Morita transport is an equivalence relation on systems of Brauer–character data, and on systems of Brauer–Clifford data: the algebra $P = K$ with the unit constraints gives reflexivity, $P \otimes_K Q$ with the composed bimodules gives transitivity, and symmetry follows from the Kan property of the section spaces. Moreover, two systems of Brauer–character data lie in the same path component of $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$ if and only if they are coherently Morita equivalent, and two systems of Brauer–Clifford data lie in the same path component of $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))$ if and only if they are coherently Morita equivalent.*

Proof. A 1-simplex of $\mathrm{Sec}_\phi(X)$ is a simplicial map

$$B_\bullet G \times \Delta^1 \to X$$

over $B_\bullet\Gamma$ restricting to the two given sections at the ends (Notation 10.1). Its restriction to the vertical edge at the base vertex lies over the degenerate edge of $B_\bullet\Gamma$, hence is a 1-cell (id, P) with P an Azumaya K-algebra. For $g \in G$ the square $g \times \Delta^1$ is the union of two triangles along the diagonal $(g, 01)$; writing $(\phi(g), T_g)$ for the image of that diagonal, the two triangles give invertible bimodules

$$A_g \in {}_{R_g \otimes_K P}\mathbf{Mod}_{T_g}, \qquad B_g \in {}_{P^{\phi(g)} \otimes_K R'_g}\mathbf{Mod}_{T_g},$$

by the composition rule of Construction 9.12. Choose an inverse $C_g \in {}_{T_g}\mathbf{Mod}_{P^{\phi(g)} \otimes_K R'_g}$ of B_g. Then

$$N_g := A_g \otimes_{T_g} C_g \in {}_{R_g \otimes_K P}\mathbf{Mod}_{P^{\phi(g)} \otimes_K R'_g}$$

is independent of the diagonal up to isomorphism. Degeneracies force $T_e \cong P$ and N_e normalized. Over a 2-simplex (g, h) of $B_\bullet G$ the prism has three 3-simplices; by Proposition 9.14 their data are, in the truncated model, the equality (61), and in the full model the isomorphisms $\Theta_{g,h}$ of (iv). The 4-simplices of the prism impose exactly (62), and there are no independent data in higher dimensions. Conversely the formulas of Proposition 10.5, applied to the interval products of $B_\bullet G \times \Delta^1$ with P inserted on the vertical edges, produce a 1-simplex of sections from a transport.

Reflexivity, transitivity and symmetry are as stated, using the canonical identifications $K \otimes_K P \cong P$, $(P \otimes_K Q)$-transport = composite of the two transports, and $P \otimes_K P^{\mathrm{op}} \simeq_{\mathrm{Mor}} K$. The projections $\mathcal{B}\mathrm{r}_\Gamma(K)_\phi \to B_\bullet G$ and $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)_\phi \to B_\bullet G$ are Kan fibrations, being pullbacks of those in Proposition 9.19, so both simplicial sets of sections are Kan complexes. Hence being joined by a single 1-simplex is already an equivalence relation on vertices, and path components coincide with single-transport classes. □

Definition 10.8. The *unit section* $s_0^{(3)} : B_\bullet G \to \mathcal{B}\mathrm{r}_\Gamma(K)$ over $B_\bullet\phi$ has every edge algebra equal to K. On a triangle with last edge label η, its bimodule is K, with left action through

$$\lambda_\eta : K^\eta \otimes_K K \xrightarrow{\cong} K, \qquad x \otimes_K y \longmapsto \eta^{-1}(x)y,$$

and the regular right K-action. Iterating these unit maps identifies both tetrahedral composites with the regular bimodule K; the tetrahedral isomorphism is the identity under these identifications; on $[g|h|\ell]$ we write $\alpha^0_{g,h,\ell}$ for it. Unit coherence makes each pentagon commute and ensures that the formulas are compatible with faces and degeneracies. Thus they define a section. Its image

$$s_0^{(2)} := q \circ s_0^{(3)} : B_\bullet G \to \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$$

has the same edge algebras and triangle classes. We write s_0 for either unit section when the target is clear.

Definition 10.9. Write

$$\mathrm{BrCliff}_{\phi}^{\mathcal{A}z}(K), \qquad \mathrm{BrChar}_{\phi}^{\mathcal{A}z}(K)$$

for the sets of coherent Morita equivalence classes of Brauer–Clifford, respectively Brauer–character, data of type ϕ, in the sense of Definition 10.6. Their base points are the classes of the unit data, with $R_g = K$ for all g all bimodules the unit constraints, and, for Brauer–Clifford data, $\alpha = \alpha^0$ of Definition 10.8. The map $\mathrm{BrCliff}_{\phi}^{\mathcal{A}z}(K) \to \mathrm{BrChar}_{\phi}^{\mathcal{A}z}(K)$ used below is induced by forgetting associators α on systems and comparison isomorphisms Θ on transports, retaining the classes $[M_{g,h}]$ and $[N_g]$, respectively.

Comparison with section spaces. By Proposition 10.5, the systems of Definition 10.2 are the vertices of the simplicial sets of sections of Notation 10.1. The projections p_{Br} and $p_{\mathrm{Br},\le 2}$ are Kan fibrations by Proposition 9.19.

The simplicial map q over $B_\bullet\Gamma$ induces the map of section simplicial sets

$$\mathrm{Sec}_{\phi}(q) : \mathrm{Sec}_{\phi}(\mathcal{B}\mathrm{r}_{\Gamma}(K)) \longrightarrow \mathrm{Sec}_{\phi}(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)), \qquad s \longmapsto q_{\phi} \circ s.$$

Proposition 10.10. *The maps induced by Proposition 10.5 are base-point preserving bijections*

$$\mathrm{BrCliff}_{\phi}^{\mathcal{A}z}(K) \xrightarrow{\sim} \pi_0\big(\mathrm{Sec}_{\phi}(\mathcal{B}\mathrm{r}_{\Gamma}(K))\big), \qquad \mathrm{BrChar}_{\phi}^{\mathcal{A}z}(K) \xrightarrow{\sim} \pi_0\big(\mathrm{Sec}_{\phi}(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big),$$

compatible with $\pi_0(\mathrm{Sec}_{\phi}(q))$ and with the forgetful map that discards the isomorphisms α and Θ.

Proof. Proposition 10.5 identifies the vertices of the two simplicial sets of sections with the two kinds of data, and Proposition 10.7 identifies their path components with the coherent Morita equivalence classes. Compatibility with q holds because q forgets the chosen α's on vertices and the chosen Θ's on 1-simplices. □

Convention for cochains with local coefficients. Use the convention dictionary. Write z for a p-cochain with values in the terminal coefficient fibres and set

$$(I_p z)(g_1, \dots, g_p) := \eta_{g_1 \cdots g_p}\big(z(g_1, \dots, g_p)\big).$$

The cochain $I_p z$ takes values in the fibre at the initial vertex, as required by our convention for left-module group cohomology. The differential on cochains with values at the terminal vertex has first term $z(g_2, \dots, g_{p+1})$, the usual alternating middle terms, and last term $\eta_{g_{p+1}}^{-1}(z(g_1, \dots, g_p))^{(-1)^{p+1}}$. Applying I_{p+1} changes the first term into $\eta_{g_1}(I_p z(g_2, \dots, g_{p+1}))$ and the last into $I_p z(g_1, \dots, g_p)^{(-1)^{p+1}}$; the middle terms are unchanged. Hence

$$I_{p+1}\partial_{\mathrm{term}} z = \partial I_p z.$$

For a terminal-fibre scalar 4-cochain c, the corresponding left group cochain is $\omega = I_4 c$.

Construction 10.11. Let a class in $\mathrm{BrChar}_{\phi}^{\mathcal{A}z}(K)$ be represented by Brauer–character data in the sense of Definition 10.2. Choose representatives $M_{g,h}$ and associativity isomorphisms $\alpha_{g,h,\ell}$ whose existence follows from (60). Require the specified unit bimodules when $g = e$ or $h = e$, and the prescribed unit comparisons whenever one argument of α is e. Unit

coherence extends the choices on nondegenerate simplices to these cases. In the notation of Construction 9.13(iv), set

$$P^{\mathrm{short}}_{g,h,\ell,m} := B^{54} \circ B^{15}, \qquad P^{\mathrm{long}}_{g,h,\ell,m} := A^{34} \circ A^{23} \circ A^{12}.$$

Thus P^{short} is the two-arrow route from C_1 to C_4, whereas P^{long} is the three-arrow route. Define the scalar $c(g,h,\ell,m) \in \mathrm{U}(K)$ by

$$P^{\mathrm{short}}_{g,h,\ell,m} = c(g,h,\ell,m) P^{\mathrm{long}}_{g,h,\ell,m},$$

as maps $\mathsf{C}_1 \longrightarrow \mathsf{C}_4$, where c multiplies the underlying bimodule map. The corresponding left group cochain is

$$\omega : G^4 \longrightarrow \mathrm{U}(K), \qquad \omega(g,h,\ell,m) := \eta_{gh\ell m}\big(c(g,h,\ell,m)\big).$$

For a normalized 3-cochain u, rescaling the associators means

$$\alpha_{g,h,\ell} \longmapsto \eta^{-1}_{gh\ell}\big(u(g,h,\ell)\big)\alpha_{g,h,\ell}.$$

Lemma 10.12. *The cochain ω of Construction 10.11 is a normalized 4-cocycle. Rescaling the associators by a normalized 3-cochain u changes it to $(\partial u)^{-1}\omega$.*

Proof. Normalization of the unit data gives $\omega = 1$ whenever one entry is e. Apply Lemma 9.23 to the labels over $[g|h|\ell|m|n]$, with $\eta_g = \phi(g)$. With the convention for cochains just specified, (59) is exactly $(\partial\omega)(g,h,\ell,m,n) = 1$ for the coboundary (94). Thus ω is a normalized 4-cocycle. If the associators are rescaled by a normalized 3-cochain u, put

$$\rho_{\mathrm{short}} = u(gh,\ell,m)u(g,h,\ell m),$$
$$\rho_{\mathrm{long}} = \eta_g(u(h,\ell,m))u(g,h\ell,m)u(g,h,\ell).$$

The underlying maps $\mathsf{C}_1 \to \mathsf{C}_4$ change by

$$(P^{\mathrm{short}})' = \eta^{-1}_{gh\ell m}(\rho_{\mathrm{short}})P^{\mathrm{short}}, \qquad (P^{\mathrm{long}})' = \eta^{-1}_{gh\ell m}(\rho_{\mathrm{long}})P^{\mathrm{long}}.$$

Taking their ratio and then applying $\eta_{gh\ell m}$ gives

$$\omega' = \rho_{\mathrm{short}}\rho^{-1}_{\mathrm{long}}\omega = (\partial u)^{-1}\omega,$$

with ∂ as in (94). The identity $\partial^2 = 1$, proved in Subsection A.4, also shows that this change preserves the cocycle identity. □

Definition 10.13. The fourth obstruction is

$$\partial_4 : \mathrm{BrChar}^{\mathcal{A}z}_{\phi}(K) \longrightarrow H^4(G, \mathrm{U}(K)_\phi), \qquad [\mathcal{B}] \longmapsto [\omega].$$

Well-definedness is proved in Proposition 10.47. For a normalized 3-cocycle u, use the rescaling assignment of Construction 10.11 on Brauer–Clifford data.

Construction 10.14. For two systems (R,M) and (R',M') put

$$(R \boxtimes R')_g := R_g \otimes_K R'_g, \qquad M''_{g,h} := M_{g,h} \otimes_K M'_{g,h}.$$

Here $M''_{g,h}$ is viewed as a bimodule over the product component algebras via the interchange isomorphism

$$\big(R_g^{\eta_h} \otimes_K R_h\big) \otimes_K \big((R'_g)^{\eta_h} \otimes_K R'_h\big) \cong \big(R_g \otimes_K R'_g\big)^{\eta_h} \otimes_K \big(R_h \otimes_K R'_h\big).$$

At the Brauer–Clifford level, put $\alpha''_{g,h,\ell} := \alpha_{g,h,\ell} \otimes \alpha'_{g,h,\ell}$. For normalization, put $(R \boxtimes R')_e := K$, take $M''_{e,g}$, $M''_{g,e}$ and every α'' with an argument equal to e to be the specified unit data,

and for $g, h \neq e$ with $gh = e$ replace $M_{g,h} \otimes_K M'_{g,h}$ by its composite with the multiplication isomorphism $K \otimes_K K \to K$, conjugating the affected α'' by the same canonical isomorphisms; apply the same normalization to N''_e and Θ'' for the product transports below. For coherent transports (P, N, Θ) and (P', N', Θ'), the product data are $P \otimes_K P'$, $N_g \otimes_K N'_g$, and $\Theta_{g,h} \otimes \Theta'_{g,h}$, with the same normalization.

Proposition 10.15. *Tensor product over K makes Brauer–Clifford and Brauer–character data of type ϕ into abelian monoids on coherent Morita classes. For Construction 10.14:*

(i) *after normalization these are again Brauer–character, respectively Brauer–Clifford, data of type ϕ;*

(ii) *the product descends to coherent Morita classes and makes* $\mathrm{BrCliff}^{\mathcal{A}z}_{\phi}(K)$ *and* $\mathrm{BrChar}^{\mathcal{A}z}_{\phi}(K)$ *abelian monoids, with identity the class of the unit data of Definition 10.9;*

(iii) *the forgetful map* $\mathrm{BrCliff}^{\mathcal{A}z}_{\phi}(K) \to \mathrm{BrChar}^{\mathcal{A}z}_{\phi}(K)$ *is a monoid homomorphism, and ∂_4 of Definition 10.13 is multiplicative: $\partial_4\big([\mathcal{B}][\mathcal{B}']\big) = \partial_4([\mathcal{B}])\,\partial_4([\mathcal{B}'])$.*

Proof. (i) The interchange isomorphism is the symmetry of $\otimes_K$, and the twist satisfies $(R \otimes_K R')^{\theta} = R^{\theta} \otimes_K R'^{\theta}$, so $M''_{g,h}$ has the source and target required in Definition 10.2(ii); it is invertible because a tensor product over K of invertible bimodules is invertible. Both composites in (60) for the product are the tensor products of the corresponding composites for the two factors, by naturality of the interchange, so $\mathsf{L}_{g,h,\ell} = \mathsf{R}_{g,h,\ell}$ holds for the product. The pentagon for α'' holds factorwise for the same reason.

(ii) Let $(P, N, \Theta) : (R, M, \alpha) \Rightarrow (S, L, \beta)$ and $(P', N', \Theta') : (R', M', \alpha') \Rightarrow (S', L', \beta')$ be coherent Morita transports as in Definition 10.6. Then $P \otimes_K P'$, with the bimodule classes $[N_g \otimes_K N'_g]$ and the isomorphisms $\Theta_{g,h} \otimes \Theta'_{g,h}$, is a coherent Morita transport between the two products: the square (61) and the cylinder (62) hold factorwise, again by naturality of the interchange, and normalization is preserved. Hence the product descends to classes. Associativity, the unit law and commutativity on classes follow from the associators, the unitors and the symmetry of $\otimes_K$, as in Proposition 8.16 one degree down.

(iii) Forgetting the chosen α and Θ commutes with $\otimes_K$, which gives the homomorphism assertion. For ∂_4, choose representatives and associators for each factor and use their tensor products for the product. The two pentagon routes of Construction 10.11 for the product are then the tensor products of the routes for the factors, so their scalar defects multiply, $c'' = cc'$, whence $\omega'' = \omega\,\omega'$ and $\partial_4([\mathcal{B}][\mathcal{B}']) = \partial_4([\mathcal{B}])\partial_4([\mathcal{B}'])$. □

Comparison with simplicial products. The monoid structures in (ii) are defined algebraically, on coherent Morita classes. The degreewise tensor product of simplices is not a simplicial monoid structure on $\mathcal{B}\mathrm{r}_{\Gamma}(K)$ or $\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K)$: it is associative and unital only up to the canonical isomorphisms, and it does not commute with degeneracies, since on a degenerate edge it replaces the unit 1-cell (id, K) by $(\mathrm{id}, K \otimes_K K)$. This is the Brauer-level counterpart of Proposition 8.16. It gives the abelian monoid structures and multiplicativity of ∂_4; Proposition 10.16 below proves invertibility, so the two Brauer terms of Theorem 10.42 are abelian groups.

Opposite duals and evaluations. For Azumaya algebras A, B and an invertible bimodule $M : A \to B$, put

$$M^{\dagger} := (M^{\vee})^{\mathrm{op}} \in {}_{A^{\mathrm{op}}}\mathbf{Mod}_{B^{\mathrm{op}}}, \qquad M^{\vee} = \mathrm{Hom}_A(M, A).$$

Here the superscript op exchanges the two actions of the B–A-bimodule $M^\vee$. Explicitly, for $a \in A$, $b \in B$, $f \in M^\vee$, and $m \in M$,

$$(a^{\mathrm{op}} \star_{A^{\mathrm{op}}} f \star_{B^{\mathrm{op}}} b^{\mathrm{op}})(m) = f(m \star_B b)a.$$

For a bimodule isomorphism $v : M \to M'$ between A–B-bimodules, define $v^\dagger := (v^\vee)^{-1} : M^\dagger \to (M')^\dagger$, where $v^\vee$ is precomposition by v. For another invertible bimodule $N : B \to C$, the composition comparison is

$$\gamma_{M,N} : M^\dagger \otimes_{B^{\mathrm{op}}} N^\dagger \xrightarrow{\sim} (M \otimes_B N)^\dagger,$$
$$f \otimes_{B^{\mathrm{op}}} \psi \longmapsto \big[m \otimes_B n \longmapsto f(m \star_B \psi(n))\big].$$

For an independently given invertible bimodule $M_1 : A_1 \to B_1$ and $\eta \in \Gamma$, there are the canonical tensor and scalar-twist identifications

$$(M \otimes_K M_1)^\dagger \cong M^\dagger \otimes_K M_1^\dagger, \qquad (M^\eta)^\dagger \cong (M^\dagger)^\eta.$$

For an Azumaya algebra A, let E_A be the underlying K-module A with, for $a, b, x \in A$, left action $(a \otimes_K b^{\mathrm{op}}) \star_{A \otimes_K A^{\mathrm{op}}} x = axb$ and its usual right K-action. Thus

$$E_A \in {}_{A \otimes_K A^{\mathrm{op}}}\mathbf{Mod}_K.$$

The evaluation comparison for $M : A \to B$ is

$$\tau_M : (M \otimes_K M^\dagger) \otimes_{B \otimes_K B^{\mathrm{op}}} E_B \xrightarrow{\sim} E_A,$$
$$(m \otimes_K f) \otimes_{B \otimes_K B^{\mathrm{op}}} b \longmapsto f(m \star_B b).$$

When twisting an evaluation module, identify its target K^η with K by $a \mapsto \eta^{-1}(a)$, using the unit maps of Construction 9.12. These identifications give $E_{A^\eta} \cong E_A^\eta$ after that target normalization, and $E_{A \otimes_K B} \cong E_A \otimes_K E_B$ after interchange and normalizing the target $K \otimes_K K$ to K. For systems, replace the dual unit representatives by the specified unit bimodules and conjugate the affected comparison maps by these identifications, as in the normalization of Construction 10.14.

Proposition 10.16. *The abelian monoids* $\mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)$ *and* $\mathrm{BrChar}_\phi^{\mathcal{A}z}(K)$ *are abelian groups. An inverse of the class of a Brauer–Clifford system* $(R_g, M_{g,h}, \alpha_{g,h,\ell})$ *is represented by the normalized opposite-dual system*

$$(R_g^{\mathrm{op}}, M_{g,h}^\dagger, \alpha_{g,h,\ell}^\dagger),$$

with the composition and tensor identifications just specified. For Brauer–character systems, retain only the classes $[M_{g,h}^\dagger]$.

Proof. First verify the dual and evaluation comparisons. The map $\gamma_{M,N}$ is the inverse-composition isomorphism for the Hom inverses, with the two factors exchanged on passing to opposite actions. For a third invertible bimodule $H : C \to D$ and $\chi \in H^\vee$, both iterated comparisons send $f \otimes \psi \otimes \chi$ to the same A-valued functional on $(M \otimes_B N) \otimes_C H$:

$$(m \otimes_B n) \otimes_C p \longmapsto f(m \star_B \psi(n \star_C \chi(p))).$$

This proves associativity of the composition comparisons. For the regular bimodule A, the identification $\mathrm{Hom}_A(A, A) \cong A$, sending f to $f(1)$, identifies $A^\dagger$ with the regular A^{op}-bimodule and gives the unit comparisons. Tensor compatibility follows from the Hom-tensor isomorphism for finitely generated projective modules; its coherence and compatibility with scalar twisting follow by evaluation on pure tensors. Consequently opposite duals preserve

the character associativity equation and the Clifford pentagon, including their normalized unit data.

The module E_A is a Morita equivalence because $A \otimes_K A^{\mathrm{op}} \cong \mathrm{End}_K(A)$ and A is a K-progenerator, as in Lemma 9.2. The map τ_M factors as the canonical identification with $M \otimes_B M^\vee$, regarded as an $(A \otimes_K A^{\mathrm{op}})$–$K$-bimodule using its two A-actions, followed by its evaluation isomorphism:

$$(m \otimes_K f) \otimes_{B\otimes_K B^{\mathrm{op}}} b \longmapsto (m \star_B b) \otimes_B f \longmapsto f(m \star_B b).$$

This proves it is a balanced bimodule isomorphism. It is compatible with composition: for $M : A \to B$ and $N : B \to C$, evaluating first with τ_N and then with τ_M gives, for $c \in C$, $f(m \star_B \psi(n \star_C c))$, the same value as $\tau_{M\otimes_B N}$ using $\gamma_{M,N}$. Its tensor compatibility is the corresponding equality of factorwise evaluations. For an isomorphism $v : M \to M'$, its naturality is the identity

$$\tau_{M'} \circ \big((v \otimes v^\dagger) \otimes 1_{E_B}\big) = \tau_M,$$

as maps $(M \otimes_K M^\dagger) \otimes_{B\otimes_K B^{\mathrm{op}}} E_B \longrightarrow E_A$. Indeed $v^\dagger(f) = f \circ v^{-1}$, so both sides send the tensor represented by (m, f, b) to $f(m \star_B b)$. For a regular bimodule τ is the canonical unit comparison. These comparisons are induced by the Hom inverse-pair evaluations of Definition 9.9, with their triangle identities.

Now form the normalized product of a system and its opposite dual. Define a transport to the unit system by taking the algebra $P = K$, the bimodules $N_g = E_{R_g}$ for $g \neq e$, and $N_e = K$ with its specified unit comparison. For a pair (g, h) put $A = R_g^{\eta_h} \otimes_K R_h$ and $B = R_{gh}$. After the product interchange, its multiplication bimodule is $M_{g,h} \otimes_K M_{g,h}^\dagger$, from $A \otimes_K A^{\mathrm{op}}$ to $B \otimes_K B^{\mathrm{op}}$. The source of $\Theta_{g,h}$ in Definition 10.6 identifies with the source of $\tau_{M_{g,h}}$. Its target identifies with E_A: the successive N_h and $N_g^{\eta_h}$ evaluate the two algebra factors, using the twisted unit maps to end at K. Take $\Theta_{g,h}$ to be $\tau_{M_{g,h}}$ under these identifications. This proves the transport square. When $g = e$ or $h = e$ it is the prescribed unit comparison. When $gh = e$, the same map is used after the target normalization $K \otimes_K K \to K$.

For the cylinder, put $A = R_g^{\eta_{h\ell}} \otimes_K R_h^{\eta_\ell} \otimes_K R_\ell$ and $B = R_{gh\ell}$, and let $L, U : A \to B$ be the two triple composite bimodules compared by $\alpha_{g,h,\ell} : L \to U$. Composition and tensor compatibility of τ identify the two horizontal pastings of Θ in (62) with τ_L and τ_U. The product associator is $\alpha_{g,h,\ell} \otimes \alpha_{g,h,\ell}^\dagger$ under the same comparisons. Thus the cylinder is precisely the naturality identity above with $v = \alpha_{g,h,\ell}$; it commutes. The evaluation transport therefore makes the product coherently Morita equivalent to the unit. For a character system the same construction, on bimodule classes, proves the square without requiring any chosen associators or pentagon. Commutativity of the monoids, already proved in Proposition 10.15, makes this inverse two-sided and proves the assertion. □

The principal statements. Notation 10.23 summarizes the algebraic data and transports at the three Postnikov stages. Theorem 10.42 gives the full sequence, with obstruction maps ∂_2, ∂_3, and ∂_4. All terms are abelian groups, including the two Brauer terms by Proposition 10.16. The final four terms are

$$H^3(G, \mathrm{U}(K)_\phi) \longrightarrow \mathrm{BrCliff}_\phi^{\mathcal{A}\mathrm{z}}(K) \longrightarrow \mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K) \xrightarrow{\partial_4} H^4(G, \mathrm{U}(K)_\phi).$$

By Theorem 10.51, this sequence is exact through $\mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K)$.

Connecting-map convention. Connecting arrows use ordered relative obstruction, with the $B_\bullet G$ coordinate first. The arrow from the fundamental group of the base to the components of the fibre is inverse to forward endpoint transport. The degree-two connecting arrow uses the ordered d_0-horn normalization. The action on rigidified lifts is specified with this convention in Construction 10.49.

Comparison with the lifting space. Proposition 10.52 identifies rigidified lifts with an $H^3(G, \mathrm{U}(K)_\phi)$-torsor and identifies the stabilizer after forgetting the rigidification with the image of the topological boundary of Construction 10.49. At the unit section that boundary corresponds to ∂_3. Theorem 10.39 supplies the section-space sequence; the following loop calculations identify its terms algebraically.

Notation 10.17. On representatives the three arrows of the Brauer row displayed above are

$$\begin{aligned} [u] &\longmapsto \text{the unit data with } \alpha_{g,h,\ell} = \eta_{gh\ell}^{-1}\big(u(g,h,\ell)\big)\alpha^0_{g,h,\ell}, \\ [(R,M,\alpha)] &\longmapsto \big[(R,[M])\big], \\ [(R,[M])] &\longmapsto [\omega], \end{aligned}$$

where u is a normalized 3-cocycle, α^0 denotes the unit associators of Definition 10.8, the middle arrow forgets the chosen α and retains the classes $[M_{g,h}]$ subject to (60), and $[\omega]$ is ∂_4 of Definition 10.13. The first arrow is the $H^3(G, \mathrm{U}(K)_\phi)$-action of Theorem 10.51(iii) applied to the unit class, and by that theorem the classes it sends to the base point are exactly $\mathrm{im}(\partial_3)$. This display is the Brauer-level counterpart of the dictionary (53) for the fixed-base row.

Multiplicativity of the unit-system map. The first arrow of Notation 10.17 is a group homomorphism, using the group structure established in Proposition 10.16. Indeed, let u, v be normalized 3-cocycles. In the external product of their unit systems, the canonical multiplication $K \otimes_K K \to K$ and the twisted unit comparisons of Definition 10.8 identify the edge algebras and triangle bimodules with the unit data. By unit coherence, the tensor product of the two unit associators becomes $\alpha^0_{g,h,\ell}$. Its scalar factor is

$$\eta_{gh\ell}^{-1}\big(u(g,h,\ell)\big)\eta_{gh\ell}^{-1}\big(v(g,h,\ell)\big) = \eta_{gh\ell}^{-1}\big((uv)(g,h,\ell)\big) \quad \text{in } \mathrm{U}(K),$$

for $g, h, \ell \in G$. Thus the normalized product represents the unit system for uv. The constant cocycle 1 gives the unit class, and the system for u^{-1} gives the inverse of the system for u. Finally, the rescaling action of Theorem 10.51 shows that cohomologous cocycles give the same class, so this multiplicativity descends to $H^3(G, \mathrm{U}(K)_\phi)$.

10.2. **Unit sections and based loops.**

Notation 10.18. By Definition A.33, the full based-loop section space has q-simplices

$$\begin{aligned} \big(\Omega_{s_0}\,\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big)_q = \big\{ H : B_\bullet G \times \Delta^1 \times \Delta^q \to \mathcal{B}\mathrm{r}_\Gamma(K) \;\big|\; p_{\mathrm{Br}} \circ H = B_\bullet\phi \circ \mathrm{pr}_{B_\bullet G}, \\ H|_{B_\bullet G\times\{0\}\times\Delta^q} = H|_{B_\bullet G\times\{1\}\times\Delta^q} = s_0 \circ \mathrm{pr}_{B_\bullet G} \big\}. \end{aligned} \tag{63}$$

Replacing $\mathcal{B}\mathrm{r}_\Gamma(K)$ by $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ gives the truncated model.

All homotopy groups of the section spaces in (65) are based at the unit section s_0 of Definition 10.8; we omit s_0 from the notation. Their algebraic identifications are given in Theorem 10.33; the comparison with the isomorphism-based equivalence relation of [Gin22, Def. 3.1] is in Remark 10.45.

The next three algebraic identifications have different purposes. Lemma 10.20 extracts a system from a specified loop. Theorem 10.33 proves completeness and independence of choices by descent. Proposition 10.35 compares the component products and boundary maps with the seven-term sequence.

Construction 10.19. For a based loop H at the unit section, use the following recipe. Definition A.33 identifies the vertex H with a simplicial map

$$H : B_\bullet G \times \Delta^1 \longrightarrow \mathcal{B}\mathrm{r}_\Gamma(K)$$

over $B_\bullet G$ whose two endpoint restrictions are s_0. The vertical edge over the unique vertex of $B_\bullet G$ has an Azumaya K-algebra R as its Street-nerve label. We use the shuffle coordinates $i^a = (i, a)$ of Definition A.2. For every $g \in G$, the product $[g] \times \Delta^1$ has the two standard shuffle triangles

$$(0^0, 1^0, 1^1), \qquad (0^0, 0^1, 1^1).$$

Let D_g be the algebra on their common diagonal, and let their bimodule labels be

$$M_g^0 \in {}_R\mathbf{Mod}_{D_g}, \qquad M_g^1 \in {}_{R^{\phi(g)}}\mathbf{Mod}_{D_g}.$$

Then

$$P_g := (M_g^0)^{-1} \circ M_g^1 \in {}_{R^{\phi(g)}}\mathbf{Mod}_R. \tag{64}$$

By Lemma 3.4, the resulting R-bimodule over k has type $\phi(g)$.

For $[g|h] \in (B_\bullet G)_2$, the three standard shuffle tetrahedra in $[g|h] \times \Delta^1$ have ordered vertices

$$(0^0, 1^0, 2^0, 2^1), \quad (0^0, 1^0, 1^1, 2^1), \quad (0^0, 0^1, 1^1, 2^1).$$

Their boundary contains the degree-one squares for g, h, and gh. Composing the tetrahedral comparisons and cancelling the diagonal bimodules gives

$$\mu_{g,h} : P_g^{\phi(h)} \otimes_{R^{\phi(h)}} P_h \xrightarrow{\sim} P_{gh}$$

as an isomorphism of $R^{\phi(gh)}$–R-bimodules. On the underlying R-bimodules over k, this is

$$\mu_{g,h} : P_g \otimes_R P_h \xrightarrow{\sim} P_{gh}.$$

For the truncated model retain only the bimodule classes, and put $\Phi_H(g) := [P_{H,g}]$. For a loop homotopy between systems with base algebras R and S, the two shuffle triangles in the interval factors give Morita morphisms $R \to D$ and $S \to D$, where D labels their common diagonal. Invert the second morphism and compose to obtain $R \to S$. The shuffle tetrahedra supply the comparisons of the extracted systems.

Lemma 10.20. *Let s_0 be the unit section.*

(i) *Every vertex*

$$H \in \big(\Omega_{s_0} \mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big)_0$$

determines an Azumaya K-algebra R, invertible bimodules

$$P_g \in {}_{R^{\phi(g)}}\mathbf{Mod}_R \quad \textit{for all } g \in G,$$

and multiplication isomorphisms

$$\mu_{g,h} : P_g \otimes_R P_h \xrightarrow{\sim} P_{gh} \quad \textit{for all } g, h \in G.$$

The unit and associativity identities make these data a monoidal functor $\underline{G} \to \boldsymbol{\mathcal{P}ic}_k(R)$ of type ϕ, hence a G-graded Clifford system extension of R in the sense of Definition 4.1.

(ii) *A* 1*-simplex between two such vertices determines an identity-central morphism between their base algebras, together with isomorphisms intertwining the two bimodule systems and their multiplication maps. Therefore the extraction in* (i) *descends to a map from*

$$\pi_0\big(\Omega_{s_0}\operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \cong \pi_1\big(\operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)), s_0\big)$$

to Morita equivalence classes (Definition 8.7) of G*-graded Clifford system extensions of type* ϕ *over Azumaya bases.*

(iii) *Let* H *be a vertex of* $\Omega_{s_0}\operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$*. Its vertical edge determines an Azumaya* K*-algebra* R*. For every* $g \in G$*, the two standard shuffle triangles over* $[g]\times\Delta^1$ *determine a class*

$$[P_{H,g}] \in \operatorname{Pic}_k(R), \qquad h_R([P_{H,g}]) = \phi(g),$$

and the standard shuffle tetrahedra over $[g|h]\times\Delta^1$ *give the equality*

$$[P_{H,g}][P_{H,h}] = [P_{H,gh}] \quad \text{for all } g,h \in G.$$

Thus

$$\Phi_H \colon G \longrightarrow \operatorname{Pic}_k(R), \qquad g \longmapsto [P_{H,g}],$$

is a generalized collective character of type ϕ*. A* 1*-simplex between based loops with endpoint data* $(R,[P_{H,g}])$ *and* $(S,[P_{H',g}])$ *induces an identity-central morphism* $L : R \to S$ *represented by* $L \in {}_R\mathbf{Mod}_S$*, and equalities*

$$[P_{H,g}\otimes_R L] = [L^{\phi(g)}\otimes_{S^{\phi(g)}} P_{H',g}] \quad \text{for all } g\in G.$$

Consequently the assignment $[H]\mapsto[R,\Phi_H]$ *gives a map from*

$$\pi_1\big(\operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)), s_0\big)$$

to Morita equivalence classes of generalized collective characters of type ϕ *over Azumaya bases.*

Proof. Apply the extraction of Construction 10.19. The degeneracy equations give the unit identities. The four standard $(3,1)$-shuffle 4-simplices in $[g|h|\ell]\times\Delta^1$ have ordered vertices

$$0^0,\dots,r^0,r^1,\dots,3^1 \quad \text{for all } 0\le r\le 3.$$

Their face equations state that

$$\mu_{gh,\ell}\circ(\mu_{g,h}\otimes \mathrm{id}_{P_\ell}) = \mu_{g,h\ell}\circ(\mathrm{id}_{P_g}\otimes\mu_{h,\ell}),$$

as maps $(P_g\otimes_R P_h)\otimes_R P_\ell \to P_{gh\ell}$, after inserting the canonical associator for balanced tensor products. Thus the extracted data are a monoidal functor. Their direct sum is the strongly G-graded algebra

$$A = \bigoplus_{g\in G} P_g, \qquad p_g p_h := \mu_{g,h}(p_g\otimes_R p_h),$$

by Theorem 4.5. This proves (i).

By (102), a 1-simplex of the based-loop simplicial set is a map on $B_\bullet G\times\Delta^1\times\Delta^1$, fixed at the two loop endpoints. Write $i^{ab}=(i,a,b)$, where i is the bar coordinate, a is the loop coordinate and b is the homotopy coordinate. The two standard triangles in the interval square are

$$\sigma_- = (00,10,11), \qquad \sigma_+ = (00,01,11),$$

where ab abbreviates (a,b). At the unique bar vertex, the edges $00 \to 10$ and $01 \to 11$ carry R and S, respectively, while the edges $00 \to 01$ and $10 \to 11$ carry K because the loop endpoints are fixed. Let D label $00 \to 11$. After applying the unit comparisons, the labels of σ_- and σ_+ are therefore invertible bimodules

$$E_- \in {}_R\mathbf{Mod}_D, \qquad E_+ \in {}_S\mathbf{Mod}_D.$$

Choose an inverse $C \in {}_D\mathbf{Mod}_S$ of E_+ and put $L := E_- \otimes_D C \in {}_R\mathbf{Mod}_S$. All these morphisms are identity-central, since the interval edges lie over the identity of Γ. Write P_g, Q_g, T_g for the component bimodules extracted from the loops with vertical algebras R, S, D, respectively.

For each $g \in G$, the three $(1,2)$-shuffle tetrahedra over each triangle have the following ordered vertices:

$$\begin{array}{c|l} \sigma_- & (0^{00}, 1^{00}, 1^{10}, 1^{11}) \\ & (0^{00}, 0^{10}, 1^{10}, 1^{11}) \\ & (0^{00}, 0^{10}, 0^{11}, 1^{11}) \\ \sigma_+ & (0^{00}, 1^{00}, 1^{01}, 1^{11}) \\ & (0^{00}, 0^{01}, 1^{01}, 1^{11}) \\ & (0^{00}, 0^{01}, 0^{11}, 1^{11}). \end{array}$$

For a triangle (w_0, w_1, w_2), consecutive tetrahedra in this list share the faces $(0^{w_0}, 1^{w_1}, 1^{w_2})$ and $(0^{w_0}, 0^{w_1}, 1^{w_2})$, respectively. Paste their comparison isomorphisms and cancel the invertible bimodules on these internal faces. The outer square faces are the extraction squares for the three edges of the interval triangle. One is a constant unit loop; its comparison disappears under the unit identifications. The other two, together with the two copies of the triangle label, give

$$P_g \otimes_R E_- \xrightarrow{\sim} E_-^{\phi(g)} \otimes_{D^{\phi(g)}} T_g,$$
$$Q_g \otimes_S E_+ \xrightarrow{\sim} E_+^{\phi(g)} \otimes_{D^{\phi(g)}} T_g.$$

Apply the inverse of the second comparison, tensoring with $C^{\phi(g)}$ on the left and C on the right, and contracting the inverse pairs. Together with the first comparison, this gives

$$\begin{aligned} \beta_g : \quad P_g \otimes_R L &\cong (P_g \otimes_R E_-) \otimes_D C \\ &\xrightarrow{\sim} E_-^{\phi(g)} \otimes_{D^{\phi(g)}} (T_g \otimes_D C) \\ &\xrightarrow{\sim} E_-^{\phi(g)} \otimes_{D^{\phi(g)}} (C^{\phi(g)} \otimes_{S^{\phi(g)}} Q_g) \\ &\cong L^{\phi(g)} \otimes_{S^{\phi(g)}} Q_g. \end{aligned}$$

Every term is an $R^{\phi(g)}$–S-bimodule; the canonical associators and twist comparisons are understood. On the underlying bimodules, tensoring on the left with $L^\vee$ and contracting $L^\vee \otimes_R L$ gives $\Xi_L(P_g) \cong Q_g$. The equations on the six $(2,2)$-shuffle 4-simplices over $[g|h]$ times each triangle say that the β_g intertwine the multiplication maps, as in Definition 8.7. Hence homotopic vertices extract Morita-equivalent systems, proving (ii).

For (iii), the map q retains the vertical algebra R and sends each triangle bimodule to its isomorphism class, by Proposition 9.17. Thus a vertex

$$\overline{H} \in \big(\Omega_{s_0} \operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big)_0$$

determines the classes $[P_{\overline{H},g}] \in \mathrm{Pic}_k(R)$ extracted from the two shuffle-triangle labels by composing the second with the inverse of the first, as in (64). The tetrahedral face relation

in $\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ is precisely

$$[P_{\overline{H},g}][P_{\overline{H},h}] = [P_{\overline{H},gh}] \quad \text{for all } g,h \in G,$$

and Lemma 3.4 gives $h_R([P_{\overline{H},g}]) = \phi(g)$. Hence these classes define the generalized collective character $\Phi_{\overline{H}}$ in the statement. For a 1-simplex between vertices with base algebras R and S, the two-factor shuffle extraction of (ii) gives an identity-central morphism $L : R \to S$ and the displayed equalities

$$[P_{\overline{H},g} \otimes_R L] = [L^{\phi(g)} \otimes_{S^{\phi(g)}} P_{\overline{H}',g}].$$

Thus the assignment is constant on components and gives the stated map. This proves (iii). □

The extracted Clifford extensions are the objects of [Gin22, Def. 4.2]. The extracted generalized collective characters are the objects of [CG01, p. 152, §2]; see also [Gin22, (4.2)]. The global converse and the independence of all coordinate choices are proved in Theorem 10.33.

10.3. **Loop models for the Brauer three-type.** For the first fibrewise Postnikov stage, use

$$\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K) := B_\bullet\big(\Gamma \ltimes \mathrm{Br}(K)\big),$$

where the group on pairs is defined by

$$(\eta,[R])(\theta,[S]) = (\eta\circ\theta,[R^\theta \otimes_K S]).$$

Projection to Γ induces

$$p_{\mathrm{Br},\le 1} : \mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K) \longrightarrow B_\bullet\Gamma.$$

This is a Kan fibration, since it is the nerve of a surjective group homomorphism, and its identity fibre is $B_\bullet \mathrm{Br}(K)$. The map

$$q_1 : \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K) \longrightarrow \mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K)$$

sends an edge (η, R) to $(\eta,[R])$. The triangle relations make this a simplicial map. On the identity fibre it is the first Postnikov truncation described in Definition 10.25. The truncation maps induce the tower of section spaces

$$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)) \longrightarrow \mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)) \longrightarrow \mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K)). \tag{65}$$

The unit section of $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))$ is $s_0^{(1)} := q_1 \circ s_0^{(2)}$, the homomorphism $g \mapsto (\phi(g),[K])$, that is, the unit datum $R_g = K$. We write $\mathrm{Br}(K)_\phi$ for $\mathrm{Br}(K)$ regarded as a left G-module by ${}^g[R] = [R^{\eta_g^{-1}}]$, as in the convention dictionary and (34); its cohomology uses (10).

Definition 10.21. A system of *Brauer data of type* ϕ consists of an Azumaya K-algebra R_g for every $g \in G$, with $R_e = K$, such that

$$[R_{gh}] = \big[R_g^{\eta_h} \otimes_K R_h\big] \qquad \text{in } \mathrm{Br}(K) \text{ for all } g,h \in G, \tag{66}$$

where $\eta_h = \phi(h)$ as in Definition 10.2. A *transport* $(R_g) \Rightarrow (R'_g)$ is an Azumaya K-algebra P with

$$\big[R_g \otimes_K P\big] = \big[P^{\eta_g} \otimes_K R'_g\big] \qquad \text{in } \mathrm{Br}(K) \text{ for all } g \in G. \tag{67}$$

Two systems are *Morita equivalent* if there is a transport between them. This is an equivalence relation: $P = K$ gives reflexivity, $P \otimes_K Q$ transitivity and P^{op} symmetry, as in Proposition 10.7 two stages up. These are the data of the first stage. Forgetting the triangle

classes $[M_{g,h}]$ sends Brauer–character data (Definition 10.2) to Brauer data, because the existence of an invertible bimodule in ${}_{R_g^{\eta_h}\otimes_K R_h}\mathbf{Mod}_{R_{gh}}$ is exactly (66); and forgetting the bimodules N_g sends a coherent Morita transport (Definition 10.6) to a transport in the present sense. For a Brauer–character system, its profile map is the assignment

$$\mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K) \longrightarrow H^1(G,\mathrm{Br}(K)_\phi), \qquad [(R,M)] \longmapsto \big[g \longmapsto [R_g^{\phi(g)^{-1}}]\big].$$

The comparison below verifies that this is well defined on classes.

Proposition 10.22. *The vertices of* $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))$ *are the Brauer profiles* $([R_g])_{g\in G}$ *of systems of Brauer data of type* ϕ*, and a* 1*-simplex is determined by the class* $[P]$ *labelling its vertical edge over the vertex of* $B_\bullet G$*, which lies over the degenerate edge of* $B_\bullet\Gamma$*; such a* 1*-simplex exists exactly when* P *is a transport. At the unit datum* $R_g = K$ *there are isomorphisms*

$$\pi_0\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) \cong H^1(G,\mathrm{Br}(K)_\phi), \qquad \pi_1\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) \cong \mathrm{Br}(K)^\phi.$$

Under the first, the Morita class of (R_g) *is the cohomology class of the* 1*-cocycle* $g \mapsto {}^g[R_g] = \big[R_g^{\eta_g^{-1}}\big]$ *of* (10)*: by* (66) *the family* $g \mapsto [R_g]$ *is a crossed homomorphism for the right action* $[R]\cdot g = [R^{\eta_g}]$*, and* $a \mapsto \big(g \mapsto {}^g a_g\big)$ *carries these bijectively to the normalized* 1*-cocycles of* (10)*. Under the same bijection* (67) *becomes the coboundary relation* ${}^g[R'_g] = {}^g[R_g]\,{}^g[P]\,[P]^{-1}$*. Under the second, the class of a self-transport of the unit datum is an invariant Brauer class. The truncation* q_1 *induces the map of component sets*

$$\mathrm{BrChar}_\phi^{\mathcal{A}\mathrm{z}}(K) \longrightarrow H^1(G,\mathrm{Br}(K)_\phi)$$

that retains only the Brauer profile.

Proof. By its definition above, $\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K)$ is the nerve of the group $\Gamma \ltimes \mathrm{Br}(K)$, so a vertex of $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))$ is a group homomorphism $G \to \Gamma \ltimes \mathrm{Br}(K)$ whose first coordinate is ϕ. Writing its second coordinate as $[R_g]$, the multiplication rule $(\eta,[R])(\theta,[S]) = (\eta\circ\theta,[R^\theta \otimes_K S])$ turns the homomorphism condition into (66), and normalization gives $[R_e] = [K]$. Thus the vertices are the normalized right crossed homomorphisms $a_{gh} = a_g^{\eta_h} a_h$ with values in $\mathrm{Br}(K)$, identified with the normalized 1-cocycles of (10) by $a \mapsto (g \mapsto {}^g a_g)$, for the left action of the convention dictionary; every such cocycle is realized, because each class in $\mathrm{Br}(K)$ is represented by an Azumaya algebra, by Definition 9.1 and Lemma 9.2. A 1-simplex over the degenerate edge is conjugation by an element of $\mathrm{Br}(K)$, say by $(1,[P])$. On $[g]\times\Delta^1$ the two shuffle triangles force the diagonal label $(\eta_g,[R_g][P]) = (1,[P])(\eta_g,[R'_g]) = (\eta_g,[P^{\eta_g}][R'_g])$, which is (67); since the nerve of a group is 2-coskeletal, higher prisms impose no further condition. Hence the components are the cocycles modulo coboundaries, which is $H^1(G,\mathrm{Br}(K)_\phi)$, and the vertex group at the unit cocycle is $\big\{[P] \mid \big[P^{\eta_g}\big] = [P] \text{ for all } g \in G\big\} = \mathrm{Br}(K)^\phi$. Compatibility with q_1 holds because q_1 sends an edge (η,R) to $(\eta,[R])$, so on vertices it retains the Brauer profile of Definition 10.2. □

Notation 10.23. The algebraic content of the three stages (65) is

	$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))$	$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$	$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))$
data	Brauer–Clifford	Brauer–character	Brauer
edges	R_g	R_g	$[R_g]$
triangles	$M_{g,h}$	$[M_{g,h}]$	—
tetrahedra	$\alpha_{g,h,\ell}$	—	—
coherence	pentagon	(60)	(66)
transport	$P,\ N_g,\ \Theta_{g,h}$	$P,\ [N_g]$	$[P]$
transport coherence	(62)	(61)	—
π_0	$\mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)$	$\mathrm{BrChar}_\phi^{\mathcal{A}z}(K)$	$H^1(G,\mathrm{Br}(K)_\phi)$

The data and transports are those of Definitions 10.6, 10.2 and 10.21, and the identifications of the last row are Propositions 10.10 and 10.22. This table records simplex labels, coherence conditions, and transports for the three section spaces, whereas the table in Theorem 10.33 records their homotopy groups. The propositions just cited identify the component sets in the last row. One nerve degree down, at the unit section, with base algebras varying along the transports of Definition 8.4, the corresponding data, transports and classes are Definitions 8.3, 8.7 and 8.9.

For a set A, write A_{disc} for the constant simplicial set with $(A_{\mathrm{disc}})_q = A$ in every degree and all faces and degeneracies equal to the identity.

Construction 10.24. We first construct the comparison with $N_\bullet\mathbf{Pic}(K)$ over $B_\bullet\Gamma$. Let $\mathcal{P} \to B_\bullet\Gamma$ be the nerve of the scalar-twist Grothendieck groupoid. Over $[\eta_1|\cdots|\eta_n]$, put $\eta_{ij} := \eta_{i+1}\circ\cdots\circ\eta_j$. Its data are invertible modules L_i and isomorphisms

$$\rho_{ij} : L_i^{\eta_{ij}} \longrightarrow L_j, \qquad \rho_{ik} = \rho_{jk}\circ\rho_{ij}^{\eta_{jk}}.$$

The identity compares maps $L_i^{\eta_{ik}} \to L_k$, using $(L_i^{\eta_{ij}})^{\eta_{jk}} \cong L_i^{\eta_{ik}}$. Repeated indices use the canonical unit data. The fibre is $N_\bullet\mathbf{Pic}(K)$. For descent along $\phi : G \to \Gamma$, the right-twist maps $\rho_g : L^{\eta_g} \to L$ and the equivariant maps $\lambda_g : {}^gL \to L$ of Definition 7.9 are related by

$$\lambda_g = (\rho_g^{\eta_g^{-1}})^{-1} : {}^gL \longrightarrow L.$$

Thus $\rho_{gh} = \rho_h\circ\rho_g^{\eta_h}$ becomes $\lambda_{gh} = \lambda_g\circ{}^g\lambda_h$ as maps ${}^{gh}L \to L$, with the canonical twist identifications. Let $\mathcal{L}_2 \to B_\bullet\Gamma$ have, over a simplex $\sigma : \Delta^n \to B_\bullet\Gamma$, the maps

$$F : \Delta^n\times\Delta^2 \longrightarrow \mathcal{B}\mathrm{r}_\Gamma(K), \qquad p_{\mathrm{Br}}\circ F = \sigma\circ\mathrm{pr}_{\Delta^n}, \qquad F|_{\Delta^n\times\partial\Delta^2} = s_0\circ\sigma\circ\mathrm{pr}_{\Delta^n}.$$

The equalities compare maps $\Delta^n\times\Delta^2 \to B_\bullet\Gamma$ and $\Delta^n\times\partial\Delta^2 \to \mathcal{B}\mathrm{r}_\Gamma(K)$, respectively. The section s_0 here is Definition 10.8 with $G = \Gamma$ and $\phi = \mathrm{id}$. We define a map $\mathcal{P} \to \mathcal{L}_2$ as follows. Write $i^a = (i,a)$, with $a \in \{0,1,2\}$. All edge algebras are K. After the unit identifications of Definition 10.8, the only nonunit labels are

$$M_{i^0j^1k^2} = L_k, \qquad \alpha_{i^0j^1k^2\ell^2} = \rho_{k\ell}.$$

In the second formula the two composites are $L_k^{\eta_{k\ell}}$ and L_ℓ. Other mixed tetrahedra containing all three levels have the identity comparison; on faces missing a level we use the unit section.

Collapse the boundary of $\Delta^1\times\Delta^1$ and its triangle $(0^0,0^1,1^1)$ to the basepoint, and send the triangle $(0^0,1^0,1^1)$ to the ordered simplex of $\Delta^2/\partial\Delta^2$.

Definition 10.25. We use the following concrete models for the two Postnikov truncations of $\mathcal{B}\mathrm{r}(K)$. The second truncation is the identity fibre of the truncated Street nerve of Construction 9.16:

$$(\tau_{\le 2}\mathcal{B}\mathrm{r}(K))_n := (p_{\mathrm{Br},\le 2,n})^{-1}(1_n).$$

Thus its edge labels are Azumaya K-algebras, its triangle labels are isomorphism classes of invertible bimodules, and on every tetrahedron the two composite bimodule classes are equal. Its face and degeneracy maps delete or repeat vertices and restrict these labels, as in Proposition 9.17. The second-truncation map is the simplicial map

$$t_2 : \mathcal{B}\mathrm{r}(K) \longrightarrow \tau_{\le 2}\mathcal{B}\mathrm{r}(K)$$

obtained by restricting the map q of Proposition 9.17 to the identity fibre: it fixes every edge algebra, sends each triangle bimodule M_{ijk} to its isomorphism class $[M_{ijk}]$, and forgets every bimodule isomorphism on a tetrahedron.

For the first truncation we use the inhomogeneous bar simplicial set

$$\tau_{\le 1}\mathcal{B}\mathrm{r}(K) := B_\bullet \mathrm{Br}(K), \qquad (\tau_{\le 1}\mathcal{B}\mathrm{r}(K))_n = \mathrm{Br}(K)^n,$$

with the operators of Definition A.3. The first-truncation map

$$t_1 : \tau_{\le 2}\mathcal{B}\mathrm{r}(K) \longrightarrow \tau_{\le 1}\mathcal{B}\mathrm{r}(K)$$

is defined in degree n by sending a simplex with edge algebras R_{ij} to

$$\big([R_{01}], [R_{12}], \dots, [R_{n-1,n}]\big) \in \mathrm{Br}(K)^n.$$

The triangle bimodules imply $[R_{ij}][R_{jk}] = [R_{ik}]$, so this formula commutes with every face and degeneracy map. On the labelled second truncation we use the unit-normalized coherent scalar-twist action of Section 9. On the first truncation it induces the action on Brauer classes.

Write $\Omega^m Y$ for the m-fold loop object of a pointed Kan complex (Y, y_0), each time based at the constant loop, using Definition A.24. Equivalently,

$$(\Omega^m Y)_q = \{F : (\Delta^1)^m \times \Delta^q \to Y \mid F|_{\partial((\Delta^1)^m)\times\Delta^q} = y_0\}.$$

The component map sends a simplex to the component of any of its vertices, giving a simplicial map to $\pi_m(Y, y_0)_{\mathrm{disc}}$.

For $Y = \mathcal{B}\mathrm{r}(K)$, the initial basepoint is its unique vertex. The first constant loop is the degenerate 1-simplex labelled by the unit algebra K. The next constant loop is represented by the unit K–K-bimodule K. For $\tau_{\le 2}\mathcal{B}\mathrm{r}(K)$ and $\tau_{\le 1}\mathcal{B}\mathrm{r}(K)$, the basepoints are the images of these basepoints under t_2 and t_1, respectively.

For the non-discrete second loop, use the comparison

$$N_\bullet \mathbf{Pic}(K) \simeq \Omega^2 \mathcal{B}\mathrm{r}(K)$$

specified over $B_\bullet\Gamma$ in Construction 10.24. It uses the ordered 2-simplex relative to its boundary and the degree-one collapse from the loop square specified there. These choices send a triangle labelled by L to $[L]$ and a scalar isomorphism m_a to a.

Write the three relevant component maps as

$$c_3 : \Omega^3 \mathcal{B}\mathrm{r}(K) \longrightarrow \mathrm{U}(K)_{\mathrm{disc}},$$
$$c_2 : \Omega^2 \tau_{\le 2}\mathcal{B}\mathrm{r}(K) \longrightarrow \mathrm{Pic}(K)_{\mathrm{disc}},$$
$$c_1 : \Omega^1 \tau_{\le 1}\mathcal{B}\mathrm{r}(K) \longrightarrow \mathrm{Br}(K)_{\mathrm{disc}},$$

using the identifications of Proposition 9.24. Thus c_3 sends a loop represented by the bimodule automorphism $m_a : K \to K$, $x \mapsto ax$, to $a \in \mathrm{U}(K)$; c_2 sends a twice-based loop represented by an invertible K-module L to $[L] \in \mathrm{Pic}(K)$; and c_1 sends a loop represented by an Azumaya algebra R to $[R] \in \mathrm{Br}(K)$. For c_1, the identification is the one induced from the algebra labels through t_1. For the inverse-labelled bars of Appendix A, the categorical identification sends $\langle b \rangle$ to b^{-1}. The map c_1 composes this with inversion, sending the bar edge $\langle [R] \rangle$ to $[R]$.

Lemma 10.26. *With successive constant-loop basepoints, there are natural weak equivalences of coherent G-diagrams of Kan simplicial sets as follows. The first comparison is a zigzag; the other arrows are the component maps of Definition 10.25 and the unique maps to a point. Unit and tensor comparisons are retained, and Proposition 10.31 supplies strict models:*

$$N_\bullet \mathbf{Pic}(K) \simeq \Omega^2 \mathcal{B}\mathrm{r}(K), \qquad \Omega^3 \mathcal{B}\mathrm{r}(K) \xrightarrow[\sim]{c_3} \mathrm{U}(K)_{\mathrm{disc}}. \tag{68}$$

$$\Omega^2 \tau_{\leq 2} \mathcal{B}\mathrm{r}(K) \xrightarrow[\sim]{c_2} \mathrm{Pic}(K)_{\mathrm{disc}}, \qquad \Omega^3 \tau_{\leq 2} \mathcal{B}\mathrm{r}(K) \xrightarrow{\sim} *. \tag{69}$$

and

$$\Omega^1 \tau_{\leq 1} \mathcal{B}\mathrm{r}(K) \xrightarrow[\sim]{c_1} \mathrm{Br}(K)_{\mathrm{disc}}, \qquad \Omega^r \tau_{\leq 1} \mathcal{B}\mathrm{r}(K) \xrightarrow{\sim} * \quad \textit{for all } r \geq 2. \tag{70}$$

On the discrete simplicial sets above, the G-actions are the actions on $\mathrm{Br}(K)$, $\mathrm{Pic}(K)$, and $\mathrm{U}(K)$ induced by scalar twisting through ϕ.

Proof. The projection $\mathcal{P} \to B_\bullet \Gamma$ of Construction 10.24 is a Kan fibration: the total category is a groupoid, and an edge of any prescribed type out of L is obtained by taking its target to be L^η. Its unit section is given by K and the scalar-twist unit maps.

The slice mapping-space axiom makes $\mathcal{L}_2$ a Kan fibration. There are six five-vertex level patterns containing all three levels. The patterns 00012, 00112, and 01112 give unit identities; 00122 and 01122 give the same ρ on both sides; and 01222 gives $\rho_{\ell m} \circ \rho_{k\ell}^{\eta_{\ell m}} = \rho_{km}$. Thus every pentagon commutes. Repeated vertices give the specified unit maps, so the construction commutes with faces and degeneracies.

On the fibre, this map sends L to the 2-simplex with triangle label L. An isomorphism $v : L \to L'$ is the tetrahedral label on $(0^0, 0^1, 0^2, 1^2)$; the other two $(1,2)$-shuffle tetrahedra have unit comparisons. This shuffle has positive sign. Consequently the map induces $[L] \mapsto [L]$ on components and $m_a \mapsto a$ on fundamental groups, with the coefficient identification of Proposition 9.24. Both fibres are 1-types, so it is a fibrewise weak equivalence.

The sphere collapse of Construction 10.24 is a degree-one weak equivalence of simplicial 2-spheres. Precomposition identifies the fibre of $\mathcal{L}_2$, up to weak equivalence, with $\Omega^2 \mathcal{B}\mathrm{r}(K)$ in the stated orientation. Since the comparison was constructed over $B_\bullet \Gamma$, it respects the coherent scalar action and its unit maps. Tensoring the labels gives its tensor comparisons; their coherence is the canonical interchange for tensor products of modules. The backward action is $L \mapsto L^{\eta^{-1}}$ and $a \mapsto \eta(a)$, as in the convention dictionary.

The simplicial set $\Omega^3 \mathcal{B}\mathrm{r}(K)$ is a Kan 0-type because $\mathcal{B}\mathrm{r}(K)$ is a 3-type. Its component represented by the 3-cell $m_a : K \to K$ corresponds to $a \in \mathrm{U}(K)$ by Proposition 9.24. Therefore the explicitly defined map c_3 induces the identity from $\pi_0(\Omega^3 \mathcal{B}\mathrm{r}(K))$ to $\mathrm{U}(K)$; both sides have no positive homotopy groups, so c_3 is a weak equivalence.

Proposition 9.24 identifies the concrete map $\mathcal{B}\mathrm{r}(K) \to \tau_{\leq 2} \mathcal{B}\mathrm{r}(K)$ of Definition 10.25 as a second Postnikov truncation: it induces isomorphisms

$$\pi_i(\mathcal{B}\mathrm{r}(K)) \to \pi_i(\tau_{\leq 2} \mathcal{B}\mathrm{r}(K))$$

for $i = 1, 2$, and $\pi_3(\tau_{\leq 2}\mathcal{B}\mathrm{r}(K)) = \{1\}$. Hence $\Omega^2\tau_{\leq 2}\mathcal{B}\mathrm{r}(K)$ is a Kan 0-type with component set $\mathrm{Pic}(K)$. The map c_2 is exactly its degreewise component map, so it induces the identity on components and is a weak equivalence. One more loop is weakly contractible, making its unique map to a point a weak equivalence.

The explicit first stage $\tau_{\leq 1}\mathcal{B}\mathrm{r}(K) = B_\bullet \mathrm{Br}(K)$ is the nerve of the group $\mathrm{Br}(K)$ and hence a Kan model of $K(\mathrm{Br}(K), 1)$. Thus $\Omega^1\tau_{\leq 1}\mathcal{B}\mathrm{r}(K)$ is a Kan 0-type with component set $\mathrm{Br}(K)$, and c_1 is its component map. It is therefore a weak equivalence. Every further loop is weakly contractible, so its unique map to a point is a weak equivalence. Finally, the unit-normalized scalar-twist functors commute with the delooping comparisons and the component maps through their specified comparisons. These are equivariant weak equivalences of coherent G-diagrams. After the rectification of Proposition 10.31, they induce the stated comparisons on homotopy fixed points. □

Definition 10.27. Let $h_1\mathbf{Az}_K$ be the groupoid with the same objects as $\mathbf{Az}_K$ and with arrows the isomorphism classes of invertible bimodules. Use the normal geometric nerve for $\mathbf{Az}_K$ and the ordinary groupoid nerve for $h_1\mathbf{Az}_K$.

Construction 10.28. We construct the comparison over $B_\bullet\Gamma$, where $\Gamma = \mathrm{Aut}_k(K)$. Let $\mathcal{E} \to B_\bullet\Gamma$ be the normal nerve of the scalar-twist Grothendieck bicategory. Over $[\eta_1|\cdots|\eta_n]$ its data are algebras R_i, invertible bimodules and isomorphisms

$$P_{ij} \in {}_{R_i^{\eta_{ij}}}\mathbf{Mod}_{R_j},$$

$$\mu_{ijk} : P_{ij}^{\eta_{jk}} \otimes_{R_j^{\eta_{jk}}} P_{jk} \xrightarrow{\sim} P_{ik},$$

where $\eta_{ij} = \eta_{i+1} \circ \cdots \circ \eta_j$. The tetrahedral equation is

$$\mu_{ik\ell} \circ (\mu_{ijk}^{\eta_{k\ell}} \otimes 1) = \mu_{ij\ell} \circ (1 \otimes \mu_{jk\ell}),$$

as maps $\left(P_{ij}^{\eta_{j\ell}} \otimes_{R_j^{\eta_{j\ell}}} P_{jk}^{\eta_{k\ell}}\right) \otimes_{R_k^{\eta_{k\ell}}} P_{k\ell} \longrightarrow P_{i\ell}$, with the canonical twist identifications and associators understood. Repeated indices insert identity bimodules and unit comparisons. Let $\overline{\mathcal{E}}$ be the ordinary nerve obtained by retaining only the classes $[P_{ij}]$. Their fibres are $N_\bullet\mathbf{Az}_K$ and $N_\bullet h_1\mathbf{Az}_K$.

For $X = \mathcal{B}\mathrm{r}_\Gamma(K)$ or $\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K)$, let $\mathcal{L}(X) \to B_\bullet\Gamma$ have, over $\sigma : \Delta^n \to B_\bullet\Gamma$, the maps

$$F : \Delta^n \times \Delta^1 \longrightarrow X, \qquad p_X \circ F = \sigma \circ \mathrm{pr}_{\Delta^n}, \qquad F|_{\Delta^n \times \{0,1\}} = s_0 \circ \sigma \circ \mathrm{pr}_{\Delta^n}.$$

The equalities compare maps $\Delta^n \times \Delta^1 \to B_\bullet\Gamma$ and $\Delta^n \times \{0,1\} \to X$, respectively. Here s_0 is the unit section for $G = \Gamma$ and $\phi = \mathrm{id}$, and p_X denotes the appropriate projection. Define $C : \mathcal{E} \to \mathcal{L}(\mathcal{B}\mathrm{r}_\Gamma(K))$ on $\Delta^n \times \Delta^1$ using the coordinates $i^a = (i, a)$. Same-level simplices carry the unit-section data. The mixed edge, triangle, and tetrahedron labels are

vertices	label
$i^0 j^1$	R_j
$i^0 j^0 k^1$	R_k
$i^0 j^1 k^1$	P_{jk}
$i^0 j^0 k^0 \ell^1$	canonical unit comparison
$i^0 j^0 k^1 \ell^1$	$1_{P_{k\ell}}$
$i^0 j^1 k^1 \ell^1$	$\mu_{jk\ell}$.

The regular bimodule R_k in the second row has left action through $K^{\eta_{jk}} \otimes_K R_k \cong R_k$. All rows use the unit identifications of Definition 10.8; when indices repeat, the labels are the corresponding unit cells.

Passing to bimodule classes in these formulas gives

$$\overline{C} : \overline{\mathcal{E}} \to \mathcal{L}(\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K)).$$

We also specify the product comparison. For two simplices $\mathcal{A} = (R_i, P_{ij}, \mu_{ijk})$ and $\mathcal{B} = (S_i, Q_{ij}, \nu_{ijk})$ over the same bar simplex, write $\mathcal{A} \boxtimes \mathcal{B}$ for their normalized tensor product. It has objects $T_i := R_i \otimes_K S_i$ and bimodules $P_{ij} \otimes_K Q_{ij}$. Its multiplication comparison is the composite

$$(P_{ij} \otimes_K Q_{ij})^{\eta_{jk}} \otimes_{T_j^{\eta_{jk}}} (P_{jk} \otimes_K Q_{jk})$$

$$\xrightarrow{\text{interchange}} (P_{ij}^{\eta_{jk}} \otimes_{R_j^{\eta_{jk}}} P_{jk}) \otimes_K (Q_{ij}^{\eta_{jk}} \otimes_{S_j^{\eta_{jk}}} Q_{jk}) \xrightarrow{\mu_{ijk} \otimes \nu_{ijk}} P_{ik} \otimes_K Q_{ik}.$$

Use the canonical regular-bimodule unit identifications for repeated indices, conjugating the multiplication comparisons by them. The product comparison is a prism

$$D_{\mathcal{A},\mathcal{B}} : \Delta^n \times \Delta^2 \longrightarrow \mathcal{B}\mathrm{r}_\Gamma(K)$$

over the given bar simplex. Its restrictions to the ordered sides 01, 12, and 02 are $C(\mathcal{A})$, $C(\mathcal{B})$, and $C(\mathcal{A} \boxtimes \mathcal{B})$, respectively, and each level vertex carries the unit section. These boundary prescriptions fix all labels using at most two levels. On a triangle with three levels put

$$i^0 j^1 k^2 : \quad P_{jk} \otimes_K S_k \in {}_{R_j^{\eta_{jk}} \otimes_K S_k}\mathbf{Mod}_{T_k}.$$

The remaining tetrahedral comparisons are

vertices	comparison
$i^0 j^0 k^1 \ell^2$	canonical unit comparison
$i^0 j^1 k^1 \ell^2$	$\mu_{jk\ell} \otimes 1_{S_\ell}$
$i^0 j^1 k^2 \ell^2$	$\mu_{jk\ell} \otimes 1_{Q_{k\ell}}$.

In the last two rows take $V = S_\ell$ and $V = Q_{k\ell}$, respectively. After the canonical interchange and unit identifications, the displayed maps have source and target

$$\mu_{jk\ell} \otimes 1_V : (P_{jk}^{\eta_{k\ell}} \otimes_{R_k^{\eta_{k\ell}}} P_{k\ell}) \otimes_K V \longrightarrow P_{j\ell} \otimes_K V,$$

All unit identifications are those of Construction 9.12. Passing to bimodule classes gives the product comparison for $\overline{C}$.

Lemma 10.29. *For the maps of Construction 10.28, there are natural weak equivalences*

$$N_\bullet \mathbf{Az}_K \simeq \Omega \mathcal{B}\mathrm{r}(K), \qquad N_\bullet h_1 \mathbf{Az}_K \simeq \Omega \tau_{\leq 2} \mathcal{B}\mathrm{r}(K).$$

These comparisons are compatible with truncation and with the coherent scalar-twist actions, including their unit and tensor comparison maps.

Proof. Use Construction 10.28. Both projections are Kan fibrations. The first total bicategory is a bigroupoid, so its normal nerve is Kan [CCG10, p. 242, Rem. 4.4]; the second is a groupoid nerve. A relative horn of dimension one is filled by scalar twisting the given algebra and taking its regular bimodule. In higher dimensions, a horn filler in the total nerve has the prescribed image because the group nerve has unique horn fillers. The unit sections have algebra K and the scalar-twist unit bimodules.

The endpoint evaluation is a Kan fibration by the slice mapping-space axiom; pulling it back along (s_0, s_0) shows that $\mathcal{L}(X) \to B_\bullet\Gamma$ is a Kan fibration. Its fibre is the based-loop space of the corresponding fibre of X.

The mixed five-vertex level patterns are 00001, 00011, 00111, and 01111. The first two give unit coherence, the third gives the same μ on both routes, and the last is exactly the displayed tetrahedral equation for μ. This checks every mixed pentagon. Higher simplices have no additional data, and restriction or repetition of vertices preserves the formulas. Thus C is a simplicial map over $B_\bullet\Gamma$. Passing to bimodule classes gives

$$\overline{C} : \overline{\mathcal{E}} \to \mathcal{L}(\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K)),$$

and the square with truncation commutes. On the unit sections both maps are constant loops.

For the product prism, five vertices with base indices i, j, k, ℓ, m and using all three levels have patterns 00012, 00112, 00122, 01112, 01122, or 01222. The first pattern gives unit coherence. The next two give the same $\mu_{k\ell m}$ on both pentagon routes, tensored respectively with 1_{S_m} and $1_{Q_{\ell m}}$, by unit and interchange naturality. The final three give the displayed tetrahedral equation for μ, tensored respectively with 1_{S_m}, $1_{Q_{\ell m}}$, and $\nu_{k\ell m}$; the last case also uses naturality of interchange with $\nu_{k\ell m}$. Patterns using at most two levels are the already checked boundary prisms, including the factorwise tensor-product comparisons. This verifies every pentagon for $D_{\mathcal{A},\mathcal{B}}$. Higher simplices have no further data. The formulas respect restriction and repetition of base indices, with the stated unit identifications, so these prisms form a simplicial family over $B_\bullet\Gamma$.

The formulas also describe the comparison on cells: an algebra R becomes the vertical edge R; a bimodule P becomes the square whose two triangles are the regular bimodule and P; and a bimodule isomorphism becomes the indicated tetrahedral comparison. At K these induce isomorphisms on the groups $\mathrm{Pic}(K)$ and $\mathrm{U}(K)$ computed in Proposition 9.24. More precisely, a scalar m_a occurs on $(0^0, 0^1, 1^1, 2^1)$, the positive $(2,1)$-shuffle, so the coefficient map is $a \mapsto a$. On components the comparison is $[R] \mapsto [R]$ in $\mathrm{Br}(K)$.

Tensoring the displayed labels commutes with C through the canonical interchange maps. The associativity and unit diagrams are the same rebracketings of balanced pure tensors on both sides. Tensoring with R^{op} and then using the evaluation Morita equivalence reduces the homotopy-group calculation on any component to that at K. The full fibres are 2-types and the truncated fibres are 1-types; hence both fibre maps are weak equivalences. The fibrewise criterion makes C and $\overline{C}$ weak equivalences over $B_\bullet\Gamma$.

The Grothendieck projections encode forward scalar twisting, with exactly the unit and tensor comparisons used above. The maps over $B_\bullet\Gamma$ therefore give the required comparisons of coherent actions; the backward action is $(-)^{\eta^{-1}}$. Pullback along $B_\bullet\phi$ gives the corresponding pointed G-comparisons, with truncation and the tensor comparisons already preserved. □

10.4. **Homotopy groups of section spaces.** Using the loop models of Subsection 10.3, we identify the homotopy groups at the unit sections. We first replace the coherent scalar-twist action by a strict action on a Kan model.

Construction 10.30. Write $\mathcal{B}\mathrm{r}_{\Gamma,\leq 3}(K) := \mathcal{B}\mathrm{r}_\Gamma(K)$ for the full third stage, with its projection to $B_\bullet\Gamma$. For $r = 1, 2, 3$ define the simplicial set

$$\widehat{\mathcal{B}\mathrm{r}}_\Gamma^{(r)}(K) := \mathrm{Map}_{B_\bullet\Gamma}(E_\bullet\Gamma, \mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K)).$$

Here an n-simplex is a map

$$E_\bullet\Gamma \times \Delta^n \to \mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K)$$

over $B_\bullet\Gamma$. Right translation $r_\eta(e) = e\eta$ on $E_\bullet\Gamma$ defines a strict left action by

$$\eta \cdot f = f \circ (r_\eta \times \mathrm{id}).$$

The evaluation map over $B_\bullet\Gamma$ is

$$\varepsilon_r : E_\bullet\Gamma \times_\Gamma \widehat{\mathcal{B}\mathrm{r}}^{(r)}_\Gamma(K) \longrightarrow \mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K), \qquad [e, f] \longmapsto f(e).$$

Proposition 10.31. *The maps ε_r of Construction 10.30 are natural weak equivalences over $B_\bullet\Gamma$. These maps preserve the unit sections and commute with truncation. Evaluation at the identity vertex of $E_\bullet\Gamma$ is a weak equivalence*

$$\widehat{\mathcal{B}\mathrm{r}}^{(r)}_\Gamma(K) \longrightarrow \tau_{\leq r}\mathcal{B}\mathrm{r}(K).$$

Thus $\widehat{\mathcal{B}\mathrm{r}}^{(r)}_\Gamma(K)$ is a model of the truncated fibre with a strict Γ-action; its Borel construction models the total space $\mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K)$.

Proof. The three projections are Kan fibrations, by Proposition 9.19 and its first-stage analogue. Since $\Delta^0 \to E_\bullet\Gamma$ is a trivial cofibration, the slice mapping-space axiom makes restriction

$$\mathrm{Map}_{B_\bullet\Gamma}(E_\bullet\Gamma, \mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K)) \longrightarrow \mathrm{Map}_{B_\bullet\Gamma}(\Delta^0, \mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K)) = \tau_{\leq r}\mathcal{B}\mathrm{r}(K)$$

a trivial fibration. In particular the source is Kan.

The action is strict: applying η after θ gives $f(e \cdot (\eta \circ \theta))$. The Borel relation identifies $(e\eta, f)$ with $(e, \eta \cdot f)$, and both have evaluation $f(e\eta)$, so ε_r is well defined and simplicial. It lies over $B_\bullet\Gamma$ because f is a map over $B_\bullet\Gamma$. On the fibre over the vertex it is the restriction map just proved to be a weak equivalence. Both its source and target are fibrations over $B_\bullet\Gamma$, so the fibrewise criterion proves that ε_r is a weak equivalence. A unit section $s : B_\bullet\Gamma \to \mathcal{B}\mathrm{r}_{\Gamma,\leq r}(K)$ gives the fixed point $s \circ (E_\bullet\Gamma \to B_\bullet\Gamma)$, proving the pointed assertion. Postcomposition with truncation proves naturality.

This construction retains the unit corrections in scalar twisting. In the Morita description, forward motion along $[\eta]$ gives the right twist Tw_η, with its unit and tensor comparisons. The backward comparison therefore gives the coherent left action $\mathrm{Tw}_{\eta^{-1}}$, and on $\pi_3(\mathcal{B}\mathrm{r}(K)) \cong \mathrm{U}(K)$ it is $u \mapsto \eta(u)$. □

Lemma 10.32. *At the unit sections, for every $r \geq 1$ there are natural weak equivalences of pointed Kan simplicial sets*

$$\Omega^r\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \simeq (\Omega^r\mathcal{B}\mathrm{r}(K))^{hG},$$

$$\Omega^r\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\leq 2}(K))\big) \simeq \big(\Omega^r\tau_{\leq 2}\mathcal{B}\mathrm{r}(K)\big)^{hG},$$

and

$$\Omega^r\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\leq 1}(K))\big) \simeq \big(\Omega^r\tau_{\leq 1}\mathcal{B}\mathrm{r}(K)\big)^{hG}.$$

The basepoint on each homotopy-fixed-point simplicial set is the constant equivariant map with value the corresponding unit simplex.

Proof. Proposition 10.31 supplies pointed weak equivalences

$$E_\bullet\Gamma \times_\Gamma \widehat{\mathcal{B}\mathrm{r}}^{(r)}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\le r}(K)$$

over $B_\bullet\Gamma$, for $r = 1, 2, 3$. Pullback and derived section spaces preserve these equivalences between fibrations. Applying Definition A.37 therefore gives basepoint-preserving weak equivalences

$$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)) \simeq \mathcal{B}\mathrm{r}(K)^{hG},$$
$$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)) \simeq (\tau_{\le 2}\mathcal{B}\mathrm{r}(K))^{hG},$$
$$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K)) \simeq (\tau_{\le 1}\mathcal{B}\mathrm{r}(K))^{hG}.$$

For a pointed G-Kan complex (X, x_0) with fixed basepoint, a q-simplex of $\Omega_{c_{x_0}}(X^{hG})$ is a map

$$F : \Delta^1 \times \Delta^q \to \mathrm{Map}_G(E_\bullet G, X)$$

satisfying

$$F|_{\{\epsilon\}\times\Delta^q} = c_{x_0} \qquad (\epsilon = 0, 1). \tag{71}$$

Each equality is between maps $\Delta^q \to \mathrm{Map}_G(E_\bullet G, X)$, with right side the constant map at c_{x_0}. The equivariant exponential adjunction (89) identifies it with a map

$$\widehat{F} : E_\bullet G \times \Delta^1 \times \Delta^q \to X$$

satisfying

$$\widehat{F}|_{E_\bullet G\times\{\epsilon\}\times\Delta^q} = x_0 \qquad (\epsilon = 0, 1). \tag{72}$$

Each equality is between maps $E_\bullet G \times \Delta^q \to X$, with right side the constant map at x_0. These are exactly the simplices of $(\Omega_{x_0}X)^{hG}$. Iteration gives

$$\Omega^r(X^{hG}, c_{x_0}) \cong (\Omega^r(X, x_0))^{hG}.$$

Applying this formula to the three basepoint-preserving equivalences above proves the lemma. □

Theorem 10.33. *At the unit section, the low homotopy groups are*

	$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))$	$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$	$\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))$
π_0	$\mathrm{BrCliff}^{\mathcal{A}\mathrm{z}}_\phi(K)$	$\mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K)$	$H^1(G, \mathrm{Br}(K)_\phi)$
π_1	$\mathrm{Cliff}^{\mathcal{A}\mathrm{z}}_\phi(K)$	$\mathrm{Char}^{\mathcal{A}\mathrm{z}}_\phi(K)$	$\mathrm{Br}(K)^\phi$
π_2	$E(\phi, \mathrm{Pic}(K))$	$\mathrm{Pic}(K)^\phi$	$\{1\}$
π_3	$\mathrm{U}(K)^\phi$	$\{1\}$	$\{1\}$.

The first two π_0-entries carry the abelian group structures of Proposition 10.16. They are the coherent Morita equivalence classes of Definition 10.9, identified with the component sets by Proposition 10.10. The first two π_1-entries are the abelian groups of Morita classes of Definition 8.9 for the class $\mathcal{A}\mathrm{z}$ of Azumaya K-algebras (Notation 9.4), and the first π_2-entry is the skew-group Picard term of Definition 8.20. The algebraic descriptions of the three columns, as data modulo transport, are tabulated in Notation 10.23. The extraction maps of Lemma 10.20 give bijections from the first two fundamental groups in the table onto the corresponding sets of Morita classes: $\mathrm{Cliff}^{\mathcal{A}\mathrm{z}}_\phi(K)$ *is naturally identified with the group of Morita classes of G-graded Clifford system extensions of type ϕ over Azumaya K-bases, and* $\mathrm{Char}^{\mathcal{A}\mathrm{z}}_\phi(K)$ *with the group of Morita classes of generalized collective characters of type ϕ*

over such bases. In the first two π_0-entries, the coherent Morita transports of Definition 10.6 have data $(P, N_g, \Theta_{g,h})$ and $(P, [N_g])$, respectively.

Proof. Use the simplicial Brauer three-type $\mathcal{B}\mathrm{r}(K)$ of Definition 9.15. By the direct cell calculation in Proposition 9.24,

$$\pi_1(\mathcal{B}\mathrm{r}(K)) = \mathrm{Br}(K), \qquad \pi_2(\mathcal{B}\mathrm{r}(K)) = \mathrm{Pic}(K), \qquad \pi_3(\mathcal{B}\mathrm{r}(K)) = \mathrm{U}(K).$$

There are no higher homotopy groups. The last identity uses automorphisms of the unit bimodule K, which are precisely multiplication by elements of $\mathrm{U}(K)$. The identification of the first two π_1-entries with the algebraic groups is proved below using the explicit comparison over $B_\bullet\Gamma$ in Lemma 10.29 and the extraction of Lemma 10.20. Lemma 10.32 reduces the calculation to homotopy fixed points of the iterated loop simplicial sets. We now apply the loop models of Lemma 10.26 before taking components. Proposition A.38 says that homotopy fixed points preserve G-equivariant weak equivalences between G-Kan complexes. It also gives, for every G-set A, the simplicial isomorphism

$$(A_{\mathrm{disc}})^{hG} \cong (A^G)_{\mathrm{disc}}. \tag{73}$$

Indeed, a simplicial map

$$E_\bullet G \times \Delta^q \to A_{\mathrm{disc}}$$

is constant, and its value defines an equivariant map precisely when it belongs to A^G.

For the second loop of the full 3-type, the result is a weak equivalence of Kan simplicial sets

$$\Omega^2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \simeq \big(N_\bullet\mathbf{Pic}(K)\big)^{hG}. \tag{74}$$

Taking components in (74) gives

$$\pi_2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \cong \pi_0\big((N_\bullet\mathbf{Pic}(K))^{hG}\big). \tag{75}$$

A vertex of $(N_\bullet\mathbf{Pic}(K))^{hG}$ is an equivariant invertible module in the sense of Definition 7.9, and its paths give the morphisms specified there. The inverse-twist conversion is given in Construction 10.24. Thus the right-hand component set in (75) is the group of isomorphism classes of these equivariant invertible modules, with product induced by $\otimes_K$. An equivariant structure $(\lambda_g)_{g\in G}$ on L is equivalent to a left $K_\phi G$-module structure on L, by the dictionary $v_g(\ell) = \lambda_g(1 \otimes_K \ell \otimes_K 1)$ of Construction 8.21. Definition 8.20 therefore identifies this group with the skew-group Picard term $E(\phi, \mathrm{Pic}(K))$.

For the two truncated stages, Lemma 10.26 and (73) give weak equivalences of simplicial sets

$$\Omega^2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) \simeq (\mathrm{Pic}(K)_{\mathrm{disc}})^{hG} \cong (\mathrm{Pic}(K)^\phi)_{\mathrm{disc}}, \qquad \Omega^2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) \simeq *. \tag{76}$$

After applying π_0, this becomes

$$\pi_2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) = H^0(G, \mathrm{Pic}(K)_\phi) = \mathrm{Pic}(K)^\phi, \qquad \pi_2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) = \{1\}.$$

The third-loop calculations take place first as weak equivalences of simplicial sets:

$$\Omega^3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \simeq (\mathrm{U}(K)_{\mathrm{disc}})^{hG} \cong (\mathrm{U}(K)^\phi)_{\mathrm{disc}},$$
$$\Omega^3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) \simeq *, \qquad \Omega^3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) \simeq *.$$

Taking components gives

$$\pi_3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) = H^0(G, \mathrm{U}(K)_\phi) = \mathrm{U}(K)^\phi,$$
$$\pi_3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) = \{1\},$$
$$\pi_3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) = \{1\}.$$

Finally, the first loop of the first truncation satisfies

$$\Omega^1\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) \simeq (\mathrm{Br}(K)_{\mathrm{disc}})^{hG} \cong (\mathrm{Br}(K)^\phi)_{\mathrm{disc}},$$

and hence

$$\pi_1\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) = H^0(G, \mathrm{Br}(K)_\phi) = \mathrm{Br}(K)^\phi.$$

For the components of the first truncation, Proposition 10.22 gives

$$\pi_0\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))\big) \cong H^1(G, \mathrm{Br}(K)_\phi).$$

The other two π_0-entries are Definition 10.9 and Proposition 10.10.

It remains to identify the two first-loop terms. Use the maps C and $\overline{C}$ over $B_\bullet\Gamma$ constructed in the Construction 10.28. Their sources and targets are Kan fibrations. Pullback along $B_\bullet\phi$ and passage to section spaces preserve these weak equivalences, by Proposition A.34. By the exponential adjunction, sections of their target pullbacks are exactly the based loops in $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))$ and $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$, respectively.

A vertex of $\mathrm{Sec}_\phi(\mathcal{E})$ is an algebra R, bimodules $P_g \in {}_{R^{\phi(g)}}\mathbf{Mod}_R$, and the normalized isomorphisms $\mu_{g,h}$ satisfying the unit and associativity equations. These are precisely Clifford data of type ϕ. For its image under C, the two triangles used in (64) have $D_g = R$, $M_g^0 = R$, and $M_g^1 = P_g$. Extraction therefore recovers P_g, and the sole nonunit shuffle tetrahedron recovers $\mu_{g,h}$. On paths the same formulas recover the compatible Morita comparison.

A path in $\mathrm{Sec}_\phi(\mathcal{E})$ has an identity-central bimodule $Q \in {}_R\mathbf{Mod}_S$ at the vertex. It represents the morphism $R \to S$ of Definition 8.4. The mixed triangle maps and tetrahedral equations give exactly the compatible comparisons of Definition 8.7. Conversely, those comparisons define a path of sections by the same prism formulas. Thus $\pi_0(\mathrm{Sec}_\phi(\mathcal{E}))$ is $\mathrm{Cliff}_\phi^{\mathcal{A}\mathrm{z}}(K)$.

Since the map on section spaces is a weak equivalence, every based loop is homotopic to one in its image. Extraction is invariant under such homotopies by Lemma 10.20(ii). Its composite with C is the identity on algebraic Morita classes, so extraction itself is the asserted bijection on components.

For $\mathrm{Sec}_\phi(\overline{\mathcal{E}})$ the argument retains only the classes $[P_g]$, with $[P_g][P_h] = [P_{gh}]$. Paths give (44), and extraction composed with $\overline{C}$ again recovers the original data. Thus $\pi_0(\mathrm{Sec}_\phi(\overline{\mathcal{E}}))$ is $\mathrm{Char}_\phi^{\mathcal{A}\mathrm{z}}(K)$.

These extraction bijections preserve products. For two sections $\mathcal{A} = (R, P_g, \mu_{g,h})$ and $\mathcal{B} = (S, Q_g, \nu_{g,h})$, the product prisms of Construction 10.28 assemble, by Lemma 10.29, into a map

$$B_\bullet G \times \Delta^2 \longrightarrow \mathcal{B}\mathrm{r}_\Gamma(K)$$

over $B_\bullet\phi$, whose three vertices are s_0. Its ordered sides are $C(\mathcal{A})$, $C(\mathcal{B})$, and $C(\mathcal{A} \boxtimes \mathcal{B})$. Thus the first loop followed by the second represents the tensor-product loop in the fundamental group; this is the defining 2-simplex relation for concatenation. The vertical edge of the third side is $R \otimes_K S$, and its two shuffle triangles are the regular bimodule

and $P_g \otimes_K Q_g$. Extraction therefore gives $P_g \otimes_K Q_g$, with multiplication obtained by the displayed interchange followed by $\mu_{g,h} \otimes \nu_{g,h}$. Here the tensor over K uses the common central K-actions on $P_g \in {}_{R^{\phi(g)}}\mathbf{Mod}_R$ and $Q_g \in {}_{S^{\phi(g)}}\mathbf{Mod}_S$. In terms of their original k-bimodules, for $p \in P_g$, $q \in Q_g$, and $a \in K$, its balancing relation is

$$(p \star_\iota a) \otimes_k q = p \otimes_k \big(\phi(g)(a) \star_\ell q\big),$$

so it is exactly $P_g \otimes_{\phi(g)} Q_g$ of Construction 8.8. The unit comparisons normalize the e-component to $R \otimes_K S$, including when $gh = e$. Passing to classes gives the same product assertion for characters, compatibly with truncation. Every based-loop class has a representative in the image of C or $\overline{C}$, so this proves the two first-loop entries as multiplicative identifications and completes the table. □

Remark 10.34. Theorem 10.33 proves the global converse and independence of choices using the fibrewise loop comparison of Lemma 10.29. Lemma 10.20 reads the algebraic data from a specified simplicial map.

Proposition 10.35. *Let s_0 be the explicit unit section of Definition 10.8. For $i = 1, 2$ there are natural isomorphisms*

$$\pi_i(\mathsf{CliffSec}_\phi(K), x_\phi^{\mathrm{Cliff}}) \cong \pi_i(\Omega_{s_0} \operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)))\,.$$

$$\pi_i(\mathsf{CharSec}_\phi(K), x_\phi^{\mathrm{char}}) \cong \pi_i(\Omega_{s_0} \operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)))\,.$$

There are also natural isomorphisms of component groups

$$\pi_0(\mathsf{CliffSp}_\phi^{\mathcal{A}z}(K)) \cong \pi_0(\Omega_{s_0} \operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)))\,,$$

$$\pi_0(\mathsf{CharSp}_\phi^{\mathcal{A}z}(K)) \cong \pi_0(\Omega_{s_0} \operatorname{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)))\,.$$

These four identifications are compatible with the truncation maps and the connecting maps. These are isomorphisms of the specified homotopy groups and component groups, not weak equivalences between the homotopy quotients and the based-loop spaces: the former have additional loops arising from Morita self-equivalences.

Proof. Use the based-loop model of Notation 10.18. Lemma 10.20 supplies the extraction and its compatibility with Morita equivalences. Theorem 10.33 identifies the components of these based-loop spaces with the algebraic Morita classes of Definition 8.9 for the full Azumaya class. By Theorem 8.27, these are also the components of the homotopy quotients in Construction 8.11, formed using the mapping spaces of Lemma 8.10. This gives the two asserted component identifications.

Product compatibility is the triangular-prism argument in the proof of Theorem 10.33: the prism has the two successive loops on sides 01 and 12, and their algebraic tensor product on side 02. Its mixed triangle and tetrahedron labels implement the interchange, so extraction gives the equal-type product of Construction 8.8, with its multiplication and unit comparisons. Hence these component bijections are group isomorphisms. The constant loop extracts the unit system. Passing to bimodule classes proves the character assertion, and truncation forgets the multiplication isomorphisms on both sides.

For the positive homotopy groups, Proposition 8.25 identifies the homotopy groups of $\mathsf{CliffSec}_\phi(K)$ and $\mathsf{CharSec}_\phi(K)$ in degrees one and two with

$$E(\phi, \mathrm{Pic}(K)), \quad \mathrm{U}(K)^\phi, \qquad \text{and} \qquad \mathrm{Pic}(K)^\phi, \quad \{1\},$$

respectively. Theorem 10.33, after using $\pi_i(\Omega X) = \pi_{i+1}(X)$, gives exactly the same four groups for the two based-loop spaces. In both constructions the maps are induced by forgetting the bimodule isomorphisms, and the connecting maps are the obstruction maps ∂_2 and ∂_3 of Propositions 8.23 and 10.46, by the representative calculations in Lemma 10.53 and Proposition 10.54, using the connecting-map convention. Here ∂_3 is the specialization of $\partial_3^{\mathcal{C}}$ to the full Azumaya class. Hence these identifications are natural and give an isomorphism of the two seven-term sequences.

The section-space fibre sequence of Theorem 10.39, transported by Remark 10.40, has fibre $\mathcal{F}^{(3)}_{\phi,K}$, with

$$\pi_i(\mathcal{F}^{(3)}_{\phi,K}) \cong H^{3-i}(G, \mathrm{U}(K)_\phi) \qquad (0 \le i \le 3).$$

Looping gives a fibre sequence with fibre $\Omega\mathcal{F}^{(3)}_{\phi,K}$. Its long exact sequence is the seven-term sequence for $\mathcal{A}\mathrm{z}$ through $\mathrm{Char}^{\mathcal{A}\mathrm{z}}_\phi(K)$, and the unlooped sequence supplies ∂_3. The original fibration then supplies the displayed continuation to $H^4(G, \mathrm{U}(K)_\phi)$. □

Comparison of algebraic sequences. Consequently the full Azumaya form of Theorem 8.26 identifies with the first seven terms of the sequence in Theorem 10.42, which continues with

$$H^3(G, \mathrm{U}(K)_\phi) \to \mathrm{BrCliff}^{\mathcal{A}\mathrm{z}}_\phi(K) \to \mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K) \to H^4(G, \mathrm{U}(K)_\phi).$$

10.5. **The exact sequence from section spaces.** We use the strict pointed action on the Kan model $\widehat{\mathcal{B}\mathrm{r}}^{(3)}_\Gamma(K)$ constructed in Proposition 10.31. Under the pointed weak equivalence with $\mathcal{B}\mathrm{r}(K)$, its third homotopy group is

$$\pi_3(\widehat{\mathcal{B}\mathrm{r}}^{(3)}_\Gamma(K)) \cong \mathrm{Aut}_{K\text{-bimod}}(K) \cong \mathrm{U}(K).$$

The action of $\eta \in \Gamma$ on this group is $u \mapsto \eta(u)$: it corresponds to the coherent left twist $\mathrm{Tw}_{\eta^{-1}}$, whose unit identification $K^{\eta^{-1}} \to K$ is η, as in Proposition 9.24. Let $E_\bullet\Gamma$ be the standard contractible free right Γ-simplicial set of Definition A.4. Thus $(E_\bullet\Gamma)_n = \Gamma^{n+1}$, faces delete a coordinate, degeneracies repeat a coordinate, and its quotient by the diagonal right Γ-action is the bar simplicial set $B_\bullet\Gamma$. Explicitly, the quotient map sends

$$(\gamma_0, \dots, \gamma_n) \longmapsto [\gamma_0 \circ \gamma_1^{-1} | \gamma_1 \circ \gamma_2^{-1} | \cdots | \gamma_{n-1} \circ \gamma_n^{-1}].$$

Notation 10.36. In the convention of Definition A.37, define the simplicial Borel fibration

$$p : \mathcal{E}_\Gamma(K) := E_\bullet\Gamma \times_\Gamma \widehat{\mathcal{B}\mathrm{r}}^{(3)}_\Gamma(K) \longrightarrow B_\bullet\Gamma$$

and its fibrewise second Postnikov truncation

$$q_{\mathrm{abs}} : \mathcal{E}_\Gamma(K) \longrightarrow \mathcal{E}_{\Gamma,\le 2}(K) := \tau^{B_\bullet\Gamma}_{\le 2}\mathcal{E}_\Gamma(K).$$

Definition 10.37. Let $\phi : G \to \Gamma$ be an action. The local Eilenberg–Mac Lane object

$$K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3) \longrightarrow B_\bullet G$$

of Definition A.29 is formed from the cocycle model of Definition A.28 and the G-module $\mathrm{U}(K)_\phi$. Define

$$\mathcal{F}^{(3)}_{\phi,K} := \mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3)\big).$$

This is the simplicial set of sections of Definition A.32; explicitly, its q-simplices are the simplicial maps

$$\sigma : B_\bullet G \times \Delta^q \longrightarrow K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3) \quad \text{over } B_\bullet G.$$

Construction 10.38. Choose a Kan-fibration representative of the relative Postnikov map q_{abs} of Notation 10.36. For $\phi : G \to \Gamma$, write q_ϕ for its pullback as in Notation 10.1. Let

$$\mathrm{Sec}_{B_\bullet G}(q_\phi) : \mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K)) \longrightarrow \mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K)), \qquad \sigma \longmapsto q_\phi \circ \sigma,$$

be the induced map on simplicial sets of sections in the sense of Definition A.32. Choose a representative of the relative top Postnikov class of q_ϕ as a map over $B_\bullet G$

$$\kappa_\phi : \mathcal{E}_{\Gamma,\leq 2}(K)_\phi \longrightarrow K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4),$$

normalized to zero along the unit section by the unit lift. For a section $s : B_\bullet G \to \mathcal{E}_{\Gamma,\leq 2}(K)_\phi$, define the pullback fibration

$$s^*\big(\mathcal{E}_\Gamma(K)_\phi\big) := B_\bullet G \times_{\mathcal{E}_{\Gamma,\leq 2}(K)_\phi} \mathcal{E}_\Gamma(K)_\phi \longrightarrow B_\bullet G.$$

Its relative top Postnikov class is represented by $\kappa_\phi \circ s$. Composition with a section therefore defines

$$o_4 : \pi_0 \mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K)) \longrightarrow \pi_0 \mathrm{Sec}_{B_\bullet G} K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4), \qquad [s] \longmapsto [\kappa_\phi \circ s].$$

This is the section-lifting obstruction of Proposition A.36.

Theorem 10.39. *The map q_{abs} may be chosen as a Kan fibration over $B_\bullet\Gamma$ whose homotopy pullback along the unit section of $\mathcal{E}_{\Gamma,\leq 2}(K)$ is identified, using the unit lift, with $K_{B_\bullet\Gamma}(\mathrm{U}(K), 3)$. More explicitly, based at those unit sections there is a homotopy fibre sequence over $B_\bullet\Gamma$*

$$K_{B_\bullet\Gamma}(\mathrm{U}(K), 3) \longrightarrow \mathcal{E}_\Gamma(K) \xrightarrow{q_{\mathrm{abs}}} \mathcal{E}_{\Gamma,\leq 2}(K).$$

For every action $\phi : G \to \Gamma$, pulling back along $B_\bullet\phi : B_\bullet G \to B_\bullet\Gamma$ gives a homotopy fibre sequence over $B_\bullet G$, with pullbacks written as in Notation 10.1:

$$K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3) \longrightarrow \mathcal{E}_\Gamma(K)_\phi \xrightarrow{q_\phi} \mathcal{E}_{\Gamma,\leq 2}(K)_\phi.$$

Here we use that pullbacks of Kan fibrations are Kan fibrations and that strict pullback along a fibration computes the homotopy pullback; see [GJ99, Ch. I, §7], [Hov99, Ch. 1], *and* [May99, Ch. 7]. *The homotopy fibre of $\mathrm{Sec}_{B_\bullet G}(q_\phi)$ over s is the simplicial set of sections*

$$\mathrm{hofib}_s \mathrm{Sec}_{B_\bullet G}(q_\phi) \simeq \mathrm{Sec}_{B_\bullet G}\big(s^*(\mathcal{E}_\Gamma(K)_\phi)\big).$$

For the unit section s_0, the distinguished unit lift and Lemma A.30 identify this pullback, up to pointed equivalence over $B_\bullet G$, with $K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3) \to B_\bullet G$. The induced section-space homotopy fibre sequence is

$$\mathcal{F}^{(3)}_{\phi,K} \longrightarrow \mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K)) \longrightarrow \mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\leq 2}(K)).$$

Consequently, we obtain the sequence of groups and pointed sets below, exact through the component set of the truncated section space:

$$\begin{aligned}
\pi_3\big(\mathcal{F}^{(3)}_{\phi,K}\big) \;\cong\; \pi_3(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) &\longrightarrow \overbrace{\pi_3(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\le 2}(K)))}^{=\{1\}} \\
\hookrightarrow \pi_2\big(\mathcal{F}^{(3)}_{\phi,K}\big) \longrightarrow \pi_2(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) &\longrightarrow \pi_2(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\le 2}(K))) \\
\hookrightarrow \pi_1\big(\mathcal{F}^{(3)}_{\phi,K}\big) \longrightarrow \pi_1(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) &\longrightarrow \pi_1(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\le 2}(K))) \\
\hookrightarrow \pi_0\big(\mathcal{F}^{(3)}_{\phi,K}\big) \longrightarrow \pi_0(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) &\longrightarrow \pi_0(\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\le 2}(K))) \xrightarrow{o_4} \\
\hookrightarrow \pi_0\,\mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4)\big).&
\end{aligned} \tag{77}$$

For a vertex $s : B_\bullet G \to \mathcal{E}_{\Gamma,\le 2}(K)_\phi$ of the truncated simplicial set of sections, the lifting problem measured by the final arrow o_4 is

$$\begin{array}{ccc}
 & & \mathcal{E}_\Gamma(K)_\phi \\
 & \overset{\widetilde{s}}{\nearrow} & \big\downarrow{\scriptstyle q_\phi} \\
B_\bullet G & \xrightarrow[s]{} & \mathcal{E}_{\Gamma,\le 2}(K)_\phi
\end{array} \tag{78}$$

in which the two solid arrows are given. A dotted simplicial map $\widetilde{s}$ exists precisely when $o_4([s])$ is the component of the zero section.

Proof. Proposition 9.24 identifies the top Postnikov coefficient system: intrinsic Brauer loops act trivially on $\mathrm{U}(K)$, and the backward action along $[\eta]$ is $a \mapsto \eta(a)$. The comparison of Proposition 10.31 gives the same coefficient system for q_{abs}. Choose a Kan-fibration model. Pullback along $B_\bullet\phi$ preserves this model and gives coefficients $\mathrm{U}(K)_\phi$.

Taking sections preserves this homotopy pullback, by the section-space fibration and invariance results in Subsection A.5. In particular,

$$\mathrm{hofib}_s\,\mathrm{Sec}_{B_\bullet G}(q_\phi) \simeq \mathrm{Sec}_{B_\bullet G}\big(s^*(\mathcal{E}_\Gamma(K)_\phi)\big).$$

This section space can be empty for a general s. At s_0, its unit lift and Lemma A.30 identify it with the section space of the local $K(\mathrm{U}(K)_\phi, 3)$. Hence the displayed homotopy fibre sequence of section complexes has fibre with homotopy groups

$$\pi_i(\mathrm{Sec}_{B_\bullet G}(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3))) \cong H^{3-i}(G, \mathrm{U}(K)_\phi) \qquad \text{for all } 0 \le i \le 3.$$

This is Theorem A.31 with $n = 3$. The simplicial set of sections $\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\le 2}(K))$ has trivial homotopy groups in degrees at least 3, because $\mathcal{E}_{\Gamma,\le 2}(K) \to B_\bullet\Gamma$ is fibrewise a 2-type. Thus the degree-three fragment of the long exact sequence is

$$\{1\} \longrightarrow \pi_3(\mathcal{F}^{(3)}_{\phi,K}) \longrightarrow \pi_3(\mathrm{Sec}_\phi(\mathcal{E}_\Gamma(K))) \longrightarrow \{1\},$$

which gives the displayed isomorphism. The remaining part of the long exact sequence gives all displayed terms through the component sets. A component of $\mathrm{Sec}_\phi(\mathcal{E}_{\Gamma,\le 2}(K))$ lifts precisely when the pulled-back top Postnikov class vanishes in $H^4(G, \mathrm{U}(K)_\phi)$, as recorded by the map o_4 of Construction 10.38. □

Remark 10.40. The Borel fibration $\mathcal{E}_\Gamma(K) = E_\bullet\Gamma \times_\Gamma \widehat{\mathcal{B}\mathrm{r}}^{(3)}_\Gamma(K)$ of Notation 10.36 and the Street nerve $\mathcal{B}\mathrm{r}_\Gamma(K)$ of Construction 9.12 present the same Postnikov tower over $B_\bullet\Gamma$. Indeed, Proposition 10.31 gives compatible weak equivalences

$$\mathcal{E}_\Gamma(K) \simeq \mathcal{B}\mathrm{r}_\Gamma(K), \qquad \mathcal{E}_{\Gamma,\le 2}(K) \simeq \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K) \quad \text{over } B_\bullet\Gamma.$$

Theorem 10.39 gives the resulting sequence of section homotopy groups, including the lifting obstruction in $H^4(G, \mathrm{U}(K)_\phi)$. Theorem 10.33 identifies those groups and components with $E(\phi, \mathrm{Pic}(K))$, $\mathrm{Cliff}^{\mathcal{A}\mathrm{z}}_\phi(K)$, $\mathrm{Char}^{\mathcal{A}\mathrm{z}}_\phi(K)$, $\mathrm{BrCliff}^{\mathcal{A}\mathrm{z}}_\phi(K)$, and $\mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K)$.

Corollary 10.41. *Let $\mathcal{F}^{(3)}_{\phi,K}$ be the local section space of Definition 10.37. At the unit section the tower (65) gives a sequence of groups, in degrees at least one, and pointed sets, exact through $\pi_0(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)))$:*

$$\begin{aligned}
&\{1\} \to \pi_2\big(\mathcal{F}^{(3)}_{\phi,K}\big) \to \pi_2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \to \pi_2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) \\
&\quad\to \pi_1\big(\mathcal{F}^{(3)}_{\phi,K}\big) \to \pi_1\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \to \pi_1\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) \\
&\quad\to \pi_0\big(\mathcal{F}^{(3)}_{\phi,K}\big) \to \pi_0\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \to \pi_0\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) \xrightarrow{o_4} \\
&\quad\to \pi_0\,\mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4)\big),
\end{aligned}$$

together with

$$\pi_3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \cong \pi_3\big(\mathcal{F}^{(3)}_{\phi,K}\big), \qquad \pi_3\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))\big) = \{1\}.$$

Here o_4 is the section-lifting obstruction of Theorem 10.39, for lifts in the sense of Definition A.32(ii)*. The terms are identified with homotopy groups and component sets of the section spaces of the tower, and the maps come from its homotopy long exact sequence, followed by the lifting obstruction o_4. The further truncation q_1 to $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 1}(K))$ contributes the two entries identified in Proposition 10.22.*

Proof. Transport Theorem 10.39 along the equivalences $\mathcal{E}_\Gamma(K) \simeq \mathcal{B}\mathrm{r}_\Gamma(K)$ and $\mathcal{E}_{\Gamma,\le 2}(K) \simeq \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$ over $B_\bullet\Gamma$ of Remark 10.40; section spaces of equivalent fibrations over $B_\bullet G$ are equivalent by Proposition A.34, compatibly with the projections and hence with the long exact sequence and with o_4. □

Theorem 10.42. *There is a sequence of abelian groups and homomorphisms, exact through $\mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K)$:*

$$\begin{aligned}
&\{1\} \to H^1(G, \mathrm{U}(K)_\phi) \to E(\phi, \mathrm{Pic}(K)) \to \mathrm{Pic}(K)^\phi \xrightarrow{\partial_2} H^2(G, \mathrm{U}(K)_\phi) \\
&\quad\to \mathrm{Cliff}^{\mathcal{A}\mathrm{z}}_\phi(K) \to \mathrm{Char}^{\mathcal{A}\mathrm{z}}_\phi(K) \xrightarrow{\partial_3} H^3(G, \mathrm{U}(K)_\phi) \\
&\quad\to \mathrm{BrCliff}^{\mathcal{A}\mathrm{z}}_\phi(K) \to \mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K) \xrightarrow{\partial_4} H^4(G, \mathrm{U}(K)_\phi).
\end{aligned}$$

The connecting maps at $\mathrm{Pic}(K)^\phi$, $\mathrm{Char}^{\mathcal{A}\mathrm{z}}_\phi(K)$, and $\mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K)$ are the obstruction maps ∂_2, ∂_3, and ∂_4 computed below; Proposition 10.54 records their formulas. Thus an invariant invertible module admits a coherent equivariant structure precisely when its ∂_2-value is

trivial, and a generalized collective character has a Clifford-system realization precisely when its ∂_3-value is trivial. Every nonempty fibre of $\mathrm{Cliff}_\phi^{\mathcal{A}z}(K) \to \mathrm{Char}_\phi^{\mathcal{A}z}(K)$ *is a transitive* $H^2(G, \mathrm{U}(K)_\phi)$*-set with stabilizer* $\mathrm{im}(\partial_2)$*. A Brauer–character class lifts to a Brauer–Clifford class precisely when its ∂_4-value is trivial. The action on the fibres of the Brauer forgetful map is described in Theorem 10.51.*

Proof. Substitute the algebraic identifications of Theorem 10.33 into Corollary 10.41. The section-space fibre $\mathcal{F}^{(3)}_{\phi,K}$ contributes $H^{3-i}(G, \mathrm{U}(K)_\phi)$ in degree i, for $0 \le i \le 3$, and the section-lifting obstruction gives the final $H^4(G, \mathrm{U}(K)_\phi)$ term. The cocycle formulas are proved in Subsections 10.6 and 10.7. The seven-term part consists of group homomorphisms by Theorem 8.26 and Proposition 10.35. The two Brauer terms are abelian groups by Proposition 10.16. Multiplicativity of the arrow from $H^3(G, \mathrm{U}(K)_\phi)$ is proved after Notation 10.17; the remaining two arrows are homomorphisms by Proposition 10.15(iii). Thus the pointed-set exactness obtained from the section spaces is exactness of groups. The $H^2(G, \mathrm{U}(K)_\phi)$-action multiplies the Clifford multiplication maps by a normalized 2-cocycle. Exactness identifies its stabilizer with $\mathrm{im}(\partial_2)$. Rigidified lift classes are the components of the homotopy fibre of the looped section map over a fixed target loop. When nonempty, they form a torsor under the component group of the looped fibre, namely $H^2(G, \mathrm{U}(K)_\phi)$. □

Proposition 10.43. *There is a natural group isomorphism*

$$\pi_2\big(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))\big) \cong E(\phi, \mathrm{Pic}(K)).$$

Under this identification, the map $E(\phi, \mathrm{Pic}(K)) \to \mathrm{Pic}(K)^\phi$ in Theorem 10.42 is the forgetful map that discards the $K_\phi G$-action. The connecting map is $[L] \mapsto [a_L]$ of Proposition 8.25.

Proof. The calculation of (75) in the proof of Theorem 10.33 identifies $\pi_2(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K)))$ with the isomorphism classes of coherent equivariant invertible K-modules. The equivalence with skew-group modules proved in Proposition 8.25 respects tensor products and the underlying K-module, which gives the asserted group isomorphism and forgetful map. The formula for the connecting map is Lemma 10.53(i). □

Remark 10.44. The comparison of isomorphism classes with Morita classes is Theorem 8.27 and Proposition 8.29; its identification with the first seven terms of the Brauer sequence is Proposition 10.35.

Remark 10.45. Let $\mathcal{C} := \mathcal{C}_{\mathrm{Az}}(K)$ (Example 8.2(iv)). Then $\mathrm{Cliff}_k^{\mathcal{C}}(\phi)$ and $\mathcal{C}_k^{\mathcal{C}}(\phi)$ are the monoids of [Gin22, §§3–4] represented by Clifford extensions and generalized collective characters with Azumaya base algebra. Using Theorem 4.5 to write extensions as Clifford data, the quotient maps keep the representatives and change only the equivalence relation:

$$\mathrm{Cliff}_k^{\mathcal{C}}(\phi) \twoheadrightarrow \mathrm{Cliff}_\phi^{\mathcal{A}z}(K), \quad [(R, M, \mu)] \longmapsto [(R, M, \mu)],$$
$$\mathcal{C}_k^{\mathcal{C}}(\phi) \twoheadrightarrow \mathrm{Char}_\phi^{\mathcal{A}z}(K), \qquad [(R, \Phi)] \longmapsto [(R, \Phi)].$$

The source relations of isomorphism and equivariance are special cases of Morita equivalence, by Definition 8.7, so the maps are well defined. Every Morita class has a representative of the displayed form, so they are surjective. Products and obstruction maps are computed on the same representatives; hence the maps are monoid homomorphisms compatible with (2) and (3). This is the Azumaya specialization of the comparison in Theorem 8.27.

10.6. **The lower obstruction maps.** The loop comparison of Proposition 10.35 identifies ∂_2 in Theorem 10.42 with the general Picard-equivariance cocycle of Proposition 8.25. We identify ∂_3 with the associativity obstruction class of a character on an Azumaya algebra.

Proposition 10.46. *Under the section-space comparison, the homotopy-fibre boundary agrees with the algebraic obstruction map*

$$\partial_3 : \mathrm{Char}_\phi^{\mathcal{A}z}(K) \longrightarrow H^3(G, \mathrm{U}(K)_\phi), \qquad [(R, \Phi)] \longmapsto T(\Phi),$$

where $T(\Phi)$ is the associativity obstruction class of Theorem 5.13, applied to the generalized collective character $\Phi : G \to \mathrm{Pic}_k(R)$.

Proof. Choose a reduced system of representative bimodules P_g and multiplication isomorphisms

$$\mu_{g,h} : P_g \otimes_R P_h \longrightarrow P_{gh}.$$

Take $P_e = R$ and $\mu_{e,g}$ and $\mu_{g,e}$ to be the canonical unit maps, as in Theorem 6.2. The associativity obstruction cocycle $t_{P,\mu}$ is characterized by

$$\mu_{gh,\ell} \circ (\mu_{g,h} \otimes \mathrm{id}) = t_{P,\mu}(g, h, \ell)\, \mu_{g,h\ell} \circ (\mathrm{id} \otimes \mu_{h,\ell}),$$

as maps $(P_g \otimes_R P_h) \otimes_R P_\ell \to P_{gh\ell}$, where tensor products are over R and the ambient associators are suppressed.

Theorem 6.2 identifies this cocycle with the associativity cochain of Theorem 5.13, so $[t_{P,\mu}] = T(\Phi)$. In particular it is normalized, satisfies $\partial t_{P,\mu} = 1$, and changes under $\mu'_{g,h} = \beta(g,h)\mu_{g,h}$ by

$$t_{P,\mu'} = t_{P,\mu}\, (\partial\beta)^{-1}.$$

The scalar action is $\eta_g = \phi(g)$, as in Corollary 3.7. Changing bimodule representatives preserves the class by Theorem 5.13(ii). A compatible morphism gives a homotopy between the corresponding based loops in $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$, so the class is also invariant under the Morita relation. Lemma 10.53(ii) identifies the value of the connecting map

$$\pi_1(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))) \to \pi_0(\mathcal{F}^{(3)}_{\phi,K}) \cong H^3(G, \mathrm{U}(K)_\phi),$$

on the based loop associated to Φ with $T(\Phi)$. This value is the obstruction to lifting that loop relative to the endpoints through the map of section spaces; hence ∂_3 is that connecting map. □

10.7. **The fourth obstruction and its lifts.** The pentagon cocycle of Construction 10.11 defines the final boundary map ∂_4. We compare it with the section-lifting obstruction and describe the lifts when its class vanishes.

Proposition 10.47. *For a Brauer–character system and its corresponding section s, the cocycle of Construction 10.11 represents $o_4([s])$ under*

$$\pi_0 \mathrm{Sec}_{B_\bullet G} K_{B_\bullet G}(\mathrm{U}(K)_\phi, 4) \cong H^4(G, \mathrm{U}(K)_\phi).$$

Its cohomology class is independent of all choices and of the coherent Morita representative.

Proof. By Lemma 10.12, $[\omega]$ is independent of the choices of associators. Changing the representatives $M_{g,h}$ or passing along a coherent Morita path changes the section by a simplicial homotopy; the obstruction class of a section-lifting problem in the sense of

Definition A.32(ii) is invariant under such homotopy. By Proposition 9.24, the homotopy fibre of

$$\mathcal{B}\mathrm{r}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$$

is $K(\mathrm{U}(K),3)$; pullback along $B_\bullet\phi$ gives the coefficient system $\mathrm{U}(K)_\phi$. Thus the obstruction to lifting a component of $\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))$ lies in $H^4(G,\mathrm{U}(K)_\phi)$ and equals $[\omega]$. Indeed, filter $B_\bullet G$ by skeleta. Over $\mathrm{sk}_3 B_\bullet G$ the section lifts: over a nondegenerate $[g|h|\ell]$ the tetrahedron of the section records (60), and a lift there is exactly a choice of $\alpha_{g,h,\ell}$, with the prescribed unit comparisons on degenerate simplices. By Construction 9.13(iv) this lift extends over a nondegenerate 4-simplex $[g|h|\ell|m]$ if and only if $c(g,h,\ell,m)=1$, and by Lemma 9.23 no further condition arises in dimensions at least 5. Hence $\omega = I_4 c$ is the primary obstruction cocycle for extending from $\mathrm{sk}_3 B_\bullet G$ to $\mathrm{sk}_4 B_\bullet G$, and changing the lift on $\mathrm{sk}_3 B_\bullet G$ changes it by $(\partial u)^{-1}$ as computed above. Identifying the primary obstruction with the pulled-back top k-invariant, with $\pi_3(\mathrm{hofib}_*(q))$ identified with automorphisms of the unit bimodule K as in Proposition 9.24, gives $o_4([s]) = [\omega]$. □

Remark 10.48. The associativity and pentagon cocycles in Propositions 10.46 and 10.47 are local-coefficient analogues of the categorical 3- and 4-cocycles of [ENO26, pp. 23–27, Eq. (4.1) and Defs. 4.2, 4.6]. For the pointed based ring $\mathbb{Z}[G]$, each product of basis elements has the single summand gh, so the based-ring cocycle equation reduces to the group-cocycle equation. In our setting its first coface acts on scalars by $\phi(g)$; for trivial ϕ, the displayed equations are exactly the constant-coefficient pointed case.

Construction 10.49. Use a Kan-fibration representative of

$$q : \mathcal{B}\mathrm{r}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$$

and pull it back along $B_\bullet\phi$. A rigidified lift of a section s is a lift together with a specified path from its image to s, as in Definition A.32. These form $\mathrm{hofib}_s(\mathrm{Sec}_\phi(q))$. After choosing a rigidified lift as basepoint, write

$$\delta_s : \pi_1(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)), s) \longrightarrow \pi_0 \,\mathrm{hofib}_s(\mathrm{Sec}_\phi(q))$$

for the signed homotopy-fibre boundary of the connecting-map convention. In the path model $(e, p : q(e) \to s)$, the fundamental group acts by appending the inverse loop to p. The chosen lift identifies the target with $\pi_0(\mathcal{F}^{(3)}_{\phi,K})$. With this lift as origin, a loop class $[\gamma]$ acts by translation by $\delta_s([\gamma])$. For a based loop in the truncated section space, rigidified loop lifts are defined by the homotopy fibre of the looped section map.

Proposition 10.50. *If the space of rigidified lifts of s is nonempty, its components form a torsor under $\pi_0(\mathcal{F}^{(3)}_{\phi,K})$. After choosing a lift as origin, the action of the fundamental group of the truncated section space is translation through δ_s. Forgetting the rigidification gives its orbit set; the stabilizer for the induced $\pi_0(\mathcal{F}^{(3)}_{\phi,K})$-action is $\mathrm{im}\,\delta_s$, viewed in this group using the chosen origin. Similarly, a nonempty space of rigidified loop lifts has components forming a torsor under $\pi_1(\mathcal{F}^{(3)}_{\phi,K})$.*

Proof. The coefficient system is pulled back from $B_\bullet G$, since intrinsic Brauer loops act trivially on scalars (Proposition 9.24). The local Eilenberg–Mac Lane path-fibration model after Theorem A.22 therefore makes the section map a principal fibration up to homotopy, with acting fibre $\mathcal{F}^{(3)}_{\phi,K}$. A chosen lift identifies its homotopy fibre with this acting fibre. The

homotopy-fibre exact sequence gives the orbit and stabilizer statements. Applying the same argument to the looped map gives the last assertion. □

Theorem 10.51. *The sequence of abelian groups and homomorphisms*

$$H^3(G,\mathrm{U}(K)_\phi) \longrightarrow \mathrm{BrCliff}_\phi^{\mathcal{A}z}(K) \longrightarrow \mathrm{BrChar}_\phi^{\mathcal{A}z}(K) \xrightarrow{\partial_4} H^4(G,\mathrm{U}(K)_\phi)$$

is exact through $\mathrm{BrChar}_\phi^{\mathcal{A}z}(K)$. *More explicitly:*

(i) *a Brauer–character class* $[\mathcal{B}]$ *lifts to a Brauer–Clifford class if and only if* $\partial_4([\mathcal{B}]) = 1$*;*
(ii) *the rescaling in Definition 10.13 defines an action of* $H^3(G,\mathrm{U}(K)_\phi)$ *on Brauer–Clifford classes. This action is multiplication by the image under the first arrow;*
(iii) *two Brauer–Clifford classes have the same Brauer–character class if and only if they differ by this action. Every point of every nonempty fibre has stabilizer* $\operatorname{im}\partial_3$*. Consequently, each nonempty fibre is a torsor under*

$$H^3(G,\mathrm{U}(K)_\phi)/\operatorname{im}\partial_3.$$

In particular, exactness at $H^3(G,\mathrm{U}(K)_\phi)$ *gives*

$$\ker\big(H^3(G,\mathrm{U}(K)_\phi) \longrightarrow \mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)\big) = \operatorname{im}\big(\partial_3 : \mathrm{Char}_\phi^{\mathcal{A}z}(K) \longrightarrow H^3(G,\mathrm{U}(K)_\phi)\big).$$

The displayed four-term sequence therefore continues the seven-term sequence for all Azumaya K*-algebras. Its three arrows are given on representatives in Notation 10.17.*

Proof. The comparison below identifies algebraic lifts and the rescaling action with the lifts and action in Proposition 10.50. Proposition 10.47 gives the lifting criterion, and Lemma 10.53(ii) identifies the unit stabilizer with $\operatorname{im}\partial_3$.

To identify the action with multiplication, let u be a normalized 3-cocycle and let x_u be the class of its unit system in Notation 10.17. For any normalized Brauer–Clifford system $\mathcal{A} = (R, M, \alpha)$, form the external product of this unit system with $\mathcal{A}$. The canonical unit comparisons identify the product edge algebras and triangle bimodules with those of $\mathcal{A}$. Under these identifications, unit coherence identifies $\alpha^0_{g,h,\ell} \otimes \alpha_{g,h,\ell}$ with $\alpha_{g,h,\ell}$, so the product associator is

$$\eta_{gh\ell}^{-1}\big(u(g,h,\ell)\big)\alpha_{g,h,\ell} :$$

$$M_{gh,\ell} \circ (M^{\eta_\ell}_{g,h} \otimes_K \mathrm{id}_{R_\ell}) \xrightarrow{\sim} M_{g,h\ell} \circ (\mathrm{id}_{R_g^{\eta_{h\ell}}} \otimes_K M_{h,\ell}).$$

Normalization is preserved because u is normalized. Thus $[u]\cdot[\mathcal{A}] = x_u[\mathcal{A}]$ in $\mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)$. By Proposition 10.16, cancellation in this group shows that $[u]$ fixes $[\mathcal{A}]$ if and only if $x_u = 1$. The unit stabilizer calculation therefore gives the same stabilizer $\operatorname{im}\partial_3$ at every point. Transitivity on each nonempty fibre then gives the asserted quotient-group torsor. □

Proposition 10.52. *Under the section-space identifications and*

$$\pi_0(\mathcal{F}^{(3)}_{\phi,K}) \cong H^3(G,\mathrm{U}(K)_\phi), \qquad \pi_1(\mathcal{F}^{(3)}_{\phi,K}) \cong H^2(G,\mathrm{U}(K)_\phi),$$

the topological action of Proposition 10.50 is the algebraic rescaling of Definition 10.13. Thus homotopy classes of rigidified section lifts form an $H^3(G,\mathrm{U}(K)_\phi)$*-torsor, and homotopy classes of rigidified loop lifts form an* $H^2(G,\mathrm{U}(K)_\phi)$*-torsor, whenever nonempty. For a chosen lift over* s*, the stabilizer after forgetting the rigidification is the image of* δ_s*, viewed in* $H^3(G,\mathrm{U}(K)_\phi)$ *through that choice. At the unit section,* δ_{s_0} *corresponds to* ∂_3*. Under Proposition 10.35, this* $H^3(G,\mathrm{U}(K)_\phi)$ *term is the final term of the full Azumaya seven-term sequence.*

Proof. Use the functorial fibrant replacement of

$$q : \mathcal{B}\mathrm{r}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$$

from Proposition 9.24, pull it back along $B_\bullet\phi : B_\bullet G \to B_\bullet\Gamma$, and take simplicial sets of sections. Section invariance is Proposition A.34. At the unit section, its distinguished unit lift, Proposition 9.24 and Lemma A.30 identify the homotopy fibre of the resulting section map with

$$\mathrm{Sec}_{B_\bullet G}\big(K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3)\big).$$

By Theorem A.31, this fibre has component set $H^3(G, \mathrm{U}(K)_\phi)$, and the corresponding next obstruction group is $H^4(G, \mathrm{U}(K)_\phi)$. The long exact sequence of the section-space fibration gives exactness at $\mathrm{BrCliff}_\phi^{\mathcal{A}z}(K)$, and the standard componentwise Postnikov lifting obstruction gives the map ∂_4 and the lift criterion.

For rigidified lifts, fix a section s with vanishing obstruction and choose one lift together with a path from its image to s. This choice identifies the homotopy fibre over s with the section space displayed above. Without a chosen lift, there is no preferred origin: its components form a torsor under $H^3(G, \mathrm{U}(K)_\phi)$. If the obstruction does not vanish, the space of lifts is empty. The coefficient system is the same for every s, since intrinsic Brauer loops act trivially on scalars by Proposition 9.24.

For the cocycle action, choose a representative lift with associators $\alpha_{g,h,\ell}$. A normalized 3-cocycle u acts by replacing

$$\alpha_{g,h,\ell} \quad \text{with} \quad \eta_{ghl}^{-1}\big(u(g,h,\ell)\big)\alpha_{g,h,\ell}.$$

This is the rescaling on the underlying bimodule map, as in the convention dictionary. The cocycle equation is exactly the condition that the new associators still satisfy the pentagon. If $u = \partial v$ is a coboundary, the resulting lift is connected to the original one by the homotopy determined by the 2-cochain v. Hence the action factors through $H^3(G, \mathrm{U}(K)_\phi)$. The exact homotopy sequence based at s contains

$$\pi_1(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)), s) \xrightarrow{\delta_s} \pi_0 \,\mathrm{hofib}_s(\mathrm{Sec}_\phi(q)) \longrightarrow \pi_0(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_\Gamma(K))) \longrightarrow \pi_0(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))).$$

By Proposition 9.24, the coefficient system of the pullback of q along $B_\bullet\phi$ is induced by $\mathrm{U}(K)_\phi$. The local Eilenberg–Mac Lane path-fibration model recalled after Theorem A.22 identifies this pullback, up to fibrewise equivalence, with a principal fibration under $K_{B_\bullet G}(\mathrm{U}(K)_\phi, 3)$, pulled back from $B_\bullet G$ to the truncated total space. Consequently, the group $\pi_1(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)), s)$ acts on $\pi_0(\mathrm{hofib}_s(\mathrm{Sec}_\phi(q)))$ by translation through δ_s. With exactness this says that the induced $H^3(G, \mathrm{U}(K)_\phi)$-action on unrigidified lift classes is transitive on the preimage of the component of s, and its stabilizer is $\mathrm{im}(\delta_s)$. At the unit section, Theorem 10.33 identifies the first group with $\mathrm{Char}_\phi^{\mathcal{A}z}(K)$, and Lemma 10.53(ii) identifies δ_{s_0} with ∂_3. □

Lemma 10.53. *At the unit section, the connecting maps of Corollary 10.41 are given on representatives by*

(i)

$$\pi_2(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))) \to \pi_1(\mathcal{F}^{(3)}_{\phi,K}) \cong H^2(G, \mathrm{U}(K)_\phi),$$
$$[L] \mapsto [a_L];$$

(ii)

$$\pi_1(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))) \to \pi_0(\mathcal{F}^{(3)}_{\phi,K}) \cong H^3(G, \mathrm{U}(K)_\phi),$$
$$[(R,\Phi)] \mapsto T(\Phi).$$

Here a_L is the cocycle of Proposition 8.25, and $T(\Phi)$ is the associativity obstruction class of Theorem 5.13.

Proof. With the connecting-map convention, for $n = 1, 2$ the connecting map is represented by the obstruction to lifting a based n-loop through q, relative to its boundary. We use the ordered fundamental class of $(\Delta^n, \partial\Delta^n)$ and the shuffle product with the $B_\bullet G$ coordinate first. Thus the relative obstruction cochain is evaluated on the oriented shuffles of $[g_1|\cdots|g_{4-n}] \times \Delta^n$. The coefficient identifications in Corollary 10.41 are taken with this ordered-simplex convention. In degree three over a point it sends m_a to a, as in Proposition 9.24. For a pentagon, the terminal scalar is the short route divided by the long route; its initial coordinate is obtained by applying $\eta_{g_1\cdots g_{4-n}}$, as in Proposition 10.47.

For (ii), represent the character by the cylinder constructed in Lemma 10.29, retaining only bimodule classes. Choose normalized representatives P_g and maps $\mu_{g,h}$ and use the same cylinder formulas to lift its 3-skeleton. Let $t_{P,\mu}$ be their associativity obstruction cocycle of Construction 5.2; its class is $T(\Phi)$. Over $[g|h|\ell] \times \Delta^1$ the four shuffles have vertices

$$\sigma_r = (0^0, \dots, r^0, r^1, \dots, 3^1), \qquad 0 \le r \le 3,$$

and signs $(-1)^{3-r}$. On σ_0 the long and short pentagon routes are, respectively,

$$P^{\mathrm{long}} = \mu_{gh,\ell} \circ (\mu^{\eta_\ell}_{g,h} \otimes 1),$$
$$P^{\mathrm{short}} = \mu_{g,h\ell} \circ (1 \otimes \mu_{h,\ell}),$$

as maps $\left(P_g^{\eta_{h\ell}} \otimes_{R^{\eta_{h\ell}}} P_h^{\eta_\ell}\right) \otimes_{R^{\eta_\ell}} P_\ell \longrightarrow P_{gh\ell}$, with the canonical twist and associativity identifications understood. On the underlying bimodules over k, their ratio is the one in (29). Hence its terminal scalar is $\eta^{-1}_{gh\ell}(t_{P,\mu}(g,h,\ell)^{-1})$. For $r = 1$ the two routes contain the same multiplication map; for $r = 2, 3$ they are canonical unit comparisons. Their scalars are therefore 1. Evaluation on the oriented prism gives

$$\eta_{gh\ell}\Big(\big[\eta^{-1}_{gh\ell}(t_{P,\mu}(g,h,\ell)^{-1})\big]^{-1}\Big) = t_{P,\mu}(g,h,\ell).$$

Its cohomology class is therefore $T(\Phi)$, proving (ii) with the stated coefficient convention.

For (i), use the ordered 2-simplex model for $[L]$ from the Construction 10.24. Choose normalized $\lambda_g : L^{\eta_{g^{-1}}} \to L$, using the identification with ${}^gL$ in the convention dictionary, and put

$$\rho_g := (\lambda_g^{\eta_g})^{-1} : L^{\eta_g} \longrightarrow L.$$

The equation defining a_L gives

$$\rho_h \circ \rho_g^{\eta_h} = \eta^{-1}_{gh}\big(a_L(g,h)^{-1}\big)\rho_{gh},$$

as maps $L^{\eta_{gh}} \to L$. Indeed, on underlying abelian groups the left side is $(\lambda_g \circ \lambda_h)^{-1}$, and moving a scalar past λ^{-1}_{gh} applies η^{-1}_{gh}.

The chosen lift on the 3-skeleton has edge algebras K, triangle label L on level pattern 012, and tetrahedral comparison ρ on pattern 0122, with the other labels as in that proof. Of the six $(2,2)$-shuffle 4-simplices, only

$$\tau = (0^0, 0^1, 0^2, 1^2, 2^2)$$

has a potentially nontrivial pentagon. Its long route is $\rho_h \circ \rho_g^{\eta_h}$ and its short route is ρ_{gh}. Thus its terminal scalar is $\eta_{gh}^{-1}(a_L(g,h))$. The shuffle τ has sign $(-1)^{2\cdot 2} = 1$; the other five have scalar 1. Passing to the initial coordinate gives $a_L(g,h)$, proving (i).

All choices are normalized on degenerate simplices. Rescaling the chosen comparisons changes these relative obstruction cochains by coboundaries, so the computed classes are the connecting maps on the stated representatives. □

Proposition 10.54. *In degrees two and three the connecting map of the tower (65), and in degree four its lifting obstruction o_4, agrees with the algebraic obstruction:*

(i) *∂_2 is the connecting map*

$$\pi_2(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))) \to \pi_1(\mathcal{F}^{(3)}_{\phi,K})$$

of Corollary 10.41, and on representatives it is $[L] \mapsto [a_L]$ of Proposition 8.25;

(ii) *∂_3 is the connecting map*

$$\pi_1(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K))) \to \pi_0(\mathcal{F}^{(3)}_{\phi,K}),$$

and on representatives it is $[R,\Phi] \mapsto T(\Phi)$ of Theorem 6.2;

(iii) *∂_4 is the section-lifting obstruction o_4 on $\pi_0(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)))$, and on representatives it is $[\mathcal{B}] \mapsto [\omega]$ of Proposition 10.47.*

Each is multiplicative, for the products of Proposition 8.16 in degrees two and three and of Proposition 10.15 in degree four, and each sends a class to the base point exactly when that class lies in the image of the preceding map of Theorem 10.42.

Proof. For (i) and (ii) the agreement with the connecting maps of the tower is Lemma 10.53; the formulas on representatives are Propositions 8.25 and 10.46. For (iii) the formula and the independence of choices are Proposition 10.47, whose last line identifies $[\omega]$ with the section-lifting obstruction, and Theorem 10.39 produces that obstruction as o_4; Corollary 10.41 places it on $\pi_0(\mathrm{Sec}_\phi(\mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)))$. Multiplicativity in degree two is the identity $a_{L\otimes_K M} = a_L a_M$ of Proposition 8.25, in degree three it is Proposition 8.16, and in degree four it is Proposition 10.15(iii). The vanishing criteria are the exactness assertions of Theorems 8.26, 10.42 and 10.51. □

The map

$$q : \mathcal{B}\mathrm{r}_\Gamma(K) \to \mathcal{B}\mathrm{r}_{\Gamma,\le 2}(K)$$

is the final nontrivial fibrewise Postnikov transition. Its homotopy fibre is weakly equivalent to $K(\mathrm{U}(K),3)$. Thus the homotopy exact sequence ends at the component terms, and ∂_4 is the obstruction to lifting a component of the truncated section space.

Proposition 10.55. *The map*

$$\partial_4 : \mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(K) \longrightarrow H^4(G, \mathrm{U}(K)_\phi)$$

is not surjective in general. In fact, let $k = K = \mathbb{C}$, let $G = C_2 \times C_2$, and let ϕ be the unique action through $\mathrm{Aut}_{\mathbb{C}}(\mathbb{C}) = \{1\}$. Then

$$\mathrm{BrChar}^{\mathcal{A}\mathrm{z}}_\phi(\mathbb{C}) = \{*\}, \qquad \mathrm{im}(\partial_4) = \{1\}, \qquad H^4(G, \mathbb{C}^\times) \cong (\mathbb{Z}/2)^2.$$

Thus the right-hand obstruction group contains classes that are not the pentagon obstruction of any Brauer–character class in this example.

Proof. Since $\Gamma = \{1\}$, the simplicial set of sections in Definition 10.9 is the mapping simplicial set from $B_\bullet G$ to $\tau_{\leq 2}\mathcal{B}\mathrm{r}(\mathbb{C})$. Every invertible $\mathbb{C}$-module is isomorphic to $\mathbb{C}$, and every finite-dimensional central simple $\mathbb{C}$-algebra is a matrix algebra. Hence

$$\mathrm{Pic}(\mathbb{C}) = \{1\}, \qquad \mathrm{Br}(\mathbb{C}) = \{1\}.$$

Proposition 9.24 therefore implies that the connected Kan 2-type $\tau_{\leq 2}\mathcal{B}\mathrm{r}(\mathbb{C})$ has no nontrivial homotopy groups. It is weakly contractible, so its mapping simplicial set from $B_\bullet G$ has one component. The unique component is represented by the unit section, which has a full lift in the sense of Definition A.32(ii); Theorem 10.51(i) therefore gives $\mathrm{im}(\partial_4) = \{1\}$.

For the target, the exponential sequence of trivial G-modules

$$\{1\} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{C} \longrightarrow \mathbb{C}^\times \longrightarrow \{1\}$$

and averaging over the finite group G give

$$H^4(G, \mathbb{C}^\times) \cong H^5(G, \mathbb{Z}).$$

The integral cohomology of BC_2 is $\mathbb{Z}$ in degree 0, $\mathbb{Z}/2$ in each positive even degree, and zero in odd degree. The cohomological Künneth exact sequence for $B(C_2 \times C_2) = BC_2 \times BC_2$ consequently gives

$$H^5(G, \mathbb{Z}) \cong \mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) \oplus \mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) \cong (\mathbb{Z}/2)^2,$$

as required. □

10.8. **Field comparison and realization.** For a field F and a discrete Galois module M, write $H^j(F, M)$ for the continuous cohomology of $\mathrm{Gal}(F^{\mathrm{sep}}/F)$ with coefficients in M. We use $H^j(F, \mathbb{G}_m)$ for $H^j(F, (F^{\mathrm{sep}})^\times)$. For étale cohomology we use the small étale site of $\mathrm{Spec}\, K$, computed on its affine basis $\mathrm{Spec}\, E$ with E an étale K-algebra.

For a central simple algebra A over a field K, write $\deg(A) = \sqrt{\dim_K A}$. The index $\mathrm{ind}(b)$ of $b \in \mathrm{Br}(K)$ is the degree of its division-algebra representative, and its period is its order in $\mathrm{Br}(K)$.

For finite Galois extensions, the fourth obstruction has the following description in terms of Brauer cocycles. Write $g \cdot [A] = [{}^g A]$ for extension of scalars along g. Thus $[A^{\eta_g}] = g^{-1} \cdot [A]$ in the right-twist convention of the convention dictionary.

Definition 10.56. Let $\mathcal{S}(A)$ be the splitting gerbe of a central simple algebra A: over an étale K-algebra E, its objects are pairs (V, θ), where V is a finite locally free E-module and

$$\theta : A \otimes_K E \xrightarrow{\sim} \mathrm{End}_E(V)$$

is an algebra isomorphism. Its arrows are module isomorphisms compatible with the specified algebra isomorphisms θ. For a finite locally free E-module V, write $V^\vee := \mathrm{Hom}_E(V, E)$. An invertible A–B bimodule M induces the band-preserving equivalence

$$\mathcal{S}(B) \longrightarrow \mathcal{S}(A), \qquad W \longmapsto M \otimes_B W.$$

The splitting gerbe $\mathcal{S}(A)$ is locally nonempty for the étale topology, and the automorphisms of every splitting module are the scalar units $\mathrm{U}(E)$. This identifies its band with $\mathbb{G}_m$.

Notation 10.57. For a finite Galois extension K/k with group G, use a G-equivariant complex $A^\bullet$ computing the derived global sections of $\mathbb{G}_m$ on $\mathrm{Spec}\, K$ and the double complex $C^p(G, A^q)$ for the Galois cover [Mil13, Thm. 14.9]. Local descent cochains may be computed on étale hypercovers, refined together with their finitely many G-translates as needed

[The26a, Tag 01H0]. Write these cochains multiplicatively. Here d is the differential of $A^\bullet$, δ is the left group-cochain differential, and $D(u) = (\delta u)(du)^{(-1)^p}$ for $u \in C^p(G, A^q)$. Write

$$d_3^{1,2} : H^1(G, \mathrm{Br}(K)) \longrightarrow H^4(G, K^\times)$$

for the Hochschild–Serre differential. The Brauer identification is the crossed-product identification $\mathrm{Br}(K) \cong H^2(K, \mathbb{G}_m)$.

Bimodules and splitting-gerbe cochains. For central simple K-algebras R, S and an invertible R–S bimodule M, put $R_E = R \otimes_K E$, $S_E = S \otimes_K E$, and $M_E = M \otimes_K E$. The functor

$$F_{M,E} : {}_{S_E}\mathbf{Mod} \longrightarrow {}_{R_E}\mathbf{Mod}, \qquad W \longmapsto M_E \otimes_{S_E} W$$

restricts to $F_M : \mathcal{S}(S) \to \mathcal{S}(R)$. An inverse bimodule $Q \in {}_S\mathbf{Mod}_R$ with the triangle-compatible evaluations of Definition 9.1 gives natural isomorphisms $F_M \circ F_Q \Rightarrow \mathrm{id}_{\mathcal{S}(R)}$ and $F_Q \circ F_M \Rightarrow \mathrm{id}_{\mathcal{S}(S)}$. Conversely, a band-preserving equivalence $F : \mathcal{S}(S) \to \mathcal{S}(R)$ gives the local bimodule

$$M_F = F(W) \otimes_E W^\vee, \qquad (f \star_{S_E} s)(w) = f(s \star_{S_E} w),$$

where W is a local splitting module for S_E and the displayed right action is on $W^\vee = \mathrm{Hom}_E(W, E)$. Under an isomorphism $u : W' \to W$, the comparison is

$$F(u) \otimes (u^{-1})^\vee : F(W') \otimes_E (W')^\vee \longrightarrow F(W) \otimes_E W^\vee.$$

Scalar factors cancel because F preserves the band, so these local bimodules descend. The evaluation isomorphism

$$\begin{aligned} (F(W) \otimes_E W^\vee) \otimes_{S_E} W &\xrightarrow{\sim} F(W), \\ (z \otimes_E f) \otimes_{S_E} w &\longmapsto f(w)z \end{aligned}$$

recovers F; the identity $W \otimes_E W^\vee \cong S_E$ recovers M from F_M. Bimodule isomorphisms give natural isomorphisms in the same direction, with $\lambda \mathrm{id}_M$ giving $\lambda \mathrm{id}_{F_M}$. These comparisons respect composition, tensor product, scalar extension, and their canonical reassociations.

Here are the local cochain conventions for this dictionary. Choose local splitting modules W_i for S and V_i for R, descent arrows $f_{ij}^S : W_j \to W_i$ and $f_{ij}^R : V_j \to V_i$, and isomorphisms $t_i : F(W_i) \to V_i$, with restrictions to overlaps understood. Define b^S, b^R, h by

$$\begin{aligned} f_{ij}^S \circ f_{jk}^S &= b_{ijk}^S f_{ik}^S, \\ f_{ij}^R \circ f_{jk}^R &= b_{ijk}^R f_{ik}^R, \\ t_i \circ F(f_{ij}^S) &= h_{ij} f_{ij}^R \circ t_j, \end{aligned}$$

as maps $W_k \to W_i$, $V_k \to V_i$, and $F(W_j) \to V_i$, respectively. Then $dh = b^S (b^R)^{-1}$; put $c = h^{-1}$. For a natural isomorphism $\tau : F \Rightarrow F'$ and choices $t_i' : F'(W_i) \to V_i$, define a_i by

$$t_i' \circ \tau_{W_i} = a_i t_i,$$

as maps $F(W_i) \to V_i$. Thus $c' c^{-1} = da$, where $(da)_{ij} = a_j a_i^{-1}$.

Construction 10.58. Let K/k be a finite Galois extension of fields with group G, and let $\beta \in Z^1(G, \mathrm{Br}(K))$ be normalized. Put

$$r(g) := g^{-1} \cdot \beta(g).$$

Choose a central division algebra R_g in class $r(g)$, with $R_e = K$, and set $S_{g,h} := R_g^{\eta_h} \otimes_K R_h$. For $g, h \neq e$, take $M_{g,h}$ to be a simple module for $S_{g,h} \otimes_K R_{gh}^{\mathrm{op}}$, regarded as an $S_{g,h}$–R_{gh} bimodule; otherwise use the specified unit bimodules. Write $\mathcal{B}_\beta$ for the data $(R_g, [M_{g,h}])$.

Proposition 10.59. *Let K/k be a finite Galois extension of fields with group G. For every normalized cocycle $\beta \in Z^1(G, \mathrm{Br}(K))$, Construction 10.58 gives normalized Brauer–character data $\mathcal{B}_\beta$ with*

$$[R_g] = g^{-1} \cdot \beta(g).$$

The algebras R_g may be chosen to be division algebras, with $R_e = K$. Since $\mathrm{Pic}(K) = \{1\}$, the intervening row in degree one vanishes in the Hochschild–Serre spectral sequence. Its terms $E_3^{1,2}$ and $E_3^{4,0}$ are therefore $H^1(G, \mathrm{Br}(K))$ and $H^4(G, K^\times)$. With the transgression convention of Notation 10.57,

$$\partial_4([\mathcal{B}_\beta]) = \big(d_3^{1,2}[\beta]\big)^{-1}. \tag{79}$$

Conversely, every Brauer–character system over K yields such a cocycle. In particular,

$$\operatorname{Im} \partial_4 = \operatorname{Im} d_3^{1,2}.$$

Proof. (a) Algebraic data from a cocycle. For r, R_g, and $M_{g,h}$ as in Construction 10.58, the left cocycle identity gives

$$r(gh) = \big(h^{-1} \cdot r(g)\big) r(h).$$

Thus $S_{g,h}$ and R_{gh} have the same Brauer class. The algebra $S_{g,h} \otimes_K R_{gh}^{\mathrm{op}}$ is therefore a matrix algebra, and its simple module $M_{g,h}$ is an invertible $S_{g,h}$–R_{gh} bimodule. The two triple composites differ by a central invertible self-bimodule of $R_{gh\ell}$. Skolem–Noether gives

$$\mathrm{Pic}_K(R_{gh\ell}) \cong \mathrm{Out}_K(R_{gh\ell}) = \{1\};$$

see [GS06, p. 40, Thm. 2.7.2]. Hence the composites are isomorphic. Choose the prescribed unit comparisons on degenerate triples and an isomorphism on every other triple. These choices give the required data.

To compare with left group cochains, set $A_g = {}^g R_g$ and twist $M_{g,h}$ by η_{gh}^{-1}. The resulting triangle bimodule is an $A_g \otimes_K {}^g A_h$–A_{gh} bimodule. The same operation on higher simplices is the conversion I_p specified before Construction 10.11. Scalar twisting preserves the periods, indices, and degrees of all labels.

(b) Splitting-gerbe cochains. Use the dictionary in the bimodule–splitting-gerbe dictionary. The triangle bimodule $\widetilde{M}_{g,h} = M_{g,h}^{\eta_{gh}^{-1}}$ induces

$$F_{\widetilde{M}_{g,h}} : \mathcal{S}(A_{gh}) \longrightarrow \mathcal{S}(A_g \otimes_K {}^g A_h).$$

Let b_g be the local splitting-module cocycle for A_g, and put $c_{g,h} = h_{g,h}^{-1}$ for this functor. The dictionary gives

$$dh_{g,h} = b_{gh}(b_g \, {}^g b_h)^{-1}, \qquad dc_{g,h} = (\delta b)_{g,h}.$$

The scalar-twisted tetrahedral isomorphism induces a natural isomorphism from the left triple composite to the right triple composite. Write $a_{g,h,\ell}$ for its local scalar in the same dictionary. Composition and tensor product of the triangle functors give

$$da_{g,h,\ell} = \frac{{}^g c_{h,\ell} c_{g,h\ell}}{c_{gh,\ell} c_{g,h}} = (\delta c)_{g,h,\ell}.$$

Consequently

$$b \in C^1(G, A^2), \qquad c \in C^2(G, A^1), \qquad a \in C^3(G, A^0),$$
$$db = 1, \qquad dc = \delta b, \qquad da = \delta c.$$

The initial-fibre short-over-long scalar is

$$\omega(g,h,\ell,m) = \frac{a_{gh,\ell,m}a_{g,h,\ell m}}{{}^g a_{h,\ell,m}a_{g,h\ell,m}a_{g,h,\ell}} = (\delta a)^{-1}(g,h,\ell,m).$$

(c) Transgression and the pentagon sign. The three cancellation equations give

$$\begin{aligned} D(bc^{-1}a^{-1}) &= (\delta b)(dc)^{-1}(\delta c)^{-1}(da)(\delta a)^{-1} \\ &= (\delta a)^{-1}. \end{aligned}$$

This global unit-valued cocycle is the initial-fibre coordinate of the short-over-long pentagon scalar, by Proposition 10.47. Changes of tetrahedral maps give coboundaries; triangle choices differ by one-dimensional K-spaces and do not change the class.

It remains to identify the sign of b under the crossed-product convention. For $A = (E/F, c)$, write $u_\sigma u_\tau = c(\sigma,\tau)u_{\sigma\tau}$. On the right E-vector space A, the semilinear maps

$$T_\sigma : A \to A,$$
$$T_\sigma(v) = vu_\sigma^{-1},$$

satisfy

$$(T_\sigma \circ T_\tau)(v) = T_{\sigma\tau}(v)c(\sigma,\tau)^{-1}.$$

Consequently the splitting-module class is the inverse of the crossed-product Brauer class. Thus b represents β^{-1}, and the staircase represents $d_3^{1,2}[\beta^{-1}] = (d_3^{1,2}[\beta])^{-1}$. This gives (79).

(d) Converse and degrees. Conversely, the triangle classes of a Brauer–character system give $r(gh) = (h^{-1}\cdot r(g))r(h)$, so $\beta(g) = g \cdot r(g)$ is a left cocycle. The same comparison applies. Since $\operatorname{im} d_3^{1,2}$ is a subgroup, (79) gives $\operatorname{im}\partial_4 = \operatorname{im} d_3^{1,2}$. Opposite algebras invert β and preserve all label degrees, so a class and its inverse are realized with the same degrees. □

For a fixed cocycle, division representatives minimize every label degree:

$$\deg(R_g) = \operatorname{ind}\big(g^{-1}\cdot\beta(g)\big) = \operatorname{ind}(\beta(g)).$$

Over a field, the required central Morita bimodules are unique up to isomorphism, and the triple composites are isomorphic, as in part (a). Thus no further obstruction is needed to construct the Brauer–character data. The remaining obstruction is the signed transgression above. Opposite algebras invert the obstruction class without changing degrees.

10.9. **The trivial group and the quaternion algebra.** The quaternion calculation below gives an explicit associator whose pentagon defect can be removed by rescaling.

Example 10.60. If $G = \{1\}$, the section space is the simplicial Brauer 3-type $\mathcal{B}\mathrm{r}(K)$ of Definition 9.15. Its homotopy groups are

$$\pi_1(\mathcal{B}\mathrm{r}(K)) \cong \mathrm{Br}(K), \qquad \pi_2(\mathcal{B}\mathrm{r}(K)) \cong \mathrm{Pic}(K), \qquad \pi_3(\mathcal{B}\mathrm{r}(K)) \cong \mathrm{U}(K).$$

For an algebraically closed field K, this reduces further to $\mathcal{B}\mathrm{r}(K) \simeq K(K^\times, 3)$, since $\mathrm{Pic}(K) = \{1\}$ and $\mathrm{Br}(K) = \{1\}$.

Example 10.61. Let $K = \mathbb{R}$, let $G = C_2 = \{1, s\}$, and take the trivial action. Then

$$\mathrm{Pic}(\mathbb{R}) = \{1\}, \qquad \mathrm{Br}(\mathbb{R}) \cong \mathbb{Z}/2,$$

with nontrivial class represented by the quaternion algebra $\mathbb{H}$. We define a system of Brauer–character data in the sense of Definition 10.2 by

$$R_1 = \mathbb{R}, \qquad R_s = \mathbb{H},$$

unit bimodules whenever one entry is 1, and the following Morita equivalence for the nonunit pair:

$$M_{s,s} \in {}_{\mathbb{H}\otimes_{\mathbb{R}}\mathbb{H}}\mathbf{Mod}_{\mathbb{R}}.$$

The bimodule $M_{s,s}$ and its pentagon. Put $A = \mathbb{H}$. Regard A as a left $(A \otimes_{\mathbb{R}} A)$-module and a right $\mathbb{R}$-module by

$$(a \otimes_{\mathbb{R}} b) \star_{A\otimes_{\mathbb{R}}A} x := axb.$$

This implements the Morita equivalence $M_{s,s}$, since $A \otimes_{\mathbb{R}} A \cong \mathrm{End}_{\mathbb{R}}(A) \cong M_4(\mathbb{R})$. After this choice, both composites associated to (s, s, s) can be identified with explicit real vector spaces, and choosing the associator $\alpha_{s,s,s}$ amounts to choosing an intertwiner between their $A^{\otimes_{\mathbb{R}}3}$-actions. Let the source and target be $L = A \otimes_{\mathbb{R}} A$ and $T = A \otimes_{\mathbb{R}} A$, respectively. In the action formulas below, superscripts distinguish the bimodules L and T.

$$\begin{aligned}
(a \otimes_{\mathbb{R}} b \otimes_{\mathbb{R}} c) \star^{L}_{A^{\otimes_{\mathbb{R}}3}} (x \otimes_{\mathbb{R}} y) &= axb \otimes_{\mathbb{R}} cy,\\
(x \otimes_{\mathbb{R}} y) \star^{L}_{A} d &= x \otimes_{\mathbb{R}} yd,\\
(a \otimes_{\mathbb{R}} b \otimes_{\mathbb{R}} c) \star^{T}_{A^{\otimes_{\mathbb{R}}3}} (x \otimes_{\mathbb{R}} y) &= ax \otimes_{\mathbb{R}} by\bar{c},\\
(x \otimes_{\mathbb{R}} y) \star^{T}_{A} d &= xd \otimes_{\mathbb{R}} y.
\end{aligned}$$

Let

$$e_0 = 1, \qquad e_1 = i, \qquad e_2 = j, \qquad e_3 = k.$$

One convenient choice of associator is

$$\alpha_{s,s,s} : L \longrightarrow T, \qquad \alpha_{s,s,s}(x \otimes_{\mathbb{R}} y) := \sum_{r=0}^{3} xe_r \otimes_{\mathbb{R}} e_r\bar{y}.$$

In the basis $e_a \otimes_{\mathbb{R}} e_b$, the determinant of $\alpha_{s,s,s}$ is 2^{16}, so $\alpha_{s,s,s}$ is an isomorphism. The identities

$$\sum_{r=0}^{3} e_rd \otimes_{\mathbb{R}} e_r = \sum_{r=0}^{3} e_r \otimes_{\mathbb{R}} e_r\bar{d}, \qquad \sum_{r=0}^{3} \bar{b}e_r \otimes_{\mathbb{R}} e_r = \sum_{r=0}^{3} e_r \otimes_{\mathbb{R}} be_r$$

checked on the basis $1, i, j, k$ show that $\alpha_{s,s,s}$ is the required $A^{\otimes_{\mathbb{R}}3}$-equivariant associator. Thus the equality (60) holds for the only nonunit triple (s, s, s).

For the single nondegenerate quadruple (s, s, s, s), put $D = A \otimes_{\mathbb{R}} A$. Let X and U be two copies of $A \otimes_{\mathbb{R}} A$, with the usual right $\mathbb{R}$-action and the following left actions:

$$\begin{aligned}
(a \otimes_{\mathbb{R}} b \otimes_{\mathbb{R}} c \otimes_{\mathbb{R}} d) \star^{X}_{A^{\otimes_{\mathbb{R}}4}} (x \otimes_{\mathbb{R}} y) &= axb \otimes_{\mathbb{R}} cy\bar{d},\\
(a \otimes_{\mathbb{R}} b \otimes_{\mathbb{R}} c \otimes_{\mathbb{R}} d) \star^{U}_{A^{\otimes_{\mathbb{R}}4}} (p \otimes_{\mathbb{R}} q) &= bp\bar{c} \otimes_{\mathbb{R}} aq\bar{d}.
\end{aligned}$$

The balanced unit isomorphisms needed to identify the five vertices are

$$\begin{aligned}
\kappa_L : (L \otimes_{\mathbb{R}} A) \otimes_D M_{s,s} &\xrightarrow{\sim} X,\\
((x \otimes_{\mathbb{R}} u) \otimes_{\mathbb{R}} v) \otimes_D w &\longmapsto x \otimes_{\mathbb{R}} uw\bar{v};
\end{aligned}$$

$$\kappa_T : (T \otimes_{\mathbb{R}} A) \otimes_D M_{s,s} \xrightarrow{\sim} U,$$
$$((p \otimes_{\mathbb{R}} q) \otimes_{\mathbb{R}} v) \otimes_D w \longmapsto q \otimes_{\mathbb{R}} pw\overline{v};$$
$$\lambda_L : (A \otimes_{\mathbb{R}} L) \otimes_D M_{s,s} \xrightarrow{\sim} U,$$
$$(u \otimes_{\mathbb{R}} (p \otimes_{\mathbb{R}} v)) \otimes_D w \longmapsto p \otimes_{\mathbb{R}} uw\overline{v};$$
$$\lambda_T : (A \otimes_{\mathbb{R}} T) \otimes_D M_{s,s} \xrightarrow{\sim} X,$$
$$(u \otimes_{\mathbb{R}} (p \otimes_{\mathbb{R}} q)) \otimes_D w \longmapsto uw\overline{p} \otimes_{\mathbb{R}} q.$$

In the labelled pentagon of Proposition 9.14, these maps identify C_1, C_4, C_5 with X and C_2, C_3 with U. The edges A^{23}, B^{15}, B^{54}, whose triples contain $ss = 1$, become identity maps. Writing $\alpha = \alpha_{s,s,s}$, the other two edges are $A^{12} = F_{123} : X \to U$ and $A^{34} = F_{234} : U \to X$, given by

$$\begin{aligned} F_{123}(x \otimes_{\mathbb{R}} y) &= \kappa_T\big((\alpha(x \otimes_{\mathbb{R}} 1) \otimes_{\mathbb{R}} 1) \otimes_D y\big) \\ &= \sum_{r=0}^{3} e_r \otimes_{\mathbb{R}} xe_ry, \\ F_{234}(p \otimes_{\mathbb{R}} q) &= \lambda_T\big((1 \otimes_{\mathbb{R}} \alpha(p \otimes_{\mathbb{R}} 1)) \otimes_D q\big) \\ &= \sum_{t=0}^{3} q\,\overline{pe_t} \otimes_{\mathbb{R}} e_t. \end{aligned}$$

Their composite is

$$(F_{234} \circ F_{123})(x \otimes_{\mathbb{R}} y) = \sum_{r,t=0}^{3} xe_ry\,\overline{e_re_t} \otimes_{\mathbb{R}} e_t.$$

A direct check on $y = 1, i, j, k$ gives

$$\sum_{r=0}^{3} e_ry\,\overline{e_re_t} = \begin{cases} 4, & e_t = y, \\ 0, & e_t \neq y, \end{cases}$$

Substituting these values into the double sum gives $4x \otimes_{\mathbb{R}} y$ for $y \in \{1, i, j, k\}$. By $\mathbb{R}$-linearity, the same identity holds for every $y \in A$. Hence

$$F_{234} \circ F_{123} = 4\ \mathrm{id}_X,$$

as maps $X \to X$. The short route is the identity, so the short-over-long pentagon scalar for this associator is $1/4$. The invariant obstruction class of Proposition 10.47 is

$$[\omega(s, s, s, s)] \in \mathbb{R}^\times/(\mathbb{R}^\times)^2 \cong \{\pm 1\},$$

and here it is trivial because $1/4$ is a square. Equivalently, replacing $\alpha_{s,s,s}$ by $\frac{1}{2}\alpha_{s,s,s}$ makes the pentagon commute. Thus the image of this Brauer–character system under

$$\partial_4 : \mathrm{BrChar}_{\phi}^{\mathcal{A}z}(\mathbb{R}) \longrightarrow H^4(C_2, \mathbb{R}^\times)$$

is trivial.

10.10. **Fourth obstructions over nonarchimedean local fields.** We apply Proposition 10.59 to finite Galois extensions of nonarchimedean local fields. For a nonarchimedean local field K, the Hasse invariant takes values in the additively written group $\mathbb{Q}/\mathbb{Z}$, so

$$\mathrm{inv}_K([A][B]) = \mathrm{inv}_K([A]) + \mathrm{inv}_K([B]).$$

Proposition 10.62. *Let K/k be a finite Galois extension of nonarchimedean local fields, with group G and its natural action ϕ. The invariant $\mathrm{inv}_K : \mathrm{Br}(K) \xrightarrow{\sim} \mathbb{Q}/\mathbb{Z}$ identifies the transgression with an isomorphism*

$$d_3^{1,2} : \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\cong} H^4(G, K^\times).$$

Every class on the right is therefore the fourth obstruction of Brauer–character data constructed in Proposition 10.59.

Proof. The invariant is unchanged by field automorphisms, so $\mathrm{Br}(K)$ is a trivial G-module. In the Hochschild–Serre spectral sequence

$$H^p\big(G, H^q(K, \mathbb{G}_m)\big) \Longrightarrow H^{p+q}(k, \mathbb{G}_m),$$

the only nontrivial rows are $q = 0, 2$; for $F = k$ and $F = K$, one has $H^1(F, \mathbb{G}_m) = \{1\}$ and $H^q(F, \mathbb{G}_m) = \{1\}$ for $q \geq 3$ [GS06, Props. 6.3.1 and 6.3.4, and Rem. 6.3.11]. The only differential out of $E_3^{1,2}$, and the only differential into $E_3^{4,0}$, is $d_3^{1,2}$. Neither term survives, since the abutment vanishes in degrees three and four. This proves the isomorphism; the realization assertion follows from Proposition 10.59.

Independently, cup product with the local fundamental class gives $H^2(G, \mathbb{Z}) \cong H^4(G, K^\times)$ [Mil20, Ch. III, Thm. 3.1], and the integral Bockstein identifies $H^2(G, \mathbb{Z})$ with $\mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$. □

10.10.1. *Reciprocity and the transgression sign.*

Construction 10.63. Let q be a prime power and $n \geq 2$. Set

$$k = \mathbb{F}_q((t)), \qquad K = \mathbb{F}_{q^n}((t)), \qquad G = \langle \sigma \rangle,$$

where $\sigma(x) = x^q$ on $\mathbb{F}_{q^n}$ and $\sigma(t) = t$; thus σ is arithmetic Frobenius. For $0 \leq i, j < n$, put

$$c(i,j) := \left\lfloor \frac{i+j}{n} \right\rfloor.$$

We use the arithmetic local-reciprocity normalization

$$\mathrm{Art}_{K/k}^{\mathrm{arith}}([t]) = \sigma, \qquad \mathrm{Art}_{K/k}^{\mathrm{arith}} : k^\times / N_{K/k}(K^\times) \xrightarrow{\cong} G. \tag{80}$$

In the crossed-product convention of Definition 6.12, the corresponding positive factor set is

$$u_t(\sigma^i, \sigma^j) := t^{c(i,j)}. \tag{81}$$

Define the character

$$\chi(\sigma^i) = \frac{i}{n} \pmod{\mathbb{Z}}$$

and its lift $\widetilde{\chi}(\sigma^i) := i/n \in \mathbb{Q}$. Let D_+ be the central division K-algebra with $\mathrm{inv}_K(D_+) = +1/n$, and define

$$\beta_+(\sigma^i) := [D_+]^i \in \mathrm{Br}(K).$$

Define also the normalized 4-cochain

$$\omega_+(\sigma^i, \sigma^j, \sigma^r, \sigma^s) = t^{c(i,j)c(r,s)}. \tag{82}$$

For the transgression calculation, use the unramified tower

$$L = \mathbb{F}_{q^{n^2}}((t)), \qquad H = \mathrm{Gal}(L/k) = C_{n^2}, \qquad N = \mathrm{Gal}(L/K) = n\mathbb{Z}/n^2\mathbb{Z}.$$

Use the homogeneous resolution $C^q = \mathrm{Map}(H^{q+1}, \mathbb{Z})^N$, with differential d deleting successive coordinates with alternating signs. The quotient G acts by simultaneous translation of the arguments, $(i \cdot f)(x_0, \dots, x_q) = f(x_0 + i, \dots, x_q + i)$; this is the action compatible with the homogeneous bar convention $[g|h] = (g + h, h, 0)$. Write δ for the horizontal bar differential. These integral cochains use additive notation, with total differential $D = \delta + (-1)^p d$ on group degree p.

Represent $x \in H$ by $0 \le x < n^2$, and put $\bar{x} = x \bmod n$ and $r(x) = \lfloor x/n \rfloor$. In rational cochains set

$$\lambda(x, y) := \frac{(r(x) - r(y)) \bmod n}{n}, \qquad b_0 := d\lambda, \qquad b_i := ib_0,$$

and

$$\begin{aligned} v_{i,j}(x) &:= \frac{j}{n} c(\bar{x}, i), \\ C_{i,j} &:= -\delta(i\lambda)_{i,j} - dv_{i,j}, \\ A_{i,j,\ell}(x) &:= c(j, \ell) c(\bar{x}, i). \end{aligned}$$

Here b, C, and A have bidegrees $(1, 2)$, $(2, 1)$, and $(3, 0)$, respectively.

Lemma 10.64. *For the data and conventions of Construction 10.63, the basis element u_σ satisfies $u_\sigma a = \sigma(a) u_\sigma$ for all $a \in K$ and $u_\sigma^n = t$. The class of (81) has Hasse invariant $+1/n$.*

With the bar coboundary of Definition A.27, the character χ has positive integral Bockstein represented by

$$(\partial\widetilde{\chi})(\sigma^i, \sigma^j) = \frac{j}{n} - \frac{i + j - nc(i, j)}{n} + \frac{i}{n} = c(i, j). \tag{83}$$

The corresponding local fundamental 2-cocycle is u_t. It represents $[t]$ in $H^2(G, K^\times)$ under (80). We use the positive cyclic periodicity isomorphism, given by cup product with $[c] \in H^2(G, \mathbb{Z})$, to denote the corresponding degree-four class by $[t]$ as well.

Then, with the crossed-product and total-complex conventions of Proposition 10.59,

$$d_3^{1,2}([\beta_+]) = [t] \quad \text{in} \quad H^4(G, K^\times). \tag{84}$$

More explicitly, a normalized 4-cocycle representing the right-hand side is ω_+ of Construction 10.63. Replacing D_+ by the division algebra of invariant $-1/n$ replaces $[t]$ by $[t]^{-1}$. The pentagon obstruction of $\mathcal{B}_{\beta_+}$ is therefore $[t]^{-1}$, by (79).

Proof. For arithmetic Frobenius, the unramified factor set in [Mil20, Ch. IV, Ex. 4.2 and Rem. 4.4(a)] is (81). Its invariant is $+1/n$, which fixes the reciprocity convention (80). Formula (83) is a direct substitution in the inhomogeneous bar differential (94). It fixes the generator $e = [c] \in H^2(G, \mathbb{Z})$ positively.

Use the tower and cochains of Construction 10.63. The equivariant map $\mathbb{Z} \to L^\times$, $1 \mapsto t$, sends the positive generator of $H^2(N, \mathbb{Z})$ to $[D_+]$. Naturality therefore reduces the calculation to the extension $N \to H \to G$ with integral coefficients. The cochain λ is N-invariant and $\lambda(x, y) \equiv (r(x) - r(y))/n \pmod{\mathbb{Z}}$, so its coboundary b_0 is integral. Its restriction to N is the positive carry cocycle, using $[g|h] = (g + h, h, 0)$. Thus $b_i = ib_0$ represents $\beta_+(\sigma^i)$ after applying $1 \mapsto t$.

Here is the integrality and staircase calculation for every n. Put $J = n\lambda$ and $T_i J(x,y) = J(x+i, y+i)$. Since $r(x+i) \equiv r(x) + c(\bar{x}, i) \pmod{n}$, the cochain

$$z_i(x,y) := \frac{T_i J(x,y) - J(x,y) - c(\bar{x}, i) + c(\bar{y}, i)}{n}$$

is integral. Substitution in the definition of C gives

$$C_{i,j} = -j z_i - c(i,j) J,$$

so C is integral as well. The carry identity

$$c(a,i) + c((a+i) \bmod n, j) = c(i,j) + c(a, (i+j) \bmod n)$$

gives

$$dC = -\delta b, \qquad (\delta v)_{i,j,\ell}(x) = \frac{\ell}{n} c(i,j) - c(j,\ell) c(\bar{x}, i), \qquad dA = \delta C.$$

Indeed, $dC = -\delta b$ follows from $d^2 v = 0$, and $\delta C = -d\delta v = dA$, since the term $\ell c(i,j)/n$ is independent of x. Consequently the successive terms of the total-complex lift are $b + C + A$, and

$$D(b + C + A) = \delta A, \qquad (\delta A)_{i,j,\ell,m}(x) = c(i,j) c(\ell, m).$$

The last equality is another substitution of the same carry identity. This last expression is independent of x. Applying $1 \mapsto t$ gives precisely (82), with positive exponent.

Finally, in the comparison from the cyclic periodic resolution to the normalized bar resolution, the degree-four generator maps to

$$\sum_{i,j=0}^{n-1} [\sigma^i \mid \sigma \mid \sigma^j \mid \sigma].$$

The exponent $c \smile c$ evaluates to 1 on this chain. Thus the resulting class under cyclic periodicity is $[t]$. Inverting β_+ negates the integral staircase and inverts the multiplicative cocycle. □

10.10.2. *Cyclic, biquadratic, and nonabelian examples.*

Construction 10.65. Let $n \geq 2$ and let q be a prime power. Set

$$k = \mathbb{F}_q((t)), \qquad K = \mathbb{F}_{q^n}((t)), \qquad G = \mathrm{Gal}(K/k) = \langle \sigma \rangle \cong C_n,$$

where σ applies $x \mapsto x^q$ to the coefficients and fixes t. Let $\phi : G \to \mathrm{Aut}_k(K)$ be this action. Choose the central division K-algebra D_- with

$$\mathrm{inv}_K(D_-) = -\frac{1}{n},$$

and define $\beta_1(\sigma^i) := [D_-]^i$. For $m \in \mathbb{Z}/n$ and $0 \leq i < n$, let $r_{m,i}$ be the representative of mi in $\{0, \ldots, n-1\}$, and put

$$R^{(m)}_{\sigma^i} := D_-^{\otimes_K r_{m,i}}, \qquad D_-^{\otimes_K 0} := K.$$

These tensor powers give explicit algebra representatives. Minimal degrees are obtained by replacing each one by its division representative; Proposition 10.59 supplies the multiplication bimodules for either choice. Their Brauer classes form the homomorphism $\beta_m := \beta_1^m : G \to \mathrm{Br}(K)$. For every pair (σ^i, σ^j), put

$$S^{(m)}_{i,j} := \left(R^{(m)}_{\sigma^i}\right)^{\eta_{\sigma^j}} \otimes_K R^{(m)}_{\sigma^j}.$$

The algebra $S_{i,j}^{(m)} \otimes_K \left(R_{\sigma^{i+j}}^{(m)}\right)^{\mathrm{op}}$ is a matrix K-algebra. For $i \neq 0$ and $j \neq 0$, define $M_{\sigma^i,\sigma^j}^{(m)}$ to be a simple left module for this matrix algebra, viewed as an $S_{i,j}^{(m)}$-$R_{\sigma^{i+j}}^{(m)}$-bimodule; for $i = 0$ or $j = 0$ use the specified regular unit bimodule. Write $\mathcal{B}_m$ for the resulting normalized data.

Proposition 10.66. *The normalized Brauer–character systems $\mathcal{B}_m$ of type ϕ from Construction 10.65 satisfy, for every $m \in \mathbb{Z}/n$,*

$$\partial_4([\mathcal{B}_m]) = [t]^m \quad \text{in} \quad H^4(G, \mathrm{U}(K)_\phi) \cong k^\times / N_{K/k}(K^\times) \cong \mathbb{Z}/n.$$

Here the last isomorphism sends $[t]$ to $1 \in \mathbb{Z}/n$, with the arithmetic-Frobenius and bar conventions of Lemma 10.64. Consequently $\mathcal{B}_m$ lifts to a Brauer–Clifford class if and only if $m = 0$.

Proof. The Hasse invariant and the restriction formula give

$$\mathrm{Br}(k) \cong \mathbb{Q}/\mathbb{Z}, \qquad \mathrm{Br}(K) \cong \mathbb{Q}/\mathbb{Z}, \qquad \mathrm{inv}_K(\mathrm{res}_{K/k}\, x) = n\, \mathrm{inv}_k(x);$$

see [GS06, p. 150, Prop. 6.3.9]. In particular, G acts trivially on $\mathrm{Br}(K)$. The cyclic-periodic resolution gives

$$H^1(G, \mathrm{Br}(K)) \cong \mathrm{Br}(K)[n] \cong \mathbb{Z}/n$$

and

$$\begin{aligned} H^4(G, \mathrm{U}(K)_\phi) \cong H^2(G, \mathrm{U}(K)_\phi) &\cong k^\times / N_{K/k}(K^\times) \\ &\cong \mathrm{Br}(K/k) = \ker\bigl(\mathrm{Br}(k) \xrightarrow{\mathrm{res}_{K/k}} \mathrm{Br}(K)\bigr) \\ &\xrightarrow[\mathrm{inv}_k]{\cong} \ker\bigl(n : \mathbb{Q}/\mathbb{Z} \longrightarrow \mathbb{Q}/\mathbb{Z}\bigr) \cong \mathbb{Z}/n. \end{aligned} \tag{85}$$

Here the periodic-resolution formulas are [Bro82, p. 58], and the relative Brauer-group identification is [GS06, p. 99, Thm. 4.4.7]. Every norm has t-adic valuation divisible by n, while $N_{K/k}(t) = t^n$; hence $[t]$ has exact order n in the quotient.

Proposition 10.62 supplies the transgression isomorphism for this extension.

The period and index of D_- are n [GS06, p. 151, Prop. 6.3.10]. Proposition 10.59 calculates the fourth obstructions of the systems $\mathcal{B}_m$ constructed above:

$$\partial_4([\mathcal{B}_m]) = \bigl(d_3^{1,2}([\beta_m])\bigr)^{-1} = \bigl(d_3^{1,2}([\beta_1])\bigr)^{-m} = [t]^m,$$

where the final equality, including its sign, is Lemma 10.64. The lifting assertion now follows from Theorem 10.51(i). Finally cyclic periodicity and Hilbert's Theorem 90 give

$$H^3(G, \mathrm{U}(K)_\phi) \cong H^1(G, K^\times) = \{1\};$$

see [GS06, p. 26, Ex. 2.3.4]. For $m = 0$, the rigidified-lift torsor of Proposition 10.52 is therefore a singleton. For $n = 2$ and $m = 1$, this is the quadratic local-field example. □

A direct order-three pentagon check. Take $n = 3$ and $m = 1$. We compute the scalar without using (79). Put $L = \mathbb{F}_{q^9}((t))$, let $\widehat{\sigma}$ be arithmetic Frobenius over k, and set $\tau = \widehat{\sigma}^3$. On $V = L^3$ with basis e_0, e_1, e_2, define

$$Ue_0 = e_1, \qquad Ue_1 = e_2, \qquad Ue_2 = te_0, \qquad U^3 = t\,\mathrm{id}_V.$$

The fixed algebra for $X \mapsto U\tau(X)U^{-1}$ on $\mathrm{End}_L(V)$ is D_-: it contains the diagonal copy $a \mapsto \mathrm{diag}(a, \tau(a), \tau^2(a))$ of L and $W = U^{-1}$, with $Wa = \tau(a)W$ and $W^3 = t^{-1}$. Thus it is the cyclic algebra $(L/K, \tau, t^{-1})$ of invariant $-1/3$.

Use left coordinates $A_i = {}^{\sigma^i}R_{\sigma^i}$ as in part (a) of Proposition 10.59, choosing $A_i = D_-^{\otimes_K i}$ for $0 \le i < 3$. Write $V_i = V^{\otimes_L i}$ and $V_0 = L$. For $x, y \in H = \mathbb{Z}/9$, the cochains λ, C, A of Construction 10.63 give splitting-module comparisons

$$F_i(x,y) = (U^{3\lambda(x,y)})^{\otimes i} : V_i(y) \longrightarrow V_i(x),$$

where $V_i(x)$ denotes a copy indexed by x. In these inverse-coefficient homogeneous coordinates an equivariant matrix cochain satisfies $Q_{x+3} = \tau^{-1}(Q_x)$, and the quotient action is $(i \cdot Q)_x = \widehat{\sigma}^i(Q_{x+i})$. For $s = (i+j) \bmod 3$, the local multiplication bimodule is $\mathrm{Hom}_L(V_s, V_i \otimes_L V_j)$. Its descent maps are

$$\mathrm{Hom}_L(V_s(y), V_i(y) \otimes_L V_j(y+i)) \longrightarrow \mathrm{Hom}_L(V_s(x), V_i(x) \otimes_L V_j(x+i)),$$
$$Q \longmapsto t^{C_{i,j}(x,y)}\big(F_i(x,y) \otimes F_j(x+i, y+i)\big) \circ Q \circ F_s(x,y)^{-1}.$$

The identity $dC = -\delta b$ is precisely their descent cocycle condition. They descend to simple invertible $(A_i \otimes_K {}^{\sigma^i}A_j)$–$A_s$ bimodules, with the regular unit bimodules when $i = 0$ or $j = 0$. Conversion by I_p gives the right-coordinate data of $\mathcal{B}_1$.

The left and right triple composites have scalar descent exponents

$$C_L = C_{i,j} + C_{(i+j) \bmod 3, \ell}, \qquad C_R = i \cdot C_{j,\ell} + C_{i,(j+\ell) \bmod 3}.$$

Their difference is $C_R - C_L = \delta C = dA$. Composition identifies the balanced tensor products of the local rectangular Hom spaces with the corresponding Hom spaces. Under these identifications, the associativity map from the left triple composite to the right one is therefore $\alpha_{i,j,\ell}(x) = t^{-A_{i,j,\ell}(x)}\mathrm{id}$. Indeed, its scalar ratio on an overlap is $t^{A_{i,j,\ell}(y) - A_{i,j,\ell}(x)}$, as required for descent. The short-over-long pentagon scalar is now obtained directly:

$$\omega(\sigma^i, \sigma^j, \sigma^\ell, \sigma^m) = t^{(\delta A)_{i,j,\ell,m}} = t^{c(i,j)c(\ell,m)}.$$

For $(i,j,\ell,m) = (2,1,2,1)$ at $x = 0$, the short route has scalar 1 and the long route has scalar t^{-1}, so their ratio is t. On the positive periodic degree-four chain displayed in Lemma 10.64, this cocycle evaluates to t. Its class has order three, since norms have valuation divisible by three. Hence the direct calculation gives $\partial_4([\mathcal{B}_1]) = [t]$, not $[t]^{-1}$. Replacing $U^3 = t$ by $U^3 = t^{-1}$ gives the invariant-$+1/3$ algebra and inverts this scalar, agreeing with the transgression sign.

Construction 10.67. Let q be odd, and set

$$k = \mathbb{F}_q((t)), \qquad K = \mathbb{F}_{q^2}((u)), \qquad u^2 = t.$$

Let σ act on coefficients by $x \mapsto x^q$ and fix u, and let τ fix coefficients and send u to $-u$. Put $G = \langle \sigma, \tau \rangle$ and let ϕ be its action on K. Choose a division K-algebra D of invariant $1/2$. Define $\beta_\sigma, \beta_\tau : G \to \mathrm{Br}(K)$ by

	σ	τ
β_σ	$[D]$	1
β_τ	1	$[D]$.

For either cocycle, take $R_g = K$ when $\beta(g) = 1$ and $R_g = D$ otherwise. Choose triangle bimodules and tetrahedral comparisons by Proposition 10.59; these choices give $\mathcal{B}_\sigma$ and $\mathcal{B}_\tau$.

Proposition 10.68. *For Construction 10.67, K/k is Galois with group*

$$G = \langle \sigma, \tau \mid \sigma^2 = \tau^2 = 1,\ \sigma\tau = \tau\sigma \rangle \cong C_2 \times C_2,$$

where σ applies $x \mapsto x^q$ to the coefficients and fixes u, whereas τ fixes $\mathbb{F}_{q^2}$ and sends u to $-u$. For the normalized systems $\mathcal{B}_\sigma$ and $\mathcal{B}_\tau$ of Brauer–character data of type $\phi : G \to \mathrm{Aut}_k(K)$ constructed in Construction 10.67, the classes

$$\partial_4([\mathcal{B}_\sigma]), \qquad \partial_4([\mathcal{B}_\tau])$$

form a basis of $H^4(G, \mathrm{U}(K)_\phi) \cong (\mathbb{Z}/2)^2$. In particular, neither system lifts to a Brauer–Clifford class.

Proof. The residue degree and ramification index are both two, and the displayed automorphisms give the asserted Galois group. Use the systems of Construction 10.67. Proposition 10.62 and (79) send the two basis characters to a basis of $H^4(G, K^\times)$, the inverse being harmless because this group is 2-torsion. The lifting criterion is Theorem 10.51(i). □

Construction 10.69. Let $q \equiv -1 \pmod 3$ be a prime power, and choose a primitive third root of unity $\zeta \in \mathbb{F}_{q^2}$. Set

$$k = \mathbb{F}_q((t)), \qquad K = \mathbb{F}_{q^2}((u)), \qquad u^3 = t.$$

Let ρ fix coefficients and send u to ζu, and let σ fix u and apply $x \mapsto x^q$ to coefficients. Put $G = \langle \rho, \sigma \rangle$ and let ϕ be its action on K. Choose a division K-algebra D of invariant $1/2$, and let $\beta_{\mathrm{sgn}} : G \to \mathrm{Br}(K)$ send ρ to 1 and σ to $[D]$. Take $R_g = K$ for $g \in \langle \rho \rangle$ and $R_g = D$ otherwise, and extend these edge algebras to $\mathcal{B}_{\mathrm{sgn}}$ by Proposition 10.59.

Proposition 10.70. *For Construction 10.69, K/k is Galois with group*

$$G = \langle \rho, \sigma \mid \rho^3 = \sigma^2 = 1,\ \sigma\rho\sigma^{-1} = \rho^{-1} \rangle \cong S_3,$$

where ρ fixes $\mathbb{F}_{q^2}$ and sends u to ζu, while σ fixes u and applies $x \mapsto x^q$ to the coefficients. The normalized system $\mathcal{B}_{\mathrm{sgn}}$ of Brauer–character data of type $\phi : G \to \mathrm{Aut}_k(K)$ constructed in Construction 10.69 satisfies

$$\partial_4([\mathcal{B}_{\mathrm{sgn}}]) \neq 1 \qquad \text{in} \qquad H^4(G, \mathrm{U}(K)_\phi) \cong \mathbb{Z}/2.$$

Consequently $\mathcal{B}_{\mathrm{sgn}}$ has no Brauer–Clifford lift.

Proof. The degree of K/k is 6: its residue-field extension has degree 2 and its ramification index is 3. The displayed maps fix t, satisfy the displayed relations, and give six distinct k-automorphisms of K. They therefore identify G with S_3.

Use $\mathcal{B}_{\mathrm{sgn}}$ of Construction 10.69. Since $\mathrm{Hom}(S_3, \mathbb{Q}/\mathbb{Z}) \cong \mathbb{Z}/2$, by (79) and Proposition 10.62, its fourth obstruction is the unique nonidentity element of $H^4(G, K^\times)$. Theorem 10.51(i) excludes a lift. □

Remark 10.71. Although this extension is nonabelian, its fourth-cohomology group only detects the abelianization $S_3^{\mathrm{ab}} \cong C_2$. Proposition 10.62 identifies $H^4(G, K^\times)$ with $\mathrm{Hom}(S_3, \mathbb{Q}/\mathbb{Z})$.

Appendix A. Background on simplicial homotopy theory

This appendix fixes the general conventions used in Sections 8 and 9–10: inverse bar labels, local coefficients, derived sections, coherent homotopy quotients, and spectral-sequence indexing. The universe and simplicial-space conventions are those of Subsection 1.4.

Our basic references are Goerss–Jardine [GJ99, Chs. I and VI] for simplicial homotopy theory and local-coefficient Postnikov towers, Hovey [Hov99, Ch. 1] for model structures,

and Bousfield–Kan [BK72, Chs. XI–XII] for homotopy limits. May [May99, Chs. 18 and 22] gives an introductory account in the simple-space setting. For weak higher categories and their nerves we use [GPS95], [CCG10, §2.4 and Prop. 7.1], and [CH14, §2 and Def. 5.1].

A.1. **Simplicial sets, bar constructions, and nerves.**

Definition A.1. The simplex category Δ consists of the finite ordinals $[n]$ and monotone maps. A simplicial set is a functor $\Delta^{\mathrm{op}} \to \mathbf{Set}_{\mathfrak{V}}$, with face maps d_i and degeneracy maps s_i. Write $\Delta^n = \mathrm{Hom}_{\Delta}(-,[n])$; thus maps $\Delta^n \to X$ are elements of X_n. Products are degreewise. See [GJ99, Ch. I, §1].

Definition A.2. For nonnegative integers n, m, q, a q-simplex of $\Delta^n \times \Delta^m$ is a pair of monotone maps

$$a : [q] \longrightarrow [n], \qquad b : [q] \longrightarrow [m].$$

It is nondegenerate exactly when $r \mapsto (a(r), b(r))$ is injective. The top-dimensional nondegenerate simplices have $q = n + m$ and correspond to monotone lattice paths from $(0,0)$ to (n,m) whose successive steps increase exactly one coordinate by one. These are the *standard (n,m)-shuffle simplices*. Equivalently, an (n,m)-shuffle is a word containing n horizontal and m vertical steps; see Kerodon [Lur18], Kerodon tag 00RH.

An edge of $\Delta^n \times \Delta^m$ is *horizontal* if its Δ^m-coordinate is constant, and *vertical* if its Δ^n-coordinate is constant. For an n-simplex $x : \Delta^n \to X$ of a simplicial set X, composing the standard shuffles with

$$x \times \mathrm{id}_{\Delta^m} : \Delta^n \times \Delta^m \longrightarrow X \times \Delta^m$$

gives the shuffle simplices over x. For $m = 1$, write $i^a = (i, a)$. The $n+1$ standard $(n,1)$-shuffle simplices have ordered vertices

$$0^0, 1^0, \ldots, r^0, r^1, (r+1)^1, \ldots, n^1 \quad \text{for all } 0 \le r \le n.$$

A simplicial map on $\Delta^n \times \Delta^m$ is determined by its values on the standard shuffles and their faces. Such assignments define a map precisely when they agree on common faces.

Definition A.3. In the terminology of Notation 2.10, let $\mathbf{B}G$ be the one-object groupoid with object $*$ and $\mathrm{Aut}_{\mathbf{B}G}(*) = G$. The *bar construction*

$$B_\bullet G := N_\bullet(\mathbf{B}G)$$

is its simplicial nerve. It has one vertex and $(B_\bullet G)_n = G^n$. We label the consecutive categorical arrows $g_1^{-1}, \ldots, g_n^{-1}$ by $[g_1|\cdots|g_n]$. In particular, the composite on the triangle $[g|h]$ is $h^{-1} \circ g^{-1} = h^{-1}g^{-1} = (gh)^{-1}$. With these labels,

$$\begin{aligned}
d_0[g_1|\cdots|g_n] &= [g_2|\cdots|g_n],\\
d_i[g_1|\cdots|g_n] &= [g_1|\cdots|g_ig_{i+1}|\cdots|g_n] \quad \text{for all } 0 < i < n,\\
d_n[g_1|\cdots|g_n] &= [g_1|\cdots|g_{n-1}],\\
s_i[g_1|\cdots|g_n] &= [g_1|\cdots|g_i|e|g_{i+1}|\cdots|g_n] \quad \text{for all } 0 \le i \le n.
\end{aligned}$$

These maps satisfy the simplicial identities because $B_\bullet G$ is the nerve of $\mathbf{B}G$: equivalently, $(B_\bullet G)_n = \mathrm{Fun}([n], \mathbf{B}G)$ and every simplicial operator is precomposition. Associativity and the unit law in G give the same direct verification. Since $\mathbf{B}G$ is a groupoid, $B_\bullet G$ is a Kan complex. Its combinatorial homotopy groups are

$$\pi_1(B_\bullet G) = G, \qquad \pi_i(B_\bullet G) = \{1\} \quad \text{for all } i > 1,$$

where $[g]$ corresponds to g^{-1} in the displayed identification of $\pi_1(B_\bullet G)$ with G. In this simplicial sense, $B_\bullet G$ is a model of $K(G,1)$.

Definition A.4. For a group G, the *homogeneous bar simplicial set* $E_\bullet G$ is defined by

$$(E_\bullet G)_n = G^{n+1}.$$

We write an n-simplex as $(g_0,\dots,g_n)$. Its face and degeneracy maps are

$$d_i(g_0,\dots,g_n) = (g_0,\dots,\widehat{g_i},\dots,g_n) \qquad \text{for all } 0\le i\le n,$$
$$s_i(g_0,\dots,g_n) = (g_0,\dots,g_i,g_i,\dots,g_n) \qquad \text{for all } 0\le i\le n.$$

The hat means that the indicated entry is omitted. These formulas satisfy the simplicial identities, so $E_\bullet G$ is a simplicial set. It has a free right G-action in each degree,

$$(g_0,\dots,g_n)\cdot h = (g_0h,\dots,g_nh),$$

and this action commutes with all face and degeneracy maps. Hence the degreewise quotient $E_\bullet G/G$ is again a simplicial set. The quotient map

$$E_\bullet G \longrightarrow B_\bullet G, \qquad (g_0,\dots,g_n)\longmapsto [g_0g_1^{-1}|g_1g_2^{-1}|\cdots|g_{n-1}g_n^{-1}]$$

is a simplicial map and identifies $E_\bullet G/G$ with the inhomogeneous bar simplicial set $B_\bullet G$ of Definition A.3. The simplicial set $E_\bullet G$ is contractible; for example the extra degeneracy $(g_0,\dots,g_n)\mapsto(e,g_0,\dots,g_n)$ gives a simplicial contraction. The same underlying simplicial set also has a free left G-action,

$$h\cdot(g_0,\dots,g_n) = (hg_0,\dots,hg_n),$$

and the quotient map

$$E_\bullet G \longrightarrow B_\bullet G, \qquad (g_0,\dots,g_n)\longmapsto [g_0^{-1}g_1|\cdots|g_{n-1}^{-1}g_n]$$

identifies $G\backslash E_\bullet G$ with $B_\bullet G$. The left-action and right-action quotient conventions are isomorphic by entrywise inversion on $E_\bullet G$.

Definition A.5. A bicategory has objects, hom-categories, horizontal composition functors, and identity 1-cells, with invertible associators and unitors satisfying the pentagon and triangle axioms [CCG10, §2.4]. We orient these constraints as

$$(h\circ g)\circ f \Longrightarrow h\circ(g\circ f), \qquad 1_B\circ f\Longrightarrow f, \qquad f\circ 1_A\Longrightarrow f.$$

A homomorphism of bicategories means a pseudofunctor.

Definition A.6. A tricategory is understood in the sense of Gordon–Power–Street [GPS95], as recalled in [CH14, §2]. It has hom-bicategories, composition homomorphisms, and identity 1-cells; associativity and units are pseudonatural equivalences, with invertible coherence modifications. A strict 3-category has identity comparison cells. For a monoidal bicategory, its one-object delooping is the corresponding tricategory.

Definition A.7. Let $(\mathcal{B},\boxtimes,\mathbf{1})$ be a monoidal bicategory. A *coherent right action* of a group Γ on $\mathcal{B}$ assigns to every $\theta\in\Gamma$ a monoidal bicategory automorphism

$$\mathrm{Tw}_\theta : \mathcal{B}\longrightarrow\mathcal{B}, \qquad X\longmapsto X^\theta,$$

together with invertible comparison transformations

$$(X^\eta)^\theta \simeq X^{\eta\theta}, \qquad X^1\simeq X, \qquad (X\boxtimes Y)^\theta\simeq X^\theta\boxtimes Y^\theta, \qquad \mathbf{1}^\theta\simeq\mathbf{1},$$

natural also in the 1-cells and 2-cells of $\mathcal{B}$. These comparisons are required to satisfy the associativity and unit coherence equations for a group action and for a monoidal bicategory automorphism.

The *semidirect tricategory* $\mathcal{B} \rtimes \Gamma$ is the one-object tricategory whose endomorphism bicategory is $\coprod_{\eta\in\Gamma} \eta\mathcal{B}$. Thus its 1-cells are pairs (η, X) with $X \in \mathrm{Ob}(\mathcal{B})$. A 2-cell $(\eta, X) \Rightarrow (\eta, Y)$ is a 1-cell $f : X \to Y$ of $\mathcal{B}$, there are no 2-cells between distinct Γ-labels, and the 3-cells are the 2-cells of $\mathcal{B}$. Horizontal composition has source and target

$$\odot : (\eta\mathcal{B}) \times (\theta\mathcal{B}) \longrightarrow (\eta\theta)\mathcal{B}$$

and is defined on 1-cells and 2-cells by

$$(\eta, X) \odot (\theta, Y) := (\eta\theta, X^\theta \boxtimes Y), \qquad f \odot g := f^\theta \boxtimes g.$$

The factors in $\odot$ are in chronological order, so the ordinary categorical composite of x followed by y is $y \circ x = x \odot y$. Its unit is $(1, \mathbf{1})$. Its associativity and unit equivalences, interchange maps, and coherence modifications are induced by those of $\mathcal{B}$ and by the displayed action comparisons. Projection is the trihomomorphism

$$\mathcal{B} \rtimes \Gamma \longrightarrow \Gamma, \qquad (\eta, X) \longmapsto \eta^{-1},$$

where Γ on the right is the one-object strict tricategory associated to the group. The arrow η^{-1} has bar label η under Definition A.3. In this paper the term *semidirect tricategory* always means this construction.

Definition A.8. The nerve of a category has composable strings of morphisms as its simplices. Street's nerve of a strict higher category makes this extension precise. If $\mathcal{O}_p$ denotes the pth *oriental*, the free strict ω-category generated by the oriented p-simplex [Rie18, pp. 65–66, Def. 3.9 and Thm. 3.14], then

$$(N_{\mathrm{St}}\mathcal{C})_p = \mathrm{Hom}_{\omega\text{-Cat}}(\mathcal{O}_p, \mathcal{C}).$$

Faces and degeneracies are induced by the corresponding maps of orientals [Str87]. In the strict case this agrees, up to the directional convention, with the Cegarra–Heredia geometric nerve [CH14, p. 2029, Def. 5.1].

For a possibly non-strict tricategory $\mathcal{T}$, the Street geometric nerve used in this paper is defined by

$$(\Delta\mathcal{T})_p = \mathrm{Rep}_u([p], \mathcal{T}).$$

Here Rep is short for *representations* of the ordinal category $[p]$ in $\mathcal{T}$. The subscript u means *unitary*: all data on identity arrows and degenerate higher simplices are the specified unit cells. Thus $\mathrm{Rep}_u([p], \mathcal{T})$ is the set of unitary representations, equivalently the normal lax functors of Definition A.9; this is the ordinal-category case of Cegarra–Heredia's unitary lax functors [CH14, pp. 2005–2006]. For a monotone map $\alpha : [q] \to [p]$, the induced simplicial operator is precomposition:

$$\alpha^* F := F \circ \alpha \in \mathrm{Rep}_u([q], \mathcal{T}).$$

Definition A.9. Cegarra–Heredia define lax functors from a category into a tricategory and their unitary, or normal, specialization in [CH14, pp. 2005–2006]. This paper uses only ordinal source categories. Let $\mathcal{T}$ be a tricategory and let $[p]$ be an ordinal category. A *unitary representation* of $[p]$ in $\mathcal{T}$, equivalently a *normal lax functor*, consists of the following data. Write $*$ for horizontal composition of 2-cells, $\cdot$ for vertical composition with the rightmost factor first, and $a_{h,g,f} : (h \circ g) \circ f \Longrightarrow h \circ (g \circ f)$ for the associator.

(i) for each vertex $0 \le i \le p$, an object X_i of $\mathcal{T}$;
(ii) for each edge $i < j$, a 1-cell $X_{ij} : X_i \to X_j$;
(iii) for each triangle $i < j < k$, a 2-cell

$$X_{ijk} : X_{jk} \circ X_{ij} \Longrightarrow X_{ik};$$

(iv) for each tetrahedron $i < j < k < \ell$, a 3-cell

$$X_{ijk\ell} : \quad X_{ij\ell} \cdot (X_{jk\ell} * 1_{X_{ij}}) \quad \Rrightarrow \quad X_{ik\ell} \cdot (1_{X_{k\ell}} * X_{ijk}) \cdot a_{X_{k\ell},X_{jk},X_{ij}}.$$

Both pastings are 2-cells

$$(X_{k\ell} \circ X_{jk}) \circ X_{ij} \Longrightarrow X_{i\ell};$$

(v) for each five-tuple $i < j < k < \ell < m$, the two pastings of the five tetrahedral 3-cells between the corresponding composites of four 1-cells, including the ambient tricategory coherence cells, are equal, as in [CH14, condition (CR1)].

The word *normal* means that the data on degenerate edges and degenerate higher simplices are the identity cells and the specified unit constraints of $\mathcal{T}$, with the unit coherence conditions [CH14, (CR2)]. For a 3-groupoid these cells are invertible.

Two special cases are used repeatedly. First, a bicategory $\mathcal{B}$ is regarded as a tricategory with identity 3-cells. Then $\mathrm{Rep}_u([p], \mathcal{B})$ consists of edge 1-cells and triangle 2-cells for which the two pasted 2-cells around every tetrahedron are equal; the five-vertex condition is then an equality between identity 3-cells.

Second, a group G is regarded as the one-object strict tricategory with only identity higher cells. A unitary representation of $[p]$ in G is then determined by the inverse labels on the consecutive arrows, so

$$\mathrm{Rep}_u([p], G) = G^p = (B_\bullet G)_p.$$

Under this identification, precomposition with monotone maps gives exactly the face and degeneracy formulas of Definition A.3.

Proposition A.10. *For every $\mathfrak{V}$-small tricategory $\mathcal{T}$, the assignment*

$$[p] \longmapsto \mathrm{Rep}_u([p], \mathcal{T}), \qquad \alpha : [q] \to [p] \longmapsto \big(F \mapsto F \circ \alpha\big),$$

is a functor $\Delta^{\mathrm{op}} \to \mathbf{Set}_{\mathfrak{V}}$ and hence a $\mathfrak{V}$-small simplicial set.

Proof. Precomposition preserves the unitary representation axioms and satisfies for $\alpha : [q] \to [p]$ and $\beta : [r] \to [q]$, $\beta^* \circ \alpha^* = (\alpha \circ \beta)^*$ as maps $\mathrm{Rep}_u([p], \mathcal{T}) \to \mathrm{Rep}_u([r], \mathcal{T})$, and $\mathrm{id}^*_{[p]} = \mathrm{id}$ on $\mathrm{Rep}_u([p], \mathcal{T})$. See [CH14, §2.2 and Def. 5.1]. □

A.2. **Horns, fibrations, and homotopy fibres.**

Definition A.11. Let $p : E \to B$ be a simplicial map.

(i) For $n \ge 1$ and $0 \le i \le n$, the *horn* $\Lambda^n_i \subseteq \Delta^n$ is the union of all faces except the ith.
(ii) A *horn-lifting problem* for p is a commutative square

$$\begin{array}{ccc} \Lambda^n_i & \longrightarrow & E \\ \downarrow & & \downarrow p \\ \Delta^n & \longrightarrow & B \end{array} \tag{86}$$

Its solution is a diagonal filler making both triangles commute; the filler extends the prescribed faces over the given base simplex.

(iii) The map p is a *Kan fibration* if every horn-lifting problem admits a solution. A simplicial set X is *Kan* when $X \to *$ is a Kan fibration.

See [GJ99, Ch. I, §3].

Definition A.12. For simplicial sets X, Y, their *mapping simplicial set* is defined degreewise by

$$\mathrm{Map}(X,Y)_q := \mathrm{Hom}_{\mathbf{sSet}}(X \times \Delta^q, Y). \tag{87}$$

Faces and degeneracies are obtained by precomposition with the coface and codegeneracy maps of Δ^q. Thus a 0-simplex is a simplicial map $X \to Y$, a 1-simplex is a simplicial homotopy $X \times \Delta^1 \to Y$, and the higher simplices are coherent higher homotopies. If Y is Kan, then $\mathrm{Map}(X,Y)$ is Kan. For every three simplicial sets A, X, Y, evaluation induces the natural bijection

$$\mathrm{Hom}_{\mathbf{sSet}}\big(A, \mathrm{Map}(X,Y)\big) \xrightarrow{\cong} \mathrm{Hom}_{\mathbf{sSet}}(X \times A, Y), \qquad F \longmapsto \widehat{F} := \mathrm{ev} \circ (\mathrm{id}_X \times F). \tag{88}$$

This is the *simplicial exponential adjunction*. If a group G acts on X and Y, let $\mathrm{Map}_G(X,Y)$ be the simplicial subset whose simplices have G-equivariant adjoints. When G acts trivially on A, restriction of (88) gives

$$\mathrm{Hom}_{\mathbf{sSet}}\big(A, \mathrm{Map}_G(X,Y)\big) \xrightarrow{\cong} \mathrm{Hom}_{G\text{-}\mathbf{sSet}}(X \times A, Y), \qquad F \longmapsto \widehat{F}. \tag{89}$$

Definition A.13. The component set $\pi_0(X)$ is the quotient of X_0 by the equivalence relation generated by the endpoints of 1-simplices. When X is Kan, horn fillers compose paths and reverse them, so two vertices are in the same component exactly when a single 1-simplex joins them.

Definition A.14. Let (X,x) be a pointed Kan complex and let $n \geq 1$. A representative of $\pi_n(X,x)$ is an n-simplex $a \in X_n$ all of whose faces are the degenerate $(n-1)$-simplex at x. Two representatives are equivalent if the corresponding maps

$$(\Delta^n, \partial\Delta^n) \longrightarrow (X,x)$$

are simplicially homotopic relative to $\partial\Delta^n$. The set of equivalence classes is $\pi_n(X,x)$. For $n = 1$ these are based edges, with homotopies fixed at both endpoints. Horn filling gives the group law, abelian for $n \geq 2$. For arbitrary X, homotopy groups mean those of a functorial Kan replacement; $\pi_0(X)$ is the component set above.

Definition A.15. We use the Kan–Quillen model structure and its simplicial slice structure on $\mathbf{sSet}/B$. Weak equivalences induce bijections of component sets and isomorphisms on every positive homotopy group at every basepoint after fibrant replacement. Cofibrations are monomorphisms and fibrations are Kan fibrations. All objects are cofibrant; the fibrant objects in the slice are Kan fibrations to B. Functorial factorization gives a trivial cofibration followed by a fibration, and the structure is right proper. The simplicial mapping axiom sends a fibration to a fibration when mapping out of a cofibrant object, and preserves weak equivalences between fibrant targets. See [GJ99, Prop. I.5.2 and Cor. II.8.6] and [Hov99, after Prop. 1.1.8]. The slice category $\mathbf{sSet}/B$ has maps $E \to B$ as objects and commutative triangles over B as morphisms. Its three classes of morphisms are detected on the underlying simplicial maps. For $u : B' \to B$, base change is the functor

$$u^* : \mathbf{sSet}/B \to \mathbf{sSet}/B', \qquad (E \to B) \longmapsto (B' \times_B E \to B').$$

It is right adjoint to composition with u. The homotopy category $\mathrm{Ho}(\mathbf{sSet})$ is obtained by formally inverting weak equivalences.

Construction A.16. Let $p : E \to B$ be a Kan fibration and let $\gamma : \Delta^1 \to B$ be an edge with vertices $b_0 = \gamma(0)$ and $b_1 = \gamma(1)$. Write

$$F_{b_j} := \{b_j\} \times_B E, \qquad E_\gamma := \Delta^1 \times_B E,$$

and let $u_j : F_{b_j} \to E_\gamma$ be the two endpoint inclusions. Right properness makes both inclusions weak equivalences, since they pull back the weak equivalences $\{j\} \to \Delta^1$ along a Kan fibration. Thus the formula

$$[\gamma_!] := [u_1]^{-1} \circ [u_0] : F_{b_0} \longrightarrow F_{b_1} \tag{90}$$

defines an isomorphism in the homotopy category $\mathrm{Ho}(\mathbf{sSet})$.

Proposition A.17. *The transport of Construction A.16 is an isomorphism in* $\mathrm{Ho}(\mathbf{sSet})$.

For a degenerate edge, $[\gamma_!]$ is the identity. If $\sigma : \Delta^2 \to B$ is a 2-simplex, then the maps induced by its three edges satisfy

$$[(d_0\sigma)_!] \circ [(d_2\sigma)_!] = [(d_1\sigma)_!]. \tag{91}$$

This is an equality of morphisms $F_{\sigma(0)} \to F_{\sigma(2)}$ in $\mathrm{Ho}(\mathbf{sSet})$. *Thus the induced map depends only on the edge-path class and is compatible with composition.*

Proof. For the composition law, pull E back along $\sigma : \Delta^2 \to B$ and call the result E_σ. If $v_j : F_{\sigma(j)} \to E_\sigma$ is the inclusion over the jth vertex, then the same right-properness argument makes every v_j a weak equivalence. Each edge inclusion $\Delta^1 \hookrightarrow \Delta^2$ is also a weak equivalence; pulling it back along $E_\sigma \to \Delta^2$ identifies the corresponding edge pullback with E_σ by a weak equivalence. Hence the maps induced by the three edges are represented in $\mathrm{Ho}(\mathbf{sSet})$ by $[v_j]^{-1} \circ [v_i] : F_{\sigma(i)} \to F_{\sigma(j)}$ for $0 \le i < j \le 2$. Hence

$$([v_2]^{-1} \circ [v_1]) \circ ([v_1]^{-1} \circ [v_0]) = [v_2]^{-1} \circ [v_0],$$

as morphisms $F_{\sigma(0)} \to F_{\sigma(2)}$ in $\mathrm{Ho}(\mathbf{sSet})$. This is exactly (91). The same calculation for a degenerate 1-simplex gives the identity. □

Definition A.18. Let $\Pi_1(B)$ be the groupoid generated by the vertices and oriented edges of B, with degenerate edges declared to be identities, edges formally inverted, and the relation

$$[d_0\sigma] \circ [d_2\sigma] = [d_1\sigma] : \sigma(0) \longrightarrow \sigma(2)$$

imposed for every 2-simplex $\sigma \in B_2$. For a Kan fibration $p : E \to B$ with fibres $F_b = \{b\} \times_B E$, Proposition A.17 defines the *functor of homotopy fibres*

$$\mathrm{Tr}_p : \Pi_1(B) \longrightarrow \mathrm{Ho}(\mathbf{sSet}), \qquad b \longmapsto F_b, \qquad [\gamma] \longmapsto [\gamma_!].$$

For $n \ge 2$, if the fibres are connected, choose a basepoint $e_b \in F_b$ in each fibre. Transport on $\pi_n(F_b, e_b)$ also requires paths comparing the transported basepoint with the chosen basepoint in the target fibre. If $\pi_1(F_b, e_b)$ acts trivially on $\pi_n(F_b, e_b)$ for every b, the transport maps are independent of these comparison paths and give a local system

$$A_p : \Pi_1(B) \longrightarrow \mathbf{Ab}, \qquad b \longmapsto \pi_n(F_b, e_b).$$

In particular this applies to the $K(A, n)$-fibres of Postnikov transitions for $n \ge 2$, without chosen basepoints or a section of the fibration. For general fibres the based homotopy groups

instead form local systems on the total space; descent to B requires the stated triviality of the fibre action.

A local Eilenberg–Mac Lane *group object* is the more specific case of pointed fibres $K(A_b, n)$ with identity-preserving fibre maps. Taking $\pi_n(K(A_b, n), 1) \cong A_b$ gives the coefficient system directly (also for $n = 1$ with A_b abelian). Its identity section is additional structure, not part of a general Postnikov fibration. Restriction to $\pi_1(B, b)$ is the *monodromy action* on A_b. For the one-vertex bar simplicial set $B_\bullet\Gamma$, the triangle $[\eta|\theta]$ gives $[\theta]_! \circ [\eta]_! = [\eta\theta]_!$ as endomorphisms of the fibre over the unique vertex of $B_\bullet\Gamma$ in $\mathrm{Ho}(\mathbf{sSet})$. Under the identification $[\eta] \mapsto \eta^{-1}$ of Definition A.3, these are the equations for a left Γ-action. Thus a left coefficient action ρ gives forward monodromy $\rho(\eta^{-1})$ along $[\eta]$, and backward coefficient map $\rho(\eta)$. Compare the construction of maps between fibres in Kerodon [Lur18], Kerodon tag 019M, and its specialization to Kan fibrations in Kerodon tag 01A2.

Definition A.19. For a map $p : E \to B$, a *fibrant replacement of p over B* is a factorization in the slice category $\mathbf{sSet}/B$

$$\begin{array}{ccc} E & \xrightarrow{\sim} & E^{\mathrm{fib}} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p^{\mathrm{fib}}} \\ B & \xrightarrow{\mathrm{id}_B} & B \end{array}$$

in which $E \to E^{\mathrm{fib}}$ is a weak equivalence and $p^{\mathrm{fib}} : E^{\mathrm{fib}} \to B$ is a Kan fibration in the sense of Definition A.11(iii); the displayed square commutes. Thus $p^{\mathrm{fib}} : E^{\mathrm{fib}} \to B$ is a fibrant object of $\mathbf{sSet}/B$. We use the functorial construction of Goerss–Jardine [GJ99, pp. 482–483, Lem. X.2.9].

Definition A.20. For $p : E \to B$ and $b \in B$, the *homotopy fibre* $\mathrm{hofib}_b(p)$ is the degreewise fibre over b of a fibrant replacement $p^{\mathrm{fib}} : E^{\mathrm{fib}} \to B$ of p in $\mathbf{sSet}/B$. Its weak-equivalence type is independent of the replacement. The *fibre homotopy groups* of p at a chosen fibre vertex e are $\pi_i(\mathrm{hofib}_b(p), e)$ for $i \geq 1$; $\pi_0(\mathrm{hofib}_b(p))$ is its component set.

If p is already a Kan fibration, its degreewise fibre computes its homotopy fibre. A pointed fibration sequence $F \to E \to B$ gives a long exact sequence

$$\begin{aligned} \cdots \to \pi_{i+1}(B) \to \pi_i(F) \to \pi_i(E) \to \pi_i(B) \to \cdots \\ \to \pi_1(F) \to \pi_1(E) \to \pi_1(B) \to \pi_0(F) \to \pi_0(E) \to \pi_0(B). \end{aligned}$$

The three π_0 terms are pointed component sets.

A.3. **Postnikov towers, truncations, and lifting obstructions.**

Definition A.21. A connected pointed Kan complex X is an *n-type* if $\pi_i(X) = \{1\}$ for $i > n$. An *nth Postnikov section* or *Postnikov truncation* of X is a map

$$e_n : X \longrightarrow P_n X = \tau_{\leq n} X$$

whose induced maps $\pi_i(X) \to \pi_i(P_n X)$ are isomorphisms for $1 \leq i \leq n$ and whose target is an n-type. A *Postnikov tower* for X consists of Postnikov sections $e_n : X \to P_n X$ and compatible maps

$$p_n : P_n X \to P_{n-1} X$$

satisfying $p_n \circ e_n = e_{n-1}$ as maps $X \to P_{n-1}X$ for every $n \geq 1$:

$$\cdots \longrightarrow P_3 X \xrightarrow{p_3} P_2 X \xrightarrow{p_2} P_1 X \xrightarrow{p_1} P_0 X = *$$

If X is an N-type, then e_N is a weak equivalence.

Here and below the displayed towers may be read in the homotopy category; after the fibrant replacements of Definition A.19 they may be represented by actual towers of Kan fibrations in the sense of Definition A.11(iii).

Theorem A.22. *Every connected pointed Kan complex X admits a simplicial Postnikov tower, natural up to weak equivalence, with the following properties:*

(i) *$p_n : P_nX \to P_{n-1}X$ may be chosen as a Kan fibration in the sense of Definition A.11*(iii)*, whose homotopy fibre in the sense of Definition A.20 is $K(\pi_n(X), n)$;*

(ii) *for $n \geq 2$, the action of $\pi_1(X)$ on $\pi_n(X)$ is the simplicial monodromy action of Definition A.18;*

(iii) *for $n \geq 2$, with its coefficient system and identification fixed, the fibration p_n is classified by a class in*

$$H^{n+1}\big(P_{n-1}X; \pi_n(X)\big),$$

where the coefficients are the local system determined by this monodromy action.

The tower is unique up to compatible weak equivalence.

Definition A.23. The class classifying p_n in Theorem A.22 is its *k-invariant*, denoted $\kappa_n(X) \in H^{n+1}(P_{n-1}X; \pi_n(X))$.

For $n \geq 1$, $K(A, n)$ is the connected Eilenberg–Mac Lane homotopy type with $\pi_n(K(A, n)) \cong A$ and all other positive homotopy groups trivial. Definition A.28 gives the simplicial model used here, including $n = 0$. The fibre assertion in Theorem A.22 follows from the long exact homotopy sequence: the homotopy fibre of p_n has $\pi_n(X)$ in degree n and trivial homotopy groups in every other degree. Existence and the twisted k-invariant description are [GJ99, Thm. VI.3.5, Prop. VI.5.1 and Cor. VI.5.2]. The latter results express each transition as a homotopy pullback of the path fibration of a local Eilenberg–Mac Lane object, with its coefficient system fixed.

The tower of a connected 3-type. If X is a connected 3-type, the entire tower is

$$X \simeq P_3X \longrightarrow P_2X \longrightarrow P_1X \simeq K(\pi_1(X), 1) \longrightarrow *.$$

The first displayed transition has fibre $K(\pi_3(X), 3)$ and is classified by a degree-4 k-invariant; the second has fibre $K(\pi_2(X), 2)$ and is classified by a degree-3 k-invariant.

Why k-invariants are lifting obstructions. For $n \geq 2$ and a map $f : Y \to P_{n-1}X$, a lift through p_n exists if and only if the pullback of the classifying k-invariant vanishes:

$$f^*\kappa_n(X) = 1 \quad \text{in} \quad H^{n+1}(Y; f^*\pi_n(X)).$$

Indeed, the twisted pullback description identifies lifts with nullhomotopies of the pulled-back k-invariant section. If one exists, comparison with a chosen nullhomotopy identifies their homotopy classes with the components of the loop section space of the local $K(f^*\pi_n(X), n+1)$-object. Local-coefficient representability [GJ99, Thms. VI.4.10–4.11] identifies these components with $H^n(Y; f^*\pi_n(X))$. Thus the lift classes form a torsor under that group, with the lower map f and its identification fixed. Allowing lower-stage homotopies requires a further orbit construction, not the same torsor assertion.

The fibrewise or Moore–Postnikov tower. For a Kan fibration $E \to B$, there is a tower of simplicial sets over B (that is, every object maps to B, and every arrow commutes with those maps),

$$E \longrightarrow \cdots \longrightarrow P^B_n E \longrightarrow P^B_{n-1} E \longrightarrow \cdots \longrightarrow B,$$

whose restriction over every $b \in B$ is the Postnikov tower of the homotopy fibre E_b. The object $P^B_n E = \tau^B_{\leq n} E$ is the *fibrewise nth Postnikov truncation*; it retains the fibre homotopy groups through degree n and the maps between them induced by paths in B, and kills the higher fibre homotopy groups. For $n \geq 2$, the transition $P^B_n E \to P^B_{n-1} E$ has Eilenberg–Mac Lane homotopy fibres in degree n. Their homotopy groups form a local system on $P^B_{n-1} E$. If this local system is pulled back from a local system A on B, write

$$K_B(A, n) \longrightarrow B$$

for the associated Eilenberg–Mac Lane group object, with its identity section. The transition is classified by a section of the pullback of $K_B(A, n+1)$ to $P^B_{n-1} E$. For $B = B_\bullet \Gamma$ with connected fibres and $n \geq 1$, comparing the long exact sequences of $P^B_n E \to B$ and $E \to B$ shows that $P^B_n E$ models the absolute Postnikov section $P_n E$, so the existence of the tower and of these classifying sections follows from Theorem A.22 applied to E.

Definition A.24. For a pointed Kan complex (Y, y_0), its based loop object is

$$\Omega_{y_0} Y = \mathrm{fib}_{(y_0, y_0)}\big(\mathrm{Map}(\Delta^1, Y) \xrightarrow{(e_0, e_1)} Y \times Y\big).$$

Its q-simplices are maps

$$\Delta^1 \times \Delta^q \to Y$$

constant at y_0 on both ends, and $\pi_i(\Omega Y) = \pi_{i+1}(Y)$ for $i \geq 0$. For a non-Kan Y, derived loops are formed after pointed Kan replacement; the strict fibre alone need not compute them.

A *connected delooping* of a pointed Kan complex X is a pointed connected Kan complex Y equipped with a pointed weak equivalence $X \simeq \Omega Y$. A homotopy-associative, unital multiplication is *group-like* when its induced monoid of components is a group.

Definition A.25. Let $\varpi \colon Y \to B$ be a Kan fibration and let $z \colon B \to Y$ be a section. The *z-based fibrewise path object* $P^z_B Y$ is the simplicial set over B whose q-simplices are pairs (b, H) consisting of a simplicial map $b \colon \Delta^q \to B$ and a simplicial map

$$H \colon \Delta^1 \times \Delta^q \to Y$$

satisfying

$$\varpi \circ H = b \circ \mathrm{pr}_2, \qquad H|_{\{0\} \times \Delta^q} = z \circ b.$$

These compare maps $\Delta^1 \times \Delta^q \to B$ and $\Delta^q \to Y$, respectively, identifying $\{0\} \times \Delta^q$ with Δ^q. Faces and degeneracies are obtained by precomposing both b and H with the corresponding operator on Δ^q. Its endpoint map is

$$e_1 \colon P^z_B Y \longrightarrow Y, \qquad (b, H) \longmapsto H|_{\{1\} \times \Delta^q}.$$

This is a map over B. The fibrewise endpoint-pair map is

$$(e_0, e_1) : \mathrm{Map}(\Delta^1, Y) \times_{\mathrm{Map}(\Delta^1, B)} B \longrightarrow Y \times_B Y,$$

where $B \to \mathrm{Map}(\Delta^1, B)$ sends a simplex to the path constant in the Δ^1 direction. This map is a Kan fibration by the simplicial mapping axiom for $\partial\Delta^1 \hookrightarrow \Delta^1$ and the fibration

ϖ. Pulling it back along $y \mapsto (z(\varpi(y)), y)$ gives the endpoint map $e_1 : P^z_B Y \to Y$, which is therefore a Kan fibration.

A.4. **Simplicial models for group cohomology.** The Eilenberg–Mac Lane coefficient object $K(A, n)$ is a *simplicial abelian group*. Cochains on $B_\bullet G$ form a *cosimplicial abelian group*, because a simplicial operator on $B_\bullet G$ acts on cochains by pullback in the opposite direction. The familiar coboundary is the alternating product of its cofaces. The section simplicial group $\mathrm{Map}_G(E_\bullet G, K(A, n))$ below is a simplicial abelian group representing this cohomology.

Definition A.26. A *simplicial group* is a simplicial set S for which every S_q is a group and every face and degeneracy map is a homomorphism. A *simplicial abelian group* is defined similarly.

Every simplicial group is Kan, equivalently fibrant as a simplicial set; see Goerss–Jardine [GJ99, Lem. I.3.4].

Definition A.27. Let A be an abelian group with a left G-action, written multiplicatively. Put

$$C^q(G, A) := \mathrm{Map}(G^q, A).$$

Define cofaces

$$\delta^i : C^q(G, A) \to C^{q+1}(G, A)$$

by

$$\begin{aligned} (\delta^0 c)(g_1, \dots, g_{q+1}) &= g_1 \cdot c(g_2, \dots, g_{q+1}), \\ (\delta^i c)(g_1, \dots, g_{q+1}) &= c(g_1, \dots, g_i g_{i+1}, \dots, g_{q+1}) \quad \text{for all } 1 \le i \le q, \\ (\delta^{q+1} c)(g_1, \dots, g_{q+1}) &= c(g_1, \dots, g_q), \end{aligned} \tag{92}$$

and codegeneracies

$$\sigma^i : C^q(G, A) \to C^{q-1}(G, A)$$

by inserting the identity:

$$(\sigma^i c)(g_1, \dots, g_{q-1}) = c(g_1, \dots, g_i, e, g_{i+1}, \dots, g_{q-1}) \qquad \text{for all } 0 \le i < q. \tag{93}$$

The group action, associativity, and unit laws give the cosimplicial identities. A cochain is normalized exactly when $\sigma^i c = 1$ for every i. For any cochain $c \in C^q(G, A)$, define its coboundary $\partial c \in C^{q+1}(G, A)$ by

$$\partial c := \prod_{i=0}^{q+1} (\delta^i c)^{(-1)^i}. \tag{94}$$

The cosimplicial identities pair the terms in $\partial(\partial c)$, so $\partial^2 c = 1$ as maps $G^{q+2} \to A$. Thus

$$Z^q(G, A) := \ker \partial, \qquad B^q(G, A) := \operatorname{im} \partial, \qquad H^q(G, A) := Z^q(G, A)/B^q(G, A).$$

Here the image defining B^q comes from degree $q - 1$, and $B^0(G, A) = \{1\}$. The normalized cochains form a subcomplex $C^\bullet_{\mathrm{norm}}(G, A) \subseteq C^\bullet(G, A)$, whose inclusion induces isomorphisms on cohomology [Bro82, Ch. I, §5]. Thus every cohomology class has a normalized cocycle representative, and a normalized coboundary in positive degree has a normalized primitive. We use normalized cocycles in the obstruction constructions, but allow arbitrary cochains in intermediate calculations. In the first two degrees the formula reads

$$(\partial a)(g) = (g \cdot a)a^{-1}, \qquad (\partial b)(g, h) = (g \cdot b(h))b(gh)^{-1}b(g).$$

Consequently $H^0(G,A) = A^G$, and the second displayed formula is (10). Formula (94) gives exactly (10)–(12).

Definition A.28. Let A be an abelian group. Define $K(A,n)_q$ to be the abelian group of normalized A-valued simplicial n-cocycles on Δ^q. Concretely, an element assigns a value

$$z(i_0,\dots,i_n) \in A$$

to every nondegenerate n-face $0 \le i_0 < \cdots < i_n \le q$, such that on every nondegenerate $(n+1)$-face

$$\prod_{r=0}^{n+1} z(i_0,\dots,\widehat{i_r},\dots,i_{n+1})^{(-1)^r} = 1. \tag{95}$$

The value on a degenerate n-face is understood to be 1. A monotone map $[p] \to [q]$ acts by pulling back the labels, giving the faces and degeneracies of $K(A,n)$.

Pullback preserves the cocycle equation and commutes with composition, so $K(A,n)$ is a simplicial abelian group and hence Kan. Its homotopy groups can be read directly from the simplices. For $n \ge 1$, there is only the identity simplex in degrees below n, while an n-simplex is freely determined by its value on the unique nondegenerate n-face of Δ^n; hence $K(A,n)_n \cong A$, and all its faces are the basepoint. If $q > n$, a q-simplex all of whose faces are the basepoint is itself the basepoint, because every n-face of Δ^q lies in a codimension-one face. Finally, (95) on Δ^{n+1} shows that two such n-simplices can be homotopic relative to their boundary only when their values agree. For $n = 0$, the construction is simply the constant simplicial abelian group A. Therefore

$$\pi_i(K(A,n)) \cong \begin{cases} A, & i = n, \\ \{1\}, & i \ne n. \end{cases} \tag{96}$$

This proves directly that the cocycle model is an Eilenberg–Mac Lane object; it is the coefficient object used below.

Definition A.29. Let $E_\bullet G$ be the homogeneous bar simplicial set of Definition A.4, equipped here with its diagonal left G-action. The quotient $G\backslash E_\bullet G$ identifies with the inhomogeneous bar construction by

$$(g_0,\dots,g_q) \longmapsto [g_0^{-1}g_1|\cdots|g_{q-1}^{-1}g_q]. \tag{97}$$

For a G-module A, define the action on $K(A,n)$ by

$$(g\cdot z)(i_0,\dots,i_n) = g\cdot z(i_0,\dots,i_n).$$

It preserves the cocycle equation and commutes with simplicial pullback. The associated local Eilenberg–Mac Lane abelian group object over $B_\bullet G$ is

$$K_{B_\bullet G}(A,n) := E_\bullet G \times_G K(A,n) \longrightarrow B_\bullet G. \tag{98}$$

Here $E_\bullet G \times_G K(A,n)$ denotes the quotient of the product by the diagonal left G-action.

Sections of (98) are equivalently G-equivariant simplicial maps

$$E_\bullet G \to K(A,n),$$

for this left-action convention.

Lemma A.30. *Let $n \geq 2$ and let $r : F \to B_\bullet G$ be a Kan fibration whose fibres are $K(A,n)$ and whose monodromy module is A* (Definition A.18)*. If r admits a section σ, there is a zigzag of weak equivalences over $B_\bullet G$ between F and $K_{B_\bullet G}(A,n)$ carrying σ to the identity section up to vertical homotopy; hence $\mathrm{Sec}(F) \simeq \mathrm{Sec}(K_{B_\bullet G}(A,n))$ as pointed Kan complexes.*

Proof. F is a connected n-type with $P_{n-1}F \simeq B_\bullet G$, so r is the transition p_n of Theorem A.22. Since σ lifts the identity, the classifying cohomology class is trivial, so $F \simeq K_{B_\bullet G}(A,n)$; fibrewise translation by the inverse of the image of σ carries that image to the identity section. □

Theorem A.31. *The simplicial set of sections of the map* (98) *is a simplicial abelian group, naturally identified with*

$$\mathrm{Map}_G(E_\bullet G, K(A,n)).$$

Its q-simplices are the equivariant simplicial maps

$$E_\bullet G \times \Delta^q \to K(A,n),$$

where G acts trivially on Δ^q. Its combinatorial homotopy groups are

$$\pi_i(\mathrm{Sec}_{B_\bullet G}(K_{B_\bullet G}(A,n))) \cong H^{n-i}(G,A) \qquad \text{for all } 0 \leq i \leq n, \tag{99}$$

and vanish for $i > n$.

Proof. A 0-simplex of the section group is an equivariant simplicial map $E_\bullet G \to K(A,n)$, hence an A-valued normalized n-cocycle on the quotient $B_\bullet G$ with its prescribed local action. Using (97), its values are functions $c(g_1,\dots,g_n)$, and the simplicial cocycle equation becomes $\partial c = 1$ with ∂ given by (92)–(94). For $n = 0$, these are precisely the invariant elements of A, and the section group is discrete. For $n \geq 1$, a simplicial path between two such sections records a normalized $(n-1)$-cochain whose coboundary is their ratio. Hence the components are $H^n(G,A)$. Iterating the loop construction of Definition A.24 lowers the cochain degree once at each step, giving (99). This is the simplicial Eilenberg–Mac Lane representability theorem; see [GJ99, Thms. VI.4.10–4.11] and [Bro82, Ch. III]. □

A.5. **Section spaces and homotopy fibres.**

Definition A.32. Let $p : E \to B$ be a simplicial map.

(i) Its *simplicial set of sections* is

$$\mathrm{Sec}_B(E)_q = \mathrm{Hom}_{\mathbf{sSet}/B}(B \times \Delta^q, E). \tag{100}$$

Operators precompose the simplex coordinate; maps over B act by postcomposition.

(ii) Let $p' : E' \to B$ be another simplicial map and let $q : E \to E'$ satisfy $p' \circ q = p$ as maps $E \to B$. A *lift* of a section $s : B \to E'$ through q is a section $\widetilde{s} : B \to E$ of p with $q \circ \widetilde{s} = s$ as maps $B \to E'$. For fibrant section spaces, a *rigidified homotopy lift* is a section $\widetilde{s}$ together with a specified vertical homotopy from $q \circ \widetilde{s}$ to s; such pairs model the homotopy fibre over s. Write

$$s^*E := B \times_{E'} E \longrightarrow B$$

for the pullback along s. If q is classified over B by $\kappa : E' \to K_B(A, n+1)$, the *obstruction class* of s is $o(s) := [\kappa \circ s] \in \pi_0 \mathrm{Sec}_B(K_B(A,n+1))$; for $B = B_\bullet G$ this group is $H^{n+1}(G,A)$ by Theorem A.31.

(iii) For sections s, t of p, a *vertical homotopy* from s to t is a map $H : B \times \Delta^1 \to E$ satisfying

$$p \circ H = \mathrm{pr}_B, \qquad H|_{B\times\{0\}} = s, \qquad H|_{B\times\{1\}} = t. \tag{101}$$

The first equality is between maps $B \times \Delta^1 \to B$; the endpoint equalities are between maps $B \to E$. Thus $\pi_0 \operatorname{Sec}_B(E)$ consists of sections modulo zigzags of vertical homotopies. If p is a Kan fibration, $\operatorname{Sec}_B(E)$ is Kan and a single vertical homotopy suffices.

(iv) For arbitrary p, define the *derived section space*

$$\operatorname{RSec}_B(E) := \operatorname{Sec}_B(E^{\mathrm{fib}})$$

using a fibrant replacement in the slice as in Definition A.19. Proposition A.34 proves independence of that choice.

Definition A.33. Let $p : E \to B$ be a *Kan fibration* with a chosen section s_0. The based loop object of its Kan section space is that of Definition A.24. By adjunction,

$$(\Omega_{s_0} \operatorname{Sec}_B(E))_q = \{\, H : B \times \Delta^1 \times \Delta^q \to E \text{ over } B \mid H|_{B\times\{i\}\times\Delta^q} = s_0 \circ \mathrm{pr}_B \text{ for } i = 0, 1 \,\}. \tag{102}$$

Each endpoint equality is between maps $B \times \Delta^q \to E$. It follows that

$$\pi_0(\Omega^n_{s_0} \operatorname{Sec}_B(E)) \cong \pi_n(\operatorname{Sec}_B(E), s_0), \qquad n \geq 1.$$

For a general map p with section s_0, first replace p in the slice, carry s_0 to s_0^{fib}, and define the derived loop-section object as $\Omega^n_{s_0^{\mathrm{fib}}} \operatorname{RSec}_B(E)$.

Proposition A.34. *Let $p : E \to B$ and $p' : E' \to B$ be Kan fibrations and let $f : E \to E'$ be a weak equivalence over B. For every simplicial map $u : B' \to B$, base change induces a weak equivalence between Kan fibrations*

$$u^* f : B' \times_B E \longrightarrow B' \times_B E'.$$

Moreover, postcomposition by f induces a weak equivalence of Kan simplicial sets

$$\operatorname{Sec}_B(f) : \operatorname{Sec}_B(E) \longrightarrow \operatorname{Sec}_B(E').$$

Consequently two fibrant replacements of the same map over B have weakly equivalent simplicial sets of sections.

Proof. In the slice category of Definition A.15, p and p' are fibrant. Composition with u does not alter an underlying morphism, so it preserves cofibrations and trivial cofibrations. Hence u^* is a right Quillen functor and preserves weak equivalences between fibrant objects. This proves the base-change assertion.

The object $\mathrm{id}_B : B \to B$ is cofibrant in $\mathbf{sSet}/B$ because every simplicial set is cofibrant. Its simplicial mapping object into $p : E \to B$ is precisely $\operatorname{Sec}_B(E)$. The simplicial model-category mapping axiom therefore says that mapping out of id_B carries the weak equivalence f between fibrant objects to the displayed weak equivalence of simplicial sets of sections. Any two fibrant replacements are connected by a zigzag of weak equivalences over B between fibrant objects; applying the section calculation to that zigzag proves the last assertion. □

Proposition A.35. *Let B be a simplicial set, and let $q : E \to E'$ be a Kan fibration in $\mathbf{sSet}/B$ between objects that are fibrant over B. Then postcomposition by q is a Kan fibration*

$$\mathrm{Sec}_B(q) : \mathrm{Sec}_B(E) \longrightarrow \mathrm{Sec}_B(E').$$

For every section $s : B \to E'$, the strict fibre of $\mathrm{Sec}_B(q)$ is $\mathrm{Sec}_B(s^*E)$, and therefore there is a homotopy fibre sequence

$$\mathrm{Sec}_B(s^*E) \longrightarrow \mathrm{Sec}_B(E) \longrightarrow \mathrm{Sec}_B(E').$$

If s admits a lift, a chosen lift supplies the basepoints and gives the long exact sequence of homotopy groups and pointed component sets. The same conclusion holds for a homotopy fibre sequence over B after the fibrant replacements of Definition A.19.

Proof. In the simplicial model category $\mathbf{sSet}/B$, the object $\mathrm{id}_B : B \to B$ is cofibrant. The simplicial mapping object from id_B to $E \to B$ is $\mathrm{Sec}_B(E)$. The simplicial mapping axiom therefore sends the fibration q between fibrant objects to the displayed Kan fibration; this is the same mapping-space argument used in Proposition A.34.

Degreewise, a j-simplex in the fibre over s is a map $H : B \times \Delta^j \to E$ over B satisfying

$$q \circ H = s \circ \mathrm{pr}_B : B \times \Delta^j \longrightarrow E'.$$

By the defining universal property of the pullback, these are exactly the j-simplices of $\mathrm{Sec}_B(s^*E)$, compatibly with every face and degeneracy. Thus the asserted identification of the strict fibre is an isomorphism of simplicial sets. Since the section map is a Kan fibration, this strict fibre computes its homotopy fibre and gives the usual long exact sequence. For an initially given homotopy fibre sequence, apply Definition A.19; independence of the chosen replacements follows from Proposition A.34. □

Proposition A.36. *Let $B = B_\bullet G$ and let $q : E \to E'$ be a Kan fibration over B. Fix a map $\kappa : E' \to K_B(A, n+1)$ over B, and suppose that, over E', the fibration q is equivalent to $E' \times_{K_B(A,n+1)} P^z_B K_B(A, n+1)$, where z is the identity section and P^z_B is the fibrewise path object of Definition A.25 and the pullback is along κ. For a section s of $E' \to B$, the obstruction class $o(s)$ of Definition A.32(ii) depends only on the vertical homotopy class of s. The section s lifts through q if and only if $o(s) = 1$. Consequently,*

$$\mathrm{im}\big(\pi_0 \mathrm{Sec}_B(E) \to \pi_0 \mathrm{Sec}_B(E')\big) = o^{-1}(1).$$

If E and E' are fibrant over B and a section $s_0 : B \to E$ is chosen, base the two section spaces at s_0 and $q \circ s_0$. The long exact sequence of Proposition A.35 then extends by

$$\pi_0 \mathrm{Sec}_B(E) \longrightarrow \pi_0 \mathrm{Sec}_B(E') \xrightarrow{o} \pi_0 \mathrm{Sec}_B(K_B(A, n+1)),$$

exact as pointed sets. The same statement holds after the fibrant replacements specified there.

Proof. A vertical nullhomotopy of $\kappa \circ s$ is a section of the pullback of the endpoint evaluation along $\kappa \circ s$; by Proposition A.34 and homotopy lifting along the Kan fibration q, such a section yields a lift of s, and a lift yields such a nullhomotopy. Vertical homotopies of s carry nullhomotopies to nullhomotopies, which gives the invariance, and the displayed exactness at $\pi_0 \mathrm{Sec}_B(E')$ is the equivalence just proved. □

A.6. **Borel constructions and homotopy fixed points.**

Definition A.37. Suppose a group G acts on a Kan simplicial set X. A standard simplicial fibration model for the Borel construction is

$$p_X : X_{hG} := E_\bullet G \times_G X \longrightarrow B_\bullet G, \qquad [e, x] \longmapsto [e].$$

Here $E_\bullet G$ is the homogeneous bar simplicial set of Definition A.4, with its right G-action, and $E_\bullet G \times_G X$ is the degreewise quotient of $E_\bullet G \times X$ by

$$(e \cdot g, x) \sim (e, gx).$$

If X is Kan, p_X is a Kan fibration, since its pullback along every simplex $\Delta^n \to B_\bullet G$ is isomorphic to $\Delta^n \times X \to \Delta^n$. For a homomorphism $\phi : G \to \Gamma$ and a Γ-simplicial set X, the map $[e, x] \mapsto ([e], [E_\bullet \phi(e), x])$ is a natural isomorphism $E_\bullet G \times_G \phi^* X \cong B_\bullet G \times_{B_\bullet \Gamma} (E_\bullet \Gamma \times_\Gamma X)$ over $B_\bullet G$. Its simplicial set of sections is equivalent to the homotopy fixed point simplicial set

$$X^{hG} := \mathrm{Map}_G(E_\bullet G, X).$$

With the quotient convention above, $\mathrm{Map}_G(E_\bullet G, X)$ means the simplicial set of maps

$$f : E_\bullet G \to X$$

satisfying $f(e \cdot g) = g^{-1} f(e)$.

Proposition A.38. *Let G be a group.*

(i) *If $f : X \to Y$ is a G-equivariant weak equivalence between Kan G-simplicial sets, then*

$$f^{hG} : X^{hG} \longrightarrow Y^{hG}$$

is a weak equivalence.

(ii) *For a G-set A, regarded as the constant simplicial set A_{disc}, evaluation at a vertex of $E_\bullet G$ gives a simplicial isomorphism*

$$(A_{\mathrm{disc}})^{hG} \cong (A^G)_{\mathrm{disc}}.$$

(iii) *If the basepoint x_0 of X is fixed by G, then at the constant homotopy fixed point $c_{x_0} : E_\bullet G \to X$ there is a natural simplicial isomorphism*

$$\Omega_{c_{x_0}}(X^{hG}) \cong (\Omega_{x_0} X)^{hG}.$$

Iterating gives $\Omega^r(X^{hG}) \cong (\Omega^r X)^{hG}$ for all $r \geq 1$.

Proof. Use the projective model structure on G-simplicial sets: weak equivalences and fibrations are detected on the underlying simplicial sets, and its generating cofibrations are $G \times \partial\Delta^n \to G \times \Delta^n$. The skeletal filtration of the free G-simplicial set $E_\bullet G$ attaches free orbits of nondegenerate simplices by these maps, so $E_\bullet G$ is cofibrant. Therefore the mapping functor $\mathrm{Map}_G(E_\bullet G, -)$ carries weak equivalences between Kan objects to weak equivalences. This proves (i); equivalently, it is the homotopy-limit invariance of homotopy fixed points [BK72, Chs. XI–XII].

For (ii), a q-simplex of the left-hand side is an equivariant simplicial map

$$E_\bullet G \times \Delta^q \longrightarrow A_{\mathrm{disc}}.$$

The source is connected and the target is discrete, so the map is constant. Its value lies in A^G, and every element of A^G gives such a constant map. This identification commutes with all faces and degeneracies.

For (iii), a q-simplex on the left is a map

$$\Delta^1 \times \Delta^q \longrightarrow \mathrm{Map}_G(E_\bullet G, X)$$

whose two endpoint restrictions are c_{x_0}. The simplicial adjunction identifies it with a G-equivariant map

$$E_\bullet G \times \Delta^1 \times \Delta^q \longrightarrow X$$

whose two Δ^1-endpoint restrictions are constant at x_0. This is exactly a q-simplex of $(\Omega_{x_0} X)^{hG}$. The identification commutes with the simplicial operators, and iteration proves the last assertion. □

Entrywise inversion on $E_\bullet G$ identifies this right-action Borel model with the diagonal left-action model of Definition A.29. Explicitly, $[e, x] \mapsto [e^{-1}, x]$ respects the quotient relations. In the right-action model the edge $(g, 1)$ with constant coordinate y joins $[g, y] = [1, gy]$ to $[1, y]$. Thus the forward fibre map along $[g]$ is $x \mapsto g^{-1}x$, while the initial-vertex convention for group cochains uses the backward map $a \mapsto ga$, as in Definition A.18.

A.7. **Prestacks and homotopy quotients.** Section 8 uses the homotopy quotient of a coherent diagram of spaces, defined below in Definition A.43. We first recall the related category-valued constructions.

We use *prestack* for a pseudofunctor on an indexing category, in the indexed-category sense of [Vis05, pp. 44–48]. The homotopy quotient retains the values of this pseudofunctor together with the maps and coherence isomorphisms attached to the indexing arrows.

Definition A.39. Let $\mathcal{I}$ be a $\mathfrak{V}$-small indexing category. A *prestack of categories* on $\mathcal{I}$ is a pseudofunctor

$$\mathcal{F} : \mathcal{I}^{\mathrm{op}} \longrightarrow \mathbf{Cat}.$$

Here **Cat** is the 2-category of $\mathfrak{V}$-small categories, formed in $\mathfrak{W}$. For $f : V \to U$ it has restriction functors

$$f^* : \mathcal{F}(U) \to \mathcal{F}(V)$$

and coherent comparisons

$$c_{f,g} : g^* \circ f^* \xRightarrow{\sim} (f \circ g)^*, \qquad u_U : \mathrm{id}_U^* \xRightarrow{\sim} \mathrm{id}_{\mathcal{F}(U)}.$$

These isomorphisms are part of the data. Explicitly, if

$$T \xrightarrow{h} W \xrightarrow{g} V \xrightarrow{f} U$$

and $x \in \mathcal{F}(U)$, associativity requires

$$c_{f\circ g,h}(x) \circ h^*\big(c_{f,g}(x)\big) = c_{f,g\circ h}(x) \circ c_{g,h}(f^*x). \tag{103}$$

Both sides are morphisms $(h^* \circ g^* \circ f^*)(x) \to (f \circ g \circ h)^*(x)$ in $\mathcal{F}(T)$. For every $f : V \to U$ and $x \in \mathcal{F}(U)$, the two unit equations are

$$c_{f,\mathrm{id}_V}(x) = u_V(f^*x), \qquad c_{\mathrm{id}_U,f}(x) = f^*(u_U(x)). \tag{104}$$

These compare morphisms $\mathrm{id}_V^*(f^*x) \to f^*x$ and $f^*(\mathrm{id}_U^*\, x) \to f^*x$, respectively, in $\mathcal{F}(V)$. Identity comparisons give a strict functor. Groupoid-valued prestacks take values in **Grpd**. For prestacks of 2-groups the arrows are strong monoidal functors and the comparisons are monoidal natural isomorphisms.

Definition A.40. For a pseudofunctor $\mathcal{F} : \mathcal{I}^{\mathrm{op}} \to \mathbf{Cat}$, its *Grothendieck construction* is the $\mathfrak{V}$-small category whose objects are pairs (U, x) with $U \in \mathcal{I}$ and $x \in \mathcal{F}(U)$. A morphism

$$(V, y) \longrightarrow (U, x)$$

is a pair (f, α) where $f : V \to U$ is a morphism in $\mathcal{I}$ and

$$\alpha : y \longrightarrow f^* x$$

is a morphism in $\mathcal{F}(V)$. Composition uses the pseudofunctorial isomorphism $c_{f,g} : g^* \circ f^* \cong (f \circ g)^*$.

Projection $(U, x) \mapsto U$ induces a nerve map to $N_\bullet \mathcal{I}$. If both the index and the values are groupoids, the total category is a groupoid and its nerve is Kan.

Definition A.41. The *objectwise nerve* of a prestack has values $N_\bullet \mathcal{F}(U)$, maps $N_\bullet(f^*)$, and the coherent homotopies induced by its comparison isomorphisms. For 2-group values, instead use their one-object classifying nerves $B_\bullet \mathcal{F}(U)$, normalizing restriction functors at the unit according to Definition A.9. The equations (103)–(104) supply the coherence, not merely the objectwise list of nerves.

Definition A.42. Let $\mathcal{M}$ be a $\mathfrak{V}$-small groupoid and $X : \mathcal{M} \to \mathbf{Set}_{\mathfrak{V}}$ a functor, with transition bijections

$$P_* : X_R \to X_S.$$

The *action groupoid* $X//\mathcal{M}$ has:

- objects (R, x) with $R \in \mathcal{M}$ and $x \in X_R$;
- a morphism $(R, x) \to (S, y)$ given by an arrow $P : R \to S$ in $\mathcal{M}$ such that $P_* x = y$.

Composition is composition in $\mathcal{M}$. Thus isomorphism classes in $X//\mathcal{M}$ are the orbits of the family X_R under the action of $\mathcal{M}$, and automorphism groups remember stabilizers.

For a groupoid-valued pseudofunctor X, an arrow also includes an isomorphism $P_* x \cong y$ in the target groupoid, with composition using the pseudofunctor comparisons. This is the Grothendieck construction, with the variance adjusted using inverses in $\mathcal{M}$. For category-valued diagrams, this isomorphism-only version uses their maximal subgroupoids; the full construction allows arbitrary comparison morphisms.

Cubical coherence and extension. Let $\mathcal{I}$ be a $\mathfrak{V}$-small category and $\mathcal{D}$ a $\mathfrak{W}$-small simplicial category enriched in $\mathfrak{V}$-small Kan complexes. Put $J = \Delta^1$, and let ∂J^r be the union of the coordinate faces of the simplicial cube J^r. A homotopy-coherent diagram is a simplicial functor $W\mathcal{I} \to \mathcal{D}$, where $W\mathcal{I}$ is the simplicial free resolution. In its cubical description, a string of m composable nonidentity arrows

$$i_0 \xrightarrow{f_1} i_1 \longrightarrow \cdots \xrightarrow{f_m} i_m$$

has a comparison

$$F_{f_m,\ldots,f_1} : J^{m-1} \longrightarrow \mathcal{D}(F(i_0), F(i_m)).$$

Its coordinate faces are prescribed by composing adjacent indexing arrows or by composing the comparisons for the two shorter strings. Strings containing identities satisfy the unit rules. Thus arrows, pairs, triples, and quadruples supply points, intervals, squares, and 3-cubes, respectively; see [CP86, §1, pp. 70–71].

Suppose every mapping complex of $\mathcal{D}$ is a 2-type. Compatible data through strings of length 3 extend to a full coherent diagram if every prescribed length-4 boundary is

nullhomotopic. In that case the space of extensions relative to these low-dimensional data is contractible. Indeed, for a Kan mapping complex H, restriction is a Kan fibration

$$\mathrm{Map}(J^r, H) \longrightarrow \mathrm{Map}(\partial J^r, H).$$

For $r \geq 2$, a prescribed boundary extends precisely when its class in $\pi_{r-1}(H)$ vanishes; after choosing an extension, its filler space is weakly equivalent to $\Omega^r H$, based at the image of a vertex. At length 4 one has $r = 3$: the obstruction can lie in $\pi_2(H)$, but a nonempty filler space is already contractible. At length $m \geq 5$ one has $r = m - 1$, so the obstruction group $\pi_{m-2}(H)$ vanishes and the filler space is contractible. Induction over the strings gives the assertion, respecting all previously prescribed faces and unit rules. The same argument is relative to fixed coherent data on a subcategory; in particular it applies to coherent comparisons indexed by $\mathcal{I} \times [1]$, with the two endpoint diagrams fixed.

Definition A.43. Let $\mathcal{M}$ be a $\mathfrak{V}$-small groupoid and X a homotopy-coherent diagram of $\mathfrak{V}$-small Kan complexes indexed by $\mathcal{M}$. We first rectify the diagram, including its specified higher comparisons, to a strict diagram

$$X^{\mathrm{str}} : \mathcal{M} \to \mathbf{sSet}$$

with a coherent comparison to X through objectwise weak equivalences. Existence and invariance under this replacement follow from rectification [CP86, Cor. 4.5, Thm. 4.7, and p. 90].

One construction uses a simplicial free resolution $W\mathcal{M} \to \mathcal{M}$, with the same objects and contractible spaces of resolutions of each morphism. Regard X as a simplicial functor on $W\mathcal{M}$. Its derived enriched left Kan extension is computed by the two-sided bar construction

$$X^{\mathrm{str}}(R) = B\big(\mathcal{M}(-, R), W\mathcal{M}, X\big).$$

In bar degree p its terms are products of $X(R_0)$, p composable mapping spaces of $W\mathcal{M}$, and $\mathcal{M}(R_p, R)$; take the diagonal of this simplicial object in simplicial sets. Postcomposition in the last factor gives a strict $\mathcal{M}$-action. Use an objectwise functorial Kan replacement if necessary. More precisely, the bar augmentation and the resolution supply a coherent zigzag of objectwise weak equivalences

$$X(R) \xleftarrow{\simeq} B\big(W\mathcal{M}(-, R), W\mathcal{M}, X\big) \xrightarrow{\simeq} B\big(\mathcal{M}(-, R), W\mathcal{M}, X\big).$$

Define

$$\left(\coprod_R X_R\right)_{h\mathcal{M}} := \operatorname{diag} B_\bullet(*, \mathcal{M}, X^{\mathrm{str}}).$$

For a strict diagram one may use X itself. Its degreewise action groupoids give the equivalent formula below. For a coherent diagram, use the rectification X^{str}. For a strict action of a group G on X, the result is weakly equivalent to $E_\bullet G \times_G X$.

In the strict case, equivalently, this homotopy quotient is the diagonal of the bisimplicial set

$$B_{p,q} := \coprod_{R_0 \xrightarrow{P_1} R_1 \xrightarrow{P_2} \cdots \xrightarrow{P_p} R_p} (X_{R_0})_q. \tag{105}$$

Thus an n-simplex is a composable chain

$$R_0 \xrightarrow{P_1} R_1 \xrightarrow{P_2} \cdots \xrightarrow{P_n} R_n$$

in $\mathcal{M}$ together with an n-simplex $x \in (X_{R_0})_n$. The diagonal face d_0 sends this to the chain beginning at R_1 and the simplex $(P_1)_*(d_0 x) \in (X_{R_1})_{n-1}$; for $0 < i < n$, d_i applies d_i to x and composes the two adjacent arrows $P_{i+1} \circ P_i$; and d_n applies d_n to x and drops the last arrow. Degeneracies apply the corresponding degeneracy to x and insert the identity arrow at the corresponding object of the chain.

A.8. **Spectra and the homotopy fixed point spectral sequence.** A connective spectrum E may be represented by a sequence of pointed Kan simplicial sets E_n with structure maps

$$E_n \longrightarrow \Omega E_{n+1}$$

that are weak equivalences. Its homotopy groups $\pi_t(E)$ vanish for $t < 0$, and its infinite loop space has homotopy groups $\pi_t(E)$ for $t \geq 0$.

Write **Sp** for the ∞-category of spectra formed from $\mathfrak{V}$-small pointed simplicial sets. A *G-spectrum* here means an ordinary spectrum with a homotopy-coherent G-action, that is, a functor

$$E : B_\bullet G \longrightarrow \mathbf{Sp}.$$

It is not a genuine equivariant spectrum. The functor assigns the action of g to the categorical arrow g; our chronological bar label $[g]$ denotes g^{-1}, as in Notation 2.10. Its *homotopy fixed point spectrum* is the limit in **Sp**,

$$E^{hG} := \lim_{B_\bullet G} E.$$

The equivariant Postnikov tower is obtained by applying the ordinary Postnikov truncations objectwise to this diagram. Thus its stages and layers retain the full coherent G-action.

Suppose a G-spectrum E has homotopy groups only in a finite interval $a \leq t \leq b$. Its equivariant Postnikov filtration has layers $\Sigma^t H(\pi_t E)$. Homotopy fixed points preserve fibre sequences, and

$$\pi_n\big((\Sigma^t H(\pi_t E))^{hG}\big) = H^{t-n}(G, \pi_t E),$$

with negative-degree cohomology interpreted as the trivial group. The exact couple of this finite filtration, reindexed by $s = t - n$, gives

$$E_2^{s,t} = H^s(G, \pi_t E) \Longrightarrow \pi_{t-s}(E^{hG}), \qquad d_r : E_r^{s,t} \longrightarrow E_r^{s+r,t+r-1}.$$

This is the Postnikov form of the homotopy fixed point spectral sequence [BK72, Chs. XI–XII]. It converges strongly: every abutment group has a finite exhaustive separated filtration whose associated graded groups are $E_\infty^{s,s+n}$. There is no infinite filtration or derived-limit ambiguity here, even when G has unbounded cohomological dimension.

For the two-row case $\pi_0 E = P$, $\pi_1 E = A$, this follows directly from the equivariant fibre sequence

$$\Sigma HA \longrightarrow E \longrightarrow HP \xrightarrow{\kappa_E} \Sigma^2 HA.$$

Comparison with group cohomology. After homotopy fixed points, the long exact sequence contains the exact segment

$$H^0(G, P) \xrightarrow{d_2^{0,0}} H^2(G, A) \longrightarrow \pi_{-1}(E^{hG}) \longrightarrow H^1(G, P) \xrightarrow{d_2^{1,0}} H^3(G, A).$$

Both labelled arrows are differentials in the homotopy fixed point spectral sequence. Only d_2 can be nontrivial and $E_3 = E_\infty$. The same finite-filtration argument applies to three-row spectra; the additional row permits d_3. The abutment can have negative homotopy groups even though E is connective.

Definition A.44. Let E be a G-spectrum with $\pi_i(E) = \{1\}$ for $i \notin \{0, 1\}$. We use the sign convention for the differential

$$d_2 : H^1(G, \pi_0(E)) \longrightarrow H^3(G, \pi_1(E))$$

in which a normalized cocycle $\xi \in Z^1(G, \pi_0(E))$ represents a section of the first stage $K_{B_\bullet G}(\pi_0(E), 1) \to B_\bullet G$ of the Borel construction for $\Omega^\infty \Sigma E$, and one has

$$d_2([\xi]) = [t_\xi],$$

where $t_\xi \in Z^3(G, \pi_1(E))$ is the normalized cochain representing the pullback of the suspended first G-equivariant Postnikov class of E along that section, valued in the local $K_{B_\bullet G}(\pi_1(E), 3)$-object. For triangle comparisons in the Picard nerve, the boundary discrepancy is oriented as in (29): the first pasting composed with the inverse of the second, an automorphism of their common long-edge object. Reversing this orientation would invert both t_ξ and $d_2([\xi])$. Normalization and the coboundary are those of Definition A.27 and (94). Thus d_2 is the obstruction class to lifting the first-stage section through the next Postnikov fibration in the sense of Definition A.32(ii).

Appendix B. Size conventions

Here we collect the details behind the standing conventions of Section 1.4. Ambient homotopy and coherent-diagram categories are formed in $\mathfrak{W}$. The free resolutions, bar constructions, and functorial Kan replacements of Subsection A.7 use $\mathfrak{V}$-small coproducts, finite products, and countably many simplicial degrees, so their individual values remain $\mathfrak{V}$-small. The same convention applies to the direct tricategorical nerves: their simplices are sets of unitary lax functors [CH14, pp. 2005–2006 and p. 2029, Def. 5.1]. The optional Gray-category comparison of Remark 9.10 takes place in $\mathfrak{W}$.

The classes in Example 8.2(ii)–(v) admit $\mathfrak{U}$-small representative families. Azumaya algebras are finite projective K-modules, whose algebra structures can be encoded on images of finite idempotent matrices. Algebras Morita equivalent to a fixed algebra A have representatives $\mathrm{End}_A(P)$ for finitely generated projective right A-module generators P. This also gives small representative families for the Morita closures of tensor powers, including the quantum-torus classes. Invertible bimodules between fixed algebras are similarly encoded by finite idempotent matrices and action maps; between algebras of cardinality at most an infinite κ, they have cardinality at most κ.

A full subgroupoid meeting every isomorphism class of objects in $\mathbf{Mor}^{\mathcal{C}}$ is an equivalent indexing groupoid and computes the same homotopy quotient. Recoding its objects and arrows by the preceding representatives gives $\mathfrak{U}$-small models whenever the algebra class admits a $\mathfrak{U}$-small representative family. The choices of representatives and comparisons in Lemma 8.13 are indexed by $\mathfrak{V}$-small sets, so set-indexed choice suffices. For the classes used here, enlargement of universes gives equivalent indexing groupoids and weakly equivalent mapping, section, and homotopy-quotient models, hence the same component sets. The algebra class, specified by its representative family or its cardinal bound κ, is held fixed in this comparison.

For the field comparison, the affine basis of the small étale site admits a $\mathfrak{U}$-small skeleton: étale algebras are finitely presented. This basis computes the same topos [The26b, Def. 34.4.8 and Lem. 34.4.12]. The splitting-gerbe groupoids have $\mathfrak{U}$-small morphism sets and equivalent $\mathfrak{U}$-small models by the same finite-projective module description. Hypercover index sets and levels may be taken in $\mathfrak{U}$, including their finitely many G-translates; countably many levels remain small since $\mathbb{N} \in \mathfrak{U}$. Separable closures, absolute Galois groups, and continuous cochain sets may likewise be taken in $\mathfrak{U}$.

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